Adaptive Threshold Estimation

Minimax laws, unbiased estimation, and rare events

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Abstract

We estimate the fraction $\theta = \mathbb{P}(P > 1/2)$ of a population of latent Bernoulli coins, charging one unit for each observed flip. Algorithms may create, revisit, and interleave coins and stop under a uniform expected budget. For any law μ of $D = |P - 1/2|$ on $(0, 1/2]$, the minimax RMS error is comparable to $\omega_\mu(B^{-1/2})$, where $\omega_\mu(h) = \mathbb{E}_\mu \min\{1, h/D\}$. One algorithm depending only on the budget attains this rate for every μ. Unbiased estimation with uniformly finite variance and expected work is possible exactly when $I_\mu = \mathbb{E}_\mu D^{-1} < \infty$; the optimal uniform variance–work product lies between $I_\mu^2/16$ and $713 I_\mu^2$.

With the additional global rarity constraint $\theta \le \rho$, the variational quantity $\Psi_{\mu,\rho}$ gives a general lower bound and characterizes the power-margin regimes below and above exponent two, but fails at the boundary. For the fixed quadratic law $F(d) = (4d)^2$, put $L = 1 + \log(1/\rho)$. The sharp RMS order is $\min\{\rho, \sqrt{\rho L(1 + \log L)/B}\}$, whereas Ψ gives only $\min\{\rho, \sqrt{\rho L/B}\}$. For every known magnitude law, rarity bound and positive budget, the risk is comparable to an entropy–tail selection functional: a selected part of the law pays its dyadic inverse-square cost entropy, and the remainder pays a truncated inverse-gap cost. Selecting all mass outside the smallest-gap quantile ρ is optimal within a factor $1 + 2\sqrt{2}$, which gives a closed formula. For finite laws that formula is also proved directly, with explicit constants. A robust stopping program has an exact four-prior dual, a factor-sixteen comparison with minimax risk, and convergent finite certificates.

The original RMS law is Lean-checked in both directions, with coefficients $1/64$ and 512 and one distribution-blind upper algorithm. The critical minimax law, exact Ψ comparison and counterexample, variational optimizer, unbiased characterization with coefficients $1/16$ and 713, confidence characterization, magnitude-oracle lower bounds, and the rare-event rate above exponent two are also Lean-checked in their stated models. The observable entropy–tail and quantile upper bounds, the deterministic quantile comparison, stopping-program duality, risk certificates, identification and genie limits are also Lean-checked. The matching entropy–tail lower bound, hence the two-sided general quantile characterization, and the remaining rare-event regimes retain their paper-proof status. This manuscript consolidates the statements, proofs, dependencies, and verification boundaries.

Research draft. Version 6 combines the consolidated version 5 with the general-law rare-event manuscript (revised v2) and the finite-gap rare-event manuscript (revision 2). No independent specialist review; agent review and formal verification have different scopes. Priority and practical superiority are not established.

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1 Introduction and contribution

Estimating the fraction of a population lying above a latent threshold is not the same task as classifying every member of that population. Observations can be concentrated on promising or ambiguous members, and biased individual classifications can be combined through telescoping estimators. These possibilities make algorithm-independent lower bounds important.

We study the following basic observation model. A fresh coin has a latent probability P drawn from a population distribution G. Conditional on P, its observations are independent Bernoulli trials. An algorithm may adapt to its entire history: it can create new coins, interleave observations of old coins, and stop at a random time. Its target is

$$\theta(G) = \mathbb{P}_G(P > 1/2). \tag{1}$$

One coin flip is one unit of work. The principal obstruction is the population's mass near $1/2$, but a single power exponent need not describe that mass accurately.

Distributional and distribution-blind optimality. The central object is the entire function

$$\omega_\mu(h) = \int \min\{1, h/d\}\, \mu(\mathrm{d}d), \qquad \mu = \mathcal{L}(|P - 1/2|). \tag{2}$$

The main theorem identifies this function as the minimax error law. Thus irregular, discrete, and logarithmically modified margin distributions are covered by one statement, rather than by separate exponent calculations. The same algorithm attains the law for every magnitude distribution, rather than requiring a distribution-dependent allocation. A second theorem characterizes unbiased feasibility by an inverse moment and proves a sharp universal lower constant for the variance–work product. Writing $F_\mu(h) = \mathbb{P}(D \le h)$ makes the two parts of the modulus explicit:

$$\omega_\mu(h) = F_\mu(h) + h\mathbb{E}_\mu[D^{-1}; D > h].$$

The first term represents unresolved near-threshold mass; the second records the accumulated contribution of the remaining difficulty scales.

The lower bound makes many difficulty scales jointly hard: each scale contributes to the target separation, while every next observation has uniformly bounded information. The upper bound is a fixed geometric randomization of capped sequential classifiers. Its conditional second moment and expected work are both controlled by the same truncated inverse margin. This common conditional bound is what permits arbitrary margin laws. An exact two-success inverse-binomial calibration then chooses the number of production samples without estimating either their mean cost or variance. The underlying inverse-sampling identity is classical [8]; its role here is to eliminate all margin knowledge from the budget-indexed rule.

1.1 Relationship to established work

Nested simulation. Uniform inner-sampling analyses include Lee–Glynn [10] and Gordy–Juneja [11]; Broadie, Du, and Moallemi [9] also identify Lee's 1998 thesis as an early analysis and introduce sequential inner allocation for loss-probability estimation. Giles and Haji-Ali [2] combine adaptive inner sampling and multilevel Monte Carlo (MLMC) to attain $O(\varepsilon^{-2}|\log\varepsilon|^2)$ work for thresholded nested expectations. Haji-Ali, Spence, and Teckentrup [12] develop a broader adaptive MLMC framework for probabilities. We do not claim these methods or the critical upper rate as new. Our observed-state construction in proposition 10.2 verifies Assumptions 2.1 and 2.4 of [2] literally and proves the matching lower rate on an unstructured subclass. It does not address models with additional smooth cross-state structure or alternative

information oracles. Giles's survey [13] discusses the high kurtosis of level corrections for discontinuous functions; our confidence amplification uses only second moments, not an empirical variance estimate or a normal approximation.

Adaptive coin mixtures and information bounds. Lee and Valiant [1] prove fully adaptive optimal bounds for mixtures separated from 1/2 by a fixed gap, including dependence on the positive fraction. Our gap-class specialization agrees with their rate in the balanced regime (corollary 8.2). The lower bounds here are driven by sign priors near 1/2; this explains why a uniform local posterior inequality and the stopped KL chain rule suffice. Their more refined rare-event analysis is not reproduced by our supremum over all sign functions. The change-of-measure step is related to the standard sequential information inequality of Kaufmann, Cappé, and Garivier [14]; it is not a new general principle. Lee–Valiant also optimize estimators for known positive and negative conditional bias laws under a per-coin depth constraint. Here the magnitude law is arbitrary, the sign function is unrestricted, and our lower bounds have no per-coin depth constraint.

Randomization and unbiased estimation. Single-term multilevel debiasing is established methodology [3, 4]. Blanchet and Glynn [15] obtain finite work-normalized variance under regularity such as local twice differentiability of the outer function. Syed and Wang [5] study repeatedly nested expectations under suitable regularity conditions. Our threshold indicator is discontinuous: the inverse-margin condition and all-algorithm converse characterize what remains possible in this setting. The inverse-binomial sampling identity used for cost calibration dates to classical work of Haldane [8]; we give a self-contained proof and do not claim that identity as new.

Other observation models and scope. Brennan, Vinayak, and Jamieson [6] estimate mass above effect-size thresholds from noisy test statistics, rather than unrestricted sequential revisiting. Chen, Naslidnyk, and Briol [7] study nested expectations with kernel quadrature and identify minimax lower bounds as a research direction. Their smooth-kernel setting differs from ours. The proposed contribution is the arbitrary-margin minimax characterization, a distribution-blind optimal-rate algorithm, and the inverse-margin unbiased converse in the precisely specified experiment. The original arbitrary-margin theorem is not an instance-optimal or rare-event theorem. Section 11 treats a global rarity constraint separately; no general solution to all nested-expectation problems is claimed.

Rare events. With a known upper bound $\theta \le \rho$, positive mass may concentrate on the most difficult magnitudes. Section 11 gives the variational lower modulus Ψ, the power-margin regimes, and the fixed quadratic law, where a scale-entropy cost contributes an extra $\sqrt{1 + \log(1 + \log(1/\rho))}$ beyond Ψ. This refutes the earlier matching-Ψ conjecture rather than assuming it. Section 13 replaces that conjecture for every known magnitude law: up to universal constants the rare-event risk is an entropy–tail selection functional, in which a selected part of the law pays its dyadic inverse-square cost entropy and the remainder pays a truncated inverse-gap cost. Section 14 evaluates the selection in closed form, by removing the smallest-gap mass ρ; appendix O proves that formula directly for finite laws. Section 15 adds a robust stopping-program dual and convergent finite certificates.

Verification boundary. Lean proves both directions of the original RMS law, with coefficients 1/64 and 512 and one distribution-blind upper algorithm. The critical minimax law, exact Ψ comparison and counterexample, and unbiased characterization with coefficients

1/16 and 713 are also proved in the observable model. One 713 estimator is fixed before every gap law. The variational optimizer has its separate formal specification. The confidence characterization is also proved in that model, with coefficients 1/64 and 8192 and one upper family depending only on budget and failure probability. The stronger revealed- and selected-magnitude RMS and confidence lower bounds are also proved, including the paper's confidence coefficient $15/(256\sqrt{2})$. The rare-event rate for power margins above exponent two is proved in the observable model by a construction different from appendix B. The entropy–tail law, the closed quantile form, the stopping-program results, and the other rare-event regimes are paper proofs. Section 18 gives the exact scope and links.

2 Model, notation, and main results

Fix $a = 1/2$. Let μ be any probability measure on $(0, a]$; in particular, μ may have atoms, gaps, or a singular continuous part. A measurable function $q : (0, a] \to [0, 1]$ specifies a population by

$$D \sim \mu, \qquad S \mid D = d \sim \mathrm{Ber}(q(d)), \qquad P = \tfrac{1}{2} + (2S - 1)D. \tag{3}$$

The variables attached to distinct new coins are independent. The population class is $\mathcal{G}(\mu) = \{G_{\mu,q} : q \text{ measurable}\}$ and $\theta(\mu, q) = \int q(d)\,\mu(\mathrm{d}d)$. Individual magnitudes and signs are not observed. The upper algorithm does not know μ either. The distribution-wise minimax benchmark may grant knowledge of μ, so attaining it simultaneously without that knowledge is the stronger statement. The lower bound remains valid with still more information. The endpoint $D = 1/2$ is allowed and corresponds to deterministic coins; $D = 0$ is excluded.

An algorithm has an independent random seed, creates finitely many coins before each observation, selects one of its coins, and receives its next Bernoulli observation. It stops after T observations and returns a real number $\widehat{\theta}$. Infinite expected work is permitted in the model, but algorithms satisfying a finite uniform budget necessarily stop almost surely. Finite internal computation between observations is implicit. We count observations, not arithmetic operations or independent random bits. Every coin actually created by the upper algorithm receives at least one observation. Charging a fixed additional amount per new outer draw therefore changes its cost by at most a constant factor.

Define

$$R_\mu(B) = \inf_{\mathcal{A}:\, \sup_q \mathbb{E}_{\mu,q} T \le B} \sup_q \big(\mathbb{E}_{\mu,q}(\widehat{\theta} - \theta)^2\big)^{1/2}, \tag{4}$$

$$r_\mu(B, \delta) = \inf_{\mathcal{A}:\, \sup_q \mathbb{E}_{\mu,q} T \le B} \inf\{r \ge 0 : \sup_q \mathbb{P}_{\mu,q}(|\widehat{\theta} - \theta| > r) \le \delta\}. \tag{5}$$

These are expected-budget criteria, not deterministic maximum-work criteria.

For $h > 0$, write

$$A_\mu(h) = \mathbb{E}_\mu \min\{h^{-1}, D^{-1}\}, \qquad \omega_\mu(h) = hA_\mu(h), \qquad I_\mu = \mathbb{E}_\mu[D^{-1}] \in [2, \infty]. \tag{6}$$

The modulus is increasing and concave, $\omega_\mu(0+) = 0$, and $\omega_\mu(h) = 1$ for $h \ge a$, while $\omega_\mu(h) \ge \min\{1, 2h\}$. In particular, for $c \ge 1$ and $0 < b \le 1$,

$$\omega_\mu(ch) \le c\omega_\mu(h), \qquad \omega_\mu(bh) \ge b\omega_\mu(h). \tag{7}$$

Theorem 2.1 (Distribution-blind arbitrary-margin minimax law)**.** *There are universal constants $0 < c < C < \infty$ such that for every probability measure μ on $(0, 1/2]$, every $B \ge 1$, and*

every $0 < \delta \le 1/4$,

$$c\omega_\mu(B^{-1/2}) \le R_\mu(B) \le C\omega_\mu(B^{-1/2}), \tag{8}$$

$$c\omega_\mu\left(\sqrt{\frac{\log(1/\delta)}{B}}\right) \le r_\mu(B,\delta) \le C\omega_\mu\left(\sqrt{\frac{\log(1/\delta)}{B}}\right). \tag{9}$$

More strongly, for each B *there is one algorithm* $\mathcal{A}_B$*, depending only on* B*, that simultaneously for all* μ, q *satisfies*

$$\mathbb{E}_{\mu,q}T \le B, \qquad \left(\mathbb{E}_{\mu,q}(\widehat{\theta}-\theta)^2\right)^{1/2} \le C\omega_\mu(B^{-1/2}). \tag{10}$$

There is likewise one algorithm depending only on (B,δ) *for the upper confidence bound. Neither algorithm observes* D *or knows* μ*. In the lower bounds one may take* $c = 1/64$ *for RMS and* $c = 15/(256\sqrt{2})$ *for confidence. The lower bounds hold even if every coin's magnitude is revealed for free, or if the algorithm selects the magnitude of each fresh coin. In the latter experiment* q *is defined on all* $(0,1/2]$ *and the target is still* $\int q\,\mathrm{d}\mu$*.*

Formal scope. The RMS law in the ordinary hidden-magnitude experiment is proved by `minimax_lower_bound` and `distribution_blind_upper`, with coefficients 1/64 and 512. The upper family is chosen before every gap law and satisfies the budget in the same experiment. The confidence law is proved by `confidence_characterization`, with coefficients 1/64 and 8192 for $B \ge 1$ and $0 < \delta \le 1/4$. Its upper family depends only on (B,δ). The stronger revealed- and selected-magnitude lower bounds are proved by `oracle_minimax_lower`, with coefficients 1/64 and $15/(256\sqrt{2})$. An exact embedding also gives the latter confidence coefficient in the hidden-magnitude experiment.

This is distribution-wise minimax optimality: the worst case is over all sign functions at a fixed magnitude law. It is not instance optimality for each particular pair (μ, q). The budget is expected work, not an almost-sure cap. The modulus in the guarantee is evaluated at the true, possibly unknown distribution; it is not an observable confidence radius.

Theorem 2.2 (Exact unbiased feasibility and variance–work scale)**.** *An estimator unbiased for every* $G \in \mathcal{G}(\mu)$ *can have both*

$$V = \sup_q \mathrm{Var}_{\mu,q}(Y) < \infty, \qquad W = \sup_q \mathbb{E}_{\mu,q}T < \infty$$

if and only if $I_\mu < \infty$*. When* $I_\mu < \infty$*,*

$$\frac{1}{16}\,I_\mu^2 \;\le\; \inf_{Y \text{ unbiased}} VW \;\le\; 713 I_\mu^2, \tag{11}$$

where the infimum is over estimators with both uniform bounds finite. A single estimator, using neither μ *nor any margin parameters, attains the upper bound simultaneously for every such magnitude law. It satisfies* $\sup_q \mathbb{E}Y^2 \le 23 I_\mu$ *and* $\sup_q \mathbb{E}T \le 31 I_\mu$*. The lower coefficient* 1/16 *is the exact universal constant: equality is attained for every fixed-gap law* $\mu = \delta_d$*. Equality is not asserted for arbitrary nondegenerate* μ*.*

Verification: Lean proves the full existence characterization, the 1/16 *lower bound, and the 713 upper construction in the same observable model; see sections 7 and 18.*

2.1 Unknown magnitude laws under an envelope

Let $F_\mu(t) = \mu((0,t])$, extended to equal one above a, and define

$$\mathcal{G}_{\preceq\mu} = \{G \text{ on } [0,1] : \mathbb{P}_G(|P - 1/2| \le t) \le F_\mu(t) \text{ for every } t > 0\}. \tag{12}$$

Here the actual magnitude law is unknown on $(0, 1/2]$. The inequality means its magnitudes are stochastically no smaller than those under μ. It also rules out an atom at $P = 1/2$.

Corollary 2.3 (Stochastic-dominance envelope version)**.** *Both conclusions of theorem 2.1 hold with the supremum taken over $\mathcal{G}_{\preceq\mu}$, with the same universal constants. The existence criterion and variance–work product bounds of theorem 2.2 also hold for this class. The finite-budget and unbiased algorithms need not know the envelope; it is used only to express a uniform guarantee.*

What distribution-blindness does not supply. The algorithm is specified without μ or an envelope. A numerical choice of budget for a prescribed accuracy, however, generally requires a bound on the unknown modulus. Neither the adaptive risk guarantee nor the unbiased estimator automatically supplies a data-computable confidence radius.

Corollary 2.4 (Critical nested-simulation obstruction)**.** *Let $U \sim \mathrm{Unif}[0,1]$, $X \in \{-1,1\}$, and $D(U) = |\mathbb{E}[X \mid U]|/2$. Consider all measurable conditional models with $D(U) \sim \mathrm{Unif}(0, 1/4]$, with no cross-state regularity. For estimating $\mathbb{P}(\mathbb{E}[X \mid U] > 0)$, the minimax expected observation cost at RMS error ε, or at error ε with a fixed failure probability $\delta_0 = 1/16$, is*

$$\asymp \varepsilon^{-2} \log^2(1/\varepsilon), \qquad \varepsilon \downarrow 0.$$

The lower bound allows arbitrary selected-state queries and revisiting. Every model satisfies Assumptions 2.1 and 2.4 of Giles–Haji-Ali [2], with uniform constants.

The continuous-magnitude construction and proof are given in proposition 10.2. This is a worst-case sample-oracle statement on a subclass satisfying those assumptions, not a claim about every structured simulator or every possible source of information.

3 A history-uniform adaptive information bound

For Bernoulli probabilities $u, v \in (0,1)$,

$$\mathrm{kl}(u,v) \le \frac{(u-v)^2}{v(1-v)}. \tag{13}$$

This follows from $\log x \le x - 1$ applied to the two likelihood ratios. We next control how an arbitrary local observation history changes two symmetric sign priors.

Lemma 3.1 (Posterior contraction at every magnitude)**.** *Fix $0 < d \le \frac{1}{2}$ and $0 \le \eta < \frac{1}{2}$. Under two experiments the sign priors are $q_\pm = \frac{1}{2} \pm \eta$, while the observation likelihoods conditional on the sign are the same. Let $L \in [0,\infty]$ be the likelihood ratio of an arbitrary local history of one coin under positive versus negative sign, let $\rho_\pm$ be its posterior sign probabilities, and let $p_\pm = \frac{1}{2} - d + 2d\rho_\pm$ be the next predictive success probabilities. If $0 < L < \infty$, then*

$$\rho_+ - \rho_- = \frac{2\eta L}{L + (\frac{1}{4} - \eta^2)(L-1)^2} \le 2\eta. \tag{14}$$

In every case,

$$\mathrm{kl}(p_+, p_-) \le \frac{32\, d^2 \eta^2}{(1 - 4\eta^2)^2}. \tag{15}$$

The same bound holds for $\mathrm{kl}(p_-, p_+)$. The constant 32 is sharp: at $L = 1$ the ratio of the two sides of (15) tends to one as $\eta \downarrow 0$.

Proof. If $L \in \{0, \infty\}$, then $\rho_+ = \rho_- \in \{0, 1\}$, so $p_+ = p_-$ and the divergence vanishes. This case cannot occur for $d < \frac{1}{2}$; for $d = \frac{1}{2}$ it occurs after the first observation. Now let $0 < L < \infty$ and write $a = \frac{1}{2} + \eta$, $b = \frac{1}{2} - \eta$. Then $a + b = 1$ and $ab = \frac{1}{4} - \eta^2 = (1 - 4\eta^2)/4$. Bayes' formula gives $\rho_+ = aL/(b + aL)$ and $\rho_- = bL/(a + bL)$, so

$$\rho_+ - \rho_- = \frac{(a^2 - b^2)L}{(b + aL)(a + bL)}, \qquad (b + aL)(a + bL) = ab(L - 1)^2 + (a + b)^2 L.$$

This proves (14). Also

$$\rho_-(1 - \rho_-) = \frac{abL}{(a + bL)^2}, \qquad \rho_+(1 - \rho_+) = \frac{abL}{(b + aL)^2}.$$

For fixed $v \in (0, 1)$, the function $u \mapsto \mathrm{kl}(u, v)$ vanishes together with its first derivative at $u = v$, and its second derivative is $1/\{u(1 - u)\}$. Since $p_- \le p_+$, Taylor's theorem gives

$$\mathrm{kl}(p_+, p_-) \le \frac{(p_+ - p_-)^2}{2 \min_{\xi \in [p_-, p_+]} \xi(1 - \xi)}.$$

The same Taylor upper bound applies with the two arguments interchanged, since its right side depends only on their squared difference and the interval between them. Thus it controls both KL directions. Every such ξ equals $\frac{1}{2} - d + 2d\rho$ for some $\rho \in [\rho_-, \rho_+] \subset (0, 1)$. Because $d \le \frac{1}{2}$,

$$\xi(1 - \xi) = \tfrac{1}{4} - d^2(2\rho - 1)^2 \ge \tfrac{1}{4}\{1 - (2\rho - 1)^2\} = \rho(1 - \rho).$$

By concavity, the minimum of $\rho(1 - \rho)$ over $[\rho_-, \rho_+]$ is attained at an endpoint. Since $p_+ - p_- = 2d(\rho_+ - \rho_-)$, the preceding displays give

$$\mathrm{kl}(p_+, p_-) \le 2d^2 \max\Big\{\frac{(\rho_+ - \rho_-)^2}{\rho_-(1 - \rho_-)}, \frac{(\rho_+ - \rho_-)^2}{\rho_+(1 - \rho_+)}\Big\} = \frac{8d^2\eta^2}{ab} \max\Big\{\frac{L}{(b + aL)^2}, \frac{L}{(a + bL)^2}\Big\}.$$

The maps $L \mapsto L/(b + aL)^2$ and $L \mapsto L/(a + bL)^2$ are maximized at $L = b/a$ and $L = a/b$ respectively, both with value $1/(4ab)$. Therefore $\mathrm{kl}(p_+, p_-) \le 2d^2\eta^2/(ab)^2$, which is (15).

For sharpness, at $L = 1$ we have $p_\pm = \frac{1}{2} \pm 2d\eta$ and

$$\mathrm{kl}(p_+, p_-) = 4d\eta \log \frac{1 + 4d\eta}{1 - 4d\eta} = 32d^2\eta^2\{1 + O(d^2\eta^2)\}. \qquad \square$$

Fix $h > 0$ and take

$$\eta_h(d) = \frac{1}{8} \min\{1, h/d\}, \qquad q_\pm(d) = \frac{1}{2} \pm \eta_h(d). \tag{16}$$

The two targets are centered at $1/2$ and separated by

$$\Delta_h = \theta_+ - \theta_- = 2\mathbb{E}_\mu \eta_h(D) = \frac{1}{4}\omega_\mu(h). \tag{17}$$

Because $d\eta_h(d) \le h/8$, every next observation has predictive KL at most $128h^2/225$ in either direction.

Proposition 3.2 (Stopped-transcript bound)**.** *For every adaptive algorithm in any of the experiments of theorem 2.1, its stopped transcript laws satisfy*

$$\mathrm{KL}(\mathsf{P}_+^{\mathcal{A}}, \mathsf{P}_-^{\mathcal{A}}) \le \frac{128}{225} h^2 \mathbb{E}_+ T. \tag{18}$$

Consequently, if it estimates both targets to radius $\varepsilon < \Delta_h/2$ with failure probability at most $\delta < 1/2$,

$$\mathbb{E}_+ T \ge \frac{225}{128h^2} \mathrm{kl}(1 - \delta, \delta). \tag{19}$$

Proof. First reveal all magnitudes; an algorithm not using them is a special case. Conditional on a realized transcript, action-selection and randomization factors are common to the two hypotheses. Conditional on the hypothesis and revealed magnitudes, the posterior of a coin's sign depends only on its own observations: the likelihood factorizes by coin. Thus the local posterior identity applies even after arbitrary revisiting and interleaving. For a fresh coin the same assertion has $L = 1$. At $d = 1/2$ the first observation has full support under both hypotheses, and after it the sign is revealed, so later observations contribute zero KL.

A newly drawn magnitude has the same law in both experiments and contributes zero KL. A selected magnitude is an algorithmic action and also contributes zero conditional KL. The chain rule, first stopped at $T \wedge m$, bounds transcript KL by $128h^2\mathbb{E}_+(T \wedge m)/225$. Padding the transcript after its stopping time and taking increasing finite prefixes gives (18) by monotone convergence of expected work and the finite-prefix characterization of relative entropy. If expected work is infinite, the bound is vacuous. Interchanging the hypotheses gives the reverse bound with $\mathbb{E}_- T$.

Thresholding $\widehat{\theta}$ at $1/2$ gives a test with both errors at most δ, since the estimation radius is strictly less than half the separation. Data processing gives $\mathrm{KL} \geq \mathrm{kl}(1-\delta, \delta)$, proving (19). □

The strict separation in this proposition is intentional. At exactly half the separation, the constant output $1/2$ is within tolerance for both members of the pair, so a test cannot be inferred without a limiting argument over strictly separated pairs.

4 A universal capped classifier

This section does not assume a magnitude law. Fix a coin with $P = 1/2 + (2S-1)d$, where $0 < d \leq 1/2$ and $S \in \{0, 1\}$. For a dyadic cap $N = 2^\ell$, examine prefixes of lengths $1, 2, 4, \ldots, N$. At a prefix $n < N$, stop when

$$\left|\overline{X}_n - \frac{1}{2}\right| > t_{n,N}, \qquad t_{n,N} = \sqrt{\frac{1 + \log(N/n)}{n}}, \tag{20}$$

and output the side of $1/2$ occupied by the sample mean. If no early stop occurs, output $Z_N = \mathbf{1}\{\overline{X}_N > 1/2\}$; a final tie is negative. Let τ_N be the number of observations used.

Set

$$m_N(d) = \min\{N, d^{-2}[1 + \log_+(Nd^2)]\}, \qquad \log_+ x = \max\{0, \log x\}. \tag{21}$$

Lemma 4.1 (Conditional error and work)**.** *For every dyadic $N \geq 1$, both signs, and $0 < d \leq 1/2$,*

$$e_N(d, S) := \mathbb{P}(Z_N \neq S \mid d, S) \leq \min\{1, (Nd^2)^{-2}\}, \tag{22}$$

$$\mathbb{E}[\tau_N \mid d, S] \leq 17 m_N(d). \tag{23}$$

Proof. A wrong early stop at a fixed prefix n requires a one-sided deviation of at least $d + t_{n,N}$. Hoeffding's inequality and a union bound over fixed prefixes, not a conditioning on the stopping event, give

$$\begin{aligned} e_N(d, S) &\leq e^{-2Nd^2} + \sum_{\substack{n<N \\ n \text{ dyadic}}} e^{-2n(d+t_{n,N})^2} \\ &\leq e^{-2Nd^2} + e^{-2} \sum_{\substack{n<N \\ n \text{ dyadic}}} (n/N)^2 e^{-2nd^2}. \end{aligned} \tag{24}$$

For every dyadic sequence of positive u values and every $v, b > 0$,

$$\sum_u u^v e^{-bu} \le \frac{\Gamma(v+1)}{(1-2^{-v})b^v}. \tag{25}$$

Indeed, the disjoint intervals $[u/2, u]$ satisfy $\int_{u/2}^{u} y^{v-1}e^{-by}\,\mathrm{d}y \ge e^{-bu}u^v(1-2^{-v})/v$. With $x = Nd^2 \ge 1$, the sum in (24) is bounded by $x^{-2}(2/3)$, and $x^2e^{-2x} \le e^{-2}$. Hence the right side is at most $(5/3)e^{-2}x^{-2} < x^{-2}$. The bound by one handles $x < 1$.

The dyadic survival identity and the fact that a surviving prefix cannot have crossed its threshold imply

$$\begin{aligned}\mathbb{E}\tau_N &= 1 + \sum_{n<N,\, n \text{ dyadic}} n\mathbb{P}(\tau_N > n) \\ &\le 1 + \sum_{n<N,\, n \text{ dyadic}} n\{\mathbf{1}[d \le 2t_{n,N}] + 2e^{-nd^2/2}\}.\end{aligned} \tag{26}$$

For $x \ge 1$, the inequality $d \le 2t_{n,N}$ implies $nd^2 \le 4(1+\log x)$: if $nd^2 < 1$ this is immediate, and otherwise $\log(N/n) = \log(x/(nd^2)) \le \log x$. The corresponding dyadic sum of n is at most $8d^{-2}(1+\log x)$. Using (25) with $v = 1, b = 1/2$ bounds the exponential part of (26) by $8d^{-2}$. Since $1 \le d^{-2}$, the total is at most $17d^{-2}(1+\log x)$. Also $\tau_N \le N$ deterministically, and $1 + \log x \le x$ for $x \ge 1$. For $x < 1$, $m_N(d) = N$, completing the proof. □

4.1 One geometric level law for every population

On a common coin stream define

$$\Delta_0 = Z_1, \qquad \Delta_\ell = Z_{2^\ell} - Z_{2^{\ell-1}} \quad (\ell \ge 1). \tag{27}$$

A coupled difference is computed by running both classifiers on the same prefix until both have stopped. Its work W_ℓ is the maximum of their stopping times and hence at most their sum. Let

$$r = 2^{-1/2}, \qquad p_\ell = (1-r)r^\ell, \qquad \mathbb{P}(J = \ell) = p_\ell. \tag{28}$$

Choose J independently of a fresh coin and its observations. For a finite cutoff L, return

$$Y_L = \frac{\Delta_J}{p_J}\mathbf{1}\{J \le L\}. \tag{29}$$

When $J > L$, no coin is sampled and the output and work are both zero. Write $\mathcal{W}_L$ for actual work. For $L = \infty$, drop the indicator and write $Y, \mathcal{W}$.

Lemma 4.2 (Truncated inverse-margin control)**.** *For every finite $L \ge 0$, $N = 2^L$, both signs, and $0 < d \le 1/2$,*

$$\mathbb{E}[Y_L \mid d, S] = \mathbb{E}[Z_N \mid d, S], \tag{30}$$

$$\mathbb{E}[Y_L^2 \mid d, S] \le 85\min\{\sqrt{N}, d^{-1}\}, \tag{31}$$

$$\mathbb{E}[\mathcal{W}_L \mid d, S] \le 149\min\{\sqrt{N}, d^{-1}\}. \tag{32}$$

Moreover,

$$\sum_{\ell \ge 0} \mathbb{E}[|\Delta_\ell| \mid d, S] \le C_0[1 + \log(1/d)] \tag{33}$$

for a universal C_0. Whenever $\mathbb{E}\log(1/D) < \infty$, Y is integrable and unbiased for θ. This logarithmic condition is sufficient here; no necessity claim for integrability alone is made. If $\mathbb{E}D^{-1} < \infty$, its second moment is at most $85\mathbb{E}D^{-1}$ and its expected work at most $149\mathbb{E}D^{-1}$.

Proof. The finite expectation telescopes. For $\ell \geq 1$, a nonzero difference requires at least one classifier to be wrong, so, with $N_\ell = 2^\ell$,

$$\mathbb{E}[\Delta_\ell^2 \mid d, S] \leq 5 \min\{1, (N_\ell d^2)^{-2}\}, \qquad \mathbb{E}[W_\ell \mid d, S] \leq 34 m_{N_\ell}(d). \tag{34}$$

The same loose inequalities cover $\ell = 0$. Consequently

$$\mathbb{E}[Y_L^2 \mid d, S] \leq \frac{5}{1-r} \sum_{\ell=0}^{L} \sqrt{N_\ell} \min\{1, (N_\ell d^2)^{-2}\}, \tag{35}$$

$$\mathbb{E}[\mathcal{W}_L \mid d, S] \leq 34(1-r) \sum_{\ell=0}^{L} N_\ell^{-1/2} m_{N_\ell}(d). \tag{36}$$

Split each sum at $N_\ell = d^{-2}$. Before that point, both summands are $\sqrt{N_\ell}$ and their sum is bounded by $(1-r)^{-1} \min\{\sqrt{N}, d^{-1}\}$. After that point, the second-moment sum is a geometric tail with ratio $2^{-3/2}$ and is at most $(1-r^3)^{-1} d^{-1}$. Its full constant is bounded by

$$\frac{5}{1-r}\left(\frac{1}{1-r} + \frac{1}{1-r^3}\right) < 85.$$

For work, the first N_ℓ after the split has $1 < N_\ell d^2 \leq 2$. Its tail is bounded by

$$d^{-1} \sum_{k \geq 0} r^k (1 + \log 2 + k \log 2).$$

Combining the prefix and tail gives the constant

$$34\left(2 + \log 2 + \frac{r \log 2}{1-r}\right) < 149.$$

If the cutoff precedes the split, only the smaller prefix bound is needed. This proves (31)–(32).

Because $|\Delta_\ell| = \Delta_\ell^2$ for $\ell \geq 1$, the number of levels before the split is $O(1 + \log(1/d))$, and the subsequent error bounds form a convergent geometric series. This proves (33). Absolute summability allows Fubini and telescoping to S, using (22). Averaging over (D, S) proves unbiasedness whenever the stated logarithmic moment is finite. The inverse-moment condition implies that condition. Finally take $L \to \infty$ in the nonnegative series for second moment and work. □

This is the standard single-term randomized multilevel identity [4]; the common conditional bounds and their distributional consequences are the key additional estimates. The estimator's output need not lie in $[0, 1]$. The interior-magnitude obstruction to nonnegative output is given in section 6.1.

5 Proof of the minimax law

5.1 Exact calibration without a known margin law

For $h = N^{-1/2}$ define

$$b_\mu(h) = \mathbb{E}_\mu \min\{1, (h/D)^4\} \leq \omega_\mu(h). \tag{37}$$

One call of the cutoff estimator returns $(Y_L, \mathcal{W}_L)$ with

$$0 \leq \mathcal{W}_L \leq N, \quad c := \mathbb{E}\mathcal{W}_L \leq 149 A_\mu(h), \quad v := \mathrm{Var}(Y_L) \leq 85 A_\mu(h), \quad |\mathbb{E}Y_L - \theta| \leq b_\mu(h). \tag{38}$$

Moreover $c > 0$, because level zero has probability $1 - r > 0$ and costs one observation. The following wrapper handles the unknown c exactly.

Lemma 5.1 (Two-success cost calibration)**.** *Suppose an independent call returns (Z, C) with $0 \le C \le N$, $c = \mathbb{E}C > 0$, $\mathbb{E}Z = \zeta$, and $\mathrm{Var}(Z) = v < \infty$. Dependence of value and cost within a call is unrestricted. Run independent pilot calls and mark each with conditional probability C_i/N, using independent randomization. Stop at the second mark, at call K. Discard all pilot values and average $M = K - 1$ fresh independent production values. If $\widehat{\zeta}$ is that average and T is the total pilot-plus-production cost, then*

$$\mathbb{E}\widehat{\zeta} = \zeta, \qquad \mathrm{Var}(\widehat{\zeta}) = \frac{vc}{N}, \qquad \mathbb{E}T = 4N - c \le 4N. \tag{39}$$

The procedure uses neither c nor v.

Proof. The pilot marks are independent Bernoulli variables with parameter $p = c/N > 0$, so K has negative-binomial probabilities

$$\mathbb{P}(K = k) = (k-1)p^2(1-p)^{k-2}, \quad k \ge 2, \qquad \mathbb{E}K = 2/p.$$

In particular $K < \infty$ almost surely, and the classical inverse-binomial identity [8] is immediate:

$$\mathbb{E}\frac{1}{K-1} = p^2 \sum_{k=2}^{\infty} (1-p)^{k-2} = p. \tag{40}$$

The event $\{K \ge i\}$ depends only on earlier calls and marks, so is independent of C_i. Tonelli's theorem yields

$$\mathbb{E}\sum_{i=1}^{K} C_i = \sum_{i \ge 1} \mathbb{P}(K \ge i)c = c\mathbb{E}K = 2N.$$

Production calls are independent of the pilot. Their mean total cost is $c\mathbb{E}(K-1) = 2N - c$, and conditional on K the production average has mean ζ and variance $v/(K-1)$. Total expectation and variance give (39). For $p = 1$, $K = 2$ deterministically and the same identities hold. □

5.2 Distribution-blind upper bounds

For $B \ge 4$, choose the largest dyadic $N \le B/4$, set $L = \log_2 N$, and apply lemma 5.1 to $(Z, C) = (Y_L, \mathcal{W}_L)$. By (38),

$$\mathbb{E}T \le 4N \le B, \qquad \mathbb{E}(\widehat{\theta} - \theta)^2 \le \omega_\mu(h)^2 + \frac{85 \cdot 149 A_\mu(h)^2}{N} = 12666\omega_\mu(h)^2. \tag{41}$$

Since $N > B/8$, (7) gives the RMS upper bound, for example with $C_R = \sqrt{8 \cdot 12666}$. For $1 \le B < 4$, output $1/2$ with no observations; then $\omega_\mu(B^{-1/2}) = 1$. This proves (10) without any knowledge of μ or q.

For confidence, choose the least odd integer $k \ge (32/9)\log(1/\delta)$; then $k < 5\log(1/\delta)$ for $0 < \delta \le 1/4$. If $B/k \ge 4$, use k independent groups, each with budget B/k, and return their median. A group's error exceeds $\sqrt{8}C_R\omega_\mu(\sqrt{k/B})$ with probability at most $1/8$, by its mean-square bound. Hoeffding's inequality bounds the probability that at least half the groups fail by $e^{-9k/32} \le \delta$. Costs add in expectation, and

$$\omega_\mu(\sqrt{k/B}) \le \sqrt{5}\,\omega_\mu\left(\sqrt{\log(1/\delta)/B}\right).$$

If $B/k < 4$, output $1/2$; then $\sqrt{\log(1/\delta)/B} > 1/\sqrt{20}$, and $\omega_\mu(\sqrt{\log(1/\delta)/B}) > 1/\sqrt{5}$. This handles the remaining case with a universal constant.

Implementation and scope. Pilot values are discarded; recycling them into the final average would invalidate the independence calculation. Value and cost may be correlated within any call. The random pilot and production counts are unbounded, so the guarantee concerns expected observations, not a run-by-run work cap. Projection of the final finite-budget estimate onto $[0, 1]$ can only reduce its error, but is not used in the variance identity.

5.3 Matching lower bounds

For RMS error use (16) with $h = (4\sqrt{B})^{-1}$. By proposition 3.2, transcript KL is at most $8/225$, so Pinsker's inequality gives total variation at most $2/15$. Thresholding an estimator at $1/2$ gives a test whose error probabilities sum to at most $8R^2/\Delta_h^2$, if its RMS error under both hypotheses is at most R. Every test has error sum at least $1 - 2/15 > 1/2$. Hence $R > \Delta_h/4 = \omega_\mu(h)/16 \geq \omega_\mu(B^{-1/2})/64$. Taking the infimum proves the stated weak inequality.

For confidence write $K_\delta = \mathrm{kl}(1-\delta, \delta)$ and take

$$h = \frac{15}{16\sqrt{2}}\sqrt{K_\delta/B}.$$

Then $225K_\delta/(128h^2) = 4B > B$; by proposition 3.2 no admissible estimator has radius strictly smaller than $\Delta_h/2 = \omega_\mu(h)/8$. On $0 < \delta \leq 1/4$,

$$K_\delta = (1-2\delta)\log\frac{1-\delta}{\delta} \geq \frac{1}{4}\log(1/\delta). \tag{42}$$

Indeed $1 - 2\delta \geq 1/2$ and $1 - \delta \geq \sqrt{\delta}$. Thus $h \geq 15\sqrt{\log(1/\delta)/B}/(32\sqrt{2})$. Apply (7) to obtain the coefficient $15/(256\sqrt{2})$. This completes the proof of theorem 2.1.

5.4 Proof for a stochastic-dominance envelope

The functions $d \mapsto \min\{1, h/d\}$, $d \mapsto \min\{1, (h/d)^4\}$, and $d \mapsto \min\{h^{-1}, d^{-1}\}$ are decreasing. Their expectations for any magnitude law in (12) are therefore bounded by their expectations under μ. The distribution-blind upper algorithm is unchanged; its risk bound is no larger than the envelope value. The lower bound follows by restricting to $\mathcal{G}(\mu)$, which is a subclass. The same reasoning with d^{-1} and theorem 2.2 proves corollary 2.3.

Remark 5.2 (Why the statement uses error as a function of budget)**.** For highly irregular μ, inverting ω_μ at two different constant multiples of ε need not preserve multiplicative constants in work. Statements such as $B \asymp \omega_\mu^{-1}(\varepsilon)^{-2}$ need a regularity qualification or explicit constants inside the inverse. The modulus bounds (8)–(9) do not have this defect. Polynomial and logarithmic examples below can be inverted with the stated asymptotic orders.

6 Proof of the unbiased characterization

Sufficiency already follows from lemma 4.2. The improved $713I_\mu^2$ bound uses the modified classifier proved in section 7. We prove the converse using only two-point inequalities and the stopped KL chain rule; no differentiation of an estimator's expectation is needed.

Lemma 6.1 (Two-point variance inequality)**.** *Let $\mathbb{P}, \mathbb{Q}$ be probability measures on a common space, let $H^2 = 1 - \int \sqrt{d\mathbb{P}\, d\mathbb{Q}}$, and let Y be a real statistic with finite second moment under both. Then $\Delta = \mathbb{E}_{\mathbb{P}} Y - \mathbb{E}_{\mathbb{Q}} Y$ satisfies*

$$\Delta^2(1 - 2H^2) \le 4H^2\big(\mathrm{Var}_{\mathbb{P}}\, Y + \mathrm{Var}_{\mathbb{Q}}\, Y\big), \tag{43}$$

Moreover, $2H^2 \le \mathrm{KL}(\mathbb{P}, \mathbb{Q})$, and the symmetric bound is

$$8H^2 \le \mathrm{KL}(\mathbb{P}, \mathbb{Q}) + \mathrm{KL}(\mathbb{Q}, \mathbb{P}). \tag{44}$$

Proof. Let p, q be densities with respect to $\nu = \mathbb{P} + \mathbb{Q}$, and let $c = (\mathbb{E}_{\mathbb{P}} Y + \mathbb{E}_{\mathbb{Q}} Y)/2$. Then $\Delta = \int (Y - c)(\sqrt{p} - \sqrt{q})(\sqrt{p} + \sqrt{q})\, d\nu$. We use three facts: $\int (\sqrt{p} - \sqrt{q})^2\, d\nu = 2H^2$; $(\sqrt{p} + \sqrt{q})^2 \le 2(p + q)$; and $\mathbb{E}_{\mathbb{P}}(Y - c)^2 = \mathrm{Var}_{\mathbb{P}}\, Y + \Delta^2/4$, with the same identity under $\mathbb{Q}$. Together with the Cauchy–Schwarz inequality they give

$$\Delta^2 \le \int (\sqrt{p} - \sqrt{q})^2\, d\nu \cdot 2 \int (Y - c)^2 (p + q)\, d\nu = 4H^2\Big(\mathrm{Var}_{\mathbb{P}}\, Y + \mathrm{Var}_{\mathbb{Q}}\, Y + \tfrac{1}{2}\Delta^2\Big).$$

Rearranging gives (43). Finally, by Jensen's inequality, $-\frac{1}{2}\mathrm{KL}(\mathbb{P}, \mathbb{Q}) = \mathbb{E}_{\mathbb{P}} \log \sqrt{q/p} \le \log(1 - H^2) \le -H^2$. For the symmetric inequality, use the pointwise bound

$$(x - y) \log(x/y) \ge 4(\sqrt{x} - \sqrt{y})^2, \qquad x, y > 0.$$

Indeed, putting $r = \sqrt{x/y}$, the difference is $2y(r - 1)(r + 1)g(r)$, where $g(r) = \log r - 2(r - 1)/(r + 1)$ has $g(1) = 0$ and $g'(r) = (r - 1)^2/[r(r + 1)^2] \ge 0$. Integrating the pointwise bound gives (44), with zero-density cases interpreted by limits. □

Converse in theorem 2.2. Suppose Y is unbiased for every q, with finite uniform variance V and expected work W. Fix $h > 0$ and $0 < t \le h/8$, and consider

$$q_{\pm}(d) = \tfrac{1}{2} \pm t\varphi_h(d), \qquad \varphi_h(d) = \min\{h^{-1}, d^{-1}\}.$$

Then $\eta(d) = t\varphi_h(d) \le t/h \le 1/8$ and $d\eta(d) \le t$. Lemma 3.1 and the stopped-transcript argument give both

$$\max\{\mathrm{KL}(\mathsf{P}_+^{\mathcal{A}}, \mathsf{P}_-^{\mathcal{A}}), \mathrm{KL}(\mathsf{P}_-^{\mathcal{A}}, \mathsf{P}_+^{\mathcal{A}})\} \le \kappa_t := \frac{32t^2 W}{(1 - 4t^2/h^2)^2}. \tag{45}$$

Each direction uses its own expected work, bounded by W. Unbiasedness gives mean separation $2tA_\mu(h)$. By (44), $H^2 \le \kappa_t/4$. Applying lemma 6.1 with both variances at most V yields

$$4t^2 A_\mu(h)^2(1 - \kappa_t/2) \le 2\kappa_t V, \qquad A_\mu(h)^2(1 - \kappa_t/2) \le \frac{16VW}{(1 - 4t^2/h^2)^2}.$$

First let $t \downarrow 0$, with h fixed, giving $A_\mu(h)^2 \le 16VW$. Then $A_\mu(h) \uparrow I_\mu$ as $h \downarrow 0$ by monotone convergence. This proves $I_\mu < \infty$ and $VW \ge I_\mu^2/16$. The envelope class contains $\mathcal{G}(\mu)$, so the same lower bound holds there. □

Sharpness of the universal coefficient. For $\mu = \delta_d$, $0 < d \le 1/2$, one fresh observation $X \in \{0, 1\}$ gives

$$Y = \frac{X - 1/2 + d}{2d}, \qquad \mathbb{E}Y = q(d), \qquad \sup_q \mathrm{Var}(Y) = \frac{1}{16d^2}, \qquad W = 1. \tag{46}$$

Since $I_\mu = 1/d$, equality holds. Hence $1/16$ cannot be increased in a bound valid for every magnitude law. This does not assert equality for every nondegenerate law, or that the parameter-free construction achieves the optimal constant for each law.

Uniformity matters. The impossibility is over the whole sign class at a fixed μ, or over the stated envelope class. It does not say that each single population with infinite inverse moment forbids every efficient unbiased estimator. For a particular known population, the target might already be known. Likewise, a small structured family of sign functions can be easier than an arbitrary measurable q.

6.1 Why a universal unbiased output must sometimes be negative

Proposition 6.2 (No nonnegative unbiased output except at a deterministic endpoint)**.** *If $\mu((0,1/2)) > 0$, no algorithm stopping almost surely for every q and always returning a nonnegative value is unbiased on $\mathcal{G}(\mu)$. For $\mu = \delta_{1/2}$ the obstruction disappears: one observation is an unbiased $[0,1]$-valued estimator with unit work.*

Proof. Under $q \equiv 0$ the target is zero, so nonnegativity and unbiasedness force the output to vanish almost surely. Compare this with $q_1(d) = \mathbf{1}\{d < 1/2\}$, whose target is $\mu((0,1/2)) > 0$. Interior coins give positive likelihood to every finite Bernoulli history under both models; endpoint coins have the same deterministic sign zero. Finite stopped transcript laws are equivalent. Intersecting with each $\{T = n\}$ and taking a countable union gives equivalence of the full stopped transcripts. Seeds and any revealed magnitudes have the same distribution. Thus the output is also zero almost surely under q_1, a contradiction. If $\mu = \delta_{1/2}$, the first observation is exactly the sign. □

This elementary support argument is not offered as a separate novelty claim. It explains the signed outputs in the construction. Clipping is allowed for a biased finite-accuracy estimator but invalidates exact unbiasedness.

7 A uniform variance–work upper constant of 713

The finite-budget construction in sections 4 and 5 uses the original threshold and constants $85, 149$. For the unbiased upper bound in theorem 2.2, change the classifier threshold to

$$t_{n,N} = \sqrt{\frac{1/4 + \log(N/n)}{n}} \qquad (n < N),$$

keeping the same dyadic checkpoints, strict crossing rule, negative final tie convention, shared-prefix increments, and independent level law $p_\ell = (1-r)r^\ell$, $r = 2^{-1/2}$. In this section $Z_N, \tau_N, \Delta_\ell, Y_L, \mathcal{W}_L$ denote this modified construction only. It observes no magnitude or sign and uses no distributional parameters.

Theorem 7.1 (Separate conditional moment bounds)**.** *For every $0 < d \le 1/2$, either sign, and finite L, $N = 2^L$,*

$$\mathbb{E}[Y_L^2 \mid d, S] \le 23 \min\{\sqrt{N}, d^{-1}\}, \qquad \mathbb{E}[\mathcal{W}_L \mid d, S] \le 31 \min\{\sqrt{N}, d^{-1}\}.$$

If $I_\mu < \infty$, the untruncated estimator is square-integrable, unbiased, and has finite expected actual work, with

$$\sup_q \mathbb{E}_{\mu,q} Y^2 \le 23 I_\mu, \qquad \sup_q \mathbb{E}_{\mu,q} \mathcal{W} \le 31 I_\mu.$$

Consequently the product of the separate variance and work suprema is at most $713 I_\mu^2$.

Error envelope. An incorrect early stop at prefix n requires a one-sided deviation of at least $d + t_{n,N}$. Hoeffding's inequality and a union bound over the fixed dyadic prefixes yield, with $x = Nd^2$,

$$e_N \le e^{-2x} + e^{-1/2}N^{-2} \sum_{n<N,\ n \text{ dyadic}} n^2 e^{-2nd^2} \le e^{-2x} + \tfrac{2}{3}e^{-1/2}x^{-2}.$$

For the second step integrate ue^{-bu} over the disjoint intervals $[n/2, n]$: each integral is at least $3n^2e^{-bn}/8$, so the sum is at most $8/(3b^2)$. Since $x^2e^{-2x} \le e^{-2}$ for every $x > 0$,

$$e_N \le \min\{1, Ax^{-2}\}, \qquad A = e^{-2} + \tfrac{2}{3}e^{-1/2} < \tfrac{1}{7} + \tfrac{5}{12} = \tfrac{47}{84} < \tfrac{3}{5}.$$

Here $e^2 > 7$ and $e^{1/2} > 8/5$ follow from their positive power series. For $\ell \ge 1$, a nonzero increment requires a fine or coarse error; the coarse scale has $x/2$. Hence

$$\mathbb{E}\Delta_\ell^2 \le \min\{1, 5Ax^{-2}\} \le \min\{1, 3x^{-2}\}. \tag{47}$$

The same envelope covers $\ell = 0$, because $d^2 \le 1/4$ and $\Delta_0 = Z_1 \in \{0, 1\}$.

Second moment. Put $d' = d/3^{1/4}$. Summing (47) divided by p_ℓ and splitting at $2^\ell(d')^2 = 1$ gives

$$\mathbb{E}Y_L^2 \le C_0 \min\{\sqrt{2^L}, 1/d'\}, \qquad C_0 = \frac{1}{1-r}\left(\frac{1}{1-r} + \frac{1}{1-r^3}\right) < 17.$$

Before the split the terms are geometric with size $\sqrt{2^\ell}$; after it their ratio is r^3. This includes cutoffs before the split. The rational comparison $r < 442/625$ and monotonicity of the displayed expression prove $C_0 < 17$ by integer arithmetic. Since $3^{1/4} < 4/3$, the conditional constant is less than $68/3 < 23$.

Expected work. Take $c = 13/7$. Survival at a dyadic prefix $n < N$ and $d > ct_{n,N}$ requires a one-sided deviation larger than $(1 - 1/c)d$. Therefore

$$\mathbb{P}(\tau_N > n) \le \mathbf{1}\{d \le ct_{n,N}\} + \exp\{-2(1 - 1/c)^2 nd^2\}.$$

Suppose $x = Nd^2 \ge 1$ and write $u = nd^2$. The indicator implies $u \le c^2[1/4 + \log(x/u)]$. If $u < 1$, then $u \le 1 + c^2 \log x$; if $u \ge 1$, use $\log u \ge 0$ and $c^2/4 = 169/196 < 1$ for the same conclusion. Its dyadic contribution is at most $2d^{-2}(1 + c^2 \log x)$. For $b > 0$, integration on $[n/2, n]$ gives $\sum ne^{-bnd^2} \le 2/(bd^2)$. The survival identity and $1 \le d^{-2}/4$ now give

$$\mathbb{E}\tau_N \le d^{-2}\left[\frac{125}{18} + \frac{338}{49}\log x\right] \le 7d^{-2}(1 + \log x).$$

Together with $\tau_N \le N$, this proves $\mathbb{E}\tau_N \le 7m_N(d)$.

On a shared stream the cap-N threshold is at least the cap-$N/2$ threshold at every prefix before $N/2$. If the fine classifier stops there, the coarse one has already stopped; at or after $N/2$ the coarse one has reached its cap. Thus $\tau_{N/2} \le \tau_N$ pathwise and the actual work of the coupled increment equals τ_N. Level zero costs one. It follows that

$$\mathbb{E}\mathcal{W}_L \le 7(1-r)\sum_{\ell=0}^{L} 2^{-\ell/2} m_{2^\ell}(d) \le 7\left(2 + \frac{\log 2}{1-r}\right)\min\{\sqrt{2^L}, d^{-1}\}.$$

To obtain the last inequality, split at $2^\ell d^2 = 1$ as in lemma 4.2. The first tail value satisfies $1 < 2^\ell d^2 \le 2$; the remaining sum is bounded by $d^{-1}\sum_{k\ge 0} r^k(1 + \log 2 + k \log 2)$. Using $r < 71/100$ and $\log 2 < 7/10$ bounds the final constant by $896/29 < 31$.

Untruncated limit and separate suprema. Integrate the conditional bounds over (D, S). Monotone convergence of the nonnegative moment and cost series gives finite second moment and work when $I_\mu < \infty$. More explicitly,

$$\mathbb{E}[(Y - Y_L)^2] = \sum_{\ell > L} \frac{\mathbb{E}\Delta_\ell^2}{p_\ell} \longrightarrow 0.$$

Thus $Y_L \to Y$ in L^2 and L^1. Finite telescoping gives $\mathbb{E}Y_L = \mathbb{E}Z_{2^L}$; the conditional error tends to zero for every $d > 0$, and bounded convergence proves $\mathbb{E}Y = \theta$. The bounds hold for every q with the same estimator and the same I_μ. Taking the two suprema separately and using $\operatorname{Var} Y \le \mathbb{E}Y^2$ proves theorem 7.1 and the improved upper constant in theorem 2.2. No interchange of a supremum with a strict inequality is needed.

Formal scope. `observable_713` realizes one geometric-seed estimator in the original observable adaptive experiment, before every gap law. Its separate extended variance and actual work bounds are $23I_\mu$ and $31I_\mu$, uniformly over every measurable sign law. The product of the two separate suprema is at most $713I_\mu^2$. Both classifiers read the same coin's observed prefix; exact execution identities transfer their output and flip count to the infinite experiment. Together with the converse, `observable_unbiased_iff` proves the full finite-variance/work existence characterization in this model.

The earlier, more general `full_bound_713` proves the untruncated estimator result in a conditional observation-kernel model, including bounded independent observations with means $1/2 \pm D$. `uniform_family_bounds` supplies the separate-supremum conclusion for a fixed kernel and estimator over a nonempty family of latent laws sharing I_μ. The finite-budget calibration wrapper for the original RMS law has its separate realization in section 18.

8 Consequences for power and logarithmic margins

Let $F(t) = \mu((0, t])$. Stieltjes integration by parts gives, for $0 < h < a$,

$$\omega_\mu(h) = F(h) + h \int_{(h,a]} \frac{1}{d}\, \mu(\mathrm{d}d) = \frac{h}{a} + h \int_h^a \frac{F(t)}{t^2}\, \mathrm{d}t. \tag{48}$$

The formula is valid with atoms. In particular,

$$I_\mu = \frac{1}{a} + \int_0^a \frac{F(t)}{t^2}\, \mathrm{d}t. \tag{49}$$

This integral, not a power exponent alone, determines exact unbiased feasibility.

8.1 The polynomial law as a corollary

For $F(t) = (t/a)^\alpha$ on $(0, a]$ and $u = h/a < 1$,

$$\omega_\mu(h) = \begin{cases} (\alpha u - u^\alpha)/(\alpha - 1), & \alpha \neq 1, \\ u(1 - \log u), & \alpha = 1. \end{cases} \tag{50}$$

Thus the main theorem immediately recovers the three regimes. The usual polynomial-margin class allows a more general constant:

$$\mathcal{G}_{\alpha,C} = \{G \text{ on } [0,1] : \mathbb{P}_G(|P - 1/2| \le t) \le Ct^\alpha, \quad 0 < t \le 1/2\}, \qquad C \ge 2^\alpha. \tag{51}$$

The last condition is necessary for nonemptiness.

Corollary 8.1 (Sharp polynomial-margin complexity)**.** *Fix $\alpha > 0$ and $C \geq 2^\alpha$. For all sufficiently small $\varepsilon > 0$ and $0 < \delta \leq 1/4$, the smallest uniform expected work for an (ε, δ)-valid estimator on $\mathcal{G}_{\alpha,C}$ is*

$$\asymp_{\alpha,C} \log(1/\delta) \begin{cases} \varepsilon^{-2/\alpha}, & 0 < \alpha < 1, \\ \varepsilon^{-2} \log^2(1/\varepsilon), & \alpha = 1, \\ \varepsilon^{-2}, & \alpha > 1. \end{cases} \tag{52}$$

There is an unbiased estimator with uniformly finite variance and expected work on this entire class if and only if $\alpha > 1$. One common estimator works for every such α, C, without knowing either parameter.

Proof. Let $\mu_{\alpha,C}$ have CDF $\min\{Ct^\alpha, 1\}$ on $(0, 1/2]$. It is supported on $(0, C^{-1/\alpha}]$, and $\mathcal{G}_{\alpha,C} = \mathcal{G}_{\preceq \mu_{\alpha,C}}$ exactly. Its modulus is (50) with a replaced by $C^{-1/\alpha}$. Apply theorem 2.1 and corollary 2.3 and invert the three orders, with constants depending on (α, C). The inverse moment is finite exactly when $\alpha > 1$, so theorem 2.2 gives both directions of the unbiased dichotomy. Every polynomial envelope also has a finite uniform logarithmic moment; thus the universal estimator is integrable and unbiased even for $\alpha \leq 1$, when finite uniform variance and work cannot both hold. □

8.2 Fixed-gap populations and rare-event scope

Corollary 8.2 (Gap classes)**.** *For $0 < \Delta \leq 1/2$, put $\mathcal{G}_\Delta = \{G : \mathbb{P}_G(|P - 1/2| < \Delta) = 0\}$. For sufficiently small ε and $0 < \delta \leq 1/4$, its minimax uniform expected work for (ε, δ) accuracy is*

$$\asymp \frac{\log(1/\delta)}{\varepsilon^2 \Delta^2},$$

with universal comparison constants and a universal small-accuracy cutoff.

Proof. Here $\mathcal{G}_\Delta = \mathcal{G}_{\preceq \delta_\Delta}$ and $\omega_{\delta_\Delta}(h) = \min\{1, h/\Delta\}$. The envelope confidence law gives the result by direct inversion, taking ε smaller than its universal lower comparison constant. □

This is the balanced-population specialization of Lee–Valiant [1]. Their finer dependence on the positive fraction ρ is important for rare events; it is outside our worst case over all q. Our hard pairs keep every sign prior within $1/8$ of $1/2$, so both targets are near $1/2$. The localized theory is developed in section 11, and section 13 characterizes it for every magnitude law.

8.3 New boundary regimes invisible to a single exponent

For $\beta \geq 0$, define an actual CDF on $(0, a]$ by

$$F_\beta(t) = \frac{t/a}{[1 + \log(a/t)]^\beta}. \tag{53}$$

It is increasing, vanishes at zero, and equals one at a. With $L_h = 1 + \log(a/h)$, its modulus is exactly

$$\omega_\beta(h) = \frac{h}{a} \begin{cases} 1 + \dfrac{L_h^{1-\beta} - 1}{1 - \beta}, & \beta \neq 1, \\ 1 + \log L_h, & \beta = 1, \end{cases} \qquad 0 < h < a. \tag{54}$$

This follows from (48) by the substitution $u = 1 + \log(a/t)$.

Magnitude law near zero	Optimal work at fixed confidence	Efficient unbiased estimator
$F(t) \asymp t^\alpha$, $\alpha < 1$	$\varepsilon^{-2/\alpha}$	No
$F(t) \asymp t$	$\varepsilon^{-2}\log^2(1/\varepsilon)$	No
$F(t) \asymp t^\alpha$, $\alpha > 1$	ε^{-2}	Yes
$F(t) \asymp t/[\log(1/t)]^\beta$, $\beta < 1$	$\varepsilon^{-2}[\log(1/\varepsilon)]^{2(1-\beta)}$	No
$F(t) \asymp t/\log(1/t)$	$\varepsilon^{-2}[\log\log(1/\varepsilon)]^2$	No
$F(t) \asymp t/[\log(1/t)]^\beta$, $\beta > 1$	ε^{-2}	Yes

Table 1: Consequences of the modulus law. "Efficient unbiased" means both variance and expected work bounded uniformly over the corresponding sign or envelope class, not a claim for every individual population. For high confidence, multiply work by $\log(1/\delta)$.

Corollary 8.3 (Logarithmically refined critical law)**.** *For the fixed magnitude law* (53)*, or its stochastic-dominance envelope class, the optimal expected work at small accuracy ε is, up to constants depending on β, a,*

$$\log(1/\delta) \times \begin{cases} \varepsilon^{-2}[\log(1/\varepsilon)]^{2(1-\beta)}, & 0 \le \beta < 1, \\ \varepsilon^{-2}[\log\log(1/\varepsilon)]^2, & \beta = 1, \\ \varepsilon^{-2}, & \beta > 1. \end{cases} \tag{55}$$

Uniformly finite variance and work for unbiased estimation are possible exactly when $\beta > 1$, in which case $\mathbb{E}D^{-1} = \beta/[a(\beta-1)]$.

Proof. Equation (54) and theorem 2.1 and corollary 2.3 give the error law. Its monotone inverse has the displayed orders: if $\omega(h) \asymp h[\log(1/h)]^\gamma$ with fixed $\gamma \ge 0$, then $h \asymp \varepsilon/[\log(1/\varepsilon)]^\gamma$; replacing the log by log log gives the boundary case. Finally,

$$\mathbb{E}D^{-1} = \frac{1}{a}\left(1 + \int_1^\infty u^{-\beta}\,\mathrm{d}u\right),$$

which is finite exactly when $\beta > 1$. Apply theorem 2.2. □

For $\beta > 1$, this class permits efficient unbiased estimation even though it satisfies no bound $F(t) \le Ct^\alpha$ with any $\alpha > 1$ and finite C. Indeed, $t^{1-\alpha}/[1+\log(a/t)]^\beta \to \infty$. The earlier exponent boundary is therefore a corollary about entire polynomial classes, not the general structural criterion.

An iterated-log example. The same calculation extends further. Set $L_1(t) = 1 + \log(a/t)$, $L_2(t) = 1 + \log L_1(t)$, and $F(t) = (t/a)/(L_1(t)L_2(t)^\gamma)$ with $\gamma \ge 0$. Then

$$\omega(h) = \frac{h}{a}\left(1 + \int_1^{L_2(h)} v^{-\gamma}\,\mathrm{d}v\right).$$

At $\gamma = 1$, the fixed-confidence work is $\asymp \varepsilon^{-2}[\log\log\log(1/\varepsilon)]^2$, for sufficiently small ε. Unbiased feasibility again changes at $\gamma > 1$. This is an immediate instance of the general integral criterion, not an additional assumption or an independently designed algorithm.

9 Persistence under fixed finite-order density smoothness

The arbitrary-sign model makes no smoothness assumption on G. We now show that the critical logarithmic penalty is not solely caused by a jump in the density at the threshold.

For an integer $s \ge 1$, write $\|f\|_{C^s} = \max_{0\le k\le s} \|f^{(k)}\|_\infty$.

Theorem 9.1 (Smooth-density robustness)**.** *For each fixed integer $s \geq 1$, there exists a finite M_s such that, for every fixed $M \geq M_s$, the class of C^∞ probability densities supported in $[1/4, 3/4]$ and satisfying $\|f\|_{C^s} \leq M$ has minimax expected complexity*

$$\asymp_{s,M} \varepsilon^{-2} \log^2(1/\varepsilon) \log(1/\delta)$$

for sufficiently small ε and $0 < \delta \leq 1/4$. No unbiased estimator has both uniformly finite variance and uniformly finite expected work on this class.

Proof. Fix the local cutoff $a_0 = 1/4$. Choose a fixed even C^∞ density g supported strictly inside $(-a_0, a_0)$, with $g \geq c_g > 0$ on $[-a_0/2, a_0/2]$. The centered coordinate is $x = p - 1/2$. For sufficiently small $h > 0$, set

$$t_h = h^{1/(s+1)}, \qquad f_\pm(1/2 + x) = g(x) \left(1 \pm \frac{2hx}{x^2 + t_h^2}\right). \tag{56}$$

The perturbation is odd, so both densities integrate to one. Its absolute size is at most $h/t_h = h^{s/(s+1)}$, which is less than $1/4$ for small h, so the densities are nonnegative. The common magnitude density is $2g(d)$ and the conditional sign perturbation is

$$\eta_h(d) = \frac{hd}{d^2 + t_h^2}, \qquad \eta_h(d) \leq \frac{h}{2t_h} \leq \frac{1}{8}, \qquad d\eta_h(d) \leq h.$$

The posterior lemma therefore gives predictive KL at most $(8192/225)h^2$ per observation, uniformly over histories and selected coins. The target separation satisfies

$$\begin{aligned} \theta_+ - \theta_- &= 4h \int_0^{a_0} g(d) \frac{d}{d^2 + t_h^2} \, \mathrm{d}d \\ &\geq 2c_g h \log \frac{(a_0/2)^2 + t_h^2}{t_h^2} \geq c_s h \log(1/h) \end{aligned}$$

for all sufficiently small h, with $c_s > 0$; for a fixed bump it can be chosen proportional to $(s+1)^{-1}$. This is not uniform as $s \to \infty$.

For $R_t(x) = x/(x^2 + t^2) = t^{-1} R_1(x/t)$,

$$\|h R_{t_h}^{(k)}\|_\infty \leq C_k h t_h^{-(k+1)} = C_k h^{(s-k)/(s+1)} \leq C_k, \qquad 0 \leq k \leq s.$$

Leibniz's rule bounds every $\|f_\pm\|_{C^s}$ by a fixed M_s independent of h. They are individually C^∞ because $t_h > 0$ and g is a smooth compactly supported function. Uniform bounds on derivatives of all orders at once are not asserted.

Choose h of order $\varepsilon / \log(1/\varepsilon)$ with a sufficiently large fixed factor so that the separation exceeds 2ε. The stopped-transcript testing argument proves the lower bound. For the upper bound, $\|f\|_\infty \leq M$ implies $\mathbb{P}(|P - 1/2| \leq t) \leq 2Mt$, so corollary 8.1 applies. The unbiased impossibility follows by averaging a putative uniform finite-variance, finite-work estimator and contradicting the lower bound at fixed confidence as $\varepsilon \downarrow 0$. □

Which smoothness is being controlled. This theorem concerns the density of the latent P. It is not a statement about smoothness of a regression function $y \mapsto \mathbb{E}[X \mid Y = y]$ in observable covariates. Smooth cross-state structure can enable information sharing, as in kernel methods [7]. Neither this theorem nor the next one rules out such improvements.

The construction does not settle analogous bounds for fixed analytic or Gevrey regularity classes; those require separate assumptions and arguments.

10 An observed-outer-state embedding

A latent coin and an observable outer state are different experiments. The following construction supplies a lower bound in the latter without silently identifying the two.

Let Y be uniform on $[0,1]$. The algorithm may observe independent outer states and may also select any state at which to request a fresh inner observation. Let $X \in \{-1,1\}$ and put $\theta = \mathbb{P}(\mathbb{E}[X \mid Y] > 0)$. There is no regularity assumption connecting the conditional means at different outer states.

Proposition 10.1 (Finite-cell observable-state lower bound)**.** *Fix a probability law μ on $(0,1/2]$, $0 < h \le 1/2$, and $0 < \delta \le 1/8$. There is a finite partition of $[0,1]$ and a family of deterministic, cellwise-constant conditional Bernoulli models such that: (a) their magnitude laws obey the envelope F_μ; (b) their construction has a scale separation $\Delta \ge \omega_\mu(h)/8$; and (c) every estimator valid to radius $\Delta/4$ and failure probability δ uniformly over this family has worst-case expected work at least*

$$\frac{225}{128h^2}\,\mathrm{kl}(1-2\delta, 2\delta). \tag{57}$$

This lower bound permits arbitrary selected-state queries and revisiting.

Proof. Round each $d \le h$ up to h. Round larger d upward to the next value in the grid $h, 2h, 4h, \ldots$, capped at a. This gives a finitely supported law μ_h on $\{d_1, \ldots, d_J\}$, with weights w_j, which is stochastically no smaller than μ. Above h the rounding increases a magnitude by at most a factor two; below h the clipped integrand stays one. Hence

$$\omega_{\mu_h}(h) \ge \tfrac{1}{2}\omega_\mu(h).$$

For each j, create M cells of lengths w_j/M and assign unknown fixed signs $s_{jk} \in \{0,1\}$. On cell (j,k) let

$$B \mid Y \sim \mathrm{Ber}\big(1/2 + (2s_{jk}-1)d_j\big), \qquad X = 2B - 1.$$

Every deterministic sign array has magnitude law μ_h, meets the envelope, and has target $\theta(s) = \sum_j (w_j/M) \sum_k s_{jk}$.

For the testing proof only, draw signs independently under two product priors with probabilities $1/2 \pm \eta_h(d_j)$ from (16). Their target means are separated by

$$\Delta = \tfrac{1}{4}\omega_{\mu_h}(h) \ge \tfrac{1}{8}\omega_\mu(h).$$

The sum of squared target weights is $\sum_j w_j^2/M \le 1/M$. Weighted Hoeffding therefore gives

$$\mathbb{P}_\pm(|\theta(s) - \mathbb{E}_\pm\theta(s)| > \Delta/8) \le 2\exp(-M\Delta^2/32).$$

Choose an integer $M \ge 32\Delta^{-2}\log(2/\delta)$. Uniform estimation error at most $\Delta/4$, together with the displayed concentration event, makes thresholding at $1/2$ a test with each error at most 2δ.

The signs are independent under either prior, so the local posterior lemma applies separately to every cell, regardless of the adaptively selected query points. Each inner observation has predictive KL at most $128h^2/225$. A uniform expected-work bound for deterministic sign arrays also bounds the prior-averaged expected work. Apply the KL chain rule and binary data processing to obtain (57). □

The partition size depends on the desired scale and confidence. This is a minimax construction in a class allowing arbitrarily fine finite partitions, not an asymptotic lower bound for one fixed finite list of cells. Since the outer distribution itself stays uniform on $[0,1]$, it establishes the relevant obstruction even with observable covariates, provided no cross-state regularity is imposed. The original upper algorithm can ignore the labels and use fresh outer states, so the envelope modulus remains the matching error scale for this unstructured class.

10.1 A continuous-magnitude subclass satisfying the nested assumptions

Proposition 10.2 (Continuous-magnitude observed-state obstruction)**.** *Fix the local cutoff* $a_0 = 1/4$. *Consider observed outer states* $U \sim \mathrm{Unif}[0,1]$ *and binary inner observations* $X \in \{-1,1\}$ *with*

$$|\mathbb{E}[X \mid U]| = 2D(U), \qquad D(U) \sim \mathrm{Unif}(0, a_0].$$

No cross-state regularity is assumed. There is a class of deterministic conditional models of this form whose minimax expected observation cost for error ε *with fixed failure probability* $1/16$ *is*

$$\asymp \varepsilon^{-2} \log^2(1/\varepsilon).$$

The lower bound allows arbitrary selected-state queries and revisiting. All models satisfy the bounded-density and normalized conditional-moment assumptions 2.1 and 2.4 of Giles–Haji-Ali [2], with constants independent of the error scale.

Proof. Fix $0 < h < a_0$ and partition $(0, a_0]$ into the first band $(0, h]$ and consecutive dyadic bands with right endpoints $2h, 4h, \ldots$, clipping the final endpoint at a_0. Let band j have endpoints ℓ_j, b_j and mass $w_j = (b_j - \ell_j)/a_0$. Partition $[0,1]$ into M equal-weight cells per band, each of length w_j/M. Within every cell in band j, let $D(U)$ increase linearly across $(\ell_j, b_j]$. Define its values at the finitely many endpoints to be positive and in the assigned band. Consequently the magnitude law is exactly uniform on $(0, a_0]$, not a discretization. The partition and the map D may be disclosed to the algorithm.

Assign each cell an unknown deterministic sign $s_{jk} \in \{0,1\}$. On that cell let $X = 2B - 1$ for $B \sim \mathrm{Ber}(1/2 + (2s_{jk} - 1)D(U))$. The target is

$$\theta(s) = \sum_j \frac{w_j}{M} \sum_{k=1}^{M} s_{jk}.$$

For the proof only, choose independent cell signs under two priors with probabilities $1/2 \pm \eta_j$, where $\eta_j = \frac{1}{8}\min\{1, h/b_j\}$. Their target means have separation

$$\Delta_h = 2\sum_j w_j \eta_j \geq \frac{1}{8}\omega_\mu(h), \qquad \mu = \mathrm{Unif}(0, a_0].$$

Indeed, within the first band the clipped integrand is one; within every other band $b_j \leq 2D(U)$. Every queried magnitude in band j satisfies $D(U)\eta_j \leq h/8$.

Repeated queries in a cell may use different magnitudes. Nevertheless, its sign has a single common likelihood ratio under the two priors, obtained by multiplying its queried sign-conditional likelihoods. The posterior identity and the predictive bound therefore still apply. Each observation contributes at most $128h^2/225$ KL, even under arbitrary query selection.

Weighted Hoeffding gives

$$\mathbb{P}_\pm\big(|\theta(s) - \mathbb{E}_\pm\theta(s)| > \Delta_h/8\big) \leq 2e^{-M\Delta_h^2/32}.$$

For $M \geq 32\Delta_h^{-2}\log(2/\delta)$, an estimator with uniform radius $\Delta_h/4$ and failure probability $\delta \leq 1/8$ gives a test of these priors with each error at most 2δ. The stopped-transcript argument gives expected work at least

$$\frac{225}{128h^2}\,\mathrm{kl}(1 - 2\delta, 2\delta).$$

For the uniform magnitude law, $\omega_\mu(h) = (h/a_0)[1 + \log(a_0/h)]$. Set h of order $\varepsilon/\log(1/\varepsilon)$ with a sufficiently large fixed constant to obtain the lower rate. The upper rate follows by ignoring state labels and using the preceding threshold estimator. The class permits arbitrarily many cells; it is not a fixed finite-partition asymptotic claim.

Finally, $\mathrm{Var}(X \mid U) = 1 - 4D(U)^2 \geq 3/4$. Their normalized magnitude is

$$Z = \frac{|\mathbb{E}[X \mid U]|}{\sqrt{\mathrm{Var}(X \mid U)}} = \frac{2D}{\sqrt{1 - 4D^2}}.$$

It has density $[2a_0(1 + z^2)^{3/2}]^{-1}$ on its support, bounded by $1/(2a_0)$. Moreover, $|X - \mathbb{E}[X \mid U]| \leq 3/2$, so for every finite $q > 2$ its standardized conditional qth absolute moment is bounded by $3^{q/2}$. These verify the two stated assumptions uniformly. A uniform RMS guarantee ε gives error at most 4ε with failure probability at most $1/16$, by Markov's inequality. Applying the fixed-confidence lower bound at 4ε yields the same RMS rate. The RMS upper bound follows from theorem 2.1. This also proves corollary 2.4. □

Interpretation. This identifies a hard subclass satisfying the explicit density and moment assumptions behind the classical adaptive nested-simulation upper rate. It does not rule out improvements from smooth cross-state regression structure, a known parametric model, or restrictions to rare-event populations. A restriction on the density of the latent magnitude is different from a restriction on the function of observed covariates.

11 Rare events and the inverse-square boundary

11.1 The rare-event experiment

The preceding minimax law takes a worst case over every sign function. For rare events, fix $0 < \rho \leq 1/4$ and restrict this class to

$$\mathcal{Q}_\rho(\mu) = \left\{ q : (0, 1/2] \to [0, 1] \text{ measurable} : \int q \, d\mu \leq \rho \right\}.$$

We write $\mathcal{Q}_\rho$ when μ is fixed. The unknown target is still $\theta(q) = \int q \, d\mu$; ρ is a known upper bound. For every $B > 0$ define

$$R_{\mu,\rho}(B) = \inf_{\mathcal{A}:\, \sup_{q \in \mathcal{Q}_\rho(\mu)} \mathbb{E}_q T_{\mathcal{A}} \leq B} \sup_{q \in \mathcal{Q}_\rho(\mu)} \left(\mathbb{E}_q(\widehat{\theta}_{\mathcal{A}} - \theta(q))^2\right)^{1/2}.$$

The observation model remains the ordinary iid fresh-coin experiment: each fresh unit samples a hidden magnitude $D \sim \mu$ and a hidden sign with conditional positive probability $q(D)$; its observations are iid Bernoulli with mean $1/2 \pm D$ according to the sign. Different units have independent latent states and observation tapes, and a unit keeps its latent state on every revisit. Individual magnitudes and signs are hidden; all creation, revisiting, interleaving, and stopping decisions use only an independent seed and the observed flips. Every requested observation costs one, including the first on a unit; units created but never observed have no statistical effect. Stopping may be unbounded, and finite expected work gives almost-sure termination under every admissible population.

The benchmark may know μ, ρ, B. In particular, the global constraint does not impose $q(d) \leq \rho$ pointwise, nor fix either conditional magnitude law given the sign. Estimates may be clipped to $[0, \rho]$ without increasing risk. Returning $\rho/2$ costs nothing and has RMS error at most $\rho/2$; randomization permits meaningful procedures even for $B < 1$. The power-law and critical results below are stated for $B \geq 1$, as in their sources; Sections 13 and 14 cover every

$B > 0$. Sections 13 and 15 and their appendices write the sign as $S \in \{0, 1\}$, with observation mean $1/2 + (2S - 1)D$; section 14 and appendix O write $S \in \{-1, +1\}$, with mean $1/2 + SD$.

Two different boundaries occur in this manuscript. The unrestricted-sign inverse-moment criterion changes at power exponent $\alpha = 1$; the rarity-dependent quadratic phenomenon below occurs at $\alpha = 2$.

11.2 A variational lower modulus and its exact optimizer

For $h > 0$ set

$$\Psi_{\mu,\rho}(h) = \sup_{\substack{v \text{ measurable, } 0 \le v \le 1 \\ \int v \, d\mu \le \rho}} \int \min\{v(d), h\sqrt{v(d)}/d\} \, \mu(dd). \tag{58}$$

There is a universal $c > 0$ such that $R_{\mu,\rho}(B) \ge c\Psi_{\mu,\rho}(B^{-1/2})$. The corresponding confidence lower bound replaces $B^{-1/2}$ by $\sqrt{\log(1/\delta)/B}$. The proof in appendix A explicitly imports the single-coin rare-prior Hellinger bound of Lee–Valiant [1, Corollary 19]; its bounded-ratio hybrid argument is written out. These statistical lower bounds are paper proofs with that published external input. The general-law lower proof in appendix E does not use that input: it derives the null-work consequence of Ψ directly in appendix H.

The variational optimization itself is fully explicit. There is a unique $t \in (0, 1/2)$ with

$$\mathbb{E}_\mu \min\{1, t^2/D^2\} = \rho.$$

For every $h > 0$, $v_*(d) = \min\{1, t^2/d^2\}$ is an optimizer. For $h \ge t$ the value is ρ; for $h < t$ it is

$$F_\mu(h) + h\mathbb{E}[D^{-1};\, h < D \le t] + ht\mathbb{E}[D^{-2};\, D > t]. \tag{59}$$

Atoms and singular laws are allowed, with no inverse-moment assumption. The Lean theorem `arbitrary_law_optimizer` proves this functional statement. It does not supply an attaining statistical algorithm. In particular,

$$\Psi_{\delta_a,\rho}(h) = \min\{\rho, h\sqrt{\rho}/a\}, \qquad \rho\omega_\mu(h/\sqrt{\rho}) \le \Psi_{\mu,\rho}(h) \le \omega_\mu(h).$$

11.3 The power-margin regimes

Let $\mu_{\alpha,a}$ have distribution function $(d/a)^\alpha$ on $(0, a]$ and put $x = (a\sqrt{B})^{-1}$. The proved RMS orders are:

Magnitude law	RMS order, with saturation included
$0 < \alpha < 1$	$\min\{\rho, x^\alpha\}$
$\alpha = 1$	$\min\{\rho, x[1 + \log_+(\rho/x)]\}$
$1 < \alpha < 2$	$\min\{\rho, \rho^{\,1-1/\alpha} x\}$
$\alpha = 2$, fixed $a = 1/4$	$\min\{\rho, \sqrt{\rho L(1 + \log L)/B}\}$
$\alpha > 2$	$\min\{\rho, \sqrt{\rho}\, x\}$

Here $L = 1 + \log(1/\rho)$. For the rows below two, $0 < a \le 1/4$; for the row above two, $0 < a \le 1/2$. Constants may depend on the fixed exponent. The quadratic row concerns one normalized law, with universal constants; it does not claim a theorem uniform in α near two. All rows hold for $B \ge 1$ and $0 < \rho \le 1/4$ in their stated domains.

For $0 < \alpha < 2$, appendix A gives full proofs, including confidence radii after replacing x by $a^{-1}\sqrt{\log(1/\delta)/B}$. The lower bound restricts the balanced hard alternatives to a region of magnitude mass ρ and does not use the imported rare-prior theorem. The upper bound starts the randomized telescope at a large enough cap that its base false-positive probability is at most ρ, then uses the same two-success calibration as section 5. It knows (α, a, ρ). It also holds under the envelope $F_\mu(d) \le \min\{(d/a)^\alpha, 1\}$.

For $\alpha > 2$, appendix B supplies an asymmetric sequential classifier: its negative-coin error is at most $\rho\varepsilon$, while its positive-coin error is at most ε. A randomized telescope then yields a one-coin unbiased output with separate bounds $\sup_q \mathbb{E}Y^2 \le (153/56)\rho$ and $\sup_q \mathbb{E}T_Y \le w_\alpha/a^2$, where $w_\alpha < \infty$ is explicit there. Independent Poisson replication calibrates every real budget, including expected replication counts below one. The upper bound is uniform over the same envelope class and knows (α, a, ρ, B). The matching exact-power lower bound uses the published fixed-gap Lee–Valiant theorem through a simulation that keeps a coin's privately sampled magnitude fixed on every revisit. Constants diverge as $\alpha \downarrow 2$. No sharp confidence characterization above or at two is asserted in this manuscript.

Both bounds of the row above two are also Lean-checked in the observable model, by a different construction: `above_two_theory` combines a single-coin family that depends only on (a, ρ), is unbiased on every power-margin law, and has second moment at most 28ρ, with fixed-count averaging and a lower bound with coefficient $1/256$ for every law supported below a. The formal lower proof does not use the Lee–Valiant input. See section 18 for the exact scope.

Remark 11.1 (Agreement with the closed quantile form)**.** Each row also follows, with constants depending on α, from Corollary 14.6, which states $R_{\mu,\rho}(B) \asymp \mathcal{Q}(B)$ for every law. Write $h = B^{-1/2} = ax$. The quantile point of $\mu_{\alpha,a}$ is $t = a\rho^{1/\alpha}$. If $h \ge t$, then $\mathcal{Q} = \rho$ and every row is at least a constant multiple of ρ. If $h < t$, substituting $d = as$ gives

$$T_h = x^\alpha + \alpha x \int_x^{\rho^{1/\alpha}} s^{\alpha-2}\, ds,$$

which is comparable to x^α for $\alpha < 1$, equals $x[1 + \log(\rho/x)]$ for $\alpha = 1$, and is at most a constant multiple of $x\rho^{1-1/\alpha}$ for $\alpha > 1$. The dyadic inverse-square costs of $\kappa = \mu|_{[t,a]}$ decrease geometrically away from t when $\alpha < 2$ and away from a when $\alpha > 2$, so their entropy is bounded and Γ_κ is comparable to $a^{-2}\rho^{1-2/\alpha}$ and a^{-2}, respectively. At $\alpha = 2$ the order-L bands meeting $[t, a]$ carry comparable costs, so $\Gamma_\kappa \asymp a^{-2}L(1 + \log L)$. Hence $h\sqrt{\rho\Gamma_\kappa}$ is comparable to $x\rho^{1-1/\alpha}$, $x\sqrt{\rho L(1 + \log L)}$ or $x\sqrt{\rho}$. Comparing the two terms of $\mathcal{Q}$ recovers each row; below exponent one, $x\rho^{1-1/\alpha} \le x^\alpha$ because $x < \rho^{1/\alpha}$. This is a consistency check between derivations, not an additional verification: the corollary rests on the paper proof of Theorem 13.2.

11.4 The full critical theorem and the correction to Ψ

Theorem 11.2 (Critical quadratic law)**.** *Fix $\mu(dd) = 32d\,dd$ on $(0, 1/4]$. There are universal $0 < c < C < \infty$ such that for all $0 < \rho \le 1/4$ and $B \ge 1$,*

$$c\min\{\rho, \sqrt{\rho LH/B}\} \le R_{\mu,\rho}(B) \le C\min\{\rho, \sqrt{\rho LH/B}\}, \qquad L = 1+\log(1/\rho), \quad H = 1+\log L.$$

The lower bound covers every observable randomized population algorithm, including nonlinear outputs, unbounded stopping, and arbitrary creation and interleaving. The upper construction may know μ, ρ, B.

The complete paper argument is in appendix C. Both bounds are Lean-checked in the original observable experiment:

`critical_minimax_lower_bound` and `critical_minimax_upper_bound`. The formal upper uses $C = 100000$ and an alternative observable construction; see section 18 for its exact scope.

The proof has three substantive transitions. First, sequential testing at many possible gaps forces a scale-entropy cost. Conditional moments, activation normalization, and repeated runs on unused flips of the *same* hidden coin transfer that fact to a small-bias one-coin estimator: its variance supremum times its expected work under the all-negative population is at least $c_0\rho LH$. Second, a finite-experiment RMS comparison and compact convex minimax argument obtain *one* population pair for every capped single-coin policy. A policy-dependent hard pair would not suffice. Third, a finite-prefix midpoint decomposition transfers this pair to interacting population algorithms, charging each flip once. The same pair works at every cap; finite expected work under both endpoints justifies passage to terminal outputs. The longer proof, including the upper construction, is in appendix C.

Relation to gap entropy. The sign-testing mechanism follows the line from Farrell [21] and Chen–Li [22] to the distributional gap-entropy bound of Chen–Li–Qiao [19, Lemma 4.1 and Remark D.4], which includes a Bernoulli extension. The finite-gap proof in appendix C supplies the required one-sided cost bound at fixed error without fair-coin termination. Agrawal–Ram–Ramdas [23] give general composite-test bounds in the Farrell regime. The recent Yao–Zhao–Zhou result [24] resolves the gap-entropy conjecture for Gaussian best-arm identification and reports Lean proofs; its target and observation model differ from the population problem here.

At the quadratic law, annular mass of order 16^{-j} balances testing cost of order 16^j. Across order L usable scales, comparable normalized weights have entropy of order $\log L$. This explains H; it does not supply a general-law theorem; section 13 obtains one by a different argument. The contribution being assessed is the null-cost obstruction and its transfer through one common hard pair to every adaptive population algorithm. No novelty is claimed for the underlying sign-entropy mechanism.

For this fixed law, the optimizer identity is

$$\rho = (t/a)^2[1+2\log(a/t)], \qquad \Psi_{\mu,\rho}(h) = \begin{cases} [2ht(1+\log(a/t)) - h^2]/a^2, & 0 < h < t, \\ \rho, & h \geq t, \end{cases} \qquad a = 1/4.$$

The exact formula is proved by `critical_psi_formula` for the actual supremum over all measurable feasible allocations. Moreover, `critical_psi_rate` proves, for $h > 0$,

$$\min\{\rho, h\sqrt{\rho L}\} \leq \Psi_{\mu,\rho}(h) \leq 16\min\{\rho, h\sqrt{\rho L}\}.$$

Thus $\Psi_{\mu,\rho}(B^{-1/2}) \asymp \min\{\rho, \sqrt{\rho L/B}\}$. For the paper's explicit sequence, taking $\rho_n = \exp(-\exp n)$ and $B_n = \rho_n^{-2}$ gives

$$\frac{R_{\mu,\rho_n}(B_n)}{\Psi_{\mu,\rho_n}(B_n^{-1/2})} \geq c'\sqrt{1+\log(1+\log(1/\rho_n))} \longrightarrow \infty.$$

Thus the proposed universal matching-Ψ upper bound is false according to theorem 11.2, even for one known law and known rarity. Independently of that particular sequence, `critical_matching_psi_refuted` proves that for every $C > 0$ there are $0 < \rho \leq 1/4$ and $B \geq 1$ with $C\Psi_{\mu,\rho}(B^{-1/2}) < R_{\mu,\rho}(B)$. Its formal witness uses $\rho = \exp(-\exp M)$ and $B = LH/\rho$; the preceding $B = \rho^{-2}$ sequence remains a paper derivation. The divergence is slow: $\sqrt{H} < 2.54$ for $\rho \geq 10^{-100}$ in the stated rarity range. Together with the loose proof constants, this precludes inferring a practically visible separation from the asymptotic claim alone. The variational optimizer remains valid. The sharp replacement for arbitrary laws, Theorem 13.2, accounts for the additional statistical difficulty across scales through cost entropy.

11.5 The formal critical lower bound

The formal lower proof starts with the following statistical finite-gap result. Let $0 < d_j \le 1/128$, $d_j/d_{j+1} \ge 2$, and $\log K \ge 32$. One measurable test with an independent seed and observable stopping time T is run on iid Bernoulli flips of mean $1/2 \pm d_j$. If it has finite expected work and error at most $1/10$ under both signs for every gap, then

$$\sum_{j=1}^{K} d_j^2 \mathbb{E}_{-d_j} T \ge \frac{K \log K}{512}.$$

`finite_gap_sign_entropy` derives this from the actual infinite Bernoulli experiment. It assumes neither a KL bound nor a packing estimate nor termination under the fair coin. This proves the finite-gap ingredient in lemma C.3.

`critical_base_cost` now proves the single-coin bound in theorem C.2: for a universal $c > 0$,

$$V\mathbb{E}_0 T \ge c\rho L(1 + \log L), \qquad 0 < \rho \le 1/4.$$

The same observable procedure and independent seed law serve every measurable prior $q \le 1/2$ with $\int q\, d\mu \le \rho$. Its uniform bias is at most $\rho/64$, its extended variance is bounded by V, and expected work is finite under each such prior. The proof derives square integrability, positive activation, the conditioned procedure, good gaps, iid restarts, and a sufficient number of scales. A raw second-moment version is also proved. Neither conditional moment bounds nor an information inequality is assumed.

The proof then constructs the compact population set, realizes finite policy mixtures, and obtains one separated pair valid for every capped policy. Exact finite-prefix simulation laws and work charging transfer the pair to arbitrary population seeds, coin creation, revisiting and interleaving. Finite expected work under each endpoint yields termination and the terminal affinity bound without a continuity assumption on the output. Thus `critical_minimax_lower_bound` proves the displayed lower rate for the full $[0, 1]$-valued rare-prior class, including saturation. Separate zero-work results prove that the budget class is nonempty and its minimax risk is at most ρ. The matching upper and explicit Ψ comparison have the separate formal proofs cited above; their constructions are described in section 18.

12 Critical sign entropy for every finite gap count

This extension supplements the preserved critical-law proof. For the same observable $1/10$-correct sign test and separated gaps $0 < d_j \le 1/128$, with finite expected work under both signs, every integer $K \ge 1$ satisfies

$$\sum_{j=1}^{K} d_j^2 \mathbb{E}_{-d_j} T \ge \frac{K \log K}{4096}.$$

The existing coefficient $1/512$ still applies when $\log K \ge 32$. Thus the restriction on the number of gaps is removed with a uniform coefficient eight times smaller. The all-K extension with coefficient $1/512$ is not established here.

The finite-prefix floor. Fix one gap d and put $n = \lfloor 1/(32d^2) \rfloor$. The negative and positive laws of the seed and first n flips have divergence at most

$$\frac{4nd^2}{(1/2 + d)(1/2 - d)} \le \frac{8}{15},$$

since $d \le 1/128 < 1/8$. Let E be the event that the test stops by time n and returns minus. Its positive-law probability is at most $1/10$. If its negative-law probability exceeded $5/8$, binary data processing and Pinsker would give divergence at least $2(5/8 - 1/10)^2 = 441/800 > 8/15$. Consequently

$$\mathbb{P}_{-d}(T > n) \ge 1 - 1/10 - 5/8 = 11/40, \qquad d^2\mathbb{E}_{-d}T \ge \frac{11}{40}\left(\frac{1}{32} - \frac{1}{16384}\right) > \frac{1}{128}.$$

Only finite expected work under the negative law is used for this floor; fair-coin termination is unnecessary. Summing gives $K/128$. For $\log K \le 32$ this dominates $K \log K/4096$; for $\log K \ge 32$ the original entropy theorem (BC2 in the preserved critical-law argument) supplies the stronger coefficient $1/512$.

12.1 Explicit base costs for the same single-coin model

Write $L = 1+\log(1/\rho)$, $H = 1+\log L$, and $W_0 = \mathbb{E}_0 T$. For any observable single-coin estimator with uniform bias at most $\rho/64$ on $\mathcal{X}_\rho$, finite work under each prior, and a uniform raw second moment bound M, the new proof gives

$$MW_0 \ge 10^{-10}\rho LH.$$

If instead V bounds the extended variance uniformly, it gives

$$VW_0 \ge \frac{\rho LH}{15 \cdot 10^9}.$$

These statements retain the original quantifiers over seed spaces, observable procedures, and priors; conditional moments and repeated sign tests are constructed inside the proof.

Here is the constant calculation. Put $P = MW_0$. Activation still gives $P \ge \rho/6$. The all-size entropy inequality and the existing good-gap selection give, whenever all selected annuli have mass at least $560P$,

$$\rho K \log K \le (21004288/3)P.$$

For $t = \log(1/\rho) \ge 16384$ choose $K = \lfloor t/(8\log 16)\rfloor$. If $P \ge e^{-t/2}$, the existing exponential bound already gives $P \ge \rho LH$. Otherwise the annulus mass condition holds and $LH \le 256\log 16\, K \log K$. Since $256\log 16\,(21004288/3) < 10^{10}$, this proves the raw-moment claim. For $t < 16384$, $LH \le 16385^2 < 10^{10}/6$, so the activation floor suffices. The variance claim follows from the existing bound $M \le 3V/2$ in the uniform-bias reduction.

The earlier constructive proof used $t_0 = 8192 + 16\log 16 + (16\log 16)^2 + 16\log 16\, e^{32}$ and a raw-moment constant about $1.29 \cdot 10^{-18}$. The new explicit constant is about $7.7 \cdot 10^7$ times larger. This numerical comparison concerns the constants produced by these proofs, not the optimal constant. No explicit constant is propagated through the common-pair and population-minimax steps here, and no practical improvement for an estimator is inferred.

The separate `CriticalAllSizes` Lean extension checks the all-size sign and estimator statements and both explicit base-cost statements. Its target and dependency snapshot was made during implementation; it does not constitute an independent specification or independent review. The recorded audit checks four exact targets and all 16 declarations in eight new modules.

13 The entropy–tail law for every magnitude law

The variational modulus Ψ of section 11 is a valid lower bound, but theorem 11.2 shows that it cannot be matched in general. This section states the replacement. The experiment is that

of section 11.1, with every budget $B > 0$ allowed. The upper procedure may know μ, ρ, B; magnitudes and signs remain hidden. No density, regular variation, positive lower support bound, finite inverse moment, or finite full-law entropy is assumed.

This section defines a scalar characterization that separates the part of the law handled by rare-event scale entropy from a part handled by the ordinary inverse-gap cost and a part left unresolved at the requested accuracy. Its observable upper bound is proved in Appendix D and its matching lower bound in Appendix E. The decomposition is analytical: the algorithm never observes a coin's component label.

13.1 A homogeneous entropy and a selection functional

For a finite submeasure $\sigma \le \mu$, use the canonical bands $I_j = (b_j/2, b_j]$, $b_j = 2^{-(j+1)}$, and put

$$a_j(\sigma) = \int_{I_j} d^{-2}\, \sigma(dd), \qquad A_\sigma = \sum_j a_j(\sigma).$$

For a finite nonnegative vector a, define

$$\mathcal{G}(a) = A + \sum_j a_j \log(A/a_j), \qquad A = \sum_j a_j, \qquad \mathcal{G}(0) = 0.$$

Zero coordinates contribute zero. The functional $\mathcal{G}$ is concave, coordinatewise increasing, homogeneous of degree one, and superadditive. Indeed, on positive coordinates its derivative is $1 + \log(A/a_j)$ and its Hessian is $A^{-1}\mathbf{1}\mathbf{1}^\top - \operatorname{diag}(a_j^{-1})$, which is negative semidefinite by Cauchy–Schwarz. Continuity gives the boundary cases, and concavity plus homogeneity gives superadditivity. Define

$$\Gamma_\sigma = \sup_J \mathcal{G}(a_0(\sigma), \dots, a_J(\sigma)). \tag{60}$$

Thus $\Gamma_0 = 0$, and, when $0 < A_\sigma < \infty$, $\Gamma_\sigma = A_\sigma[1 + \operatorname{Ent}((a_j/A_\sigma)_j)]$, with extended-value interpretation. Also write $H_\tau = \int d^{-1}\, \tau(dd)$ for any finite submeasure τ.

Let $h = B^{-1/2}$ and define

$$\boxed{\mathcal{K}_{\mu,\rho}(B) = \min\left\{\rho, \inf_{0 \le \sigma \le \mu}\left[h\sqrt{\rho\Gamma_\sigma} + \int \min\{1, h/d\}\, d(\mu - \sigma)\right]\right\}.} \tag{61}$$

Equivalently,

$$\mathcal{K}_{\mu,\rho}(B) = \min\left\{\rho, \inf_{\sigma+\tau+\zeta=\mu}\left[\zeta((0,1/2]) + \frac{\sqrt{\rho\Gamma_\sigma} + H_\tau}{\sqrt{B}}\right]\right\}. \tag{62}$$

Indeed, at fixed σ, assign each remaining magnitude to the cheaper of the costs 1 and h/d.

Lemma 13.1 (Hard assignments suffice)**.** *The infimum in* (61) *is unchanged if $\sigma = \mu|S$ for a measurable set S. In particular, for a finite atomic law a minimizing assignment can put each atom entirely in one of the three pieces in* (62)*.*

Proof. Write $f = d\sigma/d\mu \in [0,1]$ and $S_t = \{f > t\}$, so $\sigma = \int_0^1 \mu|S_t\, dt$. Concavity of the square root of every finite-fragment entropy gives

$$\sqrt{\Gamma_\sigma} \ge \int_0^1 \sqrt{\Gamma_{\mu|S_t}}\, dt$$

after monotone convergence in the fragment size. For a finite-objective input the relevant entropies are finite for $t > 0$, since $\mu|S_t \le \sigma/t$ and $\Gamma_{\mu|S_t} \le \Gamma_\sigma/t$. The remainder integral is linear, so the objective at σ is at least the average of the hard-set objectives. Their infimum is therefore no larger. The reverse inequality is immediate. For a finite law the minimum is attained among the finitely many assignments. This is a selection problem, not a claim of convex minimization over its assignments. □

The definition immediately implies

$$\mathcal{K}_{\mu,\rho}(B) \le \min\{\rho, \omega_\mu(B^{-1/2}), \sqrt{\rho\Gamma_\mu/B}\}, \qquad \omega_\mu(h) = \int \min\{1, h/d\}\, \mu(dd).$$

Infinite quantities are allowed in this display.

Theorem 13.2 (An entropy–tail law for the full joint problem)**.** *There are universal constants $0 < c < C < \infty$ such that, for every probability measure μ on $(0, 1/2]$, every $0 < \rho \le 1/4$, and every $B > 0$,*

$$\boxed{c\,\mathcal{K}_{\mu,\rho}(B) \le R_{\mu,\rho}(B) \le C\,\mathcal{K}_{\mu,\rho}(B).} \tag{63}$$

The functional has the equivalent selection and three-piece forms (61)–(62)*, and hard measurable assignments suffice. No density, regular variation, finite inverse moment, or finite full-law entropy is required.*

The upper half is Theorem 13.3; the lower half is Theorem E.1. The comparison constants in (63) are universal but conservative. They are not the factor sixteen proved for the finite convex program; these are different representations and different comparison bounds. The proof also gives the explicit conservative constants in (128).

Theorem 13.3 (Three-piece observable upper bound)**.** *There is a universal constant C such that, for every magnitude law on $(0, 1/2]$, every $0 < \rho \le 1/4$, and every $B > 0$,*

$$R_{\mu,\rho}(B) \le C\mathcal{K}_{\mu,\rho}(B). \tag{64}$$

No component label or magnitude is revealed to the algorithm.

13.2 Proof structure

The upper construction in appendix D races a likelihood bank against an ordinary capped classifier on independent observation streams of the same latent unit. Its null likelihood uses the full law μ. A mixed-work bound and finite Poisson multilevel allocation give the three-piece upper bound.

The lower proof in appendix E begins with an optimal finite-band selection. Its ordinary and unresolved pieces are controlled by a variational null-work modulus. Optimality constrains the gaps of the retained bands. Grouping comparable costs and trimming a controlled number of the smallest gaps produces annular components with admissible conditional rarities and active mass-saturation constraints. Allocating rarity proportionally to retained entropy makes their lower bounds add at every budget. Finite-prefix approximation covers laws accumulating at zero.

All specialized ingredients are proved in appendices F to H: the ordinary classifier, weighted likelihood bank, homogeneous work sum, multilevel allocation, weighted sign entropy, scale-sensitive gate, null-work transfer, and additive localization. Standard concentration, minimax, and convex-duality tools are cited where used. Section 16 distinguishes these ingredients and their antecedents from the contribution of the joint selection law.

Status. The upper direction now has a Lean proof in the original observable experiment, with coefficient 28000 and no inverse-moment or finite-support assumption. The matching statistical lower direction of Theorem 13.2 remains a paper proof, so the two-sided characterization is not yet Lean-checked. The standalone manuscript and its adversarial AI audits are preserved. Exact rational checks certify only their finite numerical instances. There has been no external specialist review. See section 18 for exact proof links and scope.

14 The closed quantile form

The selection in (61) ranges over submeasures of μ. This section shows that one explicit choice is optimal up to the factor $1+2\sqrt{2}$: remove exactly the smallest-gap mass ρ, keep its truncated inverse-gap cost, and charge the remaining core its cost entropy. For finite laws, Theorem 14.1 proves the resulting closed formula with explicit constants, by an argument that does not invoke Theorem 13.2. Corollary 14.6 extends the formula to every law through that theorem.

The direct lower proof in appendix O.1 first obtains $R(B) \geq T_h/32$ by a posterior-martingale calculation and an adaptive midpoint decomposition. It then filters the dyadic core costs. Either the tail already controls the desired rate, or the filter preserves a constant fraction of the core entropy. Grouping and trimming the surviving bands produces conditional experiments that satisfy the required rarity and saturation constraints; their lower bounds add through a common-pair argument. The upper bound in appendix O.2 applies the construction of appendix D to the quantile split as a known design choice. Appendix O.3 assembles all branches and tracks the constants. Both directions use the single-unit and transfer results of appendices F to H; no statistical selection theorem is assumed.

Status. The deterministic comparison in section 14.6, including fractional boundary atoms, is now Lean-checked for arbitrary probability laws. The original observable minimax risk satisfies $R \leq 30001Q$ in Lean on that same domain. The matching statistical lower bound and the direct finite-law lower argument remain paper proofs; thus Theorem 14.1 is not yet fully Lean-checked. The precursor received a fresh-context AI review, and an external adversarial review supplied the deterministic comparison. There has been no external specialist review. See section 18.

14.1 Persistent coins and the observation budget

This is the experiment of Section 11.1 for a finite law. Signs are written $S \in \{-1, +1\}$ in this section and in Appendix O.

Let

$$\mu = \sum_{i=1}^{n} w_i \delta_{d_i}, \qquad 0 < d_1 < \cdots < d_n \leq \tfrac{1}{2}, \qquad w_i > 0, \quad \sum_{i=1}^{n} w_i = 1,$$

be a known finite probability law. Equal gaps are merged before using this representation. A parameter is a function q on the support of μ. The complete latent pairs (D_k, S_k) of distinct fresh coins are iid, with $D_k \sim \mu$, $S_k \in \{-1, +1\}$, and $\mathbb{P}_q(S_k = +1 \mid D_k = d) = q(d)$. Each coin retains its pair on every revisit. Conditional on all latent pairs, the observation tapes of different coins are independent, and tape k consists of iid Bernoulli$(1/2 + S_k D_k)$ observations. Neither D_k nor S_k is observed.

Fix a known rarity bound $0 < \rho \leq 1/4$. The parameter class and target are

$$\mathcal{Q}_\rho = \left\{ q : \ 0 \leq q(d_i) \leq 1, \quad \theta(q) := \sum_{i=1}^{n} w_i q(d_i) \leq \rho \right\}. \tag{65}$$

This is a global constraint on the positive-sign fraction; it does not impose $q(d_i) \leq \rho$ at each gap.

An algorithm may create, revisit and interleave coins, and use an independent random seed whose distribution does not depend on q. Its choices and output are measurable functions of that seed, coin identities and purchased observations. It must terminate and return a real estimate $\widehat{\theta}$ almost surely under every $q \in \mathcal{Q}_\rho$. Each observed flip costs one, including the

first observation of a fresh coin. Let N be the total number of observations. Coin creation, arithmetic and random-number generation are not charged; creating an unobserved coin yields no information. Zero observations are allowed, and no deterministic total cap is imposed.

For a known budget $B > 0$, define the minimax root-mean-square error

$$R_{\mu,\rho}(B) = \inf_{\mathcal{A}:\ \sup_{q\in\mathcal{Q}_\rho}\mathbb{E}_q N\le B}\ \sup_{q\in\mathcal{Q}_\rho}\left\{\mathbb{E}_q(\widehat{\theta}-\theta(q))^2\right\}^{1/2}. \tag{66}$$

The budget is uniform over the entire parameter class. A procedure may depend on (μ,ρ,B), but not on q or any genuine coin's latent state. Projecting its estimate onto $[0,\rho]$ cannot increase loss or work. We abbreviate $R_{\mu,\rho}(B)$ to $R(B)$ when the inputs are fixed.

14.2 The closed quantile functional

Remove exactly the mass ρ with smallest gaps. In terms of the weights, set

$$\tau_i = \min\left\{w_i, \left(\rho - \sum_{\ell<i} w_\ell\right)_+\right\}, \qquad \nu = \sum_i \tau_i\delta_{d_i}, \qquad \kappa = \mu - \nu. \tag{67}$$

Here $x_+ = \max\{x,0\}$. Let t be the smallest support point for which $\mu((0,t]) \ge \rho$. Then ν has mass ρ and is supported on $(0,t]$, while κ has mass $1-\rho$ and is supported on $[t,1/2]$. The atom at t may be split fractionally.

For $j = 0,1,2,\ldots$, define the dyadic bands and exact core costs by

$$\begin{gathered} b_j = 2^{-j-1}, \qquad I_j = (b_j/2, b_j], \qquad a_j^* = \sum_{i:d_i\in I_j}\frac{w_i-\tau_i}{d_i^2}, \qquad A^* = \sum_j a_j^*, \\ \Gamma_\kappa = A^* + \sum_{j:a_j^*>0} a_j^*\log(A^*/a_j^*). \end{gathered} \tag{68}$$

All logarithms are natural. There are finitely many positive costs, and $A^* > 0$. Thus $\Gamma_\kappa = A^*(1+\mathrm{Ent}(a^*/A^*))$ is the inverse-square cost of the core multiplied by one plus the entropy of its distribution across dyadic scales.

With $h = B^{-1/2}$, set

$$T_h = \sum_i \tau_i \min\{1, h/d_i\}, \qquad \mathcal{Q}(B) = \min\left\{\rho,\ T_h + h\sqrt{\rho\Gamma_\kappa}\right\}. \tag{69}$$

The expression depends only on the known weights, gaps, rarity and budget. No optimization over subsets or submeasures is needed to evaluate it.

14.3 The theorem and its quantifiers

Theorem 14.1 (Closed quantile characterization)**.** *For every integer $n \ge 1$, every finite law μ described above, every $0 < \rho \le 1/4$, and every $B > 0$,*

$$10^{-20}\mathcal{Q}(B) \ \le\ R_{\mu,\rho}(B)\ \le\ 30001\mathcal{Q}(B). \tag{70}$$

More precisely, for each (μ,ρ,B) there is one observable algorithm satisfying, simultaneously for all $q \in \mathcal{Q}_\rho$,

$$\mathbb{E}_q N \le B, \qquad \left\{\mathbb{E}_q(\widehat{\theta}-\theta(q))^2\right\}^{1/2} \le 30001\mathcal{Q}(B).$$

Conversely, every algorithm with $\sup_{q\in\mathcal{Q}_\rho}\mathbb{E}_q N \le B$ has worst-case RMS error at least $10^{-20}\mathcal{Q}(B)$, including algorithms with arbitrary adaptive interleaving, nonlinear outputs and unbounded stopping.

The constants are universal and are not optimized. The claim is a comparison up to constants, not an exact formula for the minimax coefficient. The iid assumption concerns the complete latent pairs, not merely their magnitudes. The theorem concerns known finite laws and expected observation budgets; other information structures and budget conventions are not included.

14.4 Notation and testing conventions

Write $s = \mathbf{1}_{\{S=+1\}} \in \{0,1\}$ for a binary sign label; thus the conditional observation mean is $1/2 + (2s-1)d$. A single-unit procedure means a procedure on one genuine coin, with its latent state retained throughout. A null prior is $q \equiv 0$; $\mathbb{E}_0$ denotes expectation at that prior. When the two signs are compared at one specified gap, $\mathbb{E}_{-d}$ and $\mathbb{E}_{+d}$ refer to Bernoulli means $1/2 - d$ and $1/2 + d$, respectively. Fair-coin expectations are explicitly labelled $\mathbb{E}_{\text{fair}}$.

For probability laws P, Q, dominated by a common measure, write

$$\mathrm{Aff}(P,Q) = \int \sqrt{dP\,dQ}, \qquad \mathfrak{h}(P,Q) = 1 - \mathrm{Aff}(P,Q), \qquad \mathrm{TV}(P,Q) = \sup_E |P(E) - Q(E)|.$$

In particular, $\mathfrak{h}$ denotes squared Hellinger distance in the convention used here, whereas $h = B^{-1/2}$ is a budget scale. We use $\mathrm{TV} \le \sqrt{2\mathfrak{h}}$, $\mathfrak{h} \le \mathrm{KL}/2$ and Pinsker's inequality $\mathrm{TV} \le \sqrt{\mathrm{KL}/2}$, as well as KL data processing and the finite-prefix chain rule. Unbounded stopping is handled by finite caps and the stated limiting arguments. For a probability vector p, $\mathrm{Ent}(p) = \sum_{p_j>0} p_j \log(1/p_j)$; $\log_+ x = \max\{0, \log x\}$, with $\log_+ 0 = 0$. The symbols $\lesssim$, $\gtrsim$ and $\asymp$ suppress universal positive constants only.

Lemma 14.2 (Two-point RMS testing)**.** *If two admissible parameters have target separation $\Delta > 0$ and terminal transcript laws of total variation at most $1/2$, every estimator has maximum RMS error at those parameters at least $\Delta/4$.*

Proof. Threshold the estimate at the midpoint of the two target values. Each testing error incurs squared estimation loss at least $\Delta^2/4$. The sum of the two testing errors is at least $1 - \mathrm{TV} \ge 1/2$. Hence the sum of the two mean-square errors is at least $\Delta^2/8$, and their maximum is at least $\Delta^2/16$. □

14.5 Cost entropy and canonical band costs

For a finite nonnegative vector a, put $A = \sum_j a_j$ and define $\mathcal{G}(a) = A + \sum_{j:a_j>0} a_j \log(A/a_j)$, with $\mathcal{G}(0) = 0$. This continuous function is homogeneous, coordinatewise increasing and concave. On the positive coordinates, its derivative is $1 + \log(A/a_j)$ and its Hessian is $A^{-1}\mathbf{1}\mathbf{1}^\top - \mathrm{diag}(1/a_j)$. Cauchy–Schwarz makes the latter negative semidefinite, and continuity extends the inequalities to zero coordinates. Homogeneity and concavity imply superadditivity, and binary entropy gives

$$\mathcal{G}(a+b) \le (1 + \log 2)[\mathcal{G}(a) + \mathcal{G}(b)] \le 2[\mathcal{G}(a) + \mathcal{G}(b)]. \tag{71}$$

Indeed, keep and then forget the label specifying which cost vector generated a coordinate. The extra entropy is at most $\log 2$ times the total cost, which is at most $\mathcal{G}(a) + \mathcal{G}(b)$.

In the lower bound we use canonical costs

$$v_j = \kappa(I_j), \quad a_j = v_j/b_j^2, \quad A = \sum_j a_j, \quad F = \mathcal{G}(a), \quad \lambda = \sqrt{\rho/F}.$$

The core has mass $1 - \rho > 0$, so $F > 0$. Coordinatewise comparison yields

$$F \le \Gamma_\kappa \le 4F. \tag{72}$$

Only costs are rounded; the observation experiment retains its actual gaps.

14.6 Relation to the entropy–tail selection law

This subsection is deterministic: it compares the closed functional $\mathcal{Q}(B)$ with the selection functional of Section 13. It is not used in the direct proof of Theorem 14.1.

14.6.1 The selection functional

Let μ be any probability law on $(0, 1/2]$, not necessarily finite. For such μ, define $R_{\mu,\rho}(B)$ by (66), with the model of Section 14 unchanged except that $D_k \sim \mu$, q ranges over measurable functions with $0 \le q \le 1$, and $\theta(q) = \int q\, d\mu$. Use the bands $I_j = (b_j/2, b_j]$, $b_j = 2^{-j-1}$, of (68) and the functional $\mathcal{G}$ of Section 14.5. For a finite measure $\sigma \le \mu$ put $a_j(\sigma) = \int_{I_j} d^{-2}\, \sigma(dd)$ and $A_\sigma = \sum_j a_j(\sigma)$, and define $\Gamma_\sigma = \sup_J \mathcal{G}(a_0(\sigma), \dots, a_J(\sigma))$, with values in $[0, \infty]$. When only finitely many $a_j(\sigma)$ are positive this is $\mathcal{G}(a(\sigma))$; for $\sigma = \kappa$ and a finite law it is the core entropy of (68). Since $\mathcal{G}$ is coordinatewise increasing, $\sigma \le \sigma'$ implies $\Gamma_\sigma \le \Gamma_{\sigma'}$, and $\Gamma_\sigma \ge A_\sigma$. Write $H_\sigma = \int d^{-1}\, \sigma(dd)$ and, with $h = B^{-1/2}$,

$$\mathcal{K}(B) = \min\left\{\rho,\ \inf_{0 \le \sigma \le \mu}\left[h\sqrt{\rho\Gamma_\sigma} + \int \min\{1, h/d\}\,(\mu - \sigma)(dd)\right]\right\}. \tag{73}$$

This is the functional (61). Theorem 13.2 gives $R_{\mu,\rho}(B) \asymp \mathcal{K}_{\mu,\rho}(B)$ for every probability law on $(0, 1/2]$, every $0 < \rho \le 1/4$ and every $B > 0$, with explicit constants 10^{-17} and 33000 in (128). The direct proof of Theorem 14.1 does not use that theorem.

For a general law, define the quantile split as in Section 14. Let $t = \min\{s : \mu((0, s]) \ge \rho\}$, which exists and is positive. Let ν be μ on $(0, t)$ plus the fraction of the atom at t, if any, that gives ν total mass ρ, and let $\kappa = \mu - \nu$. Then ν is supported on $(0, t]$, κ is supported on $[t, 1/2]$, and only finitely many bands meet $[t, 1/2]$. Define $T_h = \int \min\{1, h/d\}\, \nu(dd)$ and $\mathcal{Q}(B) = \min\{\rho, T_h + h\sqrt{\rho\Gamma_\kappa}\}$. For a finite law these are the quantities of (67) and (69).

14.6.2 Scale-separated cost

Besides (71), the comparison uses one more entropy bound.

Lemma 14.3 (Scale-separated cost)**.** *If $t > 0$ and ξ is a finite measure on $[t, 1/2]$, then $\Gamma_\xi \le 4H_\xi/t$.*

Proof. The bound is trivial for $\xi = 0$, so let $\xi \ne 0$. Let j_0 be the band containing t, so $t \le b_{j_0}$. A band meeting $[t, 1/2]$ has index $j_0 - r$ with $r \ge 0$ and consists of gaps in $(b_{j_0}2^{r-1}, b_{j_0}2^r]$. There every gap of ξ satisfies $d \ge t$ and $d > b_{j_0}2^{r-1} \ge t2^{r-1}$, so $d^{-1} \le 2^{1-r}/t$. Writing c_r and g_r for the inverse-square and inverse costs of ξ in that band, indexed by the offset r, we get $c_r \le 2^{1-r} g_r/t$, $\sum_r c_r = A_\xi$ and $\sum_r g_r = H_\xi$. Only finitely many bands are occupied, so $\Gamma_\xi = A_\xi(1 + \mathrm{Ent}(c/A_\xi))$. Comparing the normalized costs with the probability vector $\pi_r = 2^{-(r+1)}$ gives $\mathrm{Ent}(c/A_\xi) \le \sum_r (c_r/A_\xi)\log(1/\pi_r)$, whence

$$\Gamma_\xi \le \sum_{r \ge 0} c_r[1 + (r+1)\log 2] \le \frac{1}{t}\sum_{r \ge 0} 2^{1-r}[1 + (r+1)\log 2]\, g_r.$$

The coefficient $2^{1-r}[1 + (r+1)\log 2]$ is largest at $r = 0$, where it equals $2 + 2\log 2 < 4$. □

14.6.3 The comparison

Proposition 14.4 (Quantile versus selection)**.** *For every probability law μ on $(0, 1/2]$, every $0 < \rho \le 1/4$ and every $B > 0$,*

$$\mathcal{K}(B) \le \mathcal{Q}(B) \le (1 + 2\sqrt{2})\, \mathcal{K}(B) < 4\, \mathcal{K}(B).$$

If $h \ge t$, then $\mathcal{K}(B) = \mathcal{Q}(B) = \rho$.

Proof. The measure $\sigma = \kappa$ is admissible in (73), and $\mu - \kappa = \nu$. Its objective is $h\sqrt{\rho\Gamma_\kappa} + T_h$, so $\mathcal{K}(B) \le \mathcal{Q}(B)$.

For the upper comparison fix $0 \le \sigma \le \mu$ with finite objective, let $f = d\sigma/d\mu \in [0,1]$, and write

$$x = h\sqrt{\rho\Gamma_\sigma}, \qquad y = \int \min\{1, h/d\}\,(\mu - \sigma)(dd), \qquad S = x + y.$$

Since $\nu \le \mu$, we have $(1-f)\nu \le \mu - \sigma$ and $f\nu \le \sigma$, so $\int f\,d\nu \le \rho$ and $\int d^{-2} f\,\nu(dd) \le A_\sigma \le \Gamma_\sigma$. With Cauchy–Schwarz,

$$T_h \le y + h\int d^{-1} f\,\nu(dd) \le y + h\sqrt{\rho \int d^{-2} f\,\nu(dd)} \le y + x = S.$$

As $T_h \le \rho$ always, $T_h \le \mathcal{K}(B)$. If $h \ge t$, every gap of ν has $\min\{1, h/d\} = 1$, so $T_h = \rho$ and therefore $\mathcal{K}(B) = \mathcal{Q}(B) = \rho$.

Now let $h < t$, and split $\kappa = f\kappa + (1-f)\kappa$. Both parts live on the finitely many bands meeting $[t, 1/2]$. By (71), monotonicity and Lemma 14.3,

$$\Gamma_\kappa \le 2\Gamma_{f\kappa} + 2\Gamma_{(1-f)\kappa} \le 2\Gamma_\sigma + \frac{8}{t} H_{(1-f)\kappa}.$$

Every gap of ν is at most t, so $\min\{1, h/d\} \ge h/t$ there and $h\rho/t \le T_h$. Every gap of κ is at least $t > h$, so $hH_{(1-f)\kappa} = \int (h/d)\,(1-f)\kappa(dd) \le y$. Hence

$$h\sqrt{\rho\Gamma_\kappa} \le \sqrt{2}\,x + \sqrt{8\,(h\rho/t)\,(hH_{(1-f)\kappa})} \le \sqrt{2}\,x + \sqrt{8T_h y},$$

and, with $T_h \le S$,

$$\mathcal{Q}(B) \le T_h + \sqrt{2}\,x + \sqrt{8T_h y} \le S + \sqrt{2}\,x + \sqrt{8Sy}.$$

If $S = 0$ then $\mathcal{Q}(B) = 0$. Otherwise put $u = y/S \in [0,1]$, so that $x = (1-u)S$ and

$$S + \sqrt{2}\,x + \sqrt{8Sy} = S\big[1 + \sqrt{2}\,(1 - u + 2\sqrt{u})\big] \le (1 + 2\sqrt{2})\,S,$$

because $1 - u + 2\sqrt{u} \le 2$ on $[0,1]$. Taking the infimum over σ, and using $\mathcal{Q}(B) \le \rho$, gives $\mathcal{Q}(B) \le (1+2\sqrt{2})\min\{\rho, \inf_\sigma S\} = (1+2\sqrt{2})\,\mathcal{K}(B)$. □

14.6.4 Consequences

Corollary 14.5 (Selection form for finite laws)**.** *For every finite law, rarity and budget in Theorem 14.1,*

$$10^{-20}\,\mathcal{K}(B) \le R_{\mu,\rho}(B) \le 30001(1+2\sqrt{2})\,\mathcal{K}(B) < 1.2 \cdot 10^5\,\mathcal{K}(B).$$

Proof. Combine Theorem 14.1 with Proposition 14.4. □

Corollary 14.5 is the finite-law case of Theorem 13.2; here it follows from Theorem 14.1, whose proof does not invoke Theorem 13.2. Conversely, Theorem 13.2 gives the rate $R \asymp \mathcal{Q}$ of Theorem 14.1 for finite laws through Proposition 14.4, without the separate argument of Appendix O. Its explicit constants 10^{-17} and 33000 then give $10^{-17}(1+2\sqrt{2})^{-1}\mathcal{Q} \le R \le 33000\,\mathcal{Q}$: a better lower constant than 10^{-20} and a slightly worse upper one than 30001.

The two results should be read as one rate established by two arguments, not as independent characterizations. Nor are the two arguments independent evidence: both were developed from the same research notes. They share the observable upper construction (Theorem O.4 applies the construction of Appendix D to the fixed quantile split) and Appendices F–H, and differ mainly in the lower-bound reduction.

Corollary 14.6 (The closed quantile form for every law)**.** *For every probability law μ on $(0, 1/2]$, every $0 < \rho \le 1/4$ and every $B > 0$,*

$$\frac{10^{-17}}{1+2\sqrt{2}}\,\mathcal{Q}(B) \le R_{\mu,\rho}(B) \le 33000\,\mathcal{Q}(B).$$

Proof. Combine the explicit constants (128) of Theorem 13.2 with Proposition 14.4. □

For every law, $\mathcal{Q}(B)$ remains a closed expression: κ occupies finitely many bands. Corollary 14.6 rests on the paper proof of Theorem 13.2; for finite laws, Theorem 14.1 gives the same rate by the direct argument of Appendix O.

The comparison was supplied by an external adversarial review (Astra Pro) of revision 1 of the standalone finite-gap manuscript, at repository commit `2984835`. Its proof used hard selections; the proof above works directly with the selection over submeasures in (61).

14.7 Consequences

Write $c_0 = 10^{-20}$ and $C_0 = 30001$ for the constants in Theorem 14.1.

Corollary 14.7 (Mean-square form)**.** *For every law, rarity and budget in Theorem 14.1,*

$$c_0^2 \min\{\rho^2, T_h^2 + \rho\Gamma_\kappa/B\} \le R(B)^2 \le 2C_0^2 \min\{\rho^2, T_h^2 + \rho\Gamma_\kappa/B\}.$$

If $B \ge d_1^{-2}$ and $J_\nu = \int d^{-1}\nu(dd)$, then

$$c_0^2 \min\{\rho^2, (J_\nu^2 + \rho\Gamma_\kappa)/B\} \le R(B)^2 \le 2C_0^2 \min\{\rho^2, (J_\nu^2 + \rho\Gamma_\kappa)/B\}.$$

Proof. For $x, y \ge 0$, $x^2 + y^2 \le (x+y)^2 \le 2(x^2+y^2)$. Taking the minimum with ρ^2 preserves these comparisons, with the same factor two. Apply this to $x = T_h$ and $y = h\sqrt{\rho\Gamma_\kappa}$ and square (70). If $B \ge d_1^{-2}$, then $h \le d_i$ at every atom, so $T_h = hJ_\nu$ exactly, including at equality. □

Corollary 14.8 (One-gap specialization)**.** *For $\mu = \delta_d$, $0 < d \le 1/2$,*

$$R_{\delta_d,\rho}(B) \asymp \min\{\rho, \sqrt{\rho}/(d\sqrt{B})\}.$$

Proof. Here $\nu = \rho\delta_d$, $\kappa = (1-\rho)\delta_d$, and $\Gamma_\kappa = (1-\rho)/d^2$. Put $x = h/d$. If $x \ge 1$, then $\mathcal{Q}(B) = \rho$. If $x < 1$, the sum inside the minimum defining $\mathcal{Q}$ is $x[\rho + \sqrt{\rho(1-\rho)}]$. Since $0 < \rho \le 1/4$, the bracket lies between $(\sqrt{3}/2)\sqrt{\rho}$ and $(3/2)\sqrt{\rho}$. Taking the minimum with ρ and applying the theorem proves the comparison with universal constants. □

For a general finite law the core entropy vanishes only in the normalized entropy term when a single band is occupied; its base cost A^* is always positive. The quantile split can place part of one atom in both ν and κ. These cases are included in the theorem without any generic-position assumption.

The formulas concern a uniform expected observation budget. They do not supply a deterministic total cap, a high-confidence guarantee, or an estimator that learns an unknown magnitude law. The finite support and strictly positive gaps ensure that all caps used in the upper construction exist. Extending these assumptions requires additional arguments; Corollary 14.6 records the extension to general laws, which rests on Theorem 13.2.

15 Stopping programs and finite certificates

The selection law is a comparison up to conservative universal constants. A complementary representation retains more quantitative information. The robust cap-N stopping-flow program $P_N(B)$ of appendix I has exact attained dual

$$P_N(B)^2 = \max_{q_L,q_H,q_v,q_w \in \mathcal{Q}_\rho} \frac{[\theta(q_H) - \theta(q_L)]_+^2}{4 + BK_N(q_L, q_H; q_v, q_w)}.$$

Here K_N is optimal stopped signed-likelihood information relative to q_v per expected observation under q_w. The four priors have separate endpoint, variance, and work roles. Theorem J.1 gives the identity and Proposition J.2 gives its Bellman recursion.

Reference absorption and affinity testing give the explicit comparison

$$R_{\mu,\rho}(B) \le P_{N_*}(B) \le 16 R_{\mu,\rho}(B), \qquad N_* = \left\lceil 64 \max\{B, \rho B^2\} \right\rceil, \tag{74}$$

proved in appendix K. The factor sixteen is a program guarantee, distinct from the conservative constants in Theorem 13.2.

A strengthened lower experiment reveals a unit's latent state at its cap. Its dual value G_N obeys

$$0 < G_N \uparrow P_\infty, \qquad P_N \downarrow P_\infty, \qquad G_N/8 \le R_{\mu,\rho}(B) \le P_N.$$

Thus a verified feasible upper policy U and lower dual value L with $U \le 2L$ certify a factor-sixteen approximation. For every fixed input, sufficiently accurate program solutions eventually meet that criterion. Pointwise convergence of the genie brackets assumes finite latent divergence, and appendix L includes an infinite-divergence counterexample; optimized convergence first fixes a positive regularization, takes the cap limit, and then removes the regularization. No uniform convergence rate or bit-complexity claim is made. Finite-support inputs give genuine finite conic programs. The eight rational examples in appendix N pass at caps 1, 4, or 8; independent exact checkers verify their feasibility.

The fixed-cap program also has an exact identification threshold: for m distinct positive magnitudes its infinite-budget bias floor vanishes exactly when $N \ge 2m - 1$. Infinite-support laws have a positive floor at every fixed finite cap (Theorem M.1 in appendix M). This identifies the target, not necessarily the full allocation q, and is compatible with the unrestricted problem, whose cap may increase.

Status. Lean now proves the actual observable upper realization, attained four-prior dual, sufficient-cap factor-sixteen certificate, midpoint comparison, fixed-cap identification, infinite-cap dual and minimax comparison, genie brackets whose residual converges under finite latent divergence, the finite stopping criterion, and finite-support conic formulations. It also checks the explicit infinite-divergence counterexample and optimized direct squared-risk certificate. These proofs use the original expected physical-work model. The exact rational computations retain their separate finite-instance scope. No numerical solver accuracy, uniform cap rate or bit complexity is asserted by the formal results. See section 18.

16 Related work for the rare-event results

Adaptive mixtures of coins. Lee–Valiant is the closest direct antecedent [1]. In the extended version, Sections 4–5 develop triangular single-coin estimators and single-coin-to-population lower-bound decompositions. Section 7, Theorems 4–5 and Lemma 22, optimizes

an unbiased estimator when both sign-conditional bias distributions, a nominal mixture weight, and a per-coin cap are supplied. Its linear program optimizes information per expected sample. Section 5.2 also reveals a coin's true bias after a cap in a strengthened experiment.

Knowing our marginal magnitude law does not supply those conditional distributions. For $0 < \theta(q) < 1$, the corresponding magnitude laws are

$$\mu_q^+(dd) = \frac{q(d)}{\theta(q)}\,\mu(dd), \qquad \mu_q^-(dd) = \frac{1-q(d)}{1-\theta(q)}\,\mu(dd).$$

They change with the unknown function q. Our optimization must use one observable procedure whose bias, moment, and expected work bounds hold throughout $\mathcal{Q}_\rho$. The exact dual therefore retains separate endpoint, moment-reference, and work-reference populations. The additional statements concern that robust program, its comparison with the full population experiment, and convergent upper and lower certificates. Triangular flows, information/work optimization, adaptive decomposition, and latent revelation are credited to their antecedents.

Gap entropy and sign testing. Chen–Li introduced entropy of normalized dyadic inverse-square gap costs [25, Definition 3.3]. Chen–Li–Qiao proved the weighted Sign-ξ entropy inequality [19, Lemma 4.1]; Appendix D develops its stopping-window argument, and Remark D.4 explicitly covers Bernoulli families away from their boundary. Our weighted sign-testing lemma is an adaptation of this argument. In particular, the form $A[1 + \mathrm{Ent}(\pi)]$ and the statistical origin of its entropy factor are established ideas.

The additional issue here is how much of a known magnitude law should pay that entropy cost. The optimizing set in (61) depends jointly on the law, rarity, and budget. Its complement pays a truncated inverse-gap tail cost. The upper construction implements this decomposition without observing component membership; the lower construction respects both rarity and mass saturation while combining disjoint components. These steps are needed to obtain the all-law selection formula from the sign-testing ingredient. Best-arm identification and latent population-proportion estimation have different targets and sampling access.

Gaussian best-arm identification. Aronow, Kallus and Lopatto [29] prove the gap-entropy conjecture for fixed-confidence best-arm identification among independent unit-variance Gaussian arms with means in $[0, 1]$ and a unique best arm i^*. For algorithms that identify the best arm with probability at least $1 - \delta$ on every such instance, in the range of δ treated there, the optimal expected sample count averaged over permutations of the arm labels is within constant factors of $H\,(\log(1/\delta) + \mathrm{Ent}(\pi))$. Here $H = \sum_{i \neq i^*} \gamma_i^{-2}$, γ_i is the gap from arm i to the best mean, and π gives the shares of H in dyadic gap ranges. The core term Γ_κ of (68) has the same form. That problem identifies one of finitely many labelled arms with a confidence requirement; here the target is a population fraction under RMS loss and an expected budget, with fresh unlabelled coins and unknown signs. We do not use their result and do not claim that either result implies the other. The critical-law discussion in Section 11 also notes the Yao–Zhao–Zhou result [24].

Convex estimation, testing, and stopping. Juditsky–Nemirovski give constant-factor affine estimation under their good-pair assumptions, with a Hellinger-affinity characterization [17, Theorem 3.1 and Proposition 3.1]. Polyanskiy–Wu characterize quadratic minimax risk for an affine functional over a convex prior class through a χ^2 modulus for a fixed iid observation kernel [18, Theorem 1]. These results provide the relevant estimation–testing duality. For a fixed capped one-unit policy, its transcript is a fixed observation kernel. Optimizing that policy while controlling expected work uniformly over q, and then comparing with arbitrarily

interleaved population algorithms, requires the additional arguments given here. Strong convex duality and occupation-measure formulations of optimal stopping are standard tools [27, 28]; the four-prior formula is a specific application, not a new general duality principle.

Baseline and narrower claims. Sections 2–12 and Appendices A–C contain the balanced-law benchmark, rare-event phase results, the variational Ψ lower bound, and the critical quadratic-law entropy correction. The critical proof also credits Chen–Li–Qiao for its sign-entropy mechanism. Those results preceded the general law. The specialized ingredients needed here are proved within this manuscript, with their attribution preserved.

The principal added claim is the joint, arbitrary-law entropy–tail characterization, together with a robust stopping-program formulation that admits convergent certificates. The fixed-cap odd-moment identification result is a secondary consequence of this model and elementary moment interpolation; it is not presented as a general mixture-identification principle. The targeted primary-source comparison above identifies the closest checked statements and their hypotheses. It does not certify publication priority.

17 Consequences and limits of the characterization

The selection formula recovers two immediate upper bounds by choosing the retained set to be all or none of the magnitude space:

$$R_{\mu,\rho}(B) \lesssim \min\left\{\rho, \sqrt{\rho\Gamma_\mu/B}, \int \min\{1, (d\sqrt{B})^{-1}\}\,\mu(dd)\right\}.$$

The full formula additionally allows different regions of the same law to use different treatments. Its lower proof explains why contributions from several irregular components may need to be combined.

For a single magnitude $\mu = \delta_d$, hard assignments leave two choices and give

$$R_{\delta_d,\rho}(B) \asymp \min\{\rho, \sqrt{\rho/(Bd^2)}\}.$$

For a general law, the finite-prefix selection values k_J in (112) have an explicit tail-mass error bound (113), and their limit lies between $\mathcal{K}/2$ and $\mathcal{K}$. This is a finite approximation using only known band masses. Exhaustive selection is possible, but an efficient general selection algorithm has not been established here.

The scalar characterization concerns the full minimax risk, whereas P_N bounds worst bias and worst variance separately. Accordingly, its exact dual is an identity for a certificate program, and the statistical connection is a comparison inequality. The convergent genie brackets can certify that a chosen cap suffices for a particular input without using the large universal cap. They do not give a uniform runtime bound.

The remaining quantitative questions include sharp comparison constants, efficient selection for structured laws, and rates of cap convergence. The present theorem assumes known μ, Bernoulli observations, and a uniform expected observation budget. Extensions to an unknown magnitude law, other observation kernels, or almost-sure budgets require separate arguments.

The general-law argument includes the specialized proof chain rather than relying on the earlier working notes. Separate model audits of the standalone general-law manuscript found no remaining substantive inconsistency in the reviewed chain; exact arithmetic verifies the finite numerical certificates. These checks do not constitute external refereeing or proof-assistant verification of the general theorems. The related-work comparison is targeted and does not certify publication priority.

17.1 Two scope examples

A noninterval optimal assignment. Let $\varepsilon = 1/1024$, put masses $(1-\varepsilon)/2$, ε, and $(1-\varepsilon)/2$ at $1/2$, $1/4$, and $1/8$, respectively, and take $\rho = 1/4$, $h = B^{-1/2} = 1/1024$. All three atoms lie above h. Enumerating the eight hard retained sets shows that the unique minimum retains $\{1/2, 1/8\}$ and treats $1/4$ ordinarily. The following rational enclosures for the inner cost divided by h make the strict comparison explicit:

Retained atoms	Enclosure for cost divided by h
$\varnothing$	$(4.99902, 4.99903)$
$\{1/2\}$	$(4.70676, 4.70677)$
$\{1/4\}$	$(5.05761, 5.05762)$
$\{1/2, 1/4\}$	$(4.72155, 4.72156)$
$\{1/8\}$	$(3.82997, 3.82998)$
$\{1/2, 1/8\}$	$(3.22748, 3.22749)$
$\{1/4, 1/8\}$	$(3.83271, 3.83272)$
$\{1/2, 1/4, 1/8\}$	$(3.22943, 3.22944)$

These follow by substituting the rational weights into $\mathcal{G}$. For example, after reducing each logarithm argument to $1 \le x \le 2$, use $y = (x-1)/(x+1)$ and

$$0 \le \log x - 2\sum_{k=0}^{m-1} \frac{y^{2k+1}}{2k+1} \le \frac{2y^{2m+1}}{(2m+1)(1-y^2)},$$

together with rational bounds for $\log 2$ from the same series and square-root bounds checked by squaring. Already $m = 90$ gives intervals much narrower than those displayed. The verifier `paper/rare-event/v2/checks/verify_revisions.py` performs these exact calculations. Since the outer cap is inactive, the original functional has the same optimizer. No interval can contain both retained atoms while excluding the intervening positive-mass atom.

Infinite inverse moments and a slow tail. Let $\mu\{2^{-j}\} = 1/[j(j+1)]$ for $j \ge 1$. Both inverse moments of orders one and two are infinite. At $B = 4^J$, ignore $j > J$ and treat the first J atoms ordinarily. This gives

$$\mathcal{K}(B) \le \frac{1}{J+1} + \sum_{j=1}^{J} \frac{2^{j-J}}{j(j+1)} = \frac{1}{J+1} + O(J^{-2}).$$

For fixed ρ and sufficiently large J, the ignored-tail allocation gives $\Psi(2^{-J}) \ge 1/(J+1)$. Thus the upper and lower scales match even though the full-law inverse moments diverge. This example uses only the vanishing tail probability, not an integrable full-law cost.

18 Verification, provenance, and remaining questions

The distinction between a checked theorem and a checked ingredient is part of each result's scope. Most theorem links use the completed critical-law and Psi checkpoint `eabf274c b42d4bff4daab6b58568ad6169e7f0a7`. The unbiased converse uses its later checkpoint `631886e16ddff5b8e93b73860bf6552b956128be`. The observable 713 construction and existence characterization use `15dbbda98a31e27502e3caff991d3ea21d096d23`. The rare-event theory above exponent two uses `637fb4a79d20e30415a2cb0e495752603077c0ee`. The entropy–tail upper half, the closed-quantile upper bound and the stopping-program results use `8fb16244ded0d6cebf4ba78e1ad9adae09ef293d`. All theorem links point to fixed source revisions. The preceding all-size addendum has its own linked checkpoint and audit; its stronger domain and explicit single-coin constants supplement this table.

Claim	Verification scope
Original RMS minimax lower bound	`minimax_lower_bound` proves $R_\mu(B) \geq \omega_\mu(B^{-1/2})/64$ in the hidden-magnitude iid fresh-coin model, for arbitrary measurable seeds, natural-number coin labels, revisits, interleaving, and unbounded stopping. It constructs the experiment and retains infinite risk. The separate oracle theorem below covers the stronger revealed- and selected-magnitude models.
Original RMS upper bound	`distribution_blind_upper` chooses one budget-indexed observable family before every gap law and proves $R_\mu(B) \leq 512\omega_\mu(B^{-1/2})$. Actual cost/output identities, almost-sure termination and every budget are included. The unchanged frozen specification has no additional statistical premise.
Confidence characterization	`confidence_characterization` proves both minimax bounds with coefficients $1/64$ and 8192 for $B \geq 1$ and $0 < \delta \leq 1/4$. `confidence_upper` supplies one observable family depending only on (B, δ). The next theorem sharpens the lower coefficient to $15/(256\sqrt{2})$, including in this hidden model.
Magnitude-oracle lower bounds	`oracle_minimax_lower` proves RMS coefficient $1/64$ and confidence coefficient $15/(256\sqrt{2})$ for $B \geq 1$ and $0 < \delta \leq 1/4$, both when the full iid magnitude array is revealed and when each new magnitude is selected adaptively. Fresh creation, persistent magnitudes on reuse, the actual process and expected flip budget, minimax bounds, and nonvacuity are proved.
Unbiased characterization	`unbiased_converse` proves the necessity of $I_\mu < \infty$ and $V\overline{W} \geq I_\mu^2/16$ for all uniformly unbiased observable adaptive estimators. Integrability, extended variance, and actual work are explicit. `observable_unbiased_iff` combines this converse with the next construction to prove the full existence characterization.
713 construction	`observable_713` chooses one observable algorithm before every gap law, and proves unbiasedness, extended variance at most $23I_\mu$, actual work at most $31I_\mu$, and the product of separate suprema at most $713I_\mu^2$. The earlier conditional-kernel theorem and separate-family bridge remain unchanged.

Claim	Verification scope
Variational optimizer	`arbitrary_law_optimizer` proves the optimizer and value formula for arbitrary laws. It does not assert a matching rare-event statistical upper bound.
Rare-event rates below two	The below-two rates and the general statistical Ψ lower bound remain paper theorems. The arbitrary-law Ψ lower bound in appendix A uses the cited Lee–Valiant input. The critical law has the separate formal coverage below.
Rare-event rate above two	`above_two_theory` proves both minimax bounds for the exact power law with $\alpha > 2$, $0 < a \le 1/2$, $0 < \rho \le 1/4$ and $B \ge 1$, together with law existence, a Ψ comparison and nonvacuity. One single-coin family depending only on (a, ρ) is unbiased on every power-margin law. The lower bound holds for every law supported below a and uses no Lee–Valiant input. The frozen targets assert positive constants; the proofs instantiate $1/256$ and second moment 28ρ. The paper construction of appendix B, with its constant $153/56$ and compound-Poisson calibration, is not the formal one.
Finite-gap sign entropy	`finite_gap_sign_entropy` proves the statistical bound with actual stopping and flip costs. The same test serves every specified gap and sign.
Critical single-coin base cost	`critical_base_cost` proves $V\mathbb{E}_0 T \ge c\rho L(1 + \log L)$ for one universal $c > 0$ and all $0 < \rho \le 1/4$. Extended variance, actual work and the global capped prior class are explicit. Activation and the full entropy chain are derived. It is an input to the full critical lower theorem below.
Critical minimax lower bound	`critical_minimax_lower_bound` proves the saturated lower rate in theorem 11.2 for the full rare-prior class. Statistical compactness, finite mixtures, one common pair, population transfer, endpoint termination and terminal testing are proved, including unrestricted creation and arbitrary seeds.
Critical minimax upper bound	`critical_upper_bound` constructs one rarity- and budget-indexed observable family with uniform expected work at most B and the matching RMS rate with coefficient 100000. Its actual execution and work/output identities are proved.
Exact Ψ comparison and counterexample	`critical_psi_formula` evaluates the actual variational supremum; `critical_psi_rate` proves its two-sided rate. `critical_matching_psi_refuted` rules out every universal matching-Ψ upper bound in the original minimax model. Its budget is LH/ρ, not the paper sequence ρ^{-2}.
Entropy–tail law	`entropy_tail_upper` proves the upper half of Theorem 13.2 against its locked specification, with constant 28000 in place of 33000, for every probability law on $(0, 1/2]$, $0 < \rho \le 1/4$ and $B > 0$. The construction of appendix D is realized in the original observable experiment, without an inverse-moment or finite-support assumption. The lower half (appendix E), and hence the two-sided law and its lower constant in (128), remain paper proofs with adversarial AI audits of the standalone manuscript.

Claim	Verification scope
Closed quantile form	`selection_quantile_comparison` proves the deterministic comparison proposition 14.4, $K_{\mu,\rho} \le Q \le (1+2\sqrt{2})K_{\mu,\rho}$, for every probability law, including fractional boundary atoms. `closed_quantile_upper_30001` proves the upper half of Theorem 14.1, $R \le 30001\,Q(B)$, for every law rather than only finite ones. The lower half of that theorem is a paper proof, and Corollary 14.6 rests on Theorem 13.2.
Stopping programs and certificates	For every probability law, $0 < \rho \le 1/4$, $B > 0$ and cap $N > 0$, `attained_four_prior_dual` proves the attained four-prior dual with positive value and finite optimal information ratio. `program_upper` realizes every feasible certificate in the observable experiment, so $R \le P_N$, and `finite_certificate` proves $R \le P_N \le 16R$ at $N = \lceil 64 \max(B, \rho B^2) \rceil$ without the entropy–tail lower bound. The infinite-cap dual and its factor-eight comparison are proved for $0 < \rho \le 1/2$. The genie modules prove the brackets, residual convergence under finite latent divergence, optimized convergence, finite termination of the factor-sixteen test, the finite-atom second-order cone program and the uniform-gap counterexample. `fixed_cap_identification` proves the threshold $N \ge 2m-1$ and the positive infinite-support floor. Exact rational checks additionally certify the stored finite instances.

The audits permit three standard Lean axioms: `propext`, `Classical.choice`, and `Quot.sound`. The exact specifications and model definitions are necessary to interpret that fact: absence of proof holes does not show that a statement matches an intended experiment. The finite-gap targets were introduced with their proofs and are not described as specifications frozen before the proof work. The entropy–tail specification was frozen before the proof of its upper target. The upper construction's own quantile specification records a freezing time before its proofs. The stopping-program specification was committed before the proofs of its three targets, but it was hash-locked only afterwards. The repository's verification guide gives reproduction commands and the source-specific audit records. The toolchain is Lean `v4.36.0-rc1` with Mathlib revision `9bdd0bc71aaae0fd83889f9aacd70c32fccd1bcd`.

Realizing the distribution-blind upper algorithm. The formal construction follows the two-pilot calibration strategy in section 5. A stopped cutoff call with cap $N = 2^\ell$ is padded by one unused flip when its original work is zero. Its output is unchanged, and every call advances the observed history. Two independent first-success pilots select the number of fresh production calls. The calibrated quantities have expected work at most $4N$ and mean-square error at most $12751\omega_\mu(N^{-1/2})^2$.

These estimates are transferred to a measurable controller in the original experiment by exact seed and population path identities. Positive mean marked cost proves almost-sure pilot termination. For $B \ge 4$, choosing $4N \le B < 8N$ and comparing moduli yields coefficient 512; for $1 \le B < 4$ a zero-work estimator on the same seed space suffices. The coefficient includes padding and conservative rounding and is not claimed optimal. The full audit checks the unchanged specification and all transitive axioms; the earlier component records retain their historical incomplete status.

Realizing the confidence upper algorithm. Put $m = \lceil 2\log(1/\delta) \rceil$ and $k = 2m+1$. Each of k independent calibrated batches receives expected budget B/k. Its failure probability at radius $512\omega_\mu(N^{-1/2})$ is at most $1/8$. A union bound over majority subsets puts the median's failure probability below $2^{-m} \le \delta$. The controller uses disjoint pair-numbered coin labels and independent copies of the seed. Exact finite execution identities prove that its output is the median and that actual work is the sum of the batch costs. The product law and almost-sure termination are derived in the original experiment. Since $k \le 7\log(1/\delta)$, budget rounding and modulus comparison give the coefficient 8192. A zero-work rule handles the small-budget regime. The radius is a theoretical function of μ, not a confidence interval computed from the observations alone.

The lower proof applies a binary confidence test to the same adaptive transcript information bound. It proves the conservative coefficient $1/64$; the oracle extension below also proves the paper's sharper coefficient. The four full targets, model, and transitive model dependencies were frozen before supporting proofs. The audit checks all 107 declarations and the exact targets at checkpoint `d6feb3069d1a399ab1a50de2b0e7bc13e6b21c23`.

Realizing the magnitude oracles. The revealed model grants the entire iid magnitude array, together with an arbitrary independent random seed, to the policy and output. The selected model has explicit create, revisit, and stop requests. Each creation may choose any magnitude in $(0, 1/2]$ as a measurable function of the seed and all previous flips; a revisit names a prior creation. Finite histories encode coin labels in the common process. The semantic proof shows that a fresh label has never been used on that path and that a revisited coin retains its original magnitude. Counterfactual histories do not restrict magnitudes to a deterministic discrete grid.

Both models use the canonical infinite process and its proved finite-prefix law, actual expected flip counts, and extended risks. The adversary is one measurable sign function on the full magnitude domain, including any chosen magnitudes outside the support of the target law. The estimation target remains $\int q\, d\mu$. No information inequality or termination premise is assumed by the final lower bounds; finite budget implies almost-sure stopping.

The clipped hard pair gives KL at most $(128/225)h^2B$ even with the common magnitude information revealed. The RMS argument gives coefficient $1/64$. For confidence, clipping at $h = 15\sqrt{\log(1/\delta)/B}/(32\sqrt{2})$ gives KL at most $\log(1/\delta)/16$; binary testing and modulus scaling give $15/(256\sqrt{2})$. The original hidden experiment embeds with identical law, work, risk and confidence events. Zero-work estimators establish nonvacuity. The models, eight full target types and 33 transitive model dependencies were frozen at `3474a37` before the new lower and semantic proofs. The audit covers ten modules, 54 declaration roots and all eight targets at `4a5805c771e541f557848957e0a3ffada31dcb32`.

Realizing the above-two rare-event rate. The formal upper construction uses asymmetric, stage-dependent dyadic boundaries whose wrong-positive error is scaled by ρ. Power-tail and weighted-moment estimates turn the global rarity restriction into a uniform work bound, and a geometric random-level telescope gives an integrable unbiased single-coin output with second moment at most 28ρ. Its output and work are identified in the actual observable experiment, and independent fixed-count repetitions or the zero-work estimator $\rho/2$ meet every budget. The rule is chosen before μ and q and uses α only for budget calibration. The lower proof charges a finite-path information bound to each queried label, controls gaps below $1/4$ by Bernoulli KL and the remaining range by a variation bound, and combines a quadratic rare-mixture inequality with the midpoint hybrid decomposition. The seven full targets were compiled and frozen before the supporting proofs; the audit covers 54 modules, 322 declaration

roots and all seven targets at `637fb4a79d20e30415a2cb0e495752603077c0ee`.

Realizing the entropy–tail upper algorithm. The formal construction follows appendix D: likelihood-bank classifiers raced on a shared prefix, a multilevel allocation with a finite terminal level, and marked Poisson counts realized by distinct population labels in the original experiment. Its multilevel allocation certificate is bounded by $28000(\sqrt{\rho\Gamma} + H)$ uniformly in the terminal level, and finite-band approximation removes the support cutoff. This gives $R \le 28000K'$ for the development's own functional K', whose band costs are Bochner integrals. A separate bridge proves $K' \le K_{\mu,\rho}$ for every measure and $K' = K_{\mu,\rho}$ for every finite law. No component label, hidden magnitude or sign enters the policy.

Realizing the critical upper algorithm. The formal construction uses asymmetric classifiers at dyadic flip counts, an alternative to the likelihood-bank construction in appendix C. An independent geometric level selects two adjacent classifiers on one coin's shared prefix. Their scaled difference, with the base classifier at level zero, has mean θ, extended second moment at most 28ρ, and expected actual flips at most $W = 112000000LH$. An average of $n+1$ independent fresh-coin calls therefore has MSE at most $28\rho/(n+1)$ and expected work at most $(n+1)W$.

The measurable controller requests only observed flips, stops each call from its prefix, and realizes these exact cost/output quantities in the original experiment. For $B \ge W/\rho$, it uses $\lfloor B/(2W) \rfloor + 1$ calls; otherwise it returns zero without flipping. Both regimes satisfy the budget and the RMS bound with coefficient 100000. Neither hidden magnitudes nor signs enter the policy. The full upper target was frozen before this construction. The Psi model and full types are separately hash-locked: the formula had a checked draft at freezing, while the rate and separation targets were fixed before their proofs.

The unbiased converse. The formal proof bounds the separation of two expectations by symmetric KL divergence and their variances. It uses the sign laws $q_\pm(d) = 1/2 \pm \varepsilon \min\{1, h/d\}$, charges information to actual flips in the infinite experiment, and sends ε to zero. This gives $\omega_\mu(h)^2 \le 16h^2 V\overline{W}$. Monotone convergence of truncated inverse gaps then proves both integrability and the lower product bound; finite I_μ is not assumed. The exact target has no KL or differentiability premise. The model and target were written and compiled before the supporting proofs; the dependency snapshot was recorded after the proof drafts. This unbiased converse is for the hidden-magnitude model; the new oracle results above concern the RMS and confidence lower bounds.

Realizing the 713 estimator. A geometric natural-number seed selects a fine classifier and its adjacent coarse classifier on the same coin. Monotonicity of their stopping levels means that the coarse output is already determined by the fine classifier's observed prefix. The controller requests coin zero until that stopping rule fires, then returns the scaled difference (the base output at level zero). Every seed and tape has finite actual work, bounded by its dyadic cap. No hidden magnitude, sign, or gap law enters the policy or output.

The pointwise execution proof identifies the entire stopped transcript, output and flip count. The existing population simulation law transfers the conditional Bernoulli analysis to the original adaptive experiment. Integrating over every measurable sign law gives unbiasedness and the separate bounds $23I_\mu$ and $31I_\mu$. Extended variance retains the possibility of infinite values until square integrability is proved. The two full targets and their transitive model definitions were compiled and frozen before this controller and its supporting proofs. The audit checks both targets and all 47 new declaration roots.

Numerical evidence and its limits. Exact finite certificates, floating-point recursions, and held-out simulations do not certify the continuum theorems or the iterated-log separation. Finite-grid constants do not replace 713: the grid-winning logarithmic threshold coefficient 1/2 fails the uniform error envelope at an explicit large cap, whereas this construction retains coefficient one. The finite-budget feedback advantage and practical experiments remain supporting research with their original limited scopes.

Provenance and review. This version incorporates the original law, the 713 argument, and the rare-event continuations, with full proofs in the main text and appendices A to C. It adds the general-law rare-event manuscript (revised v2) in sections 13, 15 and 16 and appendices D to N, and the finite-gap rare-event manuscript (revision 2) in section 14 and appendix O. Proofs that appeared in both rare-event manuscripts in substantially identical form appear once: the shared appendices follow the general-law version, with the finite-gap manuscript's clarifications added, and the finite-gap upper bound cites the general construction. The two proofs of the finite-law rate share that construction and those appendices, so they are not independent confirmations. The assembly script checks source identities and records editorial changes; historical sources and reviews remain unchanged. AI agents performed the work under David Iach's direction. No external specialist review or model diversity is established. One critical reviewer authored part of the argument, precluding two independent validations of the full proof. The process note records the workflow, failed approaches, and these limits.

Remaining mathematical work. The lower half of the entropy–tail law, and with it the two-sided law and closed quantile form, the rare-event regimes below two, and auxiliary envelope, smoothness, and observed-state results retain their stated paper-proof scopes; their formalization is the next verification target. Sharp comparison constants, efficient selection for structured laws, rates of cap convergence, and sharp rare-event confidence rates at and above the quadratic boundary remain open. The rare-event results assume a known magnitude law; unknown laws, other observation kernels and almost-sure budgets require separate arguments.

The model requires independent fresh coins and conditionally independent flips, without drift or additional smooth structure linking states. Expected-work optimality supplies no deterministic latency bound or data-computable confidence radius over all unknown laws. Neither useful finite critical constants nor practical superiority is established.

A Rare-event localization and margins below two

This appendix supplies the general lower bound, its optimizer, the below-two matching construction, and a general upper bracket. The earlier arbitrary-law conjecture has been replaced by its correction; section 13 gives the arbitrary-law characterization. The published rare-prior input is identified explicitly below.

A.1 Experiment, localization, and the two anchors

A fresh coin has $D \sim \mu$ on $(0, 1/2]$, $S \mid D = d \sim \mathrm{Ber}(q(d))$, and probability

$$P = \tfrac{1}{2} + (2S - 1)D.$$

Conditional on its latent probability, successive observations of that coin are independent Bernoulli trials. Distinct fresh coins are independent. An algorithm can create coins, revisit them, interleave their observations, and stop adaptively. It observes neither D nor S. One observation is one unit of work. Arithmetic and independent randomization are not charged, as in the original model.

Fix $0 < \rho \le 1/4$ and define

$$\mathcal{Q}_\rho(\mu) = \left\{ q : (0, 1/2] \to [0, 1] \text{ measurable} : \quad \theta(q) := \int q \, d\mu \le \rho \right\}. \tag{75}$$

The quantity to estimate is $\theta(q)$, not the known upper bound ρ. Replacing the inequality by the known equality $\theta = \rho$ would trivialize estimation.

Let $R_{\mu,\rho}(B)$ be the infimum, over algorithms with $\sup_{q \in \mathcal{Q}_\rho} \mathbb{E}_q T \le B$, of $\sup_q (\mathbb{E}_q |\widehat{\theta} - \theta(q)|^2)^{1/2}$. Let $r_{\mu,\rho}(B, \delta)$ be the corresponding smallest uniform error radius at failure probability at most δ. Here $B \ge 1$ and $0 < \delta \le 1/4$. These are *expected-work* criteria. The minimax benchmark may know μ, ρ. Each construction's actual parameter requirements are stated separately.

This class deliberately permits the positives to concentrate in a difficult subpopulation. It does *not* impose the stronger condition $q(d) \le \rho$ at every magnitude. Nor does it assume known conditional magnitude laws μ_+, μ_-. In the latter model the mixture weight is the main unknown; in (75), changing q changes both conditional laws. Lee–Valiant's Section 7 addresses the known-conditional-law experiment with a per-coin depth restriction [1].

A.1.1 Sanity checks

For $\mu = \delta_\Delta$, the problem is the fixed-gap mixture experiment. The Lee–Valiant bounds [1], localized to $\theta \le \rho$, give, in the nontrivial regime $\varepsilon \le c\rho$,

$$B^*(\varepsilon, \delta) \asymp \frac{\rho}{\varepsilon^2 \Delta^2} \log(1/\delta). \tag{76}$$

The corresponding error scale, saturated at the trivial radius, is

$$\min\{\rho, \sqrt{\rho \log(1/\delta)/B}/\Delta\}.$$

At fixed confidence or for RMS, omit the logarithmic factor. This is a cited anchor, not a new fixed-gap theorem.

For constant ρ, a correct localized scale must agree, up to constants, with the original

$$\omega_\mu(h) = \mathbb{E}_\mu \min\{1, h/D\}, \qquad h = B^{-1/2}$$

(and $h = \sqrt{\log(1/\delta)/B}$ for high confidence). The proposed heuristic

$$H_{\mu,\rho}(h) = \rho \, \omega_\mu(h/\sqrt{\rho}) \tag{77}$$

satisfies both anchors. The next section shows that this is insufficient.

A.2 A counterexample to the proposed rescaling

Theorem A.1. *Let* $0 < \rho \le 1/4$, $a = 1/4$, $d = \rho^2$,

$$\mu_\rho = \rho\delta_d + (1-\rho)\delta_a, \qquad B = \rho^{-4}.$$

Then $R_{\mu_\rho,\rho}(B) \ge \rho/100$, *whereas*

$$H_{\mu_\rho,\rho}(B^{-1/2}) = \rho^2 + 4(1-\rho)\rho^{5/2} = O(\rho^2).$$

The minimax error is therefore of order ρ, *and the heuristic can underestimate it by an unbounded factor.*

Proof. Put $\eta = 1/64$ and choose $q_\pm(d) = 1/2 \pm \eta$, $q_\pm(a) = 0$. Both targets lie in $[0, \rho]$ and are separated by $\Delta = 2\eta\rho = \rho/32$. Even if the algorithm is told every magnitude, only a queried coin of magnitude d is informative. The original posterior inequality gives, for every next observation and every history,

$$\mathrm{KL}_{\mathrm{next}} \le \frac{32d^2\eta^2}{(1-4\eta^2)^2}.$$

The stopped chain rule and $Bd^2 = 1$ imply $\mathrm{KL}(\mathsf{P}_+, \mathsf{P}_-) \le k_0 := 32\eta^2/(1-4\eta^2)^2 < 0.008$. Hence $\mathrm{TV} \le \sqrt{k_0/2}$. Thresholding an estimator halfway between the two targets gives error sum at most $8R^2/\Delta^2$, whereas every test has error sum at least $1 - \mathrm{TV}$. Thus

$$R \ge \Delta\sqrt{(1-\sqrt{k_0/2})/8} > \rho/100.$$

The constant estimator $\rho/2$ has error at most $\rho/2$. Finally, $(\rho B)^{-1/2} = \rho^{3/2}$ lies between d and a. Direct substitution in (77) gives the stated expression. □

The hard subpopulation contains both positive and negative near-threshold coins. Its local positive fraction is close to $1/2$, although its total population mass is only ρ. Multiplying the global modulus by ρ incorrectly treats the difficult portion as if positives were thinly spread across all magnitudes. Section A.4 also gives a failure example with one fixed, uniform magnitude law, so this is not merely an artifact of changing μ with ρ.

A.3 Two localized lower bounds for arbitrary laws

A.3.1 A balanced-region bound independent of rare-prior machinery

Define

$$\Gamma_{\mu,\rho}(h) = \sup_{A:\,\mu(A)\le\rho} \int_A \min\{1, h/d\}\,\mu(dd). \tag{78}$$

Proposition A.2. *There is a universal* $c > 0$ *such that*

$$R_{\mu,\rho}(B) \ge c\Gamma_{\mu,\rho}(B^{-1/2}), \qquad r_{\mu,\rho}(B,\delta) \ge c\Gamma_{\mu,\rho}\left(\sqrt{\log(1/\delta)/B}\right).$$

This lower bound remains valid with revealed or selected magnitudes.

Proof. For any admissible A, put $q_\pm(d) = \mathbf{1}_A(d)[1/2 \pm \eta_h(d)]$ with $\eta_h(d) = \frac{1}{8}\min(1, h/d)$. The larger target is at most $5\mu(A)/8 \le \rho$. The target separation is $\Delta_h = \frac{1}{4}\int_A \min(1, h/d)\,d\mu$. Every observation has predictive KL at most $128h^2/225$ by section 3, and observations outside A have zero divergence. Thus transcript KL is at most $128Bh^2/225$. For RMS choose $h = c_0 B^{-1/2}$

with a sufficiently small universal c_0 and use the testing argument above. For confidence use $h = c_1\sqrt{\log(1/\delta)/B}$, choosing c_1 so that transcript KL is strictly less than $\mathrm{kl}(1-\delta, \delta)$; an estimator with radius smaller than $\Delta_h/2$ would contradict data processing. The integrand satisfies $\min(1, \lambda h/d) \geq \lambda \min(1, h/d)$ for $0 < \lambda \leq 1$. Taking the supremum over A proves the claims without changing the scale. □

Atoms can prevent Γ from using a small amount of mass at a magnitude. The next functional permits fractional allocation and includes the fixed-gap rarity factor.

A.3.2 A variational lower modulus

Define

$$\Psi_{\mu,\rho}(h) = \sup_{\substack{0 \leq w \leq 1 \\ \int w\, d\mu \leq \rho}} \int \min\{w(d),\, h\sqrt{w(d)}/d\}\, \mu(dd). \tag{79}$$

The optimization variable w represents how a hard alternative allocates its available positive mass. It is not an assumed or estimated sign function of the true population.

Lemma A.3 (Published single-coin input)**.** *There are universal constants $c_*, C_* > 0$ such that, for $0 < u \leq c_*$, $0 \leq v \leq c_* u$, $0 < d \leq 1/2$, and any stopped single-coin adaptive experiment, its transcript laws under sign priors u and $u + v$ satisfy*

$$H^2(\mathsf{P}_u, \mathsf{P}_{u+v}) \leq C_* \frac{v^2 d^2}{u} \mathbb{E}_u \tau. \tag{80}$$

Here $H^2(P, Q) = 1 - \int \sqrt{dP\, dQ}$ and external, parameter-independent randomization is allowed.

Source and extension. For sufficiently small d, this is the first expected-work bound in Lee–Valiant Corollary 19 [1], with their ρ, ϵ, Δ replaced by u, v, d. Their result averages over histories and is not a history-uniform predictive-KL claim. Its proof is not reproduced here; it is an explicit external theorem used by this note. For $d \geq d_* > 0$, reveal the sign at the first requested observation. Before that request the experiment has no sign information. By data processing, the Hellinger distance is at most the probability of making the first request times the distance between $\mathrm{Ber}(u)$ and $\mathrm{Ber}(u + v)$. The latter is at most Cv^2/u when u and v/u are small. Since $\mathbb{E}_u \tau$ is at least that request probability and $d^2 \geq d_*^2$, this proves (80) on the remaining range, including deterministic coins at $d = 1/2$. Increasing a universal constant joins the two ranges. □

Lemma A.4 (Bounded-ratio decomposition)**.** *Consider two independent-coin population laws ν_0, ν_1 whose latent Radon–Nikodym ratio lies in $[1-\kappa, 1+\kappa]$ for some fixed $\kappa < 1$. For a fully adaptive experiment stopped after at most n observations, let P_0, P_1 be its transcript laws. Let P_i be the hybrid law using ν_1 only for coin i and ν_0 for every other coin. Then*

$$\mathrm{KL}(P_0, P_1) \leq C_\kappa \sum_i H^2(P_0, P_i).$$

The same conclusion holds by finite-prefix limits for almost-surely terminating experiments whenever the right-hand side is uniformly bounded along the prefixes.

Proof. Index coins in order of their first observation, so at most n indices occur. Include the algorithm's independent random seed in the transcript. For a fixed transcript, let $L_i = dP_i/dP_0$. This is the local coin likelihood ratio, marginalized over that coin's latent state, so

$1-\kappa \le L_i \le 1+\kappa$. Independence of latent coins and the parameter-independent action rule give

$$\frac{dP_1}{dP_0} = \prod_i L_i.$$

This identity holds even though the local sequence lengths were chosen using other coins' outcomes. Also $\mathbb{E}_0 L_i = 1$. On this compact ratio interval, $-\log x + (x-1) \le C_\kappa (x-1)^2$, and $(x-1)^2 \le C_\kappa(\sqrt{x}-1)^2$. Taking expectations and using $\mathbb{E}_0(\sqrt{L_i}-1)^2 = 2H^2(P_0,P_i)$ proves the finite-prefix claim. Increasing stopped-transcript sigma fields and the standard monotone limit of relative entropy prove the last assertion. Applying the argument at finite prefixes avoids any unjustified interchange of an unbounded stopping sum and a likelihood derivative. □

This lemma is the bounded-relative-change mechanism of Lee–Valiant Lemma 18, stated for the present magnitude-dependent perturbations. It is not a new general decomposition principle. Unlike an attempted uniform posterior bound, it charges informative but rare histories by their expected frequency.

Theorem A.5 (Arbitrary-law rare-event lower bound)**.** *For the ordinary iid fresh-coin experiment, there is a universal $c > 0$ such that*

$$R_{\mu,\rho}(B) \ge c\Psi_{\mu,\rho}(B^{-1/2}), \tag{81}$$

$$r_{\mu,\rho}(B,\delta) \ge c\Psi_{\mu,\rho}\left(\sqrt{\log(1/\delta)/B}\right). \tag{82}$$

The proof uses Lemma A.3; it does not assert this strengthened bound for an arbitrary selected-magnitude oracle.

Proof. Fix any feasible w in (79) and write $f_h(d) = \min\{w(d), h\sqrt{w(d)}/d\}$. Choose small universal constants $b > 0$ and $c_0 > 0$ with $c_0 \ll b$ and $b + c_0 \le 1$. Define

$$q_0(d) = bw(d), \qquad q_1(d) = bw(d) + c_0 f_h(d).$$

Both belong to (75). Their target separation is $c_0 \int f_h \, d\mu$. Where $w > 0$, put $u = bw$, $v = c_0 f_h$. The relative change v/u is uniformly small and

$$\frac{v^2 d^2}{u} \le \frac{c_0^2}{b} h^2. \tag{83}$$

Where $w = 0$ the two laws agree. The sign-prior likelihood ratios, and hence the latent coin ratios, are uniformly in $[1-\kappa, 1+\kappa]$ for a fixed $\kappa < 1$.

At a finite observation prefix, apply Lemma A.4. To bound its ith term, simulate all other coins under q_0 as external randomness, and reveal D_i only for the purpose of bounding information. Conditional on $D_i = d$, this is a single-coin stopped experiment with priors $u(d), u(d) + v(d)$. Data processing, Lemma A.3, and (83) give

$$H^2(P_0, P_i) \le Ch^2 \mathbb{E}_0 \tau_i.$$

Here the conditional distributions of all simulated coins and the algorithmic seed are the same in both worlds. Summing over coins counts each observation once. Thus

$$\mathrm{KL}(P_0, P_1) \le Ch^2 \mathbb{E}_0(T \wedge n) \le Ch^2 B.$$

Let n tend to infinity. For RMS choose a small constant multiple of $B^{-1/2}$; for confidence choose a small constant multiple of $\sqrt{\log(1/\delta)/B}$. The same two-point testing arguments as Proposition A.2 apply. Finally, $\Psi(\lambda h) \ge \lambda \Psi(h)$ for $0 < \lambda \le 1$, and taking the supremum over w yields the stated scales. □

A.3.3 An exact solution of the variational problem

Proposition A.6. *For $0 < \rho < 1$, let $t = t_{\mu,\rho} > 0$ be the unique solution of*

$$\mathbb{E}_\mu \min\{1, t^2/D^2\} = \rho. \tag{84}$$

Then $w_*(d) = \min\{1, t^2/d^2\}$ is an optimizer in (79) for every $h > 0$. If $h \geq t$, then $\Psi_{\mu,\rho}(h) = \rho$. If $0 < h < t$, then

$$\Psi_{\mu,\rho}(h) = F_\mu(h) + h\mathbb{E}_\mu[D^{-1};\, h < D \leq t] + ht\mathbb{E}_\mu[D^{-2};\, D > t]. \tag{85}$$

These identities allow atoms and singular laws.

Proof. The left side of (84) is continuous, tends to zero at $t = 0$, and is strictly increasing until it reaches one. This proves existence and uniqueness at $\rho < 1$. For $h < t$, put $\lambda = h/(2t) < 1/2$. For fixed d, the concave function $f_d(w) = \min\{w, (h/d)\sqrt{w}\}$ is linear until $w = (h/d)^2$, then square-root, on $[0, 1]$. Maximizing $f_d(w) - \lambda w$ on $[0, 1]$ gives exactly $w_*(d)$: in the square-root portion its stationary point is t^2/d^2, and otherwise the maximum is at 1. Integration and $\int w_*\, d\mu = \rho$ prove optimality. Splitting at h and t gives (85), with the endpoint contributions agreeing if there are atoms. If $h \geq t$, $f_d(w_*) = w_*$ everywhere, so the objective equals the maximum possible value ρ. □

In particular,

$$\Psi_{\delta_\Delta,\rho}(h) = \min\{\rho, h\sqrt{\rho}/\Delta\}, \qquad \rho\omega_\mu(h/\sqrt{\rho}) \leq \Psi_{\mu,\rho}(h) \leq \omega_\mu(h). \tag{86}$$

The heuristic is the feasible *constant allocation* $w \equiv \rho$ in the variational problem. Its failure is an allocation failure, not a failure of its two sanity anchors. If ρ stays bounded away from zero, (86) recovers the original modulus up to constants. The inverse-square tail in (85) describes how hard negative coins outside the unresolved core still contribute.

Correction to the earlier conjecture. The proposed arbitrary-law matching-Ψ upper bound is refuted by Theorem 11.2. The lower bound and optimizer above remain valid. The matching replacement for arbitrary laws is the entropy–tail law of section 13.

A.4 Matching rare-event rates for power margins

Let $\mu_{\alpha,a}$ have $F(t) = (t/a)^\alpha$ on $(0, a]$, with $0 < a \leq 1/4$. Write

$$\mathcal{R}_{\alpha,\rho}(x) = \begin{cases} \min\{\rho, x^\alpha\}, & 0 < \alpha < 1, \\ \min\{\rho, x[1 + \log_+(\rho/x)]\}, & \alpha = 1, \\ \min\{\rho, \rho^{1-1/\alpha}x\}, & 1 < \alpha < 2. \end{cases} \tag{87}$$

Theorem A.7. *For fixed $0 < \alpha < 2$, all $0 < a \leq 1/4$, $0 < \rho \leq 1/4$, $B \geq 1$, and $0 < \delta \leq 1/4$,*

$$R_{\mu_{\alpha,a},\rho}(B) \asymp_\alpha \mathcal{R}_{\alpha,\rho}((a\sqrt{B})^{-1}),$$
$$r_{\mu_{\alpha,a},\rho}(B, \delta) \asymp_\alpha \mathcal{R}_{\alpha,\rho}(a^{-1}\sqrt{\log(1/\delta)/B}).$$

The same upper bounds hold for every magnitude law with $F_\mu(t) \leq \min\{(t/a)^\alpha, 1\}$. The upper construction knows (α, a, ρ), not the individual magnitudes or sign function. It does not give a distribution-blind rare-event algorithm for arbitrary laws.

A.4.1 Lower proof

Take $A = (0, a\rho^{1/\alpha}]$ in Proposition A.2. Its mass is ρ. Put $t_\rho = a\rho^{1/\alpha}$. For $h \geq t_\rho$ the restricted modulus is ρ. For $h < t_\rho$ it is

$$\left(\frac{h}{a}\right)^\alpha + \frac{\alpha h}{a^\alpha} \int_h^{t_\rho} d^{\alpha-2}\, dd. \tag{88}$$

For $\alpha \neq 1$, the integral is $(t_\rho^{\alpha-1} - h^{\alpha-1})/(\alpha - 1)$; for $\alpha = 1$ the result is $(h/a)[1 + \log(a\rho/h)]$. These expressions are comparable, with constants depending only on α, to (87). This proof uses the balanced rare-region construction and does not need the imported single-coin rare-prior lemma.

A.4.2 Why the original unmodified variance is not rarity-scaled

Proposition A.8. *Fix an interior magnitude $0 < d < 1/2$ and let every coin be negative. For the original untruncated geometric single-term estimator Y, which is unbiased and square-integrable at this fixed gap,*

$$\operatorname{Var} Y \geq \frac{1/2 - d}{1 - 2^{-1/2}} > 0.$$

For the truncation at level L, the same second-moment lower bound holds, and $\operatorname{Var} Y_L \geq (1/2 - d)/(1 - 2^{-1/2}) - (\mathbb{E} Z_{2^L})^2$.

Proof. The event $J = 0$ has probability $p_0 = 1 - 2^{-1/2}$ and contributes $Y = X/p_0$ for one Bernoulli observation with success probability $1/2 - d$. Therefore $\mathbb{E}Y^2 \geq (1/2 - d)/p_0$. In the untruncated case $\mathbb{E}Y = 0$. In the truncated case telescoping gives $\mathbb{E}Y_L = \mathbb{E}Z_{2^L}$; subtract its square. □

This is an analytic obstruction to deriving a ρ factor by reusing the original second-moment estimate unchanged. At $d = 1/4$ the lower floor is approximately 0.8536, even though $\theta = 0$.

A.4.3 A shifted-start construction

Retain the original classifier Z_n with dyadic checkpoints and threshold $\sqrt{[1 + \log(n/k)]/k}$ at checkpoint $k < n$. Define

$$e_n(d) = \min\{1, (nd^2)^{-2}\}, \qquad m_n(d) = \min\{n, d^{-2}[1 + \log_+(nd^2)]\}.$$

The original conditional inequalities are $\mathbb{P}(Z_n \neq S \mid d, S) \leq e_n(d)$ and $\mathbb{E}(\tau_n \mid d, S) \leq 17 m_n(d)$ (sections 4 and 5).

Lemma A.9. *For $na^2 \geq 1$ and $0 < \alpha < 2$,*

$$e_n := \mathbb{E}_{\mu_{\alpha,a}} e_n(D) \leq K_e a^{-\alpha} n^{-\alpha/2}, \qquad K_e = \frac{4}{4 - \alpha}, \tag{89}$$

$$\mathbb{E}_{\mu_{\alpha,a}} m_n(D) \leq K_m a^{-\alpha} n^{1-\alpha/2}, \qquad K_m = 1 + \frac{\alpha}{2 - \alpha} + \frac{2\alpha}{(2 - \alpha)^2}. \tag{90}$$

The same bounds hold under the stated stochastic-dominance envelope.

Proof. Split each integral at $d = n^{-1/2}$. For (89), the first part is $(a\sqrt{n})^{-\alpha}$; extending the remaining power integral to infinity gives the factor $1 + \alpha/(4 - \alpha)$. For (90), substitute $u = d\sqrt{n}$ in the tail and extend its upper endpoint to infinity. The remaining factor is

$$1 + \alpha \int_1^\infty u^{\alpha-3}(1 + 2\log u)\, du = K_m.$$

Both integrands are nonincreasing in d, which proves the envelope statement. The restriction $\alpha < 2$ is essential to this particular work bound. □

Choose the smallest dyadic n_0 satisfying

$$n_0 \geq a^{-2}(K_e/\rho)^{2/\alpha}. \tag{91}$$

If $B < 4n_0$, return $\rho/2$. Otherwise choose $N = n_0 2^L$ as the largest such number with $4N \leq B$. Thus $B/8 < N \leq B/4$. Set $n_j = n_0 2^j$ and

$$\Delta_0 = Z_{n_0}, \qquad \Delta_j = Z_{n_j} - Z_{n_{j-1}}, \quad j \geq 1,$$

using a shared observation prefix within each increment. Sample J with $p_j = (1-r)r^j$, $r = 2^{-1/2}$, and return $Y_L = \Delta_J/p_J$ if $J \leq L$ and zero otherwise. Coins are independent between calls. The observed work C_L is at most N.

Lemma A.10. *With $S_L = \sum_{j=0}^{L} n_j^{(1-\alpha)/2}$, uniformly over the rare-event envelope class,*

$$|\mathbb{E}Y_L - \theta| \leq K_e a^{-\alpha} N^{-\alpha/2}, \tag{92}$$

$$\mathbb{E}Y_L^2 \leq C_\alpha a^{-\alpha} n_0^{-1/2} S_L, \tag{93}$$

$$\mathbb{E}C_L \leq C_\alpha a^{-\alpha} n_0^{1/2} S_L. \tag{94}$$

Proof. Telescoping proves (92). At the base, $\mathbb{E}Z_{n_0}^2 = \mathbb{P}(Z_{n_0} = 1) \leq \theta + e_{n_0} \leq 2\rho$. Dyadic rounding in (91) gives $\rho \leq 2^{\alpha/2} K_e a^{-\alpha} n_0^{-\alpha/2}$, so the base obeys the same power envelope as the increments. For $j \geq 1$,

$$\mathbb{E}\Delta_j^2 \leq e_{n_j} + e_{n_{j-1}} \leq C_\alpha a^{-\alpha} n_j^{-\alpha/2}.$$

Divide by $p_j = (1-r)\sqrt{n_0/n_j}$ and sum to obtain (93). For work, a coupled increment costs at most the sum of the two classifier costs, so its expectation is at most $34K_m a^{-\alpha} n_j^{1-\alpha/2}$; the base needs only the factor 17. Multiply by p_j and sum to obtain (94). □

A.4.4 Calibration and completion of the upper proof

Apply the original two-success wrapper to iid pairs (Y_L, C_L), with cap N. Its pilot markers have success probability C_L/N. Stop at the second success, at pilot count K, discard the pilot values, and average $K-1$ fresh production values. With $c = \mathbb{E}C_L > 0$ and $v = \operatorname{Var} Y_L$, the identities are

$$\mathbb{E}T = 4N - c \leq B, \quad \mathbb{E}\widehat{\theta} = \mathbb{E}Y_L, \quad \operatorname{Var}\widehat{\theta} = vc/N. \tag{95}$$

The identity $\mathbb{E}[1/(K-1)] = c/N$ is classical inverse-binomial estimation; the wrapper is inherited from section 5, not claimed as new [8, 16]. Independence of the fresh production samples is essential.

Equations (92)–(94) imply

$$\left(\mathbb{E}|\widehat{\theta} - \theta|^2\right)^{1/2} \leq C_\alpha \left[a^{-\alpha} N^{-\alpha/2} + a^{-\alpha} N^{-1/2} S_L\right]. \tag{96}$$

For $\alpha < 1$, $S_L \leq C_\alpha N^{(1-\alpha)/2}$. For $\alpha = 1$, $S_L = L + 1$. For $\alpha > 1$, $S_L \leq C_\alpha n_0^{(1-\alpha)/2}$. Substitution of (91) and $N \asymp B$ yields (87). In the small-budget branch, the right side of the target rate is at least a constant multiple of ρ, so returning $\rho/2$ suffices. Projecting the final estimate onto $[0, \rho]$ cannot increase squared error.

For high confidence, use an odd number $k \asymp \log(1/\delta)$ of independent groups, each with expected budget B/k, and take their median. Chebyshev gives failure at most $1/8$ at $\sqrt{8}$ times each group's RMS bound; a binomial tail bound gives median failure at most $e^{-9k/32}$. The rate follows after replacing B by B/k. If group budgets are too small, the constant estimator handles the saturated regime. Expected costs add. This completes Theorem A.7.

A.4.5 Consequences and limits of the theorem

For $0 < \varepsilon \le c_\alpha \rho$, the required expected work is, up to α-dependent factors,

Margin	Work, multiplied by $a^{-2}\log(1/\delta)$
$0 < \alpha < 1$	$\varepsilon^{-2/\alpha}$
$\alpha = 1$	$\varepsilon^{-2}[1 + \log(\rho/\varepsilon)]^2$
$1 < \alpha < 2$	$\rho^{2-2/\alpha}\varepsilon^{-2}$

At constant ρ these reduce to the corresponding original rates. The fixed-gap rate (76) is a separate cited special case; it is not obtained by extrapolating Theorem A.7 beyond its hypotheses.

For the fixed uniform law ($\alpha = 1$), set $x = (a\sqrt{B})^{-1} = \rho^2$. The proved rate is of order $\rho^2 \log(1/\rho)$, whereas the proposed heuristic is of order $\rho^{5/2}\log(1/\rho)$. Thus failure of the heuristic also occurs for a single fixed μ.

The optimizer formula predicts an additional boundary at $\alpha = 2$: in a sufficiently large-budget regime its lower modulus has scale $(h/a)\sqrt{\rho \log(1/\rho)}$, versus $(h/a)\sqrt{\rho}$ for $\alpha > 2$. The above-two rate is matched in appendix B; at the quadratic law, appendix C proves the additional scale-entropy factor. Constants in the proved theorem need not be uniform as α approaches 1 or 2.

A.5 A certified general upper bracket, not a closed characterization

For any finite dyadic sequence $n_0, \dots, n_L = N$ with $4N \le B$, let

$$e_n = \mathbb{E}_\mu e_n(D), \quad v_0 = \rho + e_{n_0}, \quad v_j = e_{n_j} + e_{n_{j-1}}, \quad b_0 = 17\mathbb{E}_\mu m_{n_0}, \quad b_j = 34\mathbb{E}_\mu m_{n_j} \ (j \ge 1).$$

All these quantities are finite at finite caps. Put $S = \sum_{j=0}^{L} \sqrt{v_j b_j}$ and choose a finite categorical law $p_j \propto \sqrt{v_j/b_j}$. Sample Δ_J/p_J and apply (95) with cap N.

Proposition A.11. *This estimator has expected work at most B and RMS error at most*

$$\sqrt{e_N^2 + S^2/N}.$$

Consequently the minimax risk is bounded above by the infimum of this expression over the finite grids, and also by $\rho/2$ and by the clipped original upper bound. The allocation here requires the stated integrals of μ or valid upper bounds for them.

Proof. The same coupling argument gives $\mathbb{E}\Delta_j^2 \le v_j$ and $\mathbb{E}W_j \le b_j$. Thus $\mathbb{E}Y^2 \le \sum v_j/p_j$ and $\mathbb{E}C \le \sum p_j b_j$. Their product equals at most S^2 for the stated choice of probabilities. Telescoping bounds the bias by e_N, and calibration bounds the variance by S^2/N. □

This bracket is constructive, but it does not remove the fixed-gap screening difficulty or establish arbitrary-law optimality. The sharp arbitrary-law replacement is theorem 13.2; the matching-Ψ upper objective is false by theorem 11.2. It would be incorrect to present the variational identity as if it supplied that algorithm.

B Asymmetric screening above exponent two

The upper proof is uniform over the stated envelope and global rarity class. The matching lower proof cites the published fixed-gap input and supplies the simulation to variable magnitudes. This appendix is a paper argument. The same rate is Lean-checked by a different observable construction and a lower bound without the published input; see section 18.

B.1 Claim and observational contract

For $\alpha > 2$ and $0 < a \le 1/2$, let $\mathcal{M}_{\alpha,a}$ be the probability laws on $(0, 1/2]$ satisfying

$$\mu((0, x]) \le (x/a)^\alpha \qquad (0 < x < a).$$

A coin has hidden $D \sim \mu$ and $S \mid D = d \sim \mathrm{Ber}(q(d))$; its successive flips are conditionally independent Bernoulli variables with parameter $1/2 + (2S - 1)D$. Fresh coins are iid. All functions q below are measurable, $0 \le q \le 1$, and $\theta = \int q \, d\mu \le \rho$, where $0 < \rho \le 1/4$. Only actual coin flips are charged. Independent random bits and arithmetic are free, as in the repository's experiment.

An explicit realization uses mutually independent variables $D_i \sim \mu$, $U_i \sim \mathrm{Unif}(0, 1)$, and $V_{i,n} \sim \mathrm{Unif}(0, 1)$, for $i, n \ge 1$, on their countable product probability space. Set $S_i = \mathbf{1}_{\{U_i \le q(D_i)\}}$ and $X_{i,n} = \mathbf{1}_{\{V_{i,n} \le 1/2 + (2S_i - 1)D_i\}}$. An additional independent seed supplies every algorithmic random choice. These are measurable variables and give the required conditional iid flip tapes and independent fresh coins. The algorithms below use only the next unused $X_{i,n}$ on a requested coin. Their likelihoods, comparisons, and stopping decisions depend only on finite observed prefixes. The random-level and Poisson choices are measurable functions of the independent seed.

Set

$$\beta = 1 - 2/\alpha, \qquad A_0 = \frac{\alpha}{\alpha - 2}, \qquad A_1 = \frac{\alpha}{(\alpha - 2)^2},$$
$$w_\alpha = \frac{3}{2}\left[A_0\left(1 + 10\log 2 + \frac{1}{e\beta}\right) + A_1\right], \qquad K = \frac{153}{56} < 3.$$

All constants here precede the choice of a, μ, ρ, q, B.

Theorem B.1 (Uniform upper theorem above two)**.** *For every $\alpha > 2$, $0 < a \le 1/2$, and $0 < \rho \le 1/4$, there is one observable single-coin estimator $Y = Y_{a,\rho}$, independent of μ and q, such that for every $\mu \in \mathcal{M}_{\alpha,a}$ and every admissible q,*

$$\mathbb{E}Y = \theta, \qquad \mathbb{E}Y^2 \le K\rho, \qquad \mathbb{E}T_Y \le w_\alpha/a^2.$$

In particular all integrability and finite-work assertions are conclusions. For every $B \ge 1$, a family of observable algorithms depending only on (α, a, ρ, B) has uniform expected work at most B and worst-case RMS at most

$$\min\left\{\frac{\rho}{2}, \sqrt{Kw_\alpha}\,\frac{\sqrt{\rho}}{a\sqrt{B}}\right\}.$$

The final budgeted estimator need not be unbiased, because clipping or a constant estimate may be used. The underlying compound-Poisson estimator is unbiased.

The theorem gives a known-law result for the exact power law $F(d) = (d/a)^\alpha$ on $(0, a]$. Its proof also gives the stated uniform upper bound for the larger margin class. This is not an arbitrary-law solution and not an estimator independent of a rarity bound. In particular constants diverge as $\alpha \downarrow 2$.

B.2 Observable asymmetric sequential classifier

For $k = 0, 1, \ldots$ define known reference gaps and weights

$$r_k = a2^{-(k+1)}, \qquad \pi_k = 2^{-(k+1)}.$$

For the observed flips $X_1, X_2, \ldots$ of one coin, let

$$L^+_{k,n} = \prod_{i=1}^{n} (1 + 2r_k)^{X_i}(1 - 2r_k)^{1-X_i}, \quad L^-_{k,n} = \prod_{i=1}^{n} (1 - 2r_k)^{X_i}(1 + 2r_k)^{1-X_i}.$$

These are likelihood ratios against the fair-coin law. Fix $0 < \varepsilon \le 1$. At the first $n \ge 1$ for which some $k \ge 0$ satisfies either

$$L^-_{k,n} \ge \frac{1}{\pi_k \varepsilon} \quad \text{or} \quad L^+_{k,n} \ge \frac{1}{\pi_k \rho \varepsilon}, \tag{B.C}$$

stop and return $Z_\varepsilon = 0$ in the former case and $Z_\varepsilon = 1$ in the latter. Fix negative priority in case both conditions first occur together. The stopping time is T_ε.

This rule does not observe D, S, or q. It also does not require infinitely many operations at any finite observation prefix. If $k \ge n \ge 1$, then $2r_k \le 2^{-k}$ and

$$L^{\pm}_{k,n} \le (1 + 2r_k)^n \le \exp(n2^{-k}) \le e^{1/2} < 2, \qquad \frac{1}{\pi_k \varepsilon} \ge 2^{k+1} \ge 4.$$

Thus only the finite set $k < n$ needs to be checked at time n. The positive threshold is even larger. The products for any newly activated index can be computed from n and the observed head count. Every action and output is therefore a measurable function of the independent seed and observable history.

Lemma B.2 (Errors and expected work)**.** *For every fixed $0 < d \le 1/2$ and $s \in \{0, 1\}$,*

$$\mathbb{P}(Z_\varepsilon = 0 \mid d, 1) \le \varepsilon, \qquad \mathbb{P}(Z_\varepsilon = 1 \mid d, 0) \le \rho\varepsilon.$$

Put $x = \min(d, a)$ and $c_0 = 1 + 2\log 2$. Then

$$\mathbb{E}(T_\varepsilon \mid d, s) \le \frac{3}{2} x^{-2} \left[c_0 + \log(a/x) + \log(1/\varepsilon) + s \log(1/\rho)\right]. \tag{B.1}$$

In particular this stopping time is finite almost surely for every allowed latent state, including $d = 1/2$.

Proof. Under a positive coin, the expected next factor of L^-_k is $1 - 4r_k d \le 1$; under a negative coin, that of L^+_k is the same. Consequently these wrong-sign processes are nonnegative supermartingales starting at one. For completeness, stopping a nonnegative supermartingale M at its first crossing of b and at a deterministic horizon N gives $b\mathbb{P}(\max_{n \le N} M_n \ge b) \le \mathbb{E}M_{\tau \wedge N} \le 1$. Letting $N \uparrow \infty$ proves the crossing bound $1/b$ without an unbounded optional-stopping assumption. Applying this to each k and summing $\sum_k \pi_k = 1$ proves the two error bounds, once almost-sure termination is established below.

Choose $k = \lceil \log_2(a/x) \rceil$. Then $x/4 < r_k \le x/2$ (the weaker $x/4 \le r_k \le x/2$ suffices), $r_k \le 1/4$, and $(k + 1)\log 2 \le \log(a/x) + 2\log 2$. Under the true sign, the logarithm of the corresponding correct-sign likelihood ratio is a sum of iid bounded increments, with mean

$$g(d, r_k) = \tfrac{1}{2}\log(1 - 4r_k^2) + d \log \frac{1 + 2r_k}{1 - 2r_k}.$$

The elementary inequalities $\log(1-z) \geq -z/(1-z)$ and $\log((1+u)/(1-u)) \geq 2u$ yield

$$g(d, r_k) \geq -\tfrac{8}{3} r_k^2 + 4dr_k \geq \tfrac{8}{3} x r_k \geq \tfrac{2}{3} x^2. \tag{B.2}$$

Every upward log increment is at most $\log(1+2r_k) < 1$. If σ is the first time this particular log process reaches

$$A = \log(1/\varepsilon) + (k+1)\log 2 + s\log(1/\rho),$$

then for every integer N, its value at $\sigma \wedge N$ is at most $A+1$. The elementary bounded-horizon identity

$$\mathbb{E} \sum_{i=1}^{\sigma \wedge N} \xi_i = \sum_{i=1}^{N} \mathbb{E}[\xi_i \mathbf{1}_{\{\sigma \geq i\}}] = g(d, r_k)\mathbb{E}(\sigma \wedge N)$$

uses only that $\{\sigma \geq i\}$ is determined before increment i. Hence $\mathbb{E}(\sigma \wedge N) \leq (A+1)/g(d, r_k)$. Monotone convergence proves $\mathbb{E}\sigma \leq (A+1)/g(d, r_k) < \infty$ and therefore $\sigma < \infty$ almost surely. Since $T_\varepsilon \leq \sigma$, equations (B.1)–(B.2) follow. This also completes the termination needed for the error assertions. □

B.3 The moment calculation that removes the logarithmic loss

The relevant distinction from symmetric high-confidence screening is that $\log(1/\rho)$ in (B.1) is paid only on positive coins.

Lemma B.3 (Margin integration)**.** *If $\mu \in \mathcal{M}_{\alpha,a}$, $x = \min(D, a)$, $\alpha > 2$, and $0 \leq q \leq 1$, $\mathbb{E}q(D) \leq \rho$, then*

$$\begin{aligned} \mathbb{E}x^{-2} &\leq A_0 a^{-2}, \\ \mathbb{E}[x^{-2}\log(a/x)] &\leq A_1 a^{-2}, \\ \mathbb{E}[q(D)x^{-2}] &\leq A_0 a^{-2}\rho^\beta. \end{aligned} \tag{B.3}$$

For the exact power law the first two inequalities are equalities, and the last is attained by $q(d) = \mathbf{1}_{\{d \leq a\rho^{1/\alpha}\}}$.

Proof. The variable x stochastically dominates the power-law variable $aU^{1/\alpha}$, where U is uniform on $(0,1)$. Integrating decreasing nonnegative functions, or equivalently applying the layer-cake identity to their sublevel events, gives the first two inequalities. Their explicit integrals are

$$\frac{\alpha}{a^\alpha}\int_0^a d^{\alpha-3}\,dd = A_0 a^{-2}, \qquad \frac{\alpha}{a^\alpha}\int_0^a d^{\alpha-3}\log(a/d)\,dd = A_1 a^{-2}.$$

For the third, Tonelli's theorem and $q \leq 1$ give

$$\mathbb{E}[qx^{-2}] = a^{-2}\mathbb{E}q + \int_{a^{-2}}^{\infty} \mathbb{E}[q\mathbf{1}_{\{x^{-2}>u\}}]\,du \leq a^{-2}\rho + \int_{a^{-2}}^{\infty} \min\{\rho, (a^2u)^{-\alpha/2}\}\,du.$$

Splitting at $a^{-2}\rho^{-2/\alpha}$ evaluates the last expression as $A_0 a^{-2}\rho^{1-2/\alpha}$. The claimed equality case follows by direct integration. These arguments concern nonnegative extended integrals; finiteness is established by the displayed evaluations. □

Combining (B.1) and (B.3), with $\theta \leq \rho$, gives the uniform bound

$$\mathbb{E}T_\varepsilon \leq \frac{3}{2a^2}\left[A_0\{c_0 + \log(1/\varepsilon)\} + A_1 + A_0\rho^\beta\log(1/\rho)\right]. \tag{B.4}$$

Here $\rho^\beta\log(1/\rho) \leq 1/(e\beta)$, obtained by maximizing $te^{-\beta t}$ over $t \geq 0$. This is where $\alpha > 2$ is essential. The logarithm is controlled uniformly in the supplied rarity bound.

B.4 Unbiased random-level estimator

The single-term randomized telescoping device is an established unbiased multilevel ingredient; see Vihola [4, Theorem 3]. The particular classifier and the rarity-sensitive moment and work calculation below are analyzed explicitly. Let $\varepsilon_j = 16^{-(j+1)}$ for $j \geq 0$ and use the above classifiers $Z_j = Z_{\varepsilon_j}$ on one common coin-flip tape. Define

$$\Delta_0 = Z_0, \qquad \Delta_j = Z_j - Z_{j-1} \quad (j \geq 1).$$

Choose an independent J with $p_j = \mathbb{P}(J = j) = 2^{-(j+1)}$, draw one fresh coin, and return $Y = \Delta_J / p_J$. Only the classifiers actually needed for this increment are evaluated. Both threshold families in (B.C) grow by the same factor 16 between successive levels, so $T_j \geq T_{j-1}$ pathwise. The increment costs exactly T_J, with the earlier decision stored from the same prefix. No assumption about independence between classifier levels is made.

The conditional error bounds give

$$\mathbb{P}(Z_j \neq S) \leq (\theta + \rho)\varepsilon_j \leq 2\rho\varepsilon_j, \qquad \mathbb{E}Z_0^2 = \mathbb{E}Z_0 \leq \theta + \rho\varepsilon_0 \leq \tfrac{17}{16}\rho.$$

For $j \geq 1$, a disagreement implies an error of at least one classifier, hence

$$\mathbb{E}\Delta_j^2 \leq 2\rho(\varepsilon_j + \varepsilon_{j-1}) = \tfrac{17}{8}\rho\, 16^{-j}.$$

All square terms are nonnegative, so Tonelli gives

$$\mathbb{E}Y^2 = \sum_{j\geq 0} \frac{\mathbb{E}\Delta_j^2}{p_j} \leq \left(\frac{17}{8} + \frac{17}{4}\sum_{j\geq 1} 8^{-j}\right)\rho = \frac{153}{56}\rho. \tag{B.5}$$

This proves square integrability, hence integrability, before using signed expectations. Alternatively $\sum_j \mathbb{E}|\Delta_j| < \infty$ directly follows from the same disagreement bounds because $|\Delta_j| = \Delta_j^2$ for $j \geq 1$. Consequently

$$\mathbb{E}Y = \sum_{j\geq 0} \mathbb{E}\Delta_j = \lim_{j\to\infty} \mathbb{E}Z_j = \mathbb{E}S = \theta. \tag{B.6}$$

Finally (B.4), $\mathbb{E}(J+1) = 2$, and $\log(1/\varepsilon_j) = 4(j+1)\log 2$ imply

$$\mathbb{E}T_Y \leq \frac{3}{2a^2}\left[A_0\{1 + 10\log 2 + 1/(e\beta)\} + A_1\right] = w_\alpha / a^2. \tag{B.7}$$

Equations (B.5)–(B.7) prove the single-coin part of Theorem B.1, with separate uniform second-moment and work bounds.

B.5 Exact expected-budget calibration and matching power-law rate

Independent Poisson replication and its variance formula are also established ingredients; see Vihola [4, Example 8]. We give the direct calculation to verify the exact expected budget, including when the expected count is below one. Set $\lambda = Ba^2/w_\alpha > 0$. Draw an independent $M \sim \text{Poisson}(\lambda)$ and run independent copies $Y_1, \ldots, Y_M$ of the single-coin procedure on fresh coins. Return

$$\widehat{\theta} = \lambda^{-1}\sum_{i=1}^{M} Y_i.$$

The empty sum is zero. Conditioning first on M, then using its mean and variance λ, proves

$$\mathbb{E}\widehat{\theta} = \theta, \quad \operatorname{Var}(\widehat{\theta}) = \lambda^{-1}\mathbb{E}Y^2 \leq \frac{Kw_\alpha\rho}{a^2 B}, \quad \mathbb{E}T \leq \lambda w_\alpha / a^2 = B.$$

All sums stop almost surely and have finite expected work. These identities are valid even when $\lambda < 1$; no rounding of the budget is hidden. Clip the result to $[0, \rho]$, which cannot increase squared error for $\theta \in [0, \rho]$. When its displayed RMS bound exceeds $\rho/2$, instead output $\rho/2$ at zero work. The choice uses only the allowed parameters. This proves the budgeted part of Theorem B.1.

Lemma B.4 (Localized fixed-gap RMS lower bound)**.** *There is a universal $c > 0$ such that the minimax RMS risk for fixed gap $0 < a \le 1/2$, prior $0 \le \vartheta \le \rho \le 1/4$, and uniform expected work at most $B \ge 1$ is at least $c \min\{\rho, \sqrt{\rho}/(a\sqrt{B})\}$. This lemma uses Lee–Valiant Corollary 19 and Lemma 18 as external inputs.*

Proof. The exact imported assertions imply the following: there are universal $a_*, c_*, C_* > 0$ such that, if $a < a_*$ and $u, v, v/u < c_*$, the KL divergence of any finite-prefix fully adaptive fixed-gap transcript under sign priors u and $u + v$ is at most $C_* v^2 a^2 \mathbb{E}_u T/u$. Indeed, Corollary 19 bounds each single-coin hybrid's squared Hellinger distance by a universal multiple of v^2a^2/u times its expected observations under u. Lemma 18 bounds joint KL by the sum of those distances. Summing coin work counts each observation once. Their Hellinger normalization is absorbed in C_*. We use the original-prior, not midpoint-prior, expectation. All their smallness hypotheses are enforced in the choices below.

For arbitrary almost-surely terminating policies with finite expected work, first stop at $T \wedge n$ and number coins by first observation. At most n coins are then observed, so the finite-prefix statement applies. This numbering preserves the paper's creation and interleaving model: an unobserved coin has provided no information, and its iid latent draw can be deferred until its first observation. A dictionary from the policy's original coin labels to the new indices preserves every revisit. Unobserved-label choices use only the seed and observed history, so this relabeling changes neither their law nor any charged observation or output. Include the independent random seed in the transcript; conditioning on it and averaging extends the deterministic statement. Relative entropy along increasing prefix sigma fields converges to full transcript relative entropy. The stopped output is measurable in this sigma field. Thus the same bound holds at unbounded stopping, without a globally fixed finite coin family.

Write $r = \min\{\rho, \sqrt{\rho}/(a\sqrt{B})\}$, and take $u = b\rho$, $v = \gamma r$ for fixed sufficiently small positive universal constants b, γ. Choose $b + \gamma \le 1$ and $\gamma/b < c_*$, so $u, u + v \in [0, \rho]$ and all prior smallness assumptions hold. Since $r^2a^2B \le \rho$, for $a < a_*$ the transcript KL is at most $C_*\gamma^2/b$.

For $a \ge a_*$, give the algorithm each coin's true sign on its first observation. This more informative experiment simulates all requested flips from that sign. Each newly observed sign has Bernoulli KL at most $2v^2/u$, using $u + v \le 1/4$ and the KL–chi-square inequality. The expected number of newly observed signs is at most $\mathbb{E}_u T \le B$, so full KL is at most $2v^2B/u \le 2\gamma^2/(ba_*^2)$. The same finite-prefix argument justifies unbounded stopping. This also covers the endpoint $a = 1/2$.

Choose γ small enough that both displayed KL bounds are at most $1/8$. Pinsker's inequality then bounds total variation by $1/4$. If an estimator has RMS at most R at both priors, thresholding halfway between them gives a test whose two error probabilities sum to at most $8R^2/v^2$. Every test has error sum at least $1 - \mathrm{TV} \ge 3/4$. Therefore $R \ge v/4$, proving the claim with $c = \gamma/4$. This proves all budget and rarity regimes, including the saturated regime, from the precise source inputs rather than an informal high-confidence asymptotic statement. □

Corollary B.5 (Exact-power regime $\alpha > 2$)**.** *Let μ have distribution function $(d/a)^\alpha$ on $(0, a]$ with $\alpha > 2$. Subject to the published fixed-gap lower bound of Lee–Valiant, there is a universal $c > 0$ and, for each $\alpha > 2$, a finite C_α such that for all $0 < a \le 1/2$, $0 < \rho \le 1/4$, and $B \ge 1$,*

$$c \min\left\{\rho, \frac{\sqrt{\rho}}{a\sqrt{B}}\right\} \le R_{\mu,\rho}(B) \le C_\alpha \min\left\{\rho, \frac{\sqrt{\rho}}{a\sqrt{B}}\right\}.$$

Moreover this risk is comparable, with constants depending only on α, to $\Psi_{\mu,\rho}(B^{-1/2})$.

Proof. The upper bound was just proved. For the lower bound it is enough to restrict to constant sign functions $q(d) = \vartheta \in [0, \rho]$. An experiment using fixed-gap a coins can simulate the variable-gap experiment at the same charged work: sample $D \sim \mu$ privately for each fresh fixed-gap coin; on each request, flip that coin once, and with probability D/a return its flip, otherwise return an independent fair bit. Conditional on the fixed coin's sign and its privately sampled D, the returned flips are iid Bernoulli$(1/2 \pm D)$. The same latent D is retained on revisits, and is not disclosed to the simulated algorithm. The simulation uses one charged flip per returned observation even when that flip is discarded. It preserves all interleaving and stopping choices. Thus a smaller variable-gap risk would contradict Lemma B.4. This reduction does not need the continuation note's arbitrary-law lower-bound theorem.

For comparison with the functional, constant allocation $w \equiv \rho$ and $D \le a$ give

$$\Psi_{\mu,\rho}(h) \ge \min\{\rho, h\sqrt{\rho}/a\}.$$

For every feasible w, Cauchy–Schwarz and $\mathbb{E}D^{-2} = A_0/a^2$ give

$$\int \min\{w, h\sqrt{w}/D\}\, d\mu \le \min\{\rho, h\sqrt{\rho A_0}/a\}.$$

Taking the supremum proves the claimed comparison, including saturation. □

B.6 Scope, edge cases, and failed approaches

The classifier and estimator operate at $D = 1/2$, with no likelihood logarithm at zero because the reference gaps satisfy $r_k \le 1/4$. At arbitrarily small $D > 0$ each conditional expected stopping time is finite; the margin assumption then proves population integrability. The level J has unbounded support and finite mean. For the admissible law $\mu = \delta_{1/2}$, $a = 1/2$, and $q = 0$, all observations are zero. At time n every correct-sign likelihood is at most $(3/2)^n$, whereas every level-j correct-sign threshold is at least $2/\varepsilon_j$. Thus each fixed n is exceeded for sufficiently large j. Since all $p_j > 0$, actual work has genuinely unbounded support and finite mean, not merely an unused unbounded seed. A finite Poisson number of such procedures also has finite expected work.

The global constraint $\int q\, d\mu \le \rho$ is used precisely in (B.3); no pointwise bound $q \le \rho$ is imposed. Its extremizer puts all positive mass on the smallest gaps. The algorithm never assumes known positive or negative conditional magnitude laws. The result is uniform jointly in B, ρ, a for each fixed $\alpha > 2$, but is not uniform as $\alpha \downarrow 2$.

Two earlier approaches exposed the obstruction addressed here. Symmetric high-confidence screening charges $\log(1/\rho)$ on all coins, retaining an unwanted logarithm. An unamplified ballot-survival estimator has positive conditional second moment of order $1/D$; placing positives near the threshold then gives an unwanted $\rho^{1-1/\alpha}$ moment scale. The asymmetric classifier followed by an unbiased random-level correction avoids both problems. The negative classifier error is $\rho\varepsilon$, whereas the positive error is only ε; the extra positive work is absorbed by the sharp global-rarity integral in (B.3).

The same proof does not settle $\alpha = 2$: its inverse-square integral diverges, and the uniform control of $\rho^\beta \log(1/\rho)$ loses meaning at $\beta = 0$. It does not characterize arbitrary laws with a finite inverse-square moment, because that moment alone does not uniformly control the logarithmically weighted moment or the rarity-weighted cost. It does not refute that larger conjecture. A direct finite-support modification gives finite bounds with support-dependent constants, but this is not the universal two-atom theorem.

C The full critical quadratic law

This appendix gives the full paper argument, including the upper bound. Both critical minimax bounds and the exact Ψ comparison are Lean-checked in the observable model stated in section 11. The formal upper uses an alternative dyadic-classifier construction, and the formal counterexample uses budget LH/ρ rather than the paper's ρ^{-2} sequence. The inherited arguments below retain their mathematical content. Existing agent review is not independent specialist verification.

C.1 Experiment, quantifiers, and result

Throughout fix $a = 1/4$ and

$$\mu(dd) = \frac{2d}{a^2}\mathbf{1}_{(0,a]}(d)\,dd.$$

For $0 < \rho \le 1/4$, let

$$\mathcal{Q}_\rho = \{q : (0,a] \to [0,1] \text{ measurable} : \int q\,d\mu \le \rho\}, \qquad \theta(q) = \int q\,d\mu.$$

Each genuine fresh coin has an independent hidden magnitude $D \sim \mu$, hidden sign $S \mid D \sim \mathrm{Ber}(q(D))$, and conditionally iid flips of mean $1/2 + (2S-1)D$. Neither latent variable is observed. One realization uses independent uniforms on $(0,1)$ for each coin: $D = a\sqrt{U_0}$, $S = \mathbf{1}_{\{U_1 \le q(D)\}}$, and flips $X_k = \mathbf{1}_{\{U_{k+1} \le 1/2+(2S-1)D\}}$ for $k \ge 1$. Measurable functions equal μ-almost everywhere define the same experiment; Borel versions may be used in this standard probability space.

The algorithm has an independent seed and sees only its seed and the observed history. At each step it stops and returns a real measurable output, revisits an observed coin, or observes a fresh coin. Every flip costs one. The output may be biased and nonlinear, and the choices may depend on all previous observations. Stopping may be unbounded. A design may know μ, ρ, B but is fixed before the unknown q is supplied. Work $T = \infty$ on nontermination. Define

$$R_{\mu,\rho}(B) = \inf_{A:\,\sup_{q\in\mathcal{Q}_\rho}\mathbb{E}_q T_A \le B} \sup_{q\in\mathcal{Q}_\rho} \left(\mathbb{E}_q(\widehat{\theta}_A - \theta(q))^2\right)^{1/2}.$$

This class permits arbitrary creation and interleaving, not a globally fixed number of coins. We number coins by their first charged observation. This is equivalent to giving names to iid unseen coins earlier: an unused coin's latent state is independent of the observed history and seed, and choosing among unused names using only that information still selects an iid fresh state. The relabeling retains the map for every later revisit. Creating or discarding an unobserved coin supplies no information. Consequently at most K different coins occur before the Kth charged observation.

Write $L = 1 + \log(1/\rho)$, $H = 1 + \log L$, and $J = \rho LH$. For the lower proof we use the convex subclass

$$\mathcal{X}_\rho = \{q \in L^\infty(\mu) : 0 \le q \le 1/2 \ \mu\text{-a.e.}, \ \int q\,d\mu \le \rho\}.$$

This pointwise cap restricts the lower-bound adversary only; the algorithmic upper and minimax definition still use the full global-rarity class.

Theorem C.1 (Full critical law)**.** *There exist universal constants $0 < c < C < \infty$ such that, for the one fixed law above, every $0 < \rho \le 1/4$ and $B \ge 1$ satisfy*

$$c\min\{\rho, \sqrt{J/B}\} \le R_{\mu,\rho}(B) \le C\min\{\rho, \sqrt{J/B}\}. \tag{C.A}$$

The lower bound holds for every observable algorithm in the stated full experiment, including arbitrary randomization, nonlinear estimation and data-dependent creation, revisiting and interleaving. The upper permits the known rarity and law; it has separately finite uniform work and squared error.

The upper construction is reproduced in the final subsection. Its source and the inherited proof identities are recorded in the v5 source manifest. The new lower proof has three steps. First, sign-testing entropy forces a small-bias variance product using *all-negative work*, not a supremum of work over populations. Second, convex estimation and minimax supply one hard pair valid for every capped single-coin observation policy. Third, a midpoint simulation charges those bounds to the actual work of an interactive population algorithm. The last step is the established Lee–Valiant Hellinger decomposition mechanism [1, Lemma 17], proved below at finite prefixes to make its observation and stopping scope explicit.

The convex-estimation step follows Juditsky–Nemirovski's saddle-point framework [17]; the passage from its exponential tails to RMS risk is written out here. Polyanskiy–Wu [18] already establish broad quadratic-risk affine comparisons and convex duality for functional estimation. Neither that general principle nor the Hellinger decomposition is claimed as new. The cost-sensitive common pair and its application to this critical hidden-coin class are the proposed contribution; priority remains unverified.

C.2 A single-coin entropy bound with all-negative reference cost

Fix $a = 1/4$ and $\mu(dd) = 2da^{-2}\mathbf{1}_{(0,a]}(d)\,dd$. Throughout this section,

$$\mathcal{X}_\rho = \left\{q : 0 \le q \le \tfrac{1}{2},\ \int q\,d\mu \le \rho\right\}, \qquad 0 < \rho \le \tfrac{1}{4}, \qquad L = 1 + \log(1/\rho), \quad H = 1 + \log L.$$

All functions defining populations are measurable. A single genuine coin has hidden $D \sim \mu$ and $S \mid D = d \sim \mathrm{Ber}(q(d))$; conditionally on (D, S) its observations are iid Bernoulli with parameter $1/2 + (2S-1)D$. A policy receives only its independent seed and observed bits. It may know μ, a, ρ, but not q, D, S.

Theorem C.2 (The critical product with null-population cost)**.** *There is a universal $c_0 > 0$ such that the following holds for every $0 < \rho \le 1/4$. Let an observable single-coin procedure have real output Y and stopping time $T \in \mathbb{N}_0 \cup \{\infty\}$, and suppose*

$$\sup_{q\in\mathcal{X}_\rho} |\mathbb{E}_q Y - \theta(q)| \le \rho/64, \qquad V := \sup_{q\in\mathcal{X}_\rho} \mathrm{Var}_q\, Y < \infty, \qquad \sup_{q\in\mathcal{X}_\rho} \mathbb{E}_q T < \infty, \qquad \theta(q) = \int q\,d\mu.$$

Writing $W_0 = \mathbb{E}_{q=0} T$, one has

$$VW_0 \ge c_0 \rho L H. \tag{C.BC1}$$

The same order lower bound holds with $M = \sup_{q\in\mathcal{X}_\rho} \mathbb{E}_q Y^2$ in place of V. The cost is the expectation under the all-negative population, rather than the supremum of expected work over the parameter class. The policy may stop before observing the coin and may have unbounded stopping.

The proof adapts the separately documented critical-two single-coin argument. We reproduce its finite-gap dependency to make this statement self-contained. Its entropy mechanism follows Chen–Li–Qiao [19, Lemma 4.1]; the Bernoulli, one-sided-cost statement used here is proved explicitly below. No novelty is claimed for that mechanism.

Lemma C.3 (Finite-gap sign entropy)**.** *There are universal $c_s > 0$ and $K_s < \infty$ with the following property. Let $d_1 > \cdots > d_K > 0$ satisfy $d_j/d_{j+1} \geq 2$ and $d_j \leq 1/128$. Suppose one observable sign-testing procedure, the same at all these gaps, has error at most $1/10$ under every Bernoulli law $1/2 \pm d_j$, and finite expected stopping time τ under both signs. For $K \geq K_s$,*

$$\sum_{j=1}^{K} d_j^2 \mathbb{E}_{-d_j} \tau \geq c_s K \log K. \tag{C.BC2}$$

Correctness and termination under the fair coin are not required. One may take $c_s = 1/512$ and $K_s = \lceil e^{32} \rceil$.

Proof. Put $b_0 = 1/1024$, $n_j = \lfloor b_0/d_j^2 \rfloor$, $u_j = \lceil 4\mathbb{E}_{-d_j}\tau \rceil$, $c_j = d_j^2 \mathbb{E}_{-d_j}\tau$, and $\bar{c} = K^{-1}\sum_j c_j$. Since $d_j \leq 1/128$, $n_j \geq b_0/(2d_j^2) \geq 1$. The event of returning minus by time n_j is measurable in the independent seed and the first n_j flips, including unused coordinates if the test has stopped. The KL divergence between these full prefixes at the minus and plus laws is at most

$$n_j \frac{(2d_j)^2}{(1/2 + d_j)(1/2 - d_j)} \leq \frac{64}{3} n_j d_j^2 \leq \frac{1}{48}.$$

Bernoulli KL is at most its chi-square divergence. Pinsker's inequality bounds the total variation by less than $1/8$. Since an early minus is an error under the plus law, it has probability at most $1/10 + 1/8$ under the minus law. Markov's inequality gives $\mathbb{P}_{-d_j}\{\tau > u_j\} \leq 1/4$. Consequently

$$A_j = \{\text{output minus and } n_j < \tau \leq u_j\} \quad \text{satisfies} \quad \mathbb{P}_{-d_j}(A_j) \geq 1 - \tfrac{1}{10} - (\tfrac{1}{10} + \tfrac{1}{8}) - \tfrac{1}{4} > \tfrac{1}{4}.$$

Let P_0 denote the fair-coin law with the same seed and policy. Comparing the deterministic prefixes of length u_j, their KL divergence from minus to fair is at most $4d_j^2 u_j \leq 16c_j + 1$. For an event of probability $p \geq 1/4$ under the minus law and probability z under the fair law, binary data processing and $\mathrm{kl}(p, z) \geq p\log(1/z) - \log 2$ therefore give

$$P_0(A_j) \geq e^{-64c_j - 8}. \tag{C.BC3}$$

Only deterministic finite prefixes have been compared; this does not invoke optional stopping under a possibly nonterminating fair law.

At least $K/2$ indices form a set G with $c_j \leq 2\bar{c}$. For them,

$$(n_j, u_j] \subset [b_0/(2d_j^2), (8\bar{c} + 1)/d_j^2].$$

The scales d_j^{-2} have successive ratios at least four. At any finite time the number of these intervals containing that time is at most $1 + \log_4((16\bar{c} + 2)/b_0)$. All events A_j refer to the same terminal time and output on a realization under the fair law; on a nonterminating path none occurs. Summing their indicators and using (C.BC3) yields

$$(K/2)e^{-128\bar{c} - 8} \leq 1 + \log_4(1024(16\bar{c} + 2)). \tag{C.BC4}$$

If $x = \log K \geq 32$ and $\bar{c} < x/512$, the left side exceeds $\frac{1}{2}e^{3x/4 - 8} \geq \frac{1}{2}e^{x/2} \geq x^2/16$. The right side is less than $1 + \log_4(32x + 2048) < 6 + x/4$, contradicting $x \geq 32$. Thus $\bar{c} \geq (\log K)/512$, as claimed. □

Proof of Theorem C.2. The bias bound implies $M \leq V + (65\rho/64)^2 < \infty$. First define the nonnegative extended conditional moments

$$M_s(d) = \mathbb{E}_{d,s} Y^2, \qquad w_s(d) = \mathbb{E}_{d,s} T.$$

The admissible populations $q = 0$ and $q \equiv \rho$ imply $\int M_- d\mu \le M$, $\int M_+ d\mu \le M/\rho$, and finite integrals of both work functions. All conditional quantities used below are thus finite outside a common null set. Define signed conditional means there, and give them arbitrary values on that null set. For $f(d) = \mathbb{E}_{d,+}Y - \mathbb{E}_{d,-}Y$, the two bias bounds imply

$$\int f \, d\mu = (\mathbb{E}_\rho Y - \mathbb{E}_0 Y)/\rho \ge 31/32.$$

Conditioning on the latent magnitude and then on its sign at the constant-prior population gives

$$V \ge \mathrm{Var}_\rho(Y) \ge \rho(1-\rho) \int f^2 \, d\mu \ge \rho(1-\rho)(31/32)^2 \ge 2\rho/3. \tag{C.BC5}$$

This uses only the bias guarantee, not a pointwise unbiased response. It also shows

$$M \le \tfrac{3}{2} V. \tag{C.BC6}$$

Normalize the first observation. The event $A = \{T \ge 1\}$ is decided by the independent seed before any observation, so its probability p is independent of the latent state. It is positive: otherwise the expectations at target values zero and ρ could not both satisfy the bias bound. The finite constant $c = \mathbb{E}[Y \mathbf{1}_{A^c}]$ is also population independent. Under the seed law conditioned on A, form the active procedure

$$Z = pY + c, \qquad T_Z = T.$$

It preserves every expectation of Y and always takes at least one observation. The conditioning is implementable by resampling independent seeds before observing the coin; its constants depend on the specified policy and known laws, not on q, D, S.

For $p < 1$, expansion, Cauchy–Schwarz on A^c, and completing the square give

$$\begin{aligned} \mathbb{E}_q Z^2 &= p\mathbb{E}_q[Y^2 \mathbf{1}_A] + 2c\mathbb{E}_q Y - c^2 \\ &\le pM + 2c\mathbb{E}_q Y - c^2/(1-p) \\ &\le pM + (1-p)(\mathbb{E}_q Y)^2 \le m := pM + 2\rho^2. \end{aligned}$$

For $p = 1$, take $c = 0$ and the final bound holds directly. Apply (C.BC5) to Z to obtain $m \ge 2\rho/3$ and $pM \ge 2\rho/3 - 2\rho^2 \ge \rho/6$. Set $P = MW_0$ and $W_{0Z} = \mathbb{E}_0 T_Z = W_0/p \ge 1$. Then

$$m \le 4pM \le 4P, \qquad mW_{0Z} \le 4P, \qquad P \ge \rho/6. \tag{C.BC7}$$

Indeed $2\rho^2 \le \rho/2 \le 3pM$. Every work quantity in (C.BC7) is all-negative work.

Select good gaps within the capped global class. Now use conditional means and moments for Z, keeping the same notation. The bad set $B_- := \{M_- > 1/64\}$ has mass at most $64m$. The set

$$B_+ := \{M_+ - M_- > 2m/\rho\}$$

has mass less than 2ρ. Otherwise nonatomicity supplies $E \subset B_+$ of mass 2ρ, and the admissible $q = \frac{1}{2}\mathbf{1}_E$ gives

$$\mathbb{E}_q Z^2 = \int M_- d\mu + \tfrac{1}{2} \int_E (M_+ - M_-) d\mu > 2m,$$

contradicting its bound. The response bad set $B_f := \{f < 3/4\}$ also has mass less than 2ρ. If not, choose such an E of mass 2ρ. Its population has target ρ, but

$$\mathbb{E}_{\mathbf{1}_E/2} Z - \mathbb{E}_0 Z = \tfrac{1}{2} \int_E f \, d\mu < 3\rho/4,$$

whereas the bias bounds force at least $31\rho/32$.

Remove these bad sets and the conditional integrability null sets. The remaining good set G has $\mu(G^c) \le 64m + 4\rho \le 70m$. On it,

$$|\mathbb{E}_- Z| \le 1/8, \qquad \mathbb{E}_+ Z \ge 5/8, \qquad M_- \le 1/64, \qquad M_+ \le 1/64 + 2m/\rho. \tag{C.BC8}$$

The use of sets of mass 2ρ is necessary for the cap $q \le 1/2$; the argument does not substitute a pointwise bound $q \le \rho$.

Build a common sign test. Let $n = \lceil 512(1 + m/\rho) \rceil \le 1282m/\rho$, the inequality using $m/\rho \ge 2/3$. On the same fixed-gap coin, restart the active procedure n times with independent seeds and consecutive unused flip blocks. Output plus if their mean exceeds $3/8$. At a fixed latent state the restarts are iid: condition on each finite stopping prefix, use the independence of its unused iid suffix, and sum over the countably many prefixes. Finite conditional expected work ensures almost-sure termination for both signs at each good gap. No new latent magnitude is drawn. The same test is used at every good gap. By (C.BC8), either mean is at least $1/4$ from the threshold, and Chebyshev bounds either error by

$$\frac{16(1/64 + 2m/\rho)}{n} < 1/10.$$

Its negative expected work at gap d is exactly $nw_-(d)$.

Average negative work over separated annuli. For $j \ge 3$, put

$$I_j = (a4^{-j}/2, a4^{-j}], \qquad \mu(I_j) = \tfrac{3}{4}16^{-j}, \qquad J = \left\lfloor \log_{16} \frac{3}{560m} \right\rfloor, \qquad K = J - 2.$$

When $J \ge 3$, every retained annulus has mass at least $140m$, so G occupies at least half of it. Choose $d_j \in G \cap I_j$ with

$$w_-(d_j) \le \frac{4}{\mu(I_j)} \int_{I_j} w_- \, d\mu.$$

This follows by averaging; the integral is positive since the active procedure always observes at least once. The choices are existential proof objects, not inputs to the policy. Successive gap ratios exceed two, every gap is at most $1/256$, and

$$\sum_{j=3}^{J} d_j^2 w_-(d_j) \le \frac{16a^2}{3} \int w_- d\mu = \frac{16a^2}{3} W_{0Z}.$$

Lemma C.3, the restart bound, and (C.BC7) imply, for $K \ge K_s$,

$$c_s K \log K \le \sum_{j=3}^{J} d_j^2 n w_-(d_j) \le \frac{16a^2}{3} \frac{1282m}{\rho} W_{0Z} \le \frac{30000a^2 P}{\rho}. \tag{C.BC9}$$

The last numerical constant is conservative, since $(16/3) \cdot 1282 \cdot 4 < 30000$.

Obtain one positive constant for every rarity. For sufficiently small ρ, if $P \ge \sqrt{\rho}$ the desired order already follows because $\sqrt{\rho} LH \to 0$. Otherwise (C.BC7) gives $m \le 4\sqrt{\rho}$ and

$$K \ge \frac{\frac{1}{2}\log(1/\rho) - \log(2240/3)}{\log 16} - 3.$$

For all sufficiently small universal ρ this exceeds K_s, is at least a positive universal multiple of L, and satisfies $\log K \geq \frac{1}{2}\log L$. Equation (C.BC9), with the fixed $a = 1/4$, gives $P \geq c\rho LH$. On the remaining compact interval of rarities, the lower bound $P \geq \rho/6$ in (C.BC7) allows the same conclusion after decreasing one positive universal constant. Finally (C.BC6) transfers the raw-moment lower bound to VW_0, proving (C.BC1). □

The all-negative cost is the strengthened part of this statement. It does not interchange separate suprema, assert a same-population product, or by itself reduce an arbitrary population algorithm to a single coin.

C.3 An RMS comparison for finite convex experiments

Let X be a finite-dimensional nonempty compact convex set and $\ell(x)$ a linear functional. Consider either of the following experiments:

1. independent finite categorical observations, whose positive probability vectors $p_i(x)$ are affine functions of x;
2. a finite vector of independent Poisson counts, whose strictly positive intensities $v_j(x)$ are affine functions of x.

In the first case let $\mathcal{F}$ be the finite-dimensional space of functions $c + \sum_i \phi_i(z_i)$. In the second let $\mathcal{F} = \{c + \sum_j \beta_j z_j\}$. Let P_x denote the whole observation law. All exponential moments of functions in $\mathcal{F}$ are finite. For fixed $\phi \in \mathcal{F}$, the function $x \mapsto \log \mathbb{E}_x e^{\phi}$ is concave: it is a sum of logarithms of positive affine forms in the categorical case, and an affine function of the intensities in the Poisson case. Moreover $\log(dP_x/dP_y) \in \mathcal{F}$.

Lemma C.4 (RMS affine comparison)**.** *For any measurable estimator G with $r_G = \sup_{x\in X}(\mathbb{E}_x(G - \ell(x))^2)^{1/2} < \infty$ and any $\eta > 0$, there exists $F \in \mathcal{F}$ such that*

$$\sup_{x\in X}(\mathbb{E}_x(F - \ell(x))^2)^{1/2} \leq 12 r_G + \eta. \tag{C.2}$$

The result permits nonattainment of the infimum and imposes no restriction on the original estimator's output range.

Proof. Put $s = \log 2$. For $\alpha > 0$ and $\phi \in \mathcal{F}$ define

$$\Phi(x, y; \phi, \alpha) = \ell(x) - \ell(y) + \alpha \log \mathbb{E}_y e^{\phi/\alpha} + \alpha \log \mathbb{E}_x e^{-\phi/\alpha} + 2\alpha s,$$
$$2S = \inf_{\phi,\alpha>0} \max_{x,y\in X} \Phi(x, y; \phi, \alpha).$$

This real-valued continuous function is convex in (ϕ, α) by the perspective transformation and concave in (x, y) by the preceding log-moment observation. Sion's minimax theorem applies because $X \times X$ is compact convex and the other domain is convex. Thus the infimum and supremum commute. For fixed x, y, Cauchy–Schwarz gives

$$\inf_{\psi\in\mathcal{F}} \left(\log \mathbb{E}_y e^{\psi} + \log \mathbb{E}_x e^{-\psi}\right) = 2 \log \mathrm{Aff}(P_x, P_y),$$

with equality at $\psi = \frac{1}{2}\log(dP_x/dP_y)$. Taking the infimum over $\alpha > 0$ after putting $\phi = \alpha\psi$ therefore proves

$$2S = \max\{\ell(x) - \ell(y) : x, y \in X,\ \mathrm{Aff}(P_x, P_y) \geq 1/2\}. \tag{C.3}$$

The constraint is closed and nonempty, so the maximum exists. Continuity of affinity is immediate from the finite categorical formula and, for Poisson vectors, from $\mathrm{Aff}(P_x, P_y) = \exp[-\frac{1}{2}\sum_j(\sqrt{v_j(x)} - \sqrt{v_j(y)})^2]$. In particular $0 \le S < \infty$.

For the maximizing pair in (C.3), total variation is at most $\sqrt{3}/2$ by the standard Cauchy–Schwarz affinity bound $\|P - Q\|_{\mathrm{TV}} \le \sqrt{1 - \mathrm{Aff}(P,Q)^2}$. Testing the two endpoints by thresholding G at their functional midpoint gives, with $\Delta = 2S$,

$$2r_G^2 \ge \frac{\Delta^2}{4}(1 - \|P_x - P_y\|_{\mathrm{TV}}), \qquad S \le \sqrt{\frac{2}{1 - \sqrt{3}/2}}\, r_G. \tag{C.4}$$

Indeed each wrong decision has squared estimation error at least $\Delta^2/4$, and the sum of the two testing errors is at least one minus total variation. The case $S = 0$ requires no division by S.

Fix $R > S$. Choose (ϕ, α) with $\max_{x,y} \Phi \le 2R$. On the diagonal, Cauchy–Schwarz gives $\Phi(x, x; \phi, \alpha) \ge 2\alpha s$, hence $\alpha \le R/s$. Set

$$U = \max_x\{\ell(x) + \alpha \log \mathbb{E}_x e^{-\phi/\alpha}\}, \qquad V = \max_x\{-\ell(x) + \alpha \log \mathbb{E}_x e^{\phi/\alpha}\}.$$

Then $U + V + 2\alpha s \le 2R$. The additive estimator $F = \phi + (U - V)/2$ satisfies, uniformly in x,

$$\mathbb{E}_x e^{(F - \ell(x))/\alpha} \le e^{R/\alpha - s}, \qquad \mathbb{E}_x e^{(\ell(x) - F)/\alpha} \le e^{R/\alpha - s}.$$

For $t \ge 0$, Markov's inequality consequently gives $\mathbb{P}_x(|F - \ell(x)| > R + t) \le e^{-t/\alpha}$, since $2e^{-s} = 1$. Integrating this tail, rather than substituting a confidence guarantee for an RMS bound, yields

$$\mathbb{E}_x(F - \ell(x))^2 \le R^2 + 2R\alpha + 2\alpha^2 \le R^2\left(1 + \frac{2}{s} + \frac{2}{s^2}\right). \tag{C.5}$$

The product of the squared constants in (C.4)–(C.5) is less than 144: $\log 2 > 2/3$ and $\sqrt{3}/2 < 7/8$ bound it by $8.5 \cdot 16 = 136$. Letting $R \downarrow S$, with the slack chosen for the given η, proves (C.2). This also proves uniform square integrability of the produced estimator, rather than assuming it. □

C.4 From a base-cost obstruction to one pair hard for every capped policy

In this section only, restrict the parameter set to

$$\mathcal{X}_\rho = \{q : 0 \le q \le 1/2,\ \int q\, d\mu \le \rho\}, \qquad J = \rho LH.$$

It is a weak-star compact convex subset of $L^\infty(\mu)$. The following implication separates its one required entropy input from the convex-experiment argument. The input is the base-cost bound proved in the accompanying base-cost section of the full-model note:

There is a universal $c_b > 0$ such that every observable single-coin estimator Y with $\sup_{q \in \mathcal{X}_\rho} |\mathbb{E}_q Y - \theta(q)| \le \rho/64$, $V = \sup_{q \in \mathcal{X}_\rho} \mathrm{Var}_q\, Y < \infty$, $\sup_{q \in \mathcal{X}_\rho} \mathbb{E}_q T < \infty$ satisfies $V\mathbb{E}_0 T \ge c_b J$. (C.BC)

The cost in (C.BC) is at the all-negative population $q = 0$, not the supremum over populations. The two lemmas below are conditional on precisely this input. No algorithm-level lower bound is inferred from a variational optimization alone. Write $h(P, Q) = 1 - \mathrm{Aff}(P, Q)$ for squared Hellinger distance in the normalization used here; it takes values in $[0, 1]$.

Exact finite observation reduction

A capped local policy A includes its independent seed and stops after at most $m < \infty$ flips of one genuine coin. Its full observed transcript contains the seed and the stopped word. Its seed distribution and policy may depend on known parameters, but not on the actual q. For each word $x \in \{0,1\}^t$, let $B_{t,x}$ be the measurable seed set that causes stopping exactly at that word. For $0 \le k \le t \le m$ put

$$a_{t,k} = \sum_{x:\,|x|=t,\,\#1(x)=k} \mathbb{P}(\text{seed} \in B_{t,x}),$$

and, with $p_s(d) = 1/2 + (2s-1)d$, put

$$m_{t,k}(q) = \int \left[(1-q(d))p_0(d)^k(1-p_0(d))^{t-k} + q(d)p_1(d)^k(1-p_1(d))^{t-k}\right] d\mu.$$

The finite statistic (T, K), where K is the number of observed heads, has probabilities $a_{t,k}m_{t,k}(q)$. Conditional on (T, K), the stopped word and seed have a distribution independent of q: their common likelihood multiplier is exactly $m_{t,k}(q)$ and cancels. This statement also follows directly by integrating any bounded transcript function over the finitely many word/seed cells, so it does not assume a latent oracle. Consequently the full transcript and this finite statistic have exactly the same Hellinger affinities for all pairs of populations.

Remove coordinates with $a_{t,k} = 0$. The remaining coordinates are positive for every q, since $p_s(d) \in [1/4, 3/4]$. They are affine weak-star continuous functions of q. Thus their probability vector together with $\theta(q)$ has a compact convex finite-dimensional range. In particular, for each capped A, its full-transcript squared Hellinger distance $h_A(q_-, q_+)$ is a continuous convex function of the pair on $\mathcal{X}_\rho^2$. Convexity follows from joint concavity of affinity for two affine probability vectors. This reduction retains the stopping time and expected cost, not just a fixed number of sampled bits.

Lemma C.5 (A hard pair for each capped policy)**.** *Assume (C.BC). There are universal constants $c_\delta > 0$ and $C_H < \infty$ such that, for every $\rho \in (0, 1/4]$, every $0 < \delta \le c_\delta \rho$, and every randomized capped local policy A, there exist $q_-, q_+ \in \mathcal{X}_\rho$ with $\theta(q_+) - \theta(q_-) \ge \delta$ and*

$$h_A(q_-, q_+) \le C_H \frac{\delta^2}{J} \mathbb{E}_0 T_A. \tag{C.6}$$

The full observed seed and stopped history are included in the Hellinger distance.

Proof. Let $w = \mathbb{E}_0 T_A$. If $w = 0$, the seed-only event of taking a first observation has probability zero, so A has the same transcript law for every q. The pair $q_- = 0$, $q_+ = \delta$ gives separation δ and $h_A = 0$. We therefore assume $w > 0$ in all the divisions below. First establish the following additive lower bound for $n \ge 1$ independent runs of A. If an additive output with bounded component functions $F = c + \sum_{i=1}^n Y_i$ has worst RMS $r < \rho/64$, put $b_i = \mathbb{E}_0 Y_i$, $b = \mathbb{E}_0 F$, and choose I uniformly in $\{1, \dots, n\}$ independently. Run just its component on one coin and return $Z = n(Y_I - b_I) + b$. This preserves every population mean and uses base cost w.

For $\theta = \theta(q)$, the laws of each component output under q and 0 have a common subprobability of mass $1-\theta$: integrate the negative-sign kernel with weight $1-q(d)$. For any two distributions sharing such a subprobability, writing each as a mixture of its normalized common part and residual part shows

$$(\mathbb{E}_q Y_i - \mathbb{E}_0 Y_i)^2 \le \frac{2\theta}{1-\theta}(\operatorname{Var}_q Y_i + \operatorname{Var}_0 Y_i). \tag{C.7}$$

Indeed the squared displacement of either mixture mean from the common mean is at most $\theta/(1-\theta)$ times its variance; use $(u-v)^2 \le 2u^2+2v^2$. The case $\theta = 0$ is identical laws. Independence and $\sup_q \operatorname{Var}_q F \le r^2$ imply

$$\sum_i (\mathbb{E}_q Y_i - b_i)^2 \le \frac{4\rho}{1-\rho} r^2, \quad \operatorname{Var}_q Z \le n \sum_i \operatorname{Var}_q Y_i + n \sum_i (\mathbb{E}_q Y_i - b_i)^2 \le \frac{7}{3} n r^2.$$

All outputs here are bounded, and stopping is capped, so (C.BC) applies to Z without an additional integrability assumption. It gives $r^2 \ge 3c_b J/(7nw)$. Including the branch $r \ge \rho/64$ establishes

$$r \ge c_a \min\{\rho, \sqrt{J/(nw)}\}, \qquad c_a = \min\{1/64, \sqrt{3c_b/7}\}. \tag{C.8}$$

For $w > 0$, let S_n be the saddle quantity S from the proof of Lemma C.4 for n independent copies of the finite statistic of A, on the parameter set $\mathcal{X}_\rho$. Equations (C.3) and (C.5) of that proof show

$$2S_n = \max\{\theta(q_+) - \theta(q_-) : \operatorname{Aff}(P_{A,q_+}, P_{A,q_-})^n \ge 1/2\}, \quad \inf_{F \text{ additive}} \sup_q \|F - \theta(q)\|_2 \le 3S_n.$$

The latter constant is valid because $1 + 2/\log 2 + 2/(\log 2)^2 < 9$. The finite statistic outputs are bounded, so (C.8) applies. Hence, for $\beta = 2c_a/3$,

$$2S_n \ge \beta \min\{\rho, \sqrt{J/(nw)}\}. \tag{C.9}$$

Take $c_\delta = \beta/2$ and $C_H = 8/\beta^2$. For $0 < \delta \le c_\delta \rho$ set $x = \beta^2 J/(4\delta^2 w)$. If $x \ge 2$, choose $n = \lfloor x \rfloor$, so $x/2 \le n \le x$. Equation (C.9) gives a pair of separation at least 2δ, with

$$h_A \le 1 - e^{-(\log 2)/n} \le \frac{\log 2}{n} \le \frac{8 \log 2}{\beta^2} \frac{\delta^2 w}{J} \le C_H \frac{\delta^2 w}{J}.$$

If $x < 2$, the pair $q_- = 0$, $q_+ = \delta$ belongs to $\mathcal{X}_\rho$, has separation δ, and $h_A \le 1$ suffices because $C_H \delta^2 w/J > 1$. □

Lemma C.6 (One pair hard for every capped policy)**.** *Assume (C.BC), and keep the universal constants of Lemma C.5. For every $0 < \rho \le 1/4$ and $0 < \delta \le c_\delta \rho$, there exist $q_-, q_+ \in \mathcal{X}_\rho$, with $\theta(q_+) - \theta(q_-) \ge \delta$, such that simultaneously for every randomized capped single-coin policy A,*

$$h_A(q_-, q_+) \le C_H \frac{\delta^2}{\rho L H} \mathbb{E}_0 T_A. \tag{C.10}$$

Each policy's seed and full stopped observable history are included. The constants and pair are chosen before the policy.

Proof. Let $\mathcal{K}_\delta = \{(q_-, q_+) \in \mathcal{X}_\rho^2 : \theta(q_+) - \theta(q_-) \ge \delta\}$. It is nonempty compact convex in the weak-star topology. Fix any finite list of capped policies $A_1, \dots, A_k$ and put $w_i = \mathbb{E}_0 T_{A_i}$ and $\lambda = C_H \delta^2/(\rho L H)$. For a simplex vector $\pi \in \Delta_k$, define the observable mixture policy that draws $i \sim \pi$, reveals this index in its independent seed, and runs A_i. This is still capped, by the largest of the finitely many caps. Its base cost is $\sum_i \pi_i w_i$ and, since the revealed labels occupy disjoint transcript components, its squared Hellinger distance is exactly $\sum_i \pi_i h_{A_i}$. Lemma C.5 therefore gives

$$\min_{z \in \mathcal{K}_\delta} \sum_i \pi_i (h_{A_i}(z) - \lambda w_i) \le 0 \qquad (\pi \in \Delta_k).$$

For the finite simplex the displayed function is continuous affine in π and continuous convex in z, by the finite observation reduction. Sion's theorem [20] now gives

$$\min_{z \in \mathcal{K}_\delta} \max_{1 \le i \le k} (h_{A_i}(z) - \lambda w_i) = \max_{\pi \in \Delta_k} \min_{z \in \mathcal{K}_\delta} \sum_i \pi_i (h_{A_i}(z) - \lambda w_i) \le 0.$$

Thus the closed subsets $\{z \in \mathcal{K}_\delta : h_A(z) \le \lambda \mathbb{E}_0 T_A\}$ have the finite intersection property as A ranges over all capped policies. Compactness of $\mathcal{K}_\delta$ gives a point in their full intersection, proving (C.10). This uses only finite revealed mixtures and a finite simplex minimax theorem; no topology on a space of all policies, unjustified exchange of arbitrary infima, or unobserved-mixture affinity identity is needed. □

C.5 From one universal pair to every population algorithm

Lemma C.7 (Finite-prefix midpoint decomposition)**.** *Suppose $q_-, q_+ \in \mathcal{X}_\rho$ have this property: every observable randomized single-coin policy with a deterministic finite cap obeys*

$$\mathsf{H}^2(P^A_{q_-}, P^A_{q_+}) \le \lambda \mathbb{E}_0 T_A, \tag{C.B}$$

where the transcript includes its independent seed and stopped observations, and $\mathsf{H}^2(P,Q) = 1 - \int \sqrt{dP\, dQ}$. Let $\bar{q} = (q_- + q_+)/2$. Then every population algorithm capped after K total flips obeys

$$\mathsf{H}^2(P^{\mathrm{pop}}_{q_-}, P^{\mathrm{pop}}_{q_+}) \le 2\lambda \mathbb{E}_{\bar{q}} T. \tag{C.C}$$

Proof. Condition first on the original independent seed. Pad the list of observed coins to K slots by their first-observation indices; an unobserved slot contributes the empty word of probability one. The policy is a finite binary decision tree, whose choices are determined by past outcomes and the fixed seed. For a leaf z, let $h_i(z)$ be the ordered word observed from slot i. Under a single fresh coin from q_r, write its word probability as $p_r(h)$, $r \in \{-,+\}$. Independence of the coin tapes gives leaf density

$$P_r(z) = \prod_{i=1}^{K} p_r(h_i(z)).$$

Interleaving changes the tree and the words, but not this factorization. Set $m_i = (p_-(h_i) + p_+(h_i))/2$ and $c_i = \sqrt{p_-(h_i)p_+(h_i)}$. Word probabilities are affine in q, so $m_i = p_{\bar{q}}(h_i)$. These quantities are positive; empty words cause no exception. Since $0 \le c_i \le m_i$, the elementary product inequality gives

$$\prod_i m_i - \prod_i c_i \le \sum_i (m_i - c_i) \prod_{j \ne i} m_j. \tag{C.D}$$

The left side sums over leaves to $\mathsf{H}^2(P^{\mathrm{pop}}_-, P^{\mathrm{pop}}_+)$, because the midpoint experiment is normalized. The ith right side sums to the squared Hellinger distance between the two hybrid population experiments having slot i drawn from q_- or q_+ and all other slots independently drawn from $\bar{q}$. Those finite experiments are normalized too. Integrating the original common seed preserves this inequality: joint Hellinger distance with a revealed common seed is the integral of conditional Hellinger distances.

For each slot i, realize the hybrid experiment by a single-coin policy A_i. Simulate all other slots using independent artificial tapes from the fixed known reference population $\bar{q}$, and supply slot i by one genuine coin. If that slot is never requested, take no genuine observations. Include the original seed and all artificial tapes in the independent auxiliary seed. This is an observable single-coin policy capped at K genuine flips. It may use the fixed reference pair as design constants, but these do not depend on the genuine coin's unknown endpoint law. The universal quantifier in (C.B) covers this policy. No artificial tape is supplied to the original population algorithm.

The hybrid population transcript is a measurable function of this larger seed and the stopped genuine observations. Hellinger data processing and (C.B) therefore bound its distance by $\lambda \mathbb{E}_0 T_{A_i}$. Since $0 \le \bar{q} \le 1/2$, the one-coin latent distribution under $\bar{q}$ dominates one half of

the all-negative distribution, with the magnitude law fixed. The same domination holds after adjoining the common auxiliary seed and taking any measurable nonnegative work functional. Hence

$$\mathbb{E}_0 T_{A_i} \le 2\mathbb{E}_{\bar{q}} T_{A_i}.$$

When the genuine coin too is from $\bar{q}$, its expected charged work is exactly the work on slot i in the original midpoint experiment. Summing over the at most K slots yields (C.C). All simulations are only devices in the lower-bound comparison; every genuine flip of the population algorithm is counted exactly once. □

Proof of the lower bound in Theorem C.1. Take the universal constants c_δ, C_H from Lemma C.6. Choose a fixed

$$0 < \gamma \le \min\{c_\delta, 1/(4\sqrt{C_H})\}, \qquad \delta = \gamma \min\{\rho, \sqrt{J/B}\}.$$

That lemma provides one pair $q_-, q_+ \in \mathcal{X}_\rho$ with target separation at least δ and property (C.B) with $\lambda = C_H\delta^2/J$. The pair is valid simultaneously for all finite caps and policies; it is not reselected after seeing a population transcript.

Let A be any full population algorithm with uniform expected work at most B. Apply Lemma C.7 to its prefix capped after K flips, recording whether it already halted and, when relevant, its terminal output as a measurable statistic. Since $\bar{q} \in \mathcal{X}_\rho \subset \mathcal{Q}_\rho$, its midpoint expected capped work is at most B. Thus every prefix has squared Hellinger distance at most $2C_H\delta^2 B/J \le 1/8$. The inequality $\mathrm{TV}(P,Q) \le \sqrt{2\mathsf{H}^2(P,Q)}$ bounds its total variation by $1/2$.

Uniform expected work implies almost-sure termination separately under both endpoints. For any terminal event E, the event $E \cap \{T \le K\}$ is determined by the capped prefix, and the omitted endpoint probabilities are at most $\mathbb{P}_{q_-}(T > K) + \mathbb{P}_{q_+}(T > K) \to 0$. Consequently the terminal transcript's total variation is also at most $1/2$. This passage assumes neither a globally bounded number of coins nor differentiability or a chain rule at an unbounded stopping time.

If the estimator has infinite worst-case squared error, the lower bound is automatic. Otherwise threshold its output halfway between the endpoint targets. With endpoint separation $\Delta \ge \delta$ and worst RMS r, Markov's inequality bounds the sum of the two testing errors by $8r^2/\Delta^2$. Every test has error sum at least $1 - \mathrm{TV} \ge 1/2$. Hence $r \ge \Delta/4 \ge \delta/4$, which proves the lower inequality in (C.A), including saturation. □

C.6 Consequence for the proposed rare-event modulus

For this exact law the variational optimizer gives a cutoff $t \in (0, a)$:

$$\rho = (t/a)^2[1 + 2\log(a/t)], \qquad \Psi_{\mu,\rho}(h) = \begin{cases} [2ht(1 + \log(a/t)) - h^2]/a^2, & 0 < h < t, \\ \rho, & h \ge t. \end{cases} \tag{C.E}$$

These are inherited variational identities, not an attaining statistical algorithm. Put $u = 1 + 2\log(a/t)$. Then $\rho = ue^{1-u}$ and $L = u - \log u$, so $L \le u \le 2L$. For $h < t$, $\Psi(h) = (ht/a^2)(u + 1 - h/t)$; its coefficient lies between u and $u + 1$. Together with the saturated branch this proves, with universal constants and fixed $a = 1/4$,

$$\Psi_{\mu,\rho}(B^{-1/2}) \asymp \min\{\rho, \sqrt{\rho L/B}\}.$$

For instance take $\rho_n = \exp(-\exp n)$ and $B_n = \rho_n^{-2}$. Both radii are eventually unsaturated, and (C.A) implies

$$\frac{R_{\mu,\rho_n}(B_n)}{\Psi_{\mu,\rho_n}(B_n^{-1/2})} \ge c'\sqrt{1 + \log(1 + \log(1/\rho_n))} \longrightarrow \infty.$$

Thus no one finite universal C can satisfy the conjectured upper bound even at this single known law, with the algorithm knowing ρ and B. The missing statistical cost is not hidden magnitudes alone: known-profile submodels can behave differently. The lower bound uses the entire convex nuisance class to obtain a common hard pair. It does not assert that the radial variational optimizer itself is that pair, nor give a closed-form expression for the pair selected by compactness. No claim for every other magnitude law follows from (C.A); section 13 treats general laws by a different argument.

C.7 Critical upper construction

Put $H = 1 + \log L$. For $j \geq 0$ choose

$$\varepsilon_j = 16^{-(j+1)}, \qquad N_j = \lceil (a^2 \rho \varepsilon_j)^{-1} \rceil, \qquad K_j = \lceil \log_2(a\sqrt{N_j}) \rceil.$$

For $0 \leq k \leq K_j$, let $r_k = a2^{-(k+1)}$ and, after n observations with h_n heads, define

$$L^+_{k,n} = (1 + 2r_k)^{h_n}(1 - 2r_k)^{n-h_n}, \qquad L^-_{k,n} = (1 - 2r_k)^{h_n}(1 + 2r_k)^{n-h_n}.$$

Before N_j, stop with output zero if a negative likelihood reaches $(K_j + 1)/\varepsilon_j$, or with output one if a positive likelihood reaches $(K_j + 1)/(\rho\varepsilon_j)$. Fix zero priority for ties. At N_j, if no earlier decision was made, return the strict majority bit. Denote this bounded, observable classifier by Z_j and its work by T_j.

For the wrong-sign likelihood, each next multiplier has mean $1 - 4r_k d \leq 1$. Stopping this nonnegative supermartingale at a crossing and a deterministic horizon, and then increasing the horizon, bounds the probability of crossing level b by $1/b$. Summing over $K_j + 1$ references gives wrong-crossing probabilities at most ε_j on a positive coin and $\rho\varepsilon_j$ on a negative coin. The wrong forced-majority event is a subset of the full-tape event, of probability at most $e^{-2N_j d^2}$. This is Hoeffding's bound: the centered Bernoulli log moment-generating function has second derivative at most $1/4$, so is at most $\lambda^2/8$; use independence and exponential Markov with $\lambda = \pm 4d$. For the exact power law,

$$\mathbb{E}_\mu e^{-2ND^2} = \frac{1 - e^{-2Na^2}}{2Na^2} \leq \frac{1}{2Na^2}.$$

No independence of a stopping event and this full-tape majority event is used. It follows, uniformly over the global-rarity class, that

$$\mathbb{P}(Z_j \neq S) \leq \tfrac{5}{2}\rho\varepsilon_j, \qquad \mathbb{P}(Z_j = 1, S = 0) \leq \tfrac{3}{2}\rho\varepsilon_j. \tag{C.U1}$$

For $d \geq N_j^{-1/2}$, choose $k = \lceil \log_2(a/d) \rceil \leq K_j$. Then $d/4 \leq r_k \leq d/2$, and the correct-sign log likelihood has mean increment at least $2d^2/3$: use

$$\tfrac{1}{2}\log(1 - 4r_k^2) + d\log\frac{1 + 2r_k}{1 - 2r_k} \geq -\tfrac{8}{3}r_k^2 + 4dr_k \geq \tfrac{2}{3}d^2.$$

Its upward increment is less than one. The bounded stopped-sum identity for the crossing time σ at a finite horizon n gives $g\mathbb{E}(\sigma \wedge n) \leq A_{j,s}$, where

$$A_{j,s} = 1 + \log(K_j + 1) + \log(1/\varepsilon_j) + s\log(1/\rho).$$

Monotone convergence proves finite expected crossing time and, including the cap and the trivial small-gap case,

$$\mathbb{E}(T_j \mid d, s) \leq \min\{N_j, (3/2)A_{j,s}/d^2\}. \tag{C.U2}$$

Write $c_0 = 3/2$. Direct integration of the negative contribution gives

$$\mathbb{E}\min\{N, c_0A/D^2\} \le (c_0A/a^2)[1 + \log_+(Na^2)].$$

For positives, layer cake and $\mathbb{E}[q\mathbf{1}_E] \le \min\{\rho, \mu(E)\}$ give

$$\mathbb{E}[q\min\{N, c_0A/D^2\}] \le \int_0^\rho \min\{N, c_0A/(a^2u)\}\,du \le (c_0A/a^2)[1 + \log_+(Na^2\rho)].$$

The exact integral is $N\rho$ when $Na^2\rho \le c_0A$, and otherwise is $(c_0A/a^2)[1 + \log(Na^2\rho/(c_0A))]$. Thus this step also enforces the global, rather than pointwise, rarity condition.

Elementary bounds from the definitions are

$$N_ja^2 \le 2/(\rho\varepsilon_j), \qquad K_j + 1 \le 5(L + j + 1),$$
$$A_{j,0} \le 7H(j + 1), \qquad A_{j,1} \le 7H(j + 1) + L,$$
$$1 + \log(N_ja^2) \le 5L(j + 1), \qquad 1 + \log(N_ja^2\rho) \le 5(j + 1).$$

For example, $\log(L+j+1) \le \log L+j+1$ and $4\log 2 < 3$ prove the bound on $A_{j,0}$. Substitution in (C.U2) and the two integrals gives

$$\mathbb{E}T_j \le (225/2)a^{-2}LH(j + 1)^2. \tag{C.U3}$$

Independently choose a level J with $p_j = 2^{-(j+1)}$. For $J = 0$ return $Y = Z_0/p_0$ on one fresh coin. For $J = j \ge 1$, run Z_j and Z_{j-1} on successive unused blocks of that same coin, set $\Delta_j = Z_j - Z_{j-1}$, and return $Y = \Delta_j/p_j$. The bounded stopping times make the iid suffix argument elementary, as in the restart argument above; both decisions share the same latent sign. A disagreement implies an error of one classifier. Hence (C.U1) gives

$$\mathbb{E}Z_0^2 \le \tfrac{35}{32}\rho, \qquad \mathbb{E}\Delta_j^2 \le \tfrac{85}{32}\rho 16^{-j} \quad (j \ge 1).$$

Tonelli and the exact geometric sum now give

$$\mathbb{E}Y^2 \le \frac{35}{16}\rho + \frac{85}{16}\rho\sum_{j\ge1} 8^{-j} = \frac{165}{56}\rho.$$

The disagreement bounds also give absolute summability of expected increments, so signed expectations may be telescoped after integrability is established: $\mathbb{E}Y = \lim_j \mathbb{E}Z_j = \theta$. Work costs at most twice (C.U3), and $\sum_j p_j(j + 1)^2 = 6$, yielding

$$\mathbb{E}T_Y \le 1350a^{-2}LH.$$

All operations are on one observable coin with independent seed; the random level is unbounded but the separate expectations are finite. This completes the upper checkpoint used in Theorem C.1. Its actual work also has genuinely unbounded support: at any fixed observation count n, all likelihoods are at most $(1+a)^n$. For sufficiently large j, both the smallest stopping boundary and N_j exceed the relevant finite thresholds, so $T_j > n$ on every tape. Since $\mathbb{P}(J = j) > 0$, actual work exceeds every finite cap with positive probability while retaining finite mean.

Proof of the upper bound in Theorem C.1. The preceding construction supplies a one-coin unbiased output with $\sup_q \mathbb{E}_qY^2 \le (165/56)\rho$ and $\sup_q \mathbb{E}_qT_Y \le W_* = 1350a^{-2}LH$. Independently select $N \sim \text{Poisson}(B/W_*)$ and return $\widehat{\theta} = (W_*/B)\sum_{i=1}^N Y_i$ from independent genuine coins. Conditioning on N proves unbiasedness, uniform expected work at most B, and variance at most $(165/56)\rho W_*/B$. All moments are finite before using these identities, by the bounds and the finite Poisson moments. Choose the zero-work constant $\rho/2$ whenever its known error bound is smaller. Both choices are observable and depend only on known parameters. With a fixed, this proves the upper inequality in (C.A). This established replication device is not claimed as new. □

D The observable entropy–tail upper bound

This appendix proves Theorem 13.3. The single-unit tools it uses are proved in Appendix F.

D.1 An uncapped reference bank with a capped observable race

Fix a decomposition $\mu = \sigma + \tau + \zeta$, put $z = \zeta((0, 1/2])$, and assume $\Gamma_\sigma, H_\tau < \infty$. It suffices to treat $z \le \rho/128$: otherwise the constant $\rho/2$ has risk at most $64z$. Write $A = A_\sigma$, $H = H_\tau$. If $\sigma \neq 0$, use all its occupied bands, with weights $\pi_j = a_j(\sigma)/A$ and references $b_j/2$ and $b_j/4$. For an internal allowance $u < \rho$, set $\epsilon = u/\rho$. Stop negative if a negative reference likelihood against fair reaches $1/(\epsilon\pi_j)$; stop positive if

$$\frac{\sum_j \pi_j P_{1/2+b_j/4,n}}{\int P_{1/2-d,n}\,\mu(dd)} \ge 1/u. \tag{97}$$

Give negative crossings priority. A zero denominator triggers a positive stop. The split $\sigma + \tau + \zeta = \mu$ specifies the references and the work analysis; membership in these submeasures is never observed. The known all-negative mixture in the denominator is the full law μ, regardless of the decomposition.

By Lemma F.3, a wrong negative crossing has conditional positive-sign probability at most ϵ, and a positive crossing has all-negative-mixture probability at most u. Thus the total probability of a wrong bank stop is at most $2u$ over $\mathcal{Q}_\rho$. Termination outside σ is not assumed.

Race this bank against the ordinary classifier of Lemma F.1, using alternating observations of the same physical coin. At an ordinary cap M, its conditional work certificate is

$$o_M(d) = \min\{M, 17d^{-2}[1 + \log_+(Md^2)]\}.$$

For any finite measure ξ, put

$$\eta_\xi(M) = \int \min\{1, (Md^2)^{-2}\}\,\xi(dd), \qquad O_\xi(M) = \int o_M(d)\,\xi(dd).$$

The ordinary error under μ is at most $\eta_\mu(M)$. Stop as soon as either component stops and return that output. On completed counterfactual streams, the physical work is at most

$$2\min\{T_{\text{bank}}, T_{\text{ordinary}}\} \le 2M.$$

A wrong output implies either a wrong bank stop or an ordinary error. When $\sigma = 0$, omit the bank and use the ordinary classifier alone.

On a σ-almost-every gap d in band j, the matching negative reference has drift at least $4d^2/3$ and log overshoot below one. Its expected crossing time is at most

$$\frac{3}{4d^2}[1 + \log(1/\epsilon) + \log(1/\pi_j)]. \tag{98}$$

On a positive gap, the event $\widehat{p}_n - 1/2 \ge d/2$ gives

$$\frac{P_{-,\mu,n}}{P_{\text{f},n}} \le \int e^{-2nx^2}\,\mu(dx) \le z + \frac{A}{2en} + \frac{H}{2\sqrt{en}}. \tag{99}$$

The last two bounds use, respectively, $\sup_x x^2 e^{-2nx^2} = 1/(2en)$ and $\sup_x x e^{-2nx^2} = 1/(2\sqrt{en})$. For $n \ge d^{-2}$ the denominator is at most $z + Ad^2 + Hd$, while the matching positive reference

contributes at least $\pi_j e^{nd^2/3}$ relative to fair. Equation (133) therefore bounds positive expected bank work by

$$\frac{6}{d^2}\left[1+\log(1/\epsilon)+\log(1/\pi_j)+\log_+\frac{z+Ad^2+Hd}{\rho}\right]. \tag{100}$$

These arguments prove finite conditional bank work on σ. On $\tau+\zeta$, the ordinary cap guarantees termination of the race.

Since $z/\rho \le 1$,

$$\log_+\frac{z+Ad^2+Hd}{\rho} \le \log 3+\log_+(Ad^2/\rho)+\log_+(Hd/\rho).$$

Using $\sup_{x\ge 1}\log x/x = 1/e$ and $\sup_{x\ge 1}\log x/x^2 = 1/(2e)$ gives

$$\int q(d)d^{-2}\log_+\frac{z+Ad^2+Hd}{\rho}\,\sigma(dd) \le (\log 3+1/e)A+\frac{H^2}{2e\rho}. \tag{101}$$

Here $\int q\,d\sigma \le \rho$ and $q\le 1$ follow from the original global rarity class; no separate conditional rarity assumption is used. Integrating (98) and (100) therefore bounds the bank work on σ by

$$6\Gamma_\sigma + 6A\log(1/\epsilon)+6(\log 3+1/e)A+\frac{3H^2}{e\rho}. \tag{102}$$

D.2 A resolved-law cap schedule

Let $\nu=\sigma+\tau$ and set

$$\delta_\ell=\rho 16^{-(\ell+1)}, \qquad u_\ell=\delta_\ell/4, \qquad \epsilon_\ell=u_\ell/\rho.$$

Choose M_ℓ as the least power of four, at least four, such that $\eta_\nu(M_\ell)\le\delta_\ell/8$. Every chosen cap has a predecessor because $\eta_\nu(4)=1-z$. The inequality $\eta_\nu(4M)\ge\eta_\nu(M)/16$ implies

$$\delta_\ell/128 < \eta_\nu(M_\ell)\le\delta_\ell/8. \tag{103}$$

Thus the selected powers of four are distinct and increasing.

If $z>0$, let J be the largest integer with $\delta_J\ge 8z$. Then

$$8z\le\delta_J<128z. \tag{104}$$

For $\ell\le J$, $\eta_\mu(M_\ell)\le\eta_\nu(M_\ell)+z\le\delta_\ell/4=u_\ell$. The raced classifier has total error at most $3u_\ell\le\delta_\ell$ and false-positive probability at most $2u_\ell\le\delta_\ell$. When $z=0$, these conclusions hold at every level.

Since $\log(1/\epsilon_\ell)=\log 64+\ell\log 16$ and $A\le\Gamma_\sigma$, (102) is at most $40(\ell+1)\Gamma_\sigma+2H^2/\rho$. A uniform raced-work certificate is consequently

$$C_\ell = 80(\ell+1)\Gamma_\sigma+4H^2/\rho+2O_\tau(M_\ell)+2zM_\ell. \tag{105}$$

It remains a valid, possibly loose, bound when $\sigma=0$ and the bank is omitted.

The binary level-zero output has second moment at most $\rho+\delta_0=17\delta_0$. An adjacent difference, run on successive unused blocks of the same coin, has second moment at most $\delta_\ell+\delta_{\ell-1}=17\delta_\ell$. Set $v_\ell=17\delta_\ell$, $k_0=C_0$, $k_\ell=C_\ell+C_{\ell-1}$ for $\ell\ge 1$, and $S_J=\sum_{\ell\le J}\sqrt{v_\ell k_\ell}$. Independent-Poisson allocation across these corrections (Lemma F.5) gives RMS at most

$$\sqrt{\delta_J^2+S_J^2/B}. \tag{106}$$

The seed may choose zero replications, and the Poisson means may be below one. Every first and subsequent observation is charged. Every such finite-level procedure has finite deterministic local caps. The reference bank may be countable; its stopping events are measurable countable unions and its positive reference is a probability mixture.

D.3 Controlling mixed work without an extra logarithm

For every finite τ, the following bounds hold:

$$O_\tau(M) \le 21H\sqrt{M}, \qquad \eta_\sigma(M) \le A/M, \qquad \eta_\tau(M) \le H/\sqrt{M}. \tag{107}$$

The latter two are pointwise inequalities before integration. For the first, put $x = \sqrt{M}\,d$. The ratio of $o_M(d)$ to $\sqrt{M}/d$ is at most one for $x \le 1$, and at most $17(1+2\log x)/x \le 34/\sqrt{e} < 21$ for $x \ge 1$. Lemma F.2 supplies the homogeneous ordinary kernel bound:

$$\sum_{k\ge 1} \sqrt{\eta_\tau(4^k) O_\tau(4^k)} \le 208H. \tag{108}$$

Its proof applies without normalization of τ: dyadic binning yields the summable kernels 2^{k-j} for $j \ge k$, and $2^{3(j-k)}$, $2^{j-k}(1+k-j)$ for $j < k$. Apply $\sqrt{xy} \le (x+y)/2$ and sum these kernels.

Split levels according to which term of $\eta_\nu = \eta_\sigma + \eta_\tau$ is at least half the total. On levels where η_τ dominates, $\delta_\ell \le 256\eta_\tau(M_\ell)$, so distinctness of the caps and (108) give a total $\sum \sqrt{\delta_\ell O_\tau(M_\ell)} \le 3328H$. On the other levels,

$$\delta_\ell \le 256\eta_\sigma(M_\ell) \le 256A/M_\ell, \qquad M_\ell \le 256A/\delta_\ell.$$

Therefore

$$\sqrt{\delta_\ell O_\tau(M_\ell)} \le 4\sqrt{21}\, H^{1/2} A^{1/4} \delta_\ell^{1/4}.$$

The sum of $\delta_\ell^{1/4}$ over all levels is $\rho^{1/4}$. The arithmetic–geometric mean inequality proves

$$\sum_{\ell \le J} \sqrt{\delta_\ell O_\tau(M_\ell)} \le C_{\mathrm{mix}}[H + \sqrt{\rho A}]. \tag{109}$$

The work on the unresolved mass is controlled separately. On levels where η_σ dominates,

$$\sum \sqrt{z\delta_\ell M_\ell} \le 16(J+1)\sqrt{zA} \le \sqrt{2\rho A}. \tag{110}$$

The last inequality uses $z \le \delta_J/8$ and $(J+1)4^{-(J+1)} \le 1/4$. On levels where η_τ dominates, $\sqrt{M_\ell} \le 256H/\delta_\ell$, so

$$\sum \sqrt{z\delta_\ell M_\ell} \le 256H\sqrt{z} \sum_{\ell=0}^{J} \delta_\ell^{-1/2} \le \frac{1024}{3} H\sqrt{z/\delta_J} \le \frac{1024}{3\sqrt{8}} H. \tag{111}$$

When $z = 0$ these contributions vanish.

Completion of Theorem 13.3. The shifted adjacent-level sum has a factor $1/4$, so

$$S_J \le \frac{5\sqrt{17}}{4} \sum_{\ell=0}^{J} \sqrt{\delta_\ell C_\ell}.$$

Use (105), (109), (110), (111), and $A \le \Gamma_\sigma$. The elementary sums $\sum_{\ell\ge 0} 4^{-\ell}\sqrt{\ell+1} \le 16/9$ and $\sum_{\ell\ge 0} 4^{-\ell} = 4/3$ then give

$$S_J \le C_{\mathrm{sum}}[\sqrt{\rho\Gamma_\sigma} + H].$$

For $z > 0$, (104) and (106) imply $R(B) \le 128z + C_{\mathrm{sum}}[\sqrt{\rho\Gamma_\sigma} + H]/\sqrt{B}$. For $z = 0$, the bound on S_J holds at every finite terminal level. Use (106) and then $\delta_J \to 0$; taking the infimum over these finite constructions gives the asserted bound. The constant estimator

covers $z > \rho/128$. Finally take the infimum over decompositions and compare with $\rho/2$. All approximations are at fixed (μ, ρ, B). The comparison constant is independent of the chosen decomposition and terminal level; no attaining infinite-dimensional decomposition is required. No infinite-algorithm interchange is required. The same infimum can be approached with finite reference banks: finite Γ_σ implies $H_\sigma < \infty$, so truncate σ to finitely many bands and move its remainder to τ. The entropy does not increase and the additional inverse-gap cost tends to zero. Then use the finite-level construction. □

E A matching scalar lower bound by entropy packing

The lower proof uses optimality of the retained set in (61). Its ordinary and ignored pieces are controlled by the variational modulus of Section 11, through the null-work obstruction proved in Appendix H. Its retained entropy can be packed into disjoint components on which the actual annular mass constraint is active. The additive localization theorem then combines these components at every budget.

Theorem E.1 (Matching entropy–tail lower bound)**.** *There is a universal $c > 0$ such that, for every magnitude law, $0 < \rho \le 1/4$, and $B > 0$,*

$$R_{\mu,\rho}(B) \ge c\,\mathcal{K}_{\mu,\rho}(B).$$

Together with Theorem 13.3, this proves Theorem 13.2.

E.1 Canonical endpoints change the functional by at most two

Put $h = B^{-1/2}$, $w_j = \mu(I_j)$, and $a_j = w_j/b_j^2$, where $b_j = 2^{-(j+1)}$ and $I_j = (b_j/2, b_j]$. For a finite prefix, define

$$k_J = \min\left\{\rho,\ \min_{I\subset\{0,\dots,J\}}\left[h\sqrt{\rho\,\mathcal{G}((a_j)_{j\in I})} + \sum_{\substack{j\le J\\ j\notin I}} w_j \min\{1, h/b_j\}\right]\right\}. \tag{112}$$

Fractional retained band masses cannot improve this finite expression: the square root of $\mathcal{G}$ is concave, the remainder is linear, and the minimum over a finite splitting box occurs at a vertex. The values k_J increase to a limit k, with

$$k_J \le k \le \min\{\rho, k_J + \sum_{j>J} w_j\}. \tag{113}$$

Restriction proves the first inequality; assigning every omitted band to ignored mass proves the second. The limit is at fixed (μ, ρ, B); no law-independent truncation depth or nested choice of minimizers is required. The omitted tail probability must be retained before the limit.

The exact and canonical functionals satisfy

$$k \le \mathcal{K}_{\mu,\rho}(B) \le 2k. \tag{114}$$

Indeed, on a band the exact inverse-square cost lies between its canonical cost and four times that cost, while

$$\min\{1, h/b_j\} \le \min\{1, h/d\} \le 2\min\{1, h/b_j\}.$$

Monotonicity and homogeneity of $\mathcal{G}$ give the factor two after taking the square root. The lower comparison applies to any fractional submeasure before minimizing the canonical splitting box. For the upper comparison, lift a canonical choice to a union of whole original bands and assign the vanishing omitted tail to ignored mass. These are comparisons of functionals. We do not replace the original observation experiment by an experiment with rounded magnitudes.

We prove $R_{\mu,\rho}(B) \ge ck_J$ uniformly in the finite prefix. The limit (113) and (114) will complete the theorem.

E.2 Two consequences of optimal finite selection

Fix a minimizer I of the inner finite objective in (112), before its outer cap by ρ. Let O consist of the complementary bands with $b_j \ge h$, and Z of those with $b_j < h$. Write

$$F = \mathcal{G}((a_j)_{j\in I}), \qquad A = \sum_{j\in I} a_j, \qquad z = \sum_{j\in Z} w_j, \qquad H_O = \sum_{j\in O} a_j b_j.$$

The minimized inner cost is $z + hH_O + h\sqrt{\rho F}$. Zero-mass bands can be omitted.

Lemma E.2 (Optimal-band inequalities)**.** *Every ordinary band $j \in O$ has $w_j \le \rho$. If $F > 0$ and $\lambda = \sqrt{\rho/F}$, every retained band satisfies*

$$b_j \ge \frac{\lambda}{2}[1 + \log(A/a_j)]. \tag{115}$$

Proof. For an ordinary band, put $x = a_j$ and compare with moving it to I. Optimality gives

$$xb_j \le \sqrt{\rho}[\sqrt{F + \Delta} - \sqrt{F}].$$

If $F = 0$, the squared bracket is x. Otherwise use $\log(1 + t) \le \sqrt{t}$ for $t \ge 0$ to obtain

$$\Delta = x + A\log(1 + x/A) + x\log(1 + A/x) \le x + 2\sqrt{Ax} \le x + 2\sqrt{Fx}.$$

Thus $\sqrt{F + \Delta} - \sqrt{F} \le \sqrt{x}$, and optimality gives $x^2 b_j^2 \le \rho x$, proving $w_j \le \rho$.

For $j \in I$, removing the band and assigning it to ordinary work is admissible even if ignoring it would be cheaper. Hence

$$a_j b_j \ge \sqrt{\rho}[\sqrt{F} - \sqrt{F_{-j}}].$$

The derivative of $\mathcal{G}$ in this coordinate is $1 + \log(A/a_j)$; it only increases as that coordinate is removed. Therefore $F - F_{-j} \ge a_j[1 + \log(A/a_j)]$. Dividing by $\sqrt{F} + \sqrt{F_{-j}} \le 2\sqrt{F}$ proves (115). □

E.3 The ordinary and ignored costs are already lower bounds

Recall the functional from Proposition H.4:

$$\Psi_{\mu,\rho}(h) = \sup_{\substack{0\le q\le 1 \\ \int q\, d\mu \le \rho}} \int \min\{q(d), h\sqrt{q(d)/d}\}\, \mu(dd).$$

Theorem H.3 and Proposition H.4 imply

$$R_{\mu,\rho}(B) \ge \Psi_{\mu,\rho}(B^{-1/2})/8192. \tag{116}$$

For an explicit deduction, take $b = \Psi(h)/64 \le \rho/64$ in the bias-indexed null obstruction in Proposition H.4. Evaluating it at $h = B^{-1/2}$ gives $J \ge (15\Psi(h)/16)^2/(192h^2)$. The bias-indexed transfer has its smaller branch at b, yielding (116).

Every ignored band has $d \le b_j < h$, so allocating on these bands gives

$$\min\{\rho, z\} \le \Psi_{\mu,\rho}(h). \tag{117}$$

For the ordinary bands we need an elementary dyadic inequality. If masses v_j satisfy $v_j \le s$ at values $f_j = b_j^{-1} = 2^{j+1}$, then

$$\sum_j v_j f_j \le 2 \sup_{\substack{0\le q_j\le 1 \\ \sum_j v_j q_j \le s}} \sum_j v_j q_j f_j. \tag{118}$$

If the total mass is at most s, this is immediate. Otherwise select the largest f_j up to mass s, allowing a fraction at the boundary index j_0. Let r be the selected mass at the boundary index. The selected mass above it is $s-r$, at values at least $2f_{j_0}$, so the selected integral is at least $(2s-r)f_{j_0}$. The excluded smaller values contribute at most sf_{j_0}, and the excluded boundary mass contributes at most $(s-r)f_{j_0}$. Thus the excluded integral is no larger than the selected integral, proving (118).

Apply it to O with $s=\rho$, using Lemma E.2. It follows that H_O is at most twice the largest ordinary-band inverse-gap integral at mass ρ. On these bands $h/b_j \le 1$ and $d \le b_j$; for every $0 \le q \le 1$, $\min\{q, h\sqrt{q}/d\} \ge qh/b_j$. Consequently

$$hH_O \le 2\Psi_{\mu,\rho}(h). \tag{119}$$

E.4 Packing entropy into comparable-cost groups

Lemma E.3 (Entropy tiers)**.** *For a finite positive vector a, let $M = \max_j a_j$ and group its coordinates by $a_j \in (M2^{-k-1}, M2^{-k}]$, $k \ge 0$. If F_k is each group's entropy cost, then*

$$\mathcal{G}(a) \le 7\sum_k F_k. \tag{120}$$

Proof. Let A_k be the cost mass in group k, n_k its cardinality, and $p_k = A_k/A$. Entropy decomposition gives $\mathcal{G}(a) = \sum_k F_k + A\,\mathrm{Ent}(p)$. Empty groups are omitted. For the comparison on $k \ge 0$, the reference probability is $q_k = 2^{-k/2}/(2+\sqrt{2})$, so $\mathrm{KL}(p\|q) \ge 0$ applies even when the nonempty tier indices are sparse. Comparison with the probability vector proportional to $2^{-k/2}$ gives

$$A\,\mathrm{Ent}(p) \le A\log(2+\sqrt{2}) + \frac{\log 2}{2}\sum_k kA_k.$$

On groups with $n_k \ge 2^{k/2}$, the second term is at most $\sum_k A_k \log n_k \le \sum_k F_k$; comparable weights imply $F_k \ge A_k[1+\log n_k - \log 2] \ge A_k \log n_k$. On the other groups, $A_k \le M2^{-k/2}$, so their contribution is at most

$$\frac{\log 2}{2} M \sum_{k\ge 0} k2^{-k/2} = \frac{(\log 2)\sqrt{2}}{2(\sqrt{2}-1)^2} M < 3M \le 3A.$$

Finally $\log(2+\sqrt{2}) < 2$ and $A \le \sum_k F_k$. Adding the initial $\sum_k F_k$ proves (120). □

Split the retained low-gap bands $j \ge 6$ by index parity and then by these cost tiers. Parity splitting loses at most a factor two in entropy, so all resulting group entropies sum to at least $F_{\mathrm{low}}/14$. Every group is a separated annular family with endpoint ratios at least four, and its cost weights lie in some interval $(c, 2c]$.

Consider a group of n bands and cost entropy at most $2cn(1+\log n)$. Since $A \ge nc$ and every group coordinate is at most $2c$, (115) implies

$$b_j \ge \frac{\lambda}{2}[1+\log(n/2)] \ge \frac{\lambda}{8}(1+\log n).$$

Remove the $t = \lceil \log_4 n \rceil$ smallest-gap bands, with $t=0$ for $n=1$. For $n \ge 2$, $t \le n/2$. The remaining entropy G_i satisfies

$$G_i \ge F_{\mathrm{group}}/16. \tag{121}$$

Indeed, at least $n/2$ coordinates remain, each at least c. Coordinate monotonicity gives entropy at least $cn'(1+\log n')$, whereas the original entropy is at most $2cn(1+\log n)$. Since

$1+\log(n/2) \geq (1-\log 2)(1+\log n)$, their ratio is at least $(1-\log 2)/4 > 1/16$. The case $n = 1$ is immediate.

Separation amplifies the smallest remaining endpoint by at least $4^t \geq n$. Hence

$$b_{\min,i} \geq \lambda n(1+\log n)/8. \tag{122}$$

Here G_i is the canonical endpoint entropy of the retained bands, which is exactly the annular quantity $\Gamma^{\circ}_{\sigma_i}$ in Appendix G. Let σ_i be the original law restricted to these remaining bands, τ_i its mass, a_i its largest endpoint, and $w_{*,i}$ its smallest unnormalized band mass. Then $\tau_i/a_i^2 \geq c$ and $w_{*,i} \geq cb_{\min,i}^2$. Together with $G_i \leq 2cn(1+\log n)$, this gives the actual annular saturation bound

$$\boxed{\frac{\tau_i w_{*,i}}{a_i^2} \geq \frac{\lambda^2 G_i^2}{256}.} \tag{123}$$

This step retains the minimum-band-mass constraint; it does not replace that constraint by an inverse-gap moment.

Allocate original rarity and conditional rarity by

$$\theta_i = \frac{\lambda^2 G_i}{1024}, \qquad r_i = \theta_i/\tau_i. \tag{124}$$

Equation (123) makes $\theta_i G_i$ at most one quarter of the saturation term. Since $G_i \geq \tau_i/a_i^2$, that saturation divided by G_i is at most $w_{*,i} \leq \tau_i$. Thus $\theta_i \leq \tau_i/4$ and $r_i \leq 1/4$. The components are disjoint, and superadditivity gives

$$\sum_i \theta_i \leq \lambda^2 F/1024 = \rho/1024.$$

Every retained component is supported on $(0, 1/128]$; parity gives endpoint ratios at least four. The normalized minimum band mass is $w_{*,i}/\tau_i$, so (123) is exactly the unnormalized saturation condition required by the annular proposition. Every component is therefore admissible, and every entropy branch of the scale-uniform annular lower bound is active.

E.5 Combining the components at every budget

Let F_{high} be the retained entropy in the first six bands and F_{low} the remainder. Entropy decomposition gives $F \leq 2(F_{\text{high}} + F_{\text{low}})$, and $F_{\text{high}} \leq 4096(1+\log 6)$. If $F_{\text{high}} \geq F_{\text{low}}$, then F is bounded by an absolute constant. The feasible constant allocation $q = \rho$ gives $\Psi(h) \geq \min\{\rho, 2h\sqrt{\rho}\}$, so

$$\min\{\rho, h\sqrt{\rho F}\} \leq C\Psi(h). \tag{125}$$

Otherwise $F_{\text{low}} > F_{\text{high}}$ and $F \leq 4F_{\text{low}}$. Lemma E.3, parity splitting, and (121) give

$$\sum_i G_i \geq F_{\text{low}}/224 \geq F/896. \tag{126}$$

Proposition G.4 and Theorem H.7 now imply

$$R_{\mu,\rho}(B) \gtrsim \sum_i \min\{\theta_i, \sqrt{\theta_i G_i/B}\}.$$

All conditional bias and annular constants here are fixed universal constants. Crucially, the ratio $\theta_i/G_i = \lambda^2/1024$ is common to every component. This remains true with the fixed

conditional-bias and annular constants retained, so all components have the same budget crossover. Therefore the displayed sum equals

$$\left(\sum_i G_i\right) \min\{\lambda^2/1024, \lambda/(32\sqrt{B})\}.$$

Using (126) and $\lambda^2 = \rho/F$ proves

$$R_{\mu,\rho}(B) \geq c \min\{\rho, h\sqrt{\rho F}\}. \tag{127}$$

No limit in B or separate lower threshold for each component is used.

Completion of Theorem E.1. If $z \geq \rho$, (117) and (116) already give $R \geq ck_J$. Otherwise (117) and (119) imply

$$k_J \leq \min\{\rho, 3\Psi(h) + h\sqrt{\rho F}\} \leq 3\Psi(h) + \min\{\rho, h\sqrt{\rho F}\}.$$

Apply (116) and either (125) or (127). If $F = 0$, the Ψ bound alone suffices. Thus $R_{\mu,\rho}(B) \geq ck_J$ with one universal constant for every finite prefix. Let J increase and use (113)–(114). This proves the theorem. The passage to every magnitude law uses only vanishing omitted tail probability and finite-band lower constructions. □

E.6 Explicit conservative comparison constants

The preceding arguments provide, without optimizing constants,

$$10^{-17}\mathcal{K}_{\mu,\rho}(B) \leq R_{\mu,\rho}(B) \leq 33000\mathcal{K}_{\mu,\rho}(B). \tag{128}$$

For completeness, the work summation in Appendix D has entropy and ordinary coefficients at most, respectively,

$$\frac{5\sqrt{17}}{4}\left(\frac{4\sqrt{80}}{9} + 2\sqrt{42} + 2\right) < 104,$$

$$\frac{5\sqrt{17}}{4}\left(\frac{2}{3} + \sqrt{2}(3328 + 2\sqrt{21}) + \frac{512}{3}\right) < 27000.$$

Together with the $128z$ bias term and the constant estimator, these give the rounded upper constant.

For the lower bound let $c_{\mathrm{ann}} = e^{-28}/2000000$ and $c_E = \sqrt{c_{\mathrm{ann}}}/(896 \cdot 128 \cdot 32)$. Before suppressing constants, the packed contribution is at least

$$\frac{\sum_i G_i}{128} \min\left\{\frac{\lambda^2}{65536}, \frac{\sqrt{c_{\mathrm{ann}}}\lambda}{32\sqrt{B}}\right\} \geq c_E \min\{\rho, h\sqrt{\rho F}\}.$$

The high-gap alternative gives $\min\{\rho, h\sqrt{\rho F}\} \leq 112\Psi(h)$ and has a stronger constant. Combining (116) with the final $3\Psi(h)$ assembly and $\mathcal{K} \leq 2k$ gives the coefficient $[2(24576 + c_E^{-1})]^{-1} > 10^{-17}$. For example, $e < 3$ and $\sqrt{2000000} < 1500$ imply $2(24576 + 896 \cdot 128 \cdot 32 \cdot 1500 \cdot 3^{14}) < 10^{17}$. These are constants for the scalar comparison, not the factor sixteen of the stopping-program certificate.

F Observable classification and finite Poisson allocation

This appendix proves the single-unit classification bounds and the finite multilevel estimator used in Appendix D and Appendix O.2. The ordinary classifier is the construction of Section 4, restated with the work bound used in the race; the weighted reference bank and multilevel allocation are proved in full. Throughout this appendix, a false-positive probability means the joint event $\mathbb{P}_q(Z = 1, S = 0)$ for the binary sign S, not the conditional probability $\mathbb{P}_q(Z = 1 \mid S = 0)$. All likelihoods refer to the actual observed binary prefix. A reference distribution is used to evaluate that prefix, and never reveals or redraws the latent state of a unit.

F.1 A capped classifier that does not know the gap

For a dyadic integer $M \geq 1$, inspect prefix lengths $1, 2, 4, \ldots, M$. At a prefix $n < M$, stop if

$$\left|\widehat{p}_n - \frac{1}{2}\right| > t_{n,M} := \sqrt{\frac{1 + \log(M/n)}{n}},$$

and output the side of $1/2$ containing $\widehat{p}_n$. At M, output $\mathbf{1}_{\{\widehat{p}_M > 1/2\}}$, with a negative output on ties. Denote the output and work by $(Z_M^{\mathrm{ord}}, T_M^{\mathrm{ord}})$.

Lemma F.1 (Ordinary conditional error and work)**.** *For every $0 < d \leq 1/2$ and either latent sign,*

$$\mathbb{P}_{d,s}(Z_M^{\mathrm{ord}} \neq s) \leq \min\{1, (Md^2)^{-2}\}, \tag{129}$$

$$\mathbb{E}_{d,s} T_M^{\mathrm{ord}} \leq o_M(d) := \min\{M, 17d^{-2}[1 + \log_+(Md^2)]\}. \tag{130}$$

Proof. For independent Bernoulli observations of mean p, Hoeffding's inequality gives $\mathbb{P}(\widehat{p}_n - p \geq t) \leq e^{-2nt^2}$ and the analogous lower-tail bound [26]. A wrong early output at a fixed inspected prefix requires a deviation of at least $d + t_{n,M}$. Applying a union bound on the underlying full tape, without conditioning on the stopping event, therefore gives

$$\mathbb{P}_{d,s}(Z_M^{\mathrm{ord}} \neq s) \leq e^{-2Md^2} + e^{-2} \sum_{\substack{n<M \\ n \text{ dyadic}}} (n/M)^2 e^{-2nd^2}.$$

For any dyadic sequence of positive u values, the intervals $[u/2, u]$ have disjoint interiors and

$$\int_{u/2}^{u} y e^{-2y}\, dy \geq \frac{3}{8} u^2 e^{-2u}.$$

Consequently $\sum_u u^2 e^{-2u} \leq 2/3$. Writing $x = Md^2 \geq 1$ and using $x^2 e^{-2x} \leq e^{-2}$, the preceding error bound is at most $(5/3)e^{-2}x^{-2} < x^{-2}$. For $x < 1$, the probability bound by one suffices.

Since the stopping time takes dyadic values,

$$\mathbb{E}_{d,s} T_M^{\mathrm{ord}} = 1 + \sum_{\substack{n<M \\ n \text{ dyadic}}} n\, \mathbb{P}_{d,s}(T_M^{\mathrm{ord}} > n).$$

If $d > 2t_{n,M}$, survival implies a deviation from the true mean of at least $d/2$. Thus

$$\mathbb{E}_{d,s} T_M^{\mathrm{ord}} \leq 1 + \sum_{\substack{n<M \\ n \text{ dyadic}}} n \left[\mathbf{1}_{\{d \leq 2t_{n,M}\}} + 2e^{-nd^2/2}\right].$$

For $x = Md^2 \geq 1$, the first indicator implies $nd^2 \leq 4(1 + \log x)$. Indeed, this is immediate if $nd^2 < 1$; otherwise $\log(M/n) = \log(x/(nd^2)) \leq \log x$. The corresponding dyadic sum is at most $8d^{-2}(1 + \log x)$. Also

$$\int_{u/2}^{u} e^{-y/2}\,dy \geq \frac{1}{2}ue^{-u/2} \quad \Longrightarrow \quad \sum_{u} ue^{-u/2} \leq 4.$$

The exponential part of the work bound is therefore at most $8d^{-2}$. Since $1 \leq d^{-2}$, total expected work is at most $17d^{-2}(1 + \log x)$. Finally $T_M^{\mathrm{ord}} \leq M$ deterministically; when $x < 1$, the displayed minimum equals M. □

For any finite measure ξ on $(0, 1/2]$, write

$$\eta_\xi(M) = \int \min\{1, (Md^2)^{-2}\}\,\xi(dd), \qquad O_\xi(M) = \int o_M(d)\,\xi(dd).$$

Both are homogeneous in ξ. The first is nonincreasing in M and the second is nondecreasing. Pointwise comparison also gives $\eta_\xi(4M) \geq \eta_\xi(M)/16$.

Lemma F.2 (A homogeneous dyadic kernel estimate)**.** *For every finite measure ξ and integer $K \geq 1$,*

$$\sum_{k=1}^{K} \sqrt{\eta_\xi(4^k)O_\xi(4^k)} \leq 208 \int \min\{2^K, d^{-1}\}\,\xi(dd).$$

In particular,

$$\sum_{k\geq 1} \sqrt{\eta_\xi(4^k)O_\xi(4^k)} \leq 208 \int d^{-1}\,\xi(dd), \tag{131}$$

with extended-value interpretation.

Proof. Use bands $(2^{-j-1}, 2^{-j}]$, $j \geq 1$, with masses w_j, and put $c_j = 2^j w_j$. Direct substitution into the two kernels yields

$$\eta_\xi(4^k) \leq 2^{-k}\left[\sum_{j\geq k} c_j 2^{k-j} + 16\sum_{j<k} c_j 2^{3(j-k)}\right] = 2^{-k}A_k,$$

$$O_\xi(4^k) \leq 2^{k}\left[\sum_{j\geq k} c_j 2^{k-j} + 136\sum_{j<k} c_j 2^{j-k}(1 + k - j)\right] = 2^{k}D_k.$$

For the second display on $j < k$, use $d^{-2} \leq 4\,2^{2j}$ and $1 + \log(4^k d^2) \leq 2(1 + k - j)$. On $j \geq k$, the deterministic cap gives the first sum exactly as an upper bound.

Apply $\sqrt{A_k D_k} \leq (A_k + D_k)/2$ and sum over $k \leq K$. For a fixed $j \leq K$, the three relevant geometric sums obey

$$\sum_{k\leq j} 2^{k-j} \leq 2, \qquad \sum_{r\geq 1} 2^{-3r} = \frac{1}{7}, \qquad \sum_{r\geq 1} 2^{-r}(1 + r) = 3.$$

The coefficient of c_j is thus at most $2 + 8/7 + 204 < 208$. For $j > K$, only the first kernels occur, with coefficient at most $2\,2^{K-j}$. Hence the result is at most

$$208\sum_{j\leq K} 2^j w_j + 2\,2^K \sum_{j>K} w_j.$$

For $j \leq K$, $\min\{2^K, d^{-1}\} \geq 2^j$ on the band; for $j > K$ it equals 2^K. This proves the finite bound. Increasing K and using monotone convergence gives (131). All rearrangements involve nonnegative terms, so normalization and moment assumptions are unnecessary. □

F.2 A weighted bank compared with the full null mixture

For a binary prefix $x_{1:n}$, let $P_{p,n}(x_{1:n})$ be its probability under iid $\mathrm{Ber}(p)$ observations, and let $P_{\mathrm{f},n} = P_{1/2,n}$. Choose any nonempty finite or countable set of canonical bands

$$I_j = (b_j/2, b_j], \qquad b_j = 2^{-j-1},$$

and positive weights π_j summing to one on these bands. Set $r_j^- = b_j/2$ and $r_j^+ = b_j/4$, and define

$$L_{j,n}^- = \frac{P_{1/2-r_j^-,n}}{P_{\mathrm{f},n}}, \qquad M_n = \frac{\sum_j \pi_j P_{1/2+r_j^+,n}}{P_{-,\mu,n}}, \qquad P_{-,\mu,n} = \int P_{1/2-d,n}\,\mu(dd).$$

For $0 < \epsilon < 1$, stop negative when any $L_{j,n}^- \geq 1/(\epsilon\pi_j)$, and stop positive when $M_n \geq 1/(\rho\epsilon)$. Give negative crossings priority. A zero denominator triggers a positive stop. Denote the bank stopping time by T_{bank}; it may be infinite outside its selected bands.

Lemma F.3 (Full-mixture likelihood bank)**.** *The probability of a wrong bank stop is at most $2\rho\epsilon$ uniformly over $q \in \mathcal{Q}_\rho$, and its false-positive probability is at most $\rho\epsilon$. These statements concern the event that the bank stops and do not assume termination at every magnitude.*

For $d \in I_j$ in a selected band,

$$\mathbb{E}_{d,-} T_{\mathrm{bank}} \leq \frac{3}{4d^2}[1 + \log(1/\epsilon) + \log(1/\pi_j)]. \tag{132}$$

Suppose a finite number $V(d) > 0$ satisfies

$$\beta_\mu(n) := \int e^{-2nx^2}\,\mu(dx) \leq V(d) \qquad \text{for every integer } n \geq d^{-2}.$$

Then

$$\mathbb{E}_{d,+} T_{\mathrm{bank}} \leq \frac{6}{d^2}\left[1 + \log(1/\epsilon) + \log(1/\pi_j) + \log_+ \frac{V(d)}{\rho}\right]. \tag{133}$$

In particular the bank has finite conditional work on every selected band whenever such an envelope is supplied.

Proof. A nonnegative supermartingale Y_n with $Y_0 = 1$ has $\mathbb{P}(\sup_n Y_n \geq a) \leq 1/a$: stopping at the first crossing or a fixed horizon gives $1 \geq \mathbb{E}Y_{T\wedge m} \geq a\,\mathbb{P}(T \leq m)$, and increasing the horizon proves the claim. This elementary maximal inequality also works when the process is a supermartingale only on supported prefixes.

Under a positive coin of gap d, the next multiplier of $L_{j,n}^-$ has mean $1 - 4r_j^- d \leq 1$. Its negative-crossing probability is at most $\epsilon\pi_j$. A countable union bound gives conditional negative-crossing probability at most ϵ, so its positive-sign contribution under any admissible prior is at most $\rho\epsilon$.

Under the full all-negative mixture, M_n is a nonnegative supermartingale on supported prefixes. If a prefix x has positive null probability, conditional expectation sums only children xu with positive null probability; their numerator probabilities sum to at most the numerator at x. This proves the supermartingale inequality even when μ has mass at $d = 1/2$. A zero-denominator prefix has null probability zero. Thus a positive crossing has null-mixture probability at most $\rho\epsilon$. Its actual negative-sign contribution is no larger, since it multiplies the conditional crossing probabilities by $1 - q(d) \leq 1$. The two error claims follow.

For negative work, fix a selected-band gap $d \in I_j$ and $r = r_j^- \in [d/2, d]$, with $r \leq 1/4$. The matching log likelihood has mean increment

$$g(d, r) = \frac{1}{2}\log(1 - 4r^2) + d\log\frac{1+2r}{1-2r} \geq -\frac{8}{3}r^2 + 4dr \geq \frac{4}{3}d^2.$$

Here $\log(1-x) \geq -x/(1-x)$ and $\log((1+x)/(1-x)) \geq 2x$ establish the first lower bound; the concave quadratic takes the value $4d^2/3$ at both endpoints $r = d/2, d$. Every upward log increment is at most $\log(3/2) < 1$. For the first correct-reference crossing τ, the stopped-sum identity at $\tau \wedge m$ gives

$$g(d,r)\mathbb{E}(\tau \wedge m) = \mathbb{E}\log L^-_{j,\tau\wedge m} \leq 1 + \log(1/(\epsilon\pi_j)).$$

Monotone convergence proves integrability of τ and the stated work bound, since $T_{\text{bank}} \leq \tau$. No unbounded stopping identity is invoked before integrability is established.

For positive work, let $E_n = \{\widehat{p}_n - 1/2 \geq d/2\}$. On this event, for $x < 1/2$,

$$\log \frac{P_{1/2-x,n}}{P_{\text{f},n}} = \frac{n}{2}\log(1-4x^2) - n(\widehat{p}_n - \tfrac{1}{2})\log\frac{1+2x}{1-2x} \leq -2nx^2.$$

For $x = 1/2$ the likelihood is zero, since E_n entails at least one head. Therefore $P_{-,\mu,n}/P_{\text{f},n} \leq \beta_\mu(n)$. The matching positive reference $r = r_j^+ \in [d/4, d/2]$ satisfies, on the same event,

$$\log \frac{P_{1/2+r,n}}{P_{\text{f},n}} \geq n\left[-\frac{8}{3}r^2 + 2dr\right] \geq nd^2/3.$$

Set

$$U = 1 + \log(1/\epsilon) + \log(1/\pi_j) + \log_+(V(d)/\rho), \qquad n_0 = \left\lceil 3d^{-2}U \right\rceil.$$

For every $n \geq n_0$, E_n implies $M_n \geq \pi_j e^{nd^2/3}/V(d) \geq 1/(\rho\epsilon)$. Thus $T_{\text{bank}} > n$ implies E_n^c, whose positive-coin probability is at most $e^{-nd^2/2}$. Summing the survival probabilities yields

$$\mathbb{E}_{d,+}T_{\text{bank}} \leq 3d^{-2}U + 2 + 2d^{-2} \leq 6d^{-2}U.$$

The last inequality uses $U \geq 1$ and $d^2 \leq 1/4$.

For a countable bank, the positive mixture is a consistent probability law and the negative stopping event is a measurable countable union. At any fixed time only finitely many negative references can cross: $L^-_{j,n} \leq (3/2)^n$ requires $\pi_j \geq 1/[\epsilon(3/2)^n]$ for a crossing. All preceding arguments therefore apply to finite and countable banks alike. □

F.3 Racing procedures and retaining a genuine unit

Lemma F.4 (Observable race)**.** *Suppose two classifiers run on the odd and even subsequences of one unit's observation tape, with independent auxiliary seeds. Stop the physical procedure when either classifier stops and use its output. On completed counterfactual subsequences its work satisfies*

$$T_{\text{race}} \leq 2\min\{T_1, T_2\}.$$

Its error event is contained in the union of the component wrong-stop events. If either component has a finite deterministic cap, the race terminates with a finite deterministic cap.

Proof. By physical time $2n$, each component has received n observations. If either component would stop after n of its own observations, the race stops no later than $2n$. This proves the pathwise work bound, including an infinite stopping time for the other component. The selected output is wrong only if the component that produces it makes a wrong stop. Conditional on the latent (D, S), both subsequences are iid Bernoulli tapes of the correct mean, so all component error and work bounds apply. The policy uses only the physical observed prefix and its seeds. □

F.4 Finite multilevel allocation at every positive budget

Lemma F.5 (Independent Poisson allocation)**.** *Let $Z_0, \dots, Z_J \in \{0,1\}$ be observable single-unit classifiers, each with finite deterministic cap. Suppose uniformly over $q \in \mathcal{Q}_\rho$ that their error probabilities are at most e_ℓ, their false-positive probabilities are at most f_ℓ, and their expected work is at most $C_\ell > 0$. Put*

$$v_0 = \rho + f_0, \qquad v_\ell = e_\ell + e_{\ell-1} \quad (\ell \geq 1), \qquad k_0 = C_0, \qquad k_\ell = C_\ell + C_{\ell-1} \quad (\ell \geq 1).$$

Assume $v_\ell > 0$ and set $A_J = \sum_{\ell=0}^{J} \sqrt{v_\ell k_\ell}$. For every $B > 0$ there is an observable population estimator of uniform expected work at most B and worst RMS error at most

$$\sqrt{e_J^2 + A_J^2/B}. \tag{134}$$

Proof. A level-zero correction uses a fresh genuine unit and returns $\Delta_0 = Z_0$. For $\ell \geq 1$, a correction draws one fresh latent unit, runs Z_ℓ on its observations, then runs $Z_{\ell-1}$ with a fresh seed on the next unused observations of the same unit, and returns $\Delta_\ell = Z_\ell - Z_{\ell-1}$.

Conditional on the latent state, the unused suffix after a finite stopping time is an iid tape with the same Bernoulli mean: for each possible stopped prefix, independence of its subsequent coordinates gives the claim, and summing over the stopped prefixes preserves it. This conditional argument is applied to successive unused blocks of the same latent unit, even though the first block has random length. The second classifier therefore has its original marginal distribution under the same latent state. No new magnitude or sign is drawn between the two classifiers. Their disagreement implies that at least one misclassifies that common sign, and hence

$$\mathbb{E}_q \Delta_\ell^2 \leq v_\ell, \qquad \mathbb{E}_q W_\ell \leq k_\ell, \qquad \sum_{\ell=0}^{J} \mathbb{E}_q \Delta_\ell = \mathbb{E}_q Z_J.$$

At level zero use $\mathbb{P}_q(Z_0 = 1) \leq \mathbb{P}_q(S = 1) + f_0 \leq \rho + f_0$.

Before drawing the genuine units and their observation tapes, draw the Poisson replication counts independently across levels, with means

$$n_\ell = \frac{B}{A_J} \sqrt{\frac{v_\ell}{k_\ell}}.$$

For each replication use an independent genuine unit, and return

$$\widehat{\theta} = \sum_{\ell=0}^{J} \frac{1}{n_\ell} \sum_{i=1}^{N_\ell} \Delta_{\ell,i}, \qquad N_\ell \sim \text{Poisson}(n_\ell).$$

Its mean is $\mathbb{E}_q Z_J$, so its absolute bias is at most e_J. The expected work is at most $\sum_\ell n_\ell k_\ell = B$.

For independent marks Δ and a Poisson count of mean n, conditioning on the count gives $\text{Var}(\sum_{i=1}^{N} \Delta_i) = n\mathbb{E}\Delta^2$. Consequently the independent level variances sum to at most

$$\sum_{\ell=0}^{J} \frac{v_\ell}{n_\ell} = \frac{A_J^2}{B}.$$

The divisor is the deterministic mean n_ℓ, including when the realized count is zero; it is not the random count N_ℓ. Count fluctuations contribute the squared mark mean, which is why the second moment rather than just the mark variance appears. The mark and its observation cost may be dependent within a replication; the count is independent of the entire sequence of these paired draws.

For any positive allocation x_ℓ with $\sum_\ell k_\ell x_\ell \le B$, Cauchy–Schwarz gives

$$\left(\sum_\ell \sqrt{v_\ell k_\ell}\right)^2 \le \left(\sum_\ell v_\ell/x_\ell\right)\left(\sum_\ell k_\ell x_\ell\right).$$

Thus the displayed allocation minimizes this second-moment certificate under its expected-work constraint. This is not a deterministic work cap. Bias squared plus variance proves (134). All counts are almost surely finite, all component caps are finite, and all first and subsequent observations are included in W_ℓ. □

The allocation may use Poisson means below one; this is valid for every $B > 0$. The bound is uniform over the original rarity class. Finite-level bounds can be optimized by taking an infimum over finite constructions without asserting the existence of a single infinite-level algorithm.

G One-unit null-work obstructions

This appendix supplies the sign-testing and annular ingredients used in Appendix E. The weighted sign-testing mechanism is an adaptation of Chen–Li–Qiao's gap-entropy argument, including their Bernoulli extension [19, Lemma 4.1 and Remark D.4]. We give the full finite-gap version needed here, followed by the scale-sensitive gate and its actual minimum-band-mass constraint.

G.1 Weighted sign testing

Lemma G.1 (Weighted sign entropy)**.** *Let $d_1 > \cdots > d_K > 0$, with $d_j/d_{j+1} \geq 2$ and $d_1 \leq 1/128$. One observable binary sign test, with the same independent seed law at all gaps, has error at most $1/10$ under every Bernoulli law $1/2 \pm d_j$. Suppose its stopping time τ has finite expectation under both signs at every such gap. Put $c_j = d_j^2 \mathbb{E}_{-d_j}\tau$. For every probability vector π on these gaps,*

$$\sum_{j=1}^{K} \pi_j c_j \geq \frac{1 + \mathrm{Ent}(\pi)}{214}. \tag{135}$$

The stopping and decision rules are the same measurable functions of the seed and observation tape at every gap; the realized gap is not supplied as an input. The test may depend on the entire known finite gap list and on the weights. Zero weights are allowed, with $0\log(1/0) = 0$. No termination or correctness assumption under the fair coin is required.

Proof. First, the ordinary stopped change-of-measure inequality gives a uniform positive floor for each c_j. The terminal sign decision has minus/plus KL divergence at least

$$\mathrm{kl}(9/10, 1/10) = \frac{4}{5}\log 9 > \frac{8}{5}.$$

One Bernoulli observation has divergence at most

$$\frac{(2d_j)^2}{(1/2 + d_j)(1/2 - d_j)} \leq \frac{64}{3} d_j^2.$$

The stopped chain rule, justified by finite expected endpoint work, implies $c_j > 3/40 > 1/16$. More explicitly, apply the chain rule first to the transcript capped at N. Encode a nonterminal decision by a third symbol. Its probability tends to zero at both endpoints, so lower semicontinuity of finite-alphabet KL and monotone convergence of $\mathbb{E}_{-d_j}(\tau \wedge N)$ justify the stopped inequality. This step uses no fair-law stopping assumption.

Set $b_0 = 1/1024$ and

$$n_j = \left\lfloor b_0/d_j^2 \right\rfloor, \qquad u_j = \left\lceil 4\mathbb{E}_{-d_j}\tau \right\rceil.$$

Since $d_j \leq 1/128$, we have $n_j \geq b_0/(2d_j^2) \geq 8$. The KL divergence between the complete prefixes of length n_j at the minus and plus laws is at most $1/48$. These prefixes include the independent seed and unused bits if the test has already stopped. Pinsker's inequality therefore gives total variation less than $1/8$. An early minus is a plus-law error, so its probability under the minus law is at most $1/10 + 1/8$. Markov's inequality gives $\mathbb{P}_{-d_j}(\tau > u_j) \leq 1/4$. Consequently the event

$$E_j = \{\text{output minus},\ n_j < \tau \leq u_j\}$$

has $\mathbb{P}_{-d_j}(E_j) > 1/4$.

Let $\mathbb{P}_0$ denote the fair-coin law with the same seed and policy. Comparing deterministic prefixes of length u_j gives KL at most $4d_j^2 u_j \le 16c_j + 1$. Binary data processing and $\mathrm{kl}(p, z) \ge p \log(1/z) - \log 2$ yield

$$\mathbb{P}_0(E_j) \ge \exp(-64c_j - 8). \tag{136}$$

Only deterministic finite prefixes have been compared with the fair law.

We next bound a weighted overlap pathwise. On a path with finite terminal time t, participation in E_j implies

$$t > n_j \ge \frac{b_0}{2d_j^2}, \qquad t \le 4\mathbb{E}_{-d_j}\tau + 1.$$

In particular $t \ge 2$ and $c_j \ge (t-1)d_j^2/4 \ge td_j^2/8$. Order the participating gaps from smallest to largest. Their squared values grow by at least a factor of four, and the smallest has $td_j^2 > b_0/2$. The corresponding costs are thus bounded below by $b_0 4^r/16 = 4^{r-7}$, for $r = 0, 1, \ldots$. Hence

$$\sum_j \frac{\mathbf{1}_{E_j}}{1 + c_j} \le \sum_{r \ge 0} \frac{1}{1 + 4^{r-7}} \le 7 + \sum_{r \ge 7} 4^{7-r} = \frac{25}{3}.$$

On a nonterminating fair path no E_j occurs, so the same inequality holds. Integrating and using (136), together with $1 + c_j \le e^{c_j}$, gives

$$Z := \sum_j e^{-65c_j} \le (25/3)e^8, \qquad \log Z < 11.$$

Gibbs' inequality for π and the normalized masses e^{-65c_j}/Z implies

$$\mathrm{Ent}(\pi) \le 65 \sum_j \pi_j c_j + \log Z.$$

Writing $\bar{c} = \sum_j \pi_j c_j$ and retaining the stronger floor $\bar{c} \ge 3/40$, we obtain

$$1 + \mathrm{Ent}(\pi) \le 65\bar{c} + 9 + \log(25/3) \le \left[65 + \frac{40}{3}\{9 + \log(25/3)\}\right] \bar{c} < 214\bar{c}.$$

The windows need not be disjoint; only the displayed weighted-overlap bound is used. □

G.2 A moment comparison on the half-prior class

Write $X_\rho = \{0 \le q \le 1/2 : \int q \, d\mu \le \rho\}$. For an integrable observable one-unit output Y with finite uniform variance and uniform bias at most $\rho/64$, put $V = \sup_{q \in X_\rho} \mathrm{Var}_q Y$ and $M = \sup_{q \in X_\rho} \mathbb{E}_q Y^2$.

Lemma G.2 (Moment comparison)**.** *For $0 < \rho \le 1/4$, one has $V \ge 2\rho/3$ and $M \le 3V/2$.*

Proof. Let $f(d) = \mathbb{E}_{d,+}Y - \mathbb{E}_{d,-}Y$. Comparing the null population with the constant prior $q = \rho$ gives $\int f \, d\mu \ge 31/32$. Conditional variance at this constant prior and Jensen's inequality yield

$$V \ge \rho(1 - \rho) \int f^2 \, d\mu \ge \rho(1 - \rho)(31/32)^2 \ge 2\rho/3.$$

Every mean has absolute value at most $65\rho/64$, so $M \le V + (65\rho/64)^2 \le 3V/2$. The constant prior also gives integrability of both fixed-sign second moments; all conditional statements may be taken outside a common null set. □

G.3 An activation gate that respects the magnitude scale

The gate argument charges activation at the information scale $a^2\mathbb{E}_0 T$. In this subsection μ is any probability law supported on $(0, a]$, $a \le 1/4$, and

$$\mathcal{X}_\rho = \{q : 0 \le q \le \tfrac{1}{2},\ \textstyle\int q\, d\mu \le \rho\}.$$

Let (Y, T) be an observable one-coin procedure with uniform bias at most $\rho/64$, finite uniform variance V, and finite expected work under every $q \in \mathcal{X}_\rho$. Write

$$M_0 = \sup_{q\in\mathcal{X}_\rho} \mathbb{E}_q Y^2, \qquad W_0 = \mathbb{E}_0 T, \qquad P = M_0 W_0.$$

Here M_0 is the uniform second-moment bound, whereas W_0 is null work; the subscript on M_0 does not restrict its supremum to the null. The moment comparison in Lemma G.2 gives $M_0 \le 3V/2$.

More explicitly, observability means that an independent seed U and the genuine observations generate $\mathcal{F}_n = \sigma(U, X_1, \dots, X_n)$, that T is a nonnegative integer-valued stopping time, and that Y is $\mathcal{F}_T$-measurable. The procedure has no access to the latent magnitude or sign. Restarts below refresh U and the unused observation block, while retaining that same latent state. These are the observation-model assumptions of Section 11.1.

Lemma G.3 (A scale-sensitive baseline and gated activation)**.** *Put*

$$C_g = e^7, \qquad \kappa = 64/3, \qquad c_g = (256\kappa C_g^4)^{-1}.$$

Every procedure above satisfies

$$P \ge \frac{\rho}{64a^2}. \tag{137}$$

If $a^2 P \le c_g$, there is an observable procedure Z on the same one-coin experiment, with moment bound m_Z, null work W_0^Z, and a measurable set $\mathcal{G}$ of good magnitudes, such that

$$\begin{gathered}\sup_{q\in\mathcal{X}_\rho} \mathbb{E}_q Z^2 \le m_Z, \qquad \rho/16 \le m_Z \le 2C_g^4 a^2 P, \\ m_Z W_0^Z \le C_g^4 P, \qquad \mu(\mathcal{G}^c) \le 144 m_Z.\end{gathered} \tag{138}$$

At every good gap, both signs have finite conditional expected work and

$$|\mathbb{E}[Z \mid d, 0]| \le \frac{1}{8}, \qquad \mathbb{E}[Z \mid d, 1] \ge \frac{1}{2}, \qquad \mathbb{E}[Z^2 \mid d, 0] \le \frac{m_Z}{\rho}, \qquad \mathbb{E}[Z^2 \mid d, 1] \le \frac{3m_Z}{\rho}. \tag{139}$$

One common sign test with both errors below $1/10$ on these gaps uses

$$n = \lceil 1024 m_Z/\rho \rceil \le 1040 m_Z/\rho \tag{140}$$

independent restarts of Z on unused observations of the same coin. Restart independence is conditional on its fixed latent magnitude and sign, not after averaging over the population. Its negative expected work is $n\,\mathbb{E}[T_Z \mid d, 0]$.

Proof. We first prove (137). Let P_- and P_+ be the full transcript laws with uniformly negative and uniformly positive signs, respectively. The priors $q = 0$ and $q \equiv \rho$ give

$$\mathbb{E}_+ Y - \mathbb{E}_- Y \ge 31/32, \qquad \mathbb{E}_- Y^2 \le M_0, \qquad \mathbb{E}_+ Y^2 \le M_0/\rho.$$

The positive moment and expected work are finite because $q \equiv \rho$ is admissible. Revealing D for the information calculation, and using the stopped chain rule, gives

$$\mathfrak{h}(P_-, P_+) \le \tfrac{1}{2}\kappa a^2 W_0,$$

where $\mathfrak{h} = 1 - \mathrm{Aff}$. Indeed the one-observation endpoint divergence is at most κd^2 for $d \le 1/4$. For any square-integrable output F,

$$|\mathbb{E}_P F - \mathbb{E}_Q F|^2 \le 4(\mathbb{E}_P F^2 + \mathbb{E}_Q F^2)\mathfrak{h}(P, Q). \tag{141}$$

Thus

$$(31/32)^2 \le (5\kappa/2)a^2 P/\rho,$$

which implies (137).

The decision to take a first observation is seed-only and precedes all data. The information gate below is instead an observable finite-prefix stopping rule; neither step reveals a latent magnitude. For the gate, on a full observation tape define

$$S_n = \sum_{i=1}^{n}(2X_i - 1), \qquad n_0 = \lceil a^{-2} \rceil, \qquad H_0 = \lceil a^{-1} \rceil,$$

$$G = \min\{n_0, \inf\{n \ge 1 : |S_n| \ge H_0\}\}, \qquad \sigma = T \wedge G, \qquad E = \{T \ge G\}. \tag{142}$$

The event E is measurable at σ, including the tie $T = G$. Include the entire independent seed in the stopped prefix. At σ, $|S_\sigma| \le H_0$ and $\sigma \le n_0$. For a fixed $d \le a$ and sign s, the likelihood relative to the fair stopped-prefix law is

$$\exp\left\{(2s-1)S_\sigma \operatorname{arctanh}(2d) + \frac{\sigma}{2}\log(1 - 4d^2)\right\}.$$

The inequalities

$$\operatorname{arctanh}(2d) \le 8a/3, \qquad -\tfrac{1}{2}\log(1 - 4d^2) \le 8a^2/3$$

show that the absolute logarithm is at most $37/6 < 7$. Every fixed-state prefix law, and the all-negative magnitude mixture, therefore has density in $[C_g^{-1}, C_g]$ relative to fair.

Let $p = \mathbb{P}_{\mathrm{fair}}(E)$ and $p_s(d) = \mathbb{P}_{d,s}(E)$. The number p is determined by the known policy, its seed law, and a bounded fair-prefix experiment. It does not depend on the unknown prior or genuine latent state. This is an existence argument; no efficient computation of arbitrary policy probabilities is claimed. No termination under the fair law beyond this bounded prefix is needed. Then

$$p/C_g \le p_s(d) \le C_g p. \tag{143}$$

Bounded fair stopping gives $\mathbb{E}_{\mathrm{fair}} S_\sigma^2 = \mathbb{E}_{\mathrm{fair}}\sigma$. A gate activation either reaches n_0 or hits H_0, so

$$p \le \left(n_0^{-1} + H_0^{-2}\right)\mathbb{E}_{\mathrm{fair}}\sigma \le 2a^2\mathbb{E}_{\mathrm{fair}}\sigma \le 2C_g a^2 W_0. \tag{144}$$

Set $Y_e = Y\mathbf{1}_{\{T<G\}}$, explicitly zero on E. This is a function of the bounded prefix. Its second moment under either fixed state is at most $C_g^2 M_0$. The stopped endpoint divergence and the same prefix domination give

$$\mathrm{KL}(P_{-d}^\sigma, P_{+d}^\sigma) \le \kappa d^2 \mathbb{E}_{-d}\sigma \le \kappa C_g^2 a^2 W_0.$$

Since $\mathfrak{h} \le \mathrm{KL}/2$, (141) implies

$$|\mathbb{E}_{+d} Y_e - \mathbb{E}_{-d} Y_e|^2 \le 4\kappa C_g^4 a^2 P. \tag{145}$$

Under $a^2 P \le c_g$, every such early mean difference is at most $1/8$ in absolute value. In this branch $p > 0$: if $p = 0$, the gate never activates under any state, so $Y = Y_e$, contradicting the integrated mean separation $31/32$ from $q = 0, q \equiv \rho$.

Here is an actual observable procedure realizing the conditioning. Keep the same genuine coin. Repeatedly run the original procedure, using a fresh original seed and unused iid flips for each trial. Reject a trial if its original stopping time is before its gate. When a trial reaches the gate, accept it, continue that original trial to its original stopping time, and return

$$Z = C_g p Y.$$

Every rejected observation is charged. Conditional on d, s, trials are iid with acceptance probability $p_s(d) > 0$. Writing $w_s(d) = \mathbb{E}[T \mid d, s]$, the renewal identity gives

$$\mathbb{E}[T_Z \mid d, s] = \frac{w_s(d)}{p_s(d)}. \tag{146}$$

This identity includes the work of all rejected trials and the final accepted trial. To verify it without assuming independence of acceptance and trial cost, let L be the first accepted trial. Conditional on (d, s),

$$\mathbb{E}\sum_{k=1}^{L} T_k = \sum_{k\geq 1} \mathbb{E}[T_k \mathbf{1}_{\{L\geq k\}}] = \sum_{k\geq 1} (1 - p_s(d))^{k-1} w_s(d) = w_s(d)/p_s(d).$$

The event $\{L \geq k\}$ involves only preceding trials, so it is independent of the current trial cost. It is not necessary that T_k be independent of its own acceptance event. Moreover,

$$\mathbb{E}[Z^2 \mid d, s] = \frac{C_g^2 p^2}{p_s(d)} \mathbb{E}[Y^2 \mathbf{1}_E \mid d, s] \leq C_g^3 p\, \mathbb{E}[Y^2 \mid d, s].$$

With $m_Z = C_g^3 p M_0$, integration and (144)–(146) give

$$\sup_q \mathbb{E}_q Z^2 \leq m_Z, \qquad m_Z \leq 2C_g^4 a^2 P, \qquad W_0^Z \leq C_g W_0/p, \qquad m_Z W_0^Z \leq C_g^4 P. \tag{147}$$

Expected work under every q is finite by the same comparison.

We establish conditional sign separation directly; conditioning has not preserved the estimator's means. Write

$$f_s(d) = \mathbb{E}[Y \mid d, s], \quad \ell_s(d) = \mathbb{E}[Y \mathbf{1}_E \mid d, s], \quad z_s(d) = \mathbb{E}[Z \mid d, s].$$

The set $\{f_+ - f_- < 3/4\}$ has mass below 2ρ. Otherwise, writing that set as S with $\mu(S) \geq 2\rho$, the prior $q = (\rho/\mu(S))\mathbf{1}_S$ lies in $\mathcal{X}_\rho$ and has total mass ρ. It would contradict the bias guarantee and the mean separation $31\rho/32$. This uses fractional prior values, so it applies unchanged when μ has atoms. Outside that set, (145) gives $\ell_+ - \ell_- \geq 5/8$. Also

$$z_s(d) = \frac{C_g p}{p_s(d)} \ell_s(d), \qquad 1 \leq \frac{C_g p}{p_s(d)} \leq C_g^2.$$

If $|z_-(d)| \leq 1/8$, then $|\ell_-(d)| \leq 1/8$, $\ell_+(d) \geq 1/2$, and $z_+(d) \geq 1/2$. The exceptional mass for $|z_-| > 1/8$ is at most $64 m_Z$. Since

$$m_Z \leq 2C_g^4 c_g = 1/(128\kappa) < 1/256,$$

this preliminary mean-good set has mass at least $1 - 2\rho - 64 m_Z \geq 1/4$. Taking $q \equiv \rho$ in the second-moment bound yields $m_Z \geq \rho/16$.

Finally, put $a_Z(d) = \mathbb{E}[Z^2 \mid d, 0]$ and $b_Z(d) = \mathbb{E}[Z^2 \mid d, 1]$. The set $\{a_Z > m_Z/\rho\}$ has mass at most ρ. The set $\{b_Z - a_Z > 2m_Z/\rho\}$ has mass below 2ρ, again by using a fractional

admissible prior on a set of mass at least 2ρ. Removing these sets and null sets of infinite conditional work gives total exceptional mass

$$5\rho + 64m_Z \le 144m_Z$$

and proves (139). Average n independent restarts of Z on unused blocks of the same coin, and threshold at $5/16$. Both mean margins are at least $3/16$, so Chebyshev gives errors at most

$$\frac{3m_Z/\rho}{n(3/16)^2} \le 1/12 < 1/10.$$

The bound on n follows from $m_Z \ge \rho/16$, and work adds, proving (140). □

G.4 A scaled annular obstruction

For a positive vector $v = (v_i)$, set

$$\mathcal{G}(v) = A[1 + \mathrm{Ent}(v/A)], \qquad A = \sum_i v_i.$$

A useful elementary identity is

$$\frac{\partial \mathcal{G}}{\partial v_i} = 1 + \log(A/v_i) \ge 1. \tag{148}$$

Thus this cost-entropy functional is coordinatewise increasing. In particular, if $u \le v_i \le U$ for K coordinates, then

$$uK(1 + \log K) \le \mathcal{G}(v) \le UK(1 + \log K). \tag{149}$$

Proposition G.4 (A gated annular lower bound)**.** *Suppose μ is supported on $(0, a]$, $a \le 1/4$. Take a nonempty finite annular family*

$$I_j = (b_j/2, b_j], \qquad b_j/b_{j+1} \ge 4, \qquad b_1 \le 1/128,$$

with positive masses $w_j = \mu(I_j)$. Define

$$w_* = \min_j w_j, \qquad A = \sum_j w_j/b_j^2, \qquad \pi_j = \frac{w_j/b_j^2}{A}, \qquad \Gamma^\circ = A[1 + \mathrm{Ent}(\pi)].$$

Every one-coin procedure with the assumptions above satisfies

$$V\mathbb{E}_0 T \ge \frac{e^{-28}}{2\,000\,000} \min\{\rho\Gamma^\circ, w_*/a^2\}. \tag{150}$$

Proof. If $a^2 P \ge c_g$, use $w_* \le 1$ and $M_0 \le 3V/2$. Otherwise apply Lemma G.3. If $m_Z > w_*/288$, then

$$P > w_*/(576C_g^4 a^2).$$

If $m_Z \le w_*/288$, at least half the mass of every annulus is good. Averaging selects $d_j \in I_j \cap \mathcal{G}$ with

$$w_-^Z(d_j) \le \frac{4}{w_j} \int_{I_j} w_-^Z(d)\, \mu(dd).$$

The selected gaps are separated by a factor greater than two and bounded by $1/128$. The representative points may depend on the procedure being bounded. This is legitimate: the weighted sign lemma holds for every such finite list. After the points have been selected, the same already-constructed restart test is run at all of them; it is not retuned to the selected

point. The weights remain the normalized band costs π_j, not the original band probabilities. Apply Lemma G.1 to the common restart test, with weights π_j:

$$\frac{1+\mathrm{Ent}(\pi)}{214} \le n\sum_j \pi_j d_j^2 w_-^Z(d_j) \le \frac{4n}{A} W_0^Z.$$

Consequently,

$$\rho\Gamma^\circ \le 856\rho n W_0^Z \le 890\,240\, m_Z W_0^Z \le 890\,240\, C_g^4 P.$$

Converting $P = M_0 W_0$ to $V W_0$ completes all branches with the displayed conservative constant. No atomlessness is used. □

The entropy branch is available when $\rho a^2 \Gamma^\circ \le w_*$. This condition is invariant under a common rescaling of all magnitudes.

H Convex comparison, common pairs, and additive localization

This appendix proves the specialized transfer inputs used by the scalar lower bound and the stopping-program comparison. The finite-experiment convex-estimation principle has antecedents in Juditsky–Nemirovski and Polyanskiy–Wu [17, 18]; the midpoint adaptive decomposition is due to Lee–Valiant [1]. The arguments needed for this model are included in full.

H.1 Finite categorical convex comparison

Lemma H.1 (Finite categorical convex comparison)**.** *Let X be a nonempty compact convex subset of a finite-dimensional real vector space. Let $\ell : X \to \mathbb{R}$ be affine, and let $p(x)$ be a strictly positive categorical probability vector with affine continuous coordinates. For $n \geq 1$ independent observations from $p(x)$, define*

$$\Omega_n = \max\{\ell(x) - \ell(y) : x, y \in X, \mathrm{Aff}(p(x), p(y))^n \geq 1/2\}.$$

For every $\eta > 0$, there is a bounded additive estimator $F = c + \sum_{i=1}^n \phi_i(Z_i)$ satisfying

$$\sup_{x \in X} \{\mathbb{E}_x (F - \ell(x))^2\}^{1/2} \leq \tfrac{3}{2}\Omega_n + \eta. \tag{151}$$

Proof. We reproduce the short argument needed for the numerical constant. Set $s = \log 2$, let $\mathcal{F}$ be the finite-dimensional space of additive functions, and define, for $\alpha > 0$,

$$\Phi(x, y; \phi, \alpha) = \ell(x) - \ell(y) + \alpha \log \mathbb{E}_y e^{\phi/\alpha} + \alpha \log \mathbb{E}_x e^{-\phi/\alpha} + 2\alpha s.$$

The function is continuous and concave in (x, y): independence writes each log moment as a sum of logarithms of positive affine forms. It is convex in (ϕ, α) by the perspective transformation of a log moment generating function. Sion's minimax theorem [20, corollary following Theorem 3.4, p. 174] therefore applies to $X \times X$ and $\mathcal{F} \times (0, \infty)$. Only $X \times X$ needs to be compact; the open, unbounded α domain is convex and causes no problem in Sion's one-compact-set form. The infimum in (ϕ, α) need not be attained, and the later argument uses a strict slack $R > S$. Write $2S = \inf_{\phi, \alpha > 0} \max_{x,y} \Phi$.

For a fixed pair, the half log likelihood ratio belongs to $\mathcal{F}$ and Cauchy–Schwarz gives

$$\inf_{\psi \in \mathcal{F}} \left[\log \mathbb{E}_y e^{\psi} + \log \mathbb{E}_x e^{-\psi}\right] = 2 \log \mathrm{Aff}(p(x)^{\otimes n}, p(y)^{\otimes n}).$$

Substituting $\phi = \alpha\psi$ and minimizing over $\alpha > 0$ gives the value $\ell(x) - \ell(y)$ when the affinity is at least $1/2$, and $-\infty$ otherwise. Consequently $2S = \Omega_n$. The constrained maximum exists because its feasible set is nonempty and compact.

For $R > S$, choose (ϕ, α) with $\max \Phi \leq 2R$. On the diagonal, Cauchy–Schwarz implies $\Phi(x, x; \phi, \alpha) \geq 2\alpha s$, hence $\alpha \leq R/s$. Set

$$U = \max_x \{\ell(x) + \alpha \log \mathbb{E}_x e^{-\phi/\alpha}\}, \qquad V = \max_x \{-\ell(x) + \alpha \log \mathbb{E}_x e^{\phi/\alpha}\}.$$

Then $U + V + 2\alpha s \leq 2R$. The estimator $F = \phi + (U - V)/2$ satisfies

$$\mathbb{E}_x e^{\pm(F - \ell(x))/\alpha} \leq e^{R/\alpha - s}.$$

Therefore $\mathbb{P}_x\{|F - \ell(x)| > R + t\} \leq e^{-t/\alpha}$ for $t \geq 0$. Tail integration yields

$$\mathbb{E}_x (F - \ell(x))^2 \leq R^2 + 2R\alpha + 2\alpha^2 \leq R^2 \left(1 + \frac{2}{\log 2} + \frac{2}{(\log 2)^2}\right) < 9R^2.$$

Let $R \downarrow S$, with the slack chosen for η. Every component is bounded because its alphabet is finite. This proves (151). □

H.2 Midpoint decomposition for interleaved observations

We use $\mathfrak{h}(P,Q) = 1 - \mathrm{Aff}(P,Q)$, so $0 \le \mathfrak{h} \le 1$ and $\mathrm{TV}(P,Q) \le \sqrt{2\mathfrak{h}(P,Q)}$. The following finite-prefix argument is the midpoint decomposition of Lee–Valiant [1, Lemmas 17–18], written in the form needed here.

Lemma H.2 (Midpoint work decomposition)**.** *Fix two endpoint priors q_-, q_+ and their midpoint $\bar{q} = (q_- + q_+)/2$. Suppose every capped randomized one-unit policy A obeys*

$$\mathfrak{h}(P_-^A, P_+^A) \le \kappa\, \mathbb{E}_{\bar{q}} T_A.$$

Then every population algorithm with a deterministic total cap satisfies

$$\mathfrak{h}(P_-^{\mathrm{pop}}, P_+^{\mathrm{pop}}) \le \kappa\, \mathbb{E}_{\bar{q}} T_{\mathrm{pop}}.$$

It suffices that the local hypothesis hold up to that same total cap. If instead $q_\pm \le 1/2$ and the local bound uses null work $\lambda \mathbb{E}_0 T_A$, the conclusion holds with $\kappa = 2\lambda$. The statement also holds for finitely many observable component types, each with its own endpoint pair and midpoint, when the same coefficient bounds every local policy for every type. Types may be selected adaptively, but each fresh unit's type is fixed before its first observation and retained on every revisit.

Proof. Fix total cap K and condition on the algorithm's independent seed. Defer latent draws until a unit is first observed, which changes no observable law. Index units by that first observation. A consistent global leaf z contains local binary words $h_i(z)$, with empty words for unused slots. The endpoint leaf densities factor as

$$p_\pm^{\mathrm{pop}}(z) = \prod_{i=1}^{K} p_\pm(h_i(z)).$$

The common policy-consistency indicator is implicit. Put $m_i = (p_-(h_i) + p_+(h_i))/2 = p_{\bar{q}}(h_i)$ and $c_i = \sqrt{p_-(h_i) p_+(h_i)}$. Then

$$\prod_i m_i - \prod_i c_i \le \sum_i (m_i - c_i) \prod_{j \ne i} m_j.$$

The integral of $\prod_i m_i$ over consistent leaves is one, since it is the population law in which every fresh unit has midpoint prior. Integrating the inequality therefore bounds population squared Hellinger by a sum of hybrid squared Hellinger distances. In hybrid i only unit i has an endpoint prior; all other units have midpoint priors.

Simulate all other units from independent midpoint tapes included in an auxiliary seed, and run the population rule with one genuine unit in slot i. This is a capped local policy. Its full transcript, including that seed, determines the hybrid global transcript. Data processing bounds hybrid distance by local distance. The expected midpoint local works sum to $\mathbb{E}_{\bar{q}} T_{\mathrm{pop}}$ because each purchased observation belongs to exactly one slot. This proves the first assertion, including after integrating the original seed. The charged expectation is under the midpoint population, admissible by convexity. It is not replaced by the average endpoint work, which can be much smaller under an adaptive stopping rule.

If $q_\pm \le 1/2$, the midpoint latent law dominates half the null latent law. The same measure inequality holds after passing through any observation policy and auxiliary seed. Thus $\mathbb{E}_0 T_A \le 2\mathbb{E}_{\bar{q}} T_A$.

For observable types, the type selected for a slot is a function of the history before its first observation. Use the resulting type-specific kernel in each leaf product. To implement

hybrid slot i, first generate all observations before that slot under the componentwise midpoint law. Conditional on this prehistory and its selected type, the genuine unit has that type's prescribed endpoint law and is fresh, independently of the remaining auxiliary randomness. This is conditional independence: the latent law may depend on the type contained in the seed.

Retain the already generated tapes of other units on every revisit. For later slots, an independent array of potential tapes indexed by slot and type can be placed in the auxiliary seed; choose the appropriate unused tape only after that slot's type has been selected. The seed's distribution is the same under the two endpoint laws for the genuine unit. Conditional on the prehistory, the local hypothesis for its selected type applies. If slot i is never reached, the genuine local work is zero. Summing and integrating counts every purchased observation once under the actual componentwise midpoint population. No latent magnitude or sign of the genuine unit is revealed to the local policy. □

For an unbounded stopping rule, apply the conclusion to every bounded global prefix. Total variation of the full transcript is the increasing limit of the prefix total variations: bounded cylinder functions generate the transcript sigma-field, or equivalently one applies conditional expectation to the likelihood densities under the dominating sum law. Finite expected endpoint work ensures almost-sure finite termination. A common pair valid for every cap therefore gives the same terminal testing bound. Equivalently, couple the stopped experiment with its cap-L truncation under each endpoint. Their total coupling error is at most $\mathbb{P}_-(T > L) + \mathbb{P}_+(T > L)$, which tends to zero. This passage requires no common deterministic bound on stopping.

For unbounded stopping in the budgeted applications, the same endpoint pair works at every cap, and the original uniform budget gives expected work at most B at both endpoints and at their admissible midpoint. Let N be original purchased work, and cap after L observations, recording an overflow flag if needed. Any terminal event restricted to $\{N \le L\}$ is measurable from this capped transcript. Therefore

$$\mathrm{TV}(P_-, P_+) \le \mathrm{TV}(P_-^{(L)}, P_+^{(L)}) + \mathbb{P}_-(N > L) + \mathbb{P}_+(N > L) \le \sqrt{2\kappa B} + 2B/L.$$

Letting $L \to \infty$ gives the terminal testing bound. For an oracle simulation, cap at the original observation count; its genuine work is no larger, and the same argument applies. This avoids exchanging a cap limit with a separately chosen pair.

H.3 The half-prior class and null work

Put

$$X_\rho = \left\{ q : 0 \le q \le \tfrac{1}{2}, \quad \theta(q) := \int q \, d\mu \le \rho \right\}, \qquad 0 < \rho \le \tfrac{1}{4}.$$

For a capped observable one-coin procedure with bounded real output Y, write

$$V = \sup_{q \in X_\rho} \mathrm{Var}_q \, Y, \qquad M = \sup_{q \in X_\rho} \mathbb{E}_q Y^2, \qquad W_0 = \mathbb{E}_0 T,$$

and define

$$\mathfrak{J}_0(b) = \inf \left\{ VW_0 : \sup_{q \in X_\rho} |\mathbb{E}_q Y - \theta(q)| \le b, \ T \text{ capped}, \ Y \text{ bounded} \right\}. \tag{152}$$

An empty infimum is infinity. The function is nonincreasing in b, and $\mathfrak{J}_0(\rho/2) = 0$.

H.4 Transfer at an arbitrary bias tolerance

Theorem H.3 (Bias-indexed common pairs and population lower bounds)**.** *Suppose that every capped bounded one-coin output with uniform bias at most $b > 0$ over X_ρ satisfies $VW_0 \geq J$, where $0 < J < \infty$. Then, for every $0 < \delta \leq b/3$, there is one pair $q_-, q_+ \in X_\rho$ with target separation at least δ such that simultaneously for all capped randomized local policies A,*

$$\mathfrak{h}(P^A_{q_-}, P^A_{q_+}) \leq 42\frac{\delta^2}{J}\mathbb{E}_0 T_A. \tag{153}$$

Consequently,

$$R_{\mu,\rho}(B) \geq \frac{1}{128}\min\{b, \sqrt{J/B}\}. \tag{154}$$

The conclusion includes fully adaptive creation, revisiting, interleaving, nonlinear outputs and unbounded stopping.

Proof. The premise forces $b < \rho/2$, since the zero-work constant $\rho/2$ would otherwise contradict it. First fix a finite strictly positive categorical statistic of a capped policy of null work $w > 0$. For n independent runs, let $F = c + \sum_{i=1}^n Y_i$ be a bounded additive estimator of worst RMS r. Put $b_i = \mathbb{E}_0 Y_i$, $b_F = \mathbb{E}_0 F$, and choose an independent uniform index I. Running just its component and returning $Z = n(Y_I - b_I) + b_F$ preserves all population means at null work w.

The laws under q and zero share a subprobability of mass $1 - \theta(q)$, obtained by weighting the negative-sign kernel by $1 - q(d)$. The elementary common-mixture variance inequality gives

$$(\mathbb{E}_q Y_i - b_i)^2 \leq \frac{2\theta(q)}{1-\theta(q)}(\operatorname{Var}_q Y_i + \operatorname{Var}_0 Y_i).$$

Indeed, either mean's displacement from the normalized common part is at most the square root of $\theta/(1-\theta)$ times its standard deviation; square the sum of the two displacements. Independence in F now yields

$$\sup_q \operatorname{Var}_q Z \leq nr^2\left(1 + \frac{4\rho}{1-\rho}\right) \leq \tfrac{7}{3}nr^2.$$

If $r < b$, apply the assumed obstruction to Z; if $r \geq b$, retain that inequality. In both cases,

$$r \geq \min\{b, \sqrt{3J/(7nw)}\}.$$

Lemma H.1 gives an additive estimator of RMS at most $(3/2)\Omega_n + \eta$, where Ω_n is the maximum target separation whose product affinity is at least $1/2$. Letting $\eta \downarrow 0$ therefore gives

$$\Omega_n \geq \tfrac{2}{3}\min\{b, \sqrt{3J/(7nw)}\}.$$

Set $x = J/(21\delta^2 w)$. If $x \geq 2$, take $n = \lfloor x \rfloor$; then $\Omega_n \geq 2\delta$, and the corresponding pair satisfies

$$\mathfrak{h} \leq \frac{\log 2}{n} \leq 42\log 2\,\frac{\delta^2 w}{J} \leq 42\frac{\delta^2 w}{J}.$$

If $x < 2$, the last right side exceeds one, so any separated pair suffices. If $w = 0$, the seed-only first-observation event has probability zero; all local laws agree and $q_- = 0, q_+ = \delta$ suffices.

We now obtain one pair simultaneously for every capped local policy. For an arbitrary capped policy, partition its full transcript into finitely many measurable cells, then replace the cell independently by a uniform cell with probability $0 < \varepsilon < 1$. Run the policy even on replacement outcomes, so its null work is unchanged. The resulting strictly positive category

probabilities and the target have compact convex finite-dimensional range, so the preceding comparison applies. For any finite collection of these experiments, independently mix them and reveal the chosen index. Categories with zero mixture weight can be dropped, leaving a strictly positive finite alphabet. For a fixed one-unit policy, its categorical law is affine in q: condition on the one genuine latent state first. This affineness is not being asserted for an entire adaptive population transcript.

Set $k = 42\delta^2/J$ and use the oriented, weak-star compact convex pair set

$$K_\delta = \{(q_-, q_+) \in X_\rho^2 : \theta(q_+) - \theta(q_-) \geq \delta\}.$$

For experiment i, let w_i be its null work and $f_i(q_-, q_+) = \mathfrak{h}(P_{q_-}^{A_i}, P_{q_+}^{A_i}) - kw_i$. Each f_i is continuous and convex on K_δ. For any mixture weights λ, the revealed index gives exactly $\mathfrak{h}_{\mathrm{mix}} = \sum_i \lambda_i \mathfrak{h}_i$ and $w_{\mathrm{mix}} = \sum_i \lambda_i w_i$. The preceding one-experiment argument therefore implies

$$\min_{z \in K_\delta} \sum_i \lambda_i f_i(z) \leq 0 \qquad \text{for every } \lambda \text{ in the finite simplex.}$$

Applying convex minimax in its indicated direction gives

$$\min_{z \in K_\delta} \max_i f_i(z) = \max_\lambda \min_{z \in K_\delta} \sum_i \lambda_i f_i(z) \leq 0.$$

The left minimum is attained. Thus the finite family has a single feasible pair; separate pairs for the individual experiments would not suffice. All its constraint sets are closed in K_δ. The finite intersection property supplies one pair for all experiments. For fixed b and δ, this pair is chosen before the adaptive population algorithm and its total-observation cap. The same pair works for every capped local policy and every cap. The intersection ranges over all caps, finite partitions and positive noise levels at once, so this pair does not depend on the stopping cap. The pair is fixed while $\varepsilon \downarrow 0$; finite categorical Hellinger distance is continuous even when some limiting cells have zero mass. Full squared Hellinger distance is the supremum over finite measurable partitions. For completeness, put $R = P + Q$ and $t = dP/dR$. Partition the range of t into intervals of width at most $1/m$. Since $g(t) = \sqrt{t(1-t)}$ satisfies $|g(u) - g(v)| \leq \sqrt{|u - v|}$, the full and coarsened affinities differ by at most $2/\sqrt{m}$. Thus the supremum statement holds without a countable-generation assumption on the transcript space. This proves (153) for arbitrary transcript spaces. The quantifier is one pair for every policy and cap at fixed $(\mu, \rho, b, J, \delta)$. The pair may change when δ, and hence the chosen budget-dependent separation, changes.

The midpoint decomposition in Lemma H.2 bounds population squared Hellinger by $84\delta^2 B/J$: the local null cost is at most twice the midpoint cost, and summing midpoint costs counts each observation once. Set

$$\delta = \frac{1}{32} \min\{b, \sqrt{J/B}\}.$$

The distance is at most $84/1024 < 1/8$, hence TV is at most $1/2$. The same pair works for every deterministic observation cap. Finite expected endpoint work and increasing capped prefixes give the same terminal TV bound for unbounded stopping. Thresholding an estimator at the target midpoint gives RMS at least $\delta/4$, proving (154). A budget over the full rarity class implies the required budget over X_ρ. □

H.5 A direct variational null-work obstruction

For Bernoulli observations, recall the variational modulus of Section 11

$$\Psi_{\mu,\rho}(h) = \sup_{\substack{0 \leq w \leq 1 \\ \int w \, d\mu \leq \rho}} \int \min\{w(d), h\sqrt{w(d)}/d\} \, \mu(dd). \tag{155}$$

The proof below establishes the exact null-work input required here directly from fixed-sign comparisons. No optimizer formula is needed.

Proposition H.4 (Variational accuracy-dependent null obstruction)**.** *For every probability measure μ on $(0, 1/2]$ and every $0 \le b \le \rho/64$,*

$$\mathfrak{J}_0(b) \ge \sup_{h>0} \frac{(\Psi_{\mu,\rho}(h) - 4b)_+^2}{192h^2}. \tag{156}$$

No nonatomicity, finite support, or separation of magnitudes is assumed.

Proof. Fix an eligible (Y, T), and define its conditional response, moments, and negative work by

$$f(d) = \mathbb{E}_{d,+}Y - \mathbb{E}_{d,-}Y, \qquad A_s(d) = \mathbb{E}_{d,s}Y^2, \qquad w_-(d) = \mathbb{E}_{d,-}T.$$

We first show

$$f(d)^2 \le 64d^2 w_-(d)[A_-(d) + A_+(d)]. \tag{157}$$

For $d \le 1/4$, the stopped KL chain rule gives $\mathrm{KL}(P^A_{d,-}, P^A_{d,+}) \le (64/3)d^2 w_-(d)$. For any square-integrable output under two laws,

$$|\mathbb{E}_P Y - \mathbb{E}_Q Y|^2 \le 4(\mathbb{E}_P Y^2 + \mathbb{E}_Q Y^2)\mathfrak{h}(P, Q), \qquad \mathfrak{h}(P, Q) \le \tfrac{1}{2}\,\mathrm{KL}(P, Q).$$

These prove (157) with coefficient $128/3 < 64$. For $d > 1/4$, the conditional laws agree whenever the seed requests no observation. Their Hellinger distance is therefore at most the activation probability, hence at most $w_-(d)$. The same mean inequality gives $f(d)^2 \le 4w_-(d)(A_- + A_+) \le 64d^2 w_-(d)(A_- + A_+)$. This also handles $d = 1/2$ without a singular endpoint KL comparison.

For every feasible w in (155), the allocation $q = w/2$ belongs to X_ρ, and

$$\int w(A_- + A_+)\, d\mu \le 2\int [(1 - w/2)A_- + (w/2)A_+]\, d\mu \le 2M.$$

Weighted Cauchy–Schwarz and (157) yield

$$\left(\int \frac{\sqrt{w}\,|f|}{d}\, d\mu\right)^2 \le W_0 \int \frac{wf^2}{d^2 w_-}\, d\mu \le 128 M W_0. \tag{158}$$

On $\{w_- = 0\}$, the response inequality forces $f = 0$, and the quotient is zero. Truncation justifies Cauchy–Schwarz before passing to the limit if an integral is initially extended-valued.

Put $v_h = \min\{w, h\sqrt{w}/d\}$. Comparing the mean under the admissible allocation $q = v_h/2$ with the null mean, and using both bias bounds, gives

$$\int v_h f\, d\mu \ge \int v_h\, d\mu - 4b.$$

Since $v_h \le h\sqrt{w}/d$, (158) implies

$$MW_0 \ge \frac{(\int v_h\, d\mu - 4b)_+^2}{128h^2}.$$

Take the supremum over w. To convert the second moment to variance, the null and constant-prior populations force the all-positive and all-negative means to differ by at least $31/32$. The mixture with constant sign prior ρ consequently has variance at least $\rho(1-\rho)(31/32)^2 \ge 2\rho/3$. Also every admissible mean has absolute value at most $65\rho/64$. Thus $M \le V + (65\rho/64)^2 \le 3V/2$, proving (156). □

This product bound accounts for accuracy-dependent tail difficulty in the same framework as the annular scale-entropy obstruction. At a fixed gap d, the directly evaluated identity $\Psi(h) = \min\{\rho, h\sqrt{\rho}/d\}$ recovers product scale ρ/d^2 as $b \downarrow 0$. For the quadratic law it supplies the inverse-square logarithm; the additional scale-entropy factor still requires the annular argument.

The next proposition restates the optimizer of Section 11, whose value formula is Lean-checked there; the short proof is retained because the argument above uses its form.

Proposition H.5 (An optimizer for the variational modulus)**.** *For $0 < \rho < 1$, there is a unique $\alpha > 0$ satisfying*

$$\int \min\{1, \alpha^2/d^2\}\, \mu(dd) = \rho.$$

The allocation $w_*(d) = \min\{1, \alpha^2/d^2\}$ maximizes $\Psi_{\mu,\rho}(h)$ simultaneously for every $h > 0$. In particular, $\Psi(h) = \rho$ when $h \geq \alpha$. Fractional allocation at an atom is permitted and may be essential; this is distinct from the hard assignment statement for the entropy-selection infimum.

Proof. The mass on the left is continuous in α, tends to zero at zero and to one at infinity, and is strictly increasing while it is below one. These assertions follow by dominated convergence and the positive mass of unsaturated magnitudes. Thus the stated root exists uniquely. If $h \geq \alpha$, the allocation attains the upper bound ρ. If $h < \alpha$, put $\lambda = h/(2\alpha) < 1/2$. The function

$$\phi_d(w) = \min\{w, (h/d)\sqrt{w}\}, \qquad 0 \leq w \leq 1,$$

is concave. For $d > \alpha$, the point $w_* = (\alpha/d)^2$ is in its square-root branch and has derivative λ. For $h < d \leq \alpha$, the left derivative at $w_* = 1$ is $h/(2d) \geq \lambda$; for $d \leq h$, $\phi_d(w) = w$. Hence w_* maximizes $\phi_d(w) - \lambda w$ pointwise. Integrating and using its mass ρ proves optimality over all allocations of mass at most ρ, including atomic laws. □

Remark H.6 (The bias subtraction cannot be uniformly reduced)**.** Let $\rho = 1/8$, $\varepsilon = 2^{-10}$, and $\mu = (1-\varepsilon)\delta_{1/2} + \varepsilon\delta_d$. The one-flip output $Y = X - b$, with $b = \varepsilon(1-2d)/4$, has bias at most b on X_ρ, null work one, and variance at most $1/4$. Indeed its bias is $\varepsilon(1/2-d)(1-2q(d)) - b \in [-b, b]$. Yet $\Psi(d) \geq \varepsilon$. Replacing $4b$ by cb with any fixed $c < 4$ in (156) would make its right-hand side diverge as $d \downarrow 0$. The coefficient four cannot be reduced in this form of the bound, even if its denominator constant is changed.

H.6 A direct sum of conditional experiments

Let $\sigma_1, \ldots, \sigma_m$ be nonzero finite submeasures with $\sum_i \sigma_i \leq \mu$. Write $\tau_i = \sigma_i((0, 1/2])$ and $\nu_i = \sigma_i/\tau_i$. Disjoint restrictions are a special case; fractional restrictions of atoms are also allowed. Choose

$$0 < r_i \leq 1/4, \qquad \sum_i \tau_i r_i \leq \rho.$$

Within component i, use the half-prior subclass

$$X_{r_i}(\nu_i) = \{0 \leq q \leq 1/2 : \ \textstyle\int q\, d\nu_i \leq r_i\}.$$

The original population lower bound is valid on this subclass of the full rarity class.

Theorem H.7 (Additive localization)**.** *Suppose that, for each i, every capped bounded observable one-unit output with uniform bias at most $b_i > 0$ over $X_{r_i}(\nu_i)$ satisfies*

$$\left(\sup_q \operatorname{Var}_q Y\right) \mathbb{E}_{i,0} T \geq J_i > 0.$$

Then, for every $B > 0$,

$$\boxed{R_{\mu,\rho}(B) \geq \frac{1}{128} \sum_{i=1}^{m} \tau_i \min\{b_i, \sqrt{J_i/B}\}.} \tag{159}$$

The same total budget B occurs in every summand. The component label is chosen before the first observation and retained on every revisit; its conditional magnitude and sign remain hidden.

Proof. Theorem H.3 applies within every component. Set

$$\delta_i = \frac{1}{32} \min\{b_i, \sqrt{J_i/B}\}.$$

It supplies a pair $q_{i,-}, q_{i,+} \in X_{r_i}(\nu_i)$, oriented to have conditional target separation at least δ_i, such that simultaneously for every capped local policy A,

$$\mathfrak{h}(P^A_{i,-}, P^A_{i,+}) \leq 42 \frac{\delta_i^2}{J_i} \mathbb{E}_{i,0} T_A \leq \frac{42}{1024B} \mathbb{E}_{i,0} T_A. \tag{160}$$

Fix an arbitrary original algorithm with uniform expected budget B. Consider an oracle experiment that can select a component at no cost and then receive a genuine coin with magnitude law ν_i. Its component label is visible, while its magnitude and sign remain hidden. The algorithm can adaptively select, revisit, and interleave these coins. Apply the argument to the simulation of the fixed original algorithm described below. At the componentwise midpoint it induces the original parameter $\bar{q} = (q_- + q_+)/2 \in \mathcal{Q}_\rho$, so its genuine midpoint work is at most the original work and hence at most B. Under the conditional midpoint prior, $q \leq 1/2$, so the midpoint law dominates half the null law. Lemma H.2 therefore bounds terminal squared Hellinger by $84/1024 < 1/8$: the local coefficient in (160) is common to all components, and the summed conditional midpoint costs count each purchased observation once. Increasing capped prefixes handles unbounded stopping. The total target separation is at least $\sum_i \tau_i \delta_i$.

Every original algorithm can be simulated in this stronger experiment. Privately choose the component label with probabilities τ_i whenever it creates a fresh coin. On the remaining probability mass, simulate the known all-negative residual law for free. Keep the chosen label when the coin is revisited. For fractional submeasures, integrating this private mixture label gives the original admissible parameters

$$q_\pm(d) = \sum_i \frac{d\sigma_i}{d\mu}(d) q_{i,\pm}(d), \qquad 0 \leq q_\pm \leq 1/2, \qquad \int q_\pm \, d\mu \leq \sum_i \tau_i r_i \leq \rho.$$

The number of genuine oracle observations does not exceed the original number. Thus the oracle lower bound applies to the original experiment. The Hellinger bound gives TV at most $1/2$, and midpoint testing gives RMS at least one quarter of the target separation. This is (159). □

For a direct explicit input, suppose component i is supported in $(0, a_i]$, $a_i \leq 1/4$, and select a finite separated family of small-gap annuli. Let Γ_i° and $w_{*,i}$ denote its conditional cost entropy and minimum band mass. Proposition G.4 gives

$$b_i = r_i/64, \qquad J_i = c_{\mathrm{ann}} \min\{r_i \Gamma_i^\circ, w_{*,i}/a_i^2\}, \qquad c_{\mathrm{ann}} = \frac{e^{-28}}{2{,}000{,}000}.$$

Write the corresponding unnormalized quantities with respect to σ_i as $\Gamma^\circ_{\sigma_i} = \tau_i \Gamma_i^\circ$ and $w_{*,\sigma_i} = \tau_i w_{*,i}$. The conversion uses the exact identity, with $\theta = \tau r$,

$$\tau^2 \min\{r\Gamma^\circ, w_*/a^2\} = \min\{\theta \Gamma^\circ_\sigma, \tau w_{*,\sigma}/a^2\}.$$

In particular the second branch retains the factor τ. Consequently,

$$R_{\mu,\rho}(B) \gtrsim \sum_i \min\left\{\tau_i r_i, \sqrt{\frac{\min\{\tau_i r_i \Gamma^\circ_{\sigma_i}, \tau_i w_{*,\sigma_i}/a_i^2\}}{B}}\right\}. \tag{161}$$

For clarity, the squared component mass is essential before cancellation:

$$\tau_i^2 J_i = c_{\mathrm{ann}} \min\{\tau_i r_i \Gamma^\circ_{\sigma_i}, \tau_i w_{*,\sigma_i}/a_i^2\}.$$

This explains the extra τ_i in the saturation term of (161); it must not be dropped when passing between conditional and unnormalized quantities. Supremizing this expression over finite decompositions and rarity allocations gives an explicit lower functional. The bias-indexed form (159) also permits each component's accuracy scale to be chosen separately. Appendix E applies this expression to components selected by the entropy–tail optimizer, and Appendix O.1 applies it to the components constructed by the quantile-core filter and packing.

H.7 A universal risk floor

Lemma H.8 (Bernoulli-label floor)**.** *For every magnitude law,* $0 < \rho \le 1/4$*, and* $B > 0$*,*

$$R_{\mu,\rho}(B) \ge \frac{1}{16} \min\{\rho, \sqrt{\rho/B}\}.$$

Proof. Restrict to constant sign allocations. Give an oracle the exact sign of a unit when its first observation is requested, for one charged observation. It can sample $D \sim \mu$ independently and simulate all future flips for free, so every original algorithm can be simulated with no greater work. The informative labels are iid Bernoulli with the unknown constant prior.

Take $p_0 = \rho/2$ and $p_1 = p_0 + \delta$, where $\delta = \min\{\rho/2, \sqrt{\rho/B}/4\}$. Both are admissible and $p_1(1-p_1) \ge 3\rho/8$. The stopped KL chain rule, first at bounded prefixes and then by monotone convergence, gives

$$\mathrm{KL}(P_0, P_1) \le B\ \mathrm{kl}(p_0, p_1) \le \frac{8B\delta^2}{3\rho} \le \frac{1}{6}.$$

Thus $\mathrm{TV}(P_0, P_1) \le 1/2$. Thresholding any estimator at the target midpoint implies worst RMS at least $\delta/4$. Finally $\delta/4 = \min\{\rho/8, \sqrt{\rho/B}/16\} \ge \min\{\rho, \sqrt{\rho/B}\}/16$. □

I Representation and realization of the stopping flow

This section justifies the normalized flow cone and robust program in (168)–(169). Triangular encodings of stopped coin experiments have direct antecedents in Lee–Valiant [1]; occupation-measure formulations of optimal stopping are also classical [28]. The proofs below give the representation and estimator realization needed for this particular robust program.

I.1 The support function of the rarity class

For bounded measurable $f : (0, 1/2] \to \mathbb{R}$, put

$$Q_{\mu,\rho}(f) = \sup_{q \in \mathcal{Q}_\rho} \int q(d) f(d)\, \mu(dd).$$

Lemma I.1 (A fractional upper-tail formula)**.** *The support function satisfies the attained identity*

$$Q_{\mu,\rho}(f) = \min_{\ell \geq 0} \left\{ \rho\ell + \int (f - \ell)_+ \, d\mu \right\}. \tag{162}$$

The minimum can be restricted to $[0, \|f_+\|_\infty]$. The supremum on the left is attained by selecting the largest positive values of f, allowing a fractional selection on the cutoff level set.

Proof. For every $0 \leq q \leq 1$ and $\ell \geq 0$, $qf \leq q\ell + (f - \ell)_+$. Integration and $\int q \, d\mu \leq \rho$ prove the upper bound by the right side.

If $\mu(f > 0) \leq \rho$, take $q = \mathbf{1}_{\{f>0\}}$ and $\ell = 0$. Otherwise choose a positive cutoff ℓ satisfying

$$\mu(f > \ell) \leq \rho \leq \mu(f \geq \ell).$$

Such a cutoff exists by the monotonicity and one-sided continuity of the tail distribution. Take $q = 1$ on $\{f > \ell\}$, $q = 0$ on $\{f < \ell\}$, and choose a constant fraction in $[0, 1]$ on $\{f = \ell\}$ so that its total mass is ρ. The resulting value is

$$\int_{\{f>\ell\}} f \, d\mu + \ell[\rho - \mu(f > \ell)] = \rho\ell + \int (f - \ell)_+ \, d\mu.$$

This proves equality and attainment, including atoms of μ or atoms in the distribution of f. Cutoffs larger than $\|f_+\|_\infty$ cannot improve on the endpoint. □

For an array $u = (u_{nk})$, define

$$L_u^{\pm}(d) = \sum_{n,k} u_{nk} \binom{n}{k} (1/2 \pm d)^k (1/2 \mp d)^{n-k}, \qquad I_u^- = \int L_u^- \, d\mu, \qquad F_u = L_u^+ - L_u^-.$$

The root or final-depth terms are included only when the corresponding array permits them. The convention $0^0 = 1$ covers the deterministic Bernoulli endpoints. By the definition of $m(q)$,

$$\sup_{q \in \mathcal{Q}_\rho} \langle m(q), u \rangle = I_u^- + Q_{\mu,\rho}(F_u). \tag{163}$$

In particular, the two bias suprema in (169) are

$$c_0 + I_z^- + Q_{\mu,\rho}(F_z - 1), \qquad -c_0 - I_z^- + Q_{\mu,\rho}(1 - F_z).$$

The moment and work suprema use (163) with $u = t$ and $u = c$. Thus the robust program requires only integration against the known magnitude law and a scalar cutoff optimization.

I.2 Arbitrary seeds and ordered histories aggregate to the cone

Proposition I.2 (Normalized prefix aggregation)**.** *Every capped one-unit policy with* $1 \le T \le N$*, arbitrary independent auxiliary randomization, and bounded terminal output has normalized arrays in the cone* (168) *with root intensity one. Its mean, raw second moment, and expected work under any* q *are respectively*

$$\langle m(q), z\rangle, \qquad \langle m(q), t\rangle, \qquad \langle m(q), c\rangle.$$

Multiplying the four arrays by $\lambda \ge 0$ *gives the corresponding intensity-*λ *representation. The stopping representation alone requires no output or bounded-output assumption.*

Proof. Place all auxiliary randomization in one seed r, drawn from a law ν independent of the latent unit and its observations. Once the seed is fixed, all decisions and outputs are deterministic functions of the ordered observed prefix.

For each ordered binary prefix x of length n, let A_x be the set of seeds for which the policy reaches x and stops there; let C_x be the set for which it reaches x and continues. These sets are disjoint. For $n \ge 1$ their union is exactly C_{x^-}, where x^- is the prefix with its last symbol removed. Write

$$\alpha_x = \nu(A_x), \quad \gamma_x = \nu(C_x), \quad \zeta_x = \int_{A_x} Y(r,x)\,\nu(dr), \quad \tau_x = \int_{A_x} Y(r,x)^2\,\nu(dr).$$

At the root, $\gamma_\varnothing = 1$ and $\alpha_\varnothing = 0$; at depth N, $\gamma_x = 0$.

For $s = (n,k)$, sum these quantities over all prefixes with k heads and divide by $\binom{n}{k}$:

$$a_{nk} = \frac{\sum_{|x|=n,\,\#1(x)=k} \alpha_x}{\binom{n}{k}}, \qquad c_{nk} = \frac{\sum_{|x|=n,\,\#1(x)=k} \gamma_x}{\binom{n}{k}},$$

and analogously define z_{nk} and t_{nk} from ζ_x, τ_x. Aggregating $\alpha_x + \gamma_x = \gamma_{x^-}$ gives

$$a_{nk} + c_{nk} = \frac{n-k}{n} c_{n-1,k} + \frac{k}{n} c_{n-1,k-1}.$$

Cauchy–Schwarz on the disjoint union of the prefix-indexed stopping-seed sets gives $z_{nk}^2 \le a_{nk} t_{nk}$. Hence the arrays belong to the stated cone.

Conditional on Bernoulli mean p, every ordered prefix in this count state has the same probability $p^k(1-p)^{n-k}$. Summing its output integrals therefore gives the stated moment identities after mixing over the latent state. Expected work is the sum of the probabilities of continuing at all prefixes, including the root: each continuation requests exactly one charged observation. This proves the work identity and accounts for the first observation.

Multiplication by λ preserves the linear flow identities and the homogeneous perspective inequalities. The argument also applies to outputs with finite first and second seed aggregates. For the bounded outputs used in finite constructions, these are automatic. □

The normalized arrays encode probabilities over the independent seed for fixed observation prefixes. They are not the unconditional probabilities of reaching a count state. This distinction is the reason the Pascal coefficients in (168) are $(n-k)/n$ and k/n.

I.3 Every feasible point is an observable estimator

Proposition I.3 (Compound-Poisson realization)**.** *Every feasible point of* (169) *constructs an observable population estimator with uniform expected work at most* B *and worst RMS error at most* $\sqrt{b^2 + v}$*. In particular,*

$$R_{\mu,\rho}(B) \le P_N(B).$$

Proof. Suppose first that $\lambda > 0$. Draw a Poisson number of independent genuine units with mean λ. Each unit first requests one observation. At an observed state (n, k), stop with probability $a_{nk}/(a_{nk} + c_{nk})$ and continue otherwise. At a stopping state with $a_{nk} > 0$, return z_{nk}/a_{nk}. Unreachable states may have arbitrary actions, and outputs at zero stopping flow can be set to zero. The constraint $c_{Nk} = 0$ enforces the cap.

For a fixed Bernoulli mean p, the expected number of units reaching (n, k) is

$$(a_{nk} + c_{nk})\binom{n}{k}p^k(1-p)^{n-k}.$$

This follows by induction from the normalized Pascal recursion. For example, a continuation at $(n-1, k)$ followed by a zero contributes $c_{n-1,k}\binom{n-1}{k}p^k(1-p)^{n-k}$, and the other parent contributes the analogous term with $k-1$. Their sum is exactly the displayed expression. Multiplying by the stopping or continuation probability gives the same formula with a_{nk} or c_{nk} alone.

Return c_0 plus the sum of all terminal unit outputs. Independence of the genuine units and the compound-Poisson moment identities give

$$\mathbb{E}_q\widehat{\theta} = c_0 + \langle m(q), z\rangle, \tag{164}$$

$$\mathrm{Var}_q(\widehat{\theta}) = \left\langle m(q), \frac{z^2}{a}\right\rangle \le \langle m(q), t\rangle, \tag{165}$$

$$\mathbb{E}_q T = \langle m(q), c\rangle. \tag{166}$$

Set $z_{nk}^2/a_{nk} = 0$ where $a_{nk} = 0$; the perspective constraint then also gives $z_{nk} = 0$. Every sum is finite, and each nonzero terminal output is a finite real number. The root contribution to (166) is λ, since $m_{00}(q) = 1$. The work constraint therefore charges every first observation and implies $\lambda \le B$.

The bias, moment, and work inequalities in the program prove the claimed uniform guarantees. When $\lambda = 0$, the flow recursion forces $a = c = 0$ and the perspective constraint forces $z = 0$. The deterministic estimator c_0 realizes the required bound; any unused nonnegative moment slack is harmless. □

Allowing zero-observation marks would not improve the attainable Poisson guarantee. Indeed, the event that a one-unit procedure stops before its first observation depends only on its independent seed. Poisson thinning separates those marks from the active units. Replace their total random contribution by its deterministic expectation. This preserves every population mean and the work, while removing their nonnegative variance contribution. The remaining active intensity, together with the adjusted offset c_0, has precisely the root convention in (168).

I.4 Finite-support laws give finite second-order cone programs

Suppose $\mu = \sum_{i=1}^m w_i\delta_{d_i}$ is supplied by its positive masses and distinct positive magnitudes. For each of the four support functions in the two bias, moment, and work constraints, introduce one cutoff $\ell \ge 0$ and nonnegative epigraph variables r_i with

$$r_i \ge f(d_i) - \ell.$$

The corresponding robust term is represented by $\rho\ell + \sum_i w_i r_i$, by Lemma I.1. All $f(d_i)$ are affine functions of the relevant flow or output array. Thus these epigraph constraints are linear.

The perspective constraints are second-order cones:

$$z_{nk}^2 \le a_{nk} t_{nk} \quad \Longleftrightarrow \quad \|(2z_{nk}, a_{nk} - t_{nk})\|_2 \le a_{nk} + t_{nk}, \qquad a_{nk}, t_{nk} \ge 0.$$

One additional cone represents an epigraph variable $u \ge b^2$, for example $\|(2b, u-1)\|_2 \le u+1$, and the objective is $u + v$.

Consequently the finite-support version has $O(N^2 + m)$ scalar variables and cone or linear constraints, with a dense coefficient description of size $O(mN^2)$ after the Bernoulli kernels are evaluated. This is a dimension count at a specified cap, without a uniform claim about the sufficient cap, numerical conditioning, or bit complexity.

J An exact dual for the robust stopping program

Triangular stopping representations and their adaptive comparison have close antecedents in Lee–Valiant [1]. Here the sign allocation ranges over the full convex rarity class, and bias, variance, and work are robust constraints. We derive the exact four-prior dual of this program and absorb its variance and work references into the endpoints. Appendix I proves the stopping-flow representation and its observable realization in full.

Throughout this section, μ is any probability measure on $(0, 1/2]$, $0 < \rho \le 1/4$, $B > 0$, and $N \ge 1$ is an integer. Put

$$\mathcal{Q}_\rho = \{q \in L^\infty(\mu) : 0 \le q \le 1,\ \theta(q) = \textstyle\int q\, d\mu \le \rho\}.$$

For a state $s = (n, k)$, $0 \le k \le n \le N$, define

$$m_{nk}(q) = \int \binom{n}{k} \big[(1 - q(d))(1/2 - d)^k (1/2 + d)^{n-k} + q(d)(1/2 + d)^k (1/2 - d)^{n-k}\big]\, \mu(dd). \tag{167}$$

In particular $m_{00}(q) = 1$. These are probabilities of count states at fixed depth, before imposing a stopping rule.

J.1 The primal and its four-prior dual

For clarity, the finite program being dualized is recalled in compact form. Its nonnegative stopping and continuation arrays a, c, real output array z, and nonnegative moment array t obey

$$\begin{aligned} &a_{00} = z_{00} = t_{00} = 0, \qquad c_{00} = \lambda \ge 0, \qquad c_{Nk} = 0, \\ &a_{nk} + c_{nk} = \frac{n-k}{n} c_{n-1,k} + \frac{k}{n} c_{n-1,k-1}, \qquad 1 \le n \le N, \\ &z_{nk}^2 \le a_{nk} t_{nk}. \end{aligned} \tag{168}$$

Missing indices are zero. These are normalized prefix-weight flows, not unconditional state probabilities. Write $\mathcal{C}_N$ for this closed convex cone. The squared program value is

$$\begin{aligned} P_N(B)^2 = \min\ & b^2 + v \quad \text{over } (a, c, z, t, \lambda) \in \mathcal{C}_N,\ c_0 \in \mathbb{R},\ b, v \ge 0, \\ \text{subject to }\ & \sup_{q \in \mathcal{Q}_\rho} \big|c_0 + \langle m(q), z\rangle - \theta(q)\big| \le b, \\ & \sup_q \langle m(q), t\rangle \le v, \qquad \sup_q \langle m(q), c\rangle \le B. \end{aligned} \tag{169}$$

Stopping sums exclude the root and continuation sums exclude depth N. Every feasible point realizes a Poisson estimator with uniform RMS at most $\sqrt{b^2 + v}$ and expected work at most B (Proposition I.3).

Use the closed perspective

$$\mathfrak{r}(x, y) = \begin{cases} x^2/y, & y > 0, \\ 0, & x = y = 0, \\ +\infty, & y = 0, x \ne 0. \end{cases}$$

Given four priors $q_L, q_H, q_v, q_w \in \mathcal{Q}_\rho$, put $d_s = m_s(q_H) - m_s(q_L)$, $v_s = m_s(q_v)$, $w_s = m_s(q_w)$, and define

$$K_N(q_L, q_H; q_v, q_w) = \sup_{A:\, 1 \le T_A \le N} \frac{\sum_s a_s^A \mathfrak{r}(d_s, v_s)}{\sum_s c_s^A w_s}. \tag{170}$$

Here A is a one-unit policy that always requests its first observation; its normalized flow has root intensity one. A zero stopping coefficient contributes zero even when the perspective is infinite. The denominator is $\mathbb{E}_{q_w} T_A \in [1, N]$.

For a signed measure $P_H - P_L$, write

$$\mathcal{I}(P_H, P_L; P_v) = \int \left(\frac{d(P_H - P_L)}{dP_v} \right)^2 dP_v,$$

with value infinity unless the signed measure is absolutely continuous. The numerator in (170) equals $\mathcal{I}(P_H^A, P_L^A; P_v^A)$. Include the independent seed in the transcript: its likelihood ratio depends only on the terminal count state, so this identity covers order-dependent stopping and randomized policies as well.

Theorem J.1 (Exact four-prior dual)**.** *The minimum in* (169) *is attained, and*

$$\boxed{P_N(B)^2 = \max_{q_L, q_H, q_v, q_w \in \mathcal{Q}_\rho} \frac{[\theta(q_H) - \theta(q_L)]_+^2}{4 + BK_N(q_L, q_H; q_v, q_w)}.} \tag{171}$$

The ratio is zero when $K_N = \infty$. The maximum is attained with positive value, positive target separation, and finite K_N. Swapping the endpoints allows the positive part to be replaced by an absolute value.

Proof. Compactness and constraint qualification. The finite moment image $\{(\theta(q), m(q)) : q \in \mathcal{Q}_\rho\}$ is compact and convex: $\mathcal{Q}_\rho$ is weak-star compact, and every coordinate is a bounded linear functional of q. All robust functions in the program are thus finite continuous convex functions.

The zero-flow output $\rho/2$ gives objective $\rho^2/4$. On this objective sublevel, $b \le \rho/2$, $v \le \rho^2/4$, and the root work charge gives $\lambda \le B$. The normalized recursion implies $0 \le a_s, c_s \le B$. For the constant prior $q_* = \rho/2$, $m_s(q_*) > 0$ unless state s is impossible under every prior. At every nonnull stopping state,

$$t_s \le \frac{\rho^2}{4 m_s(q_*)}, \qquad |z_s| \le \sqrt{B t_s}.$$

At universally null states set $z_s = t_s = 0$, without changing any mean, moment, or cost. The bias constraint at $q = 0$ then bounds c_0. Consequently this feasible sublevel, with null output coordinates removed, is compact. The primal minimum is attained.

There is also a strict relative-interior feasible point. Choose $0 < \lambda < B/4$, continue with probability $1/2$ at each nonterminal observed state, and stop at depth N. All free stopping and continuation flows are positive and expected work is less than $2\lambda < B$. Set $z = 0$, take all free $t_s > 0$, and choose sufficiently large b, v. Every perspective inequality and all four robust inequalities are strict. Finite-dimensional convex Slater duality therefore gives no duality gap, attained inequality multipliers, and KKT conditions at a primal optimum.

Reduction to four priors. Split the absolute bias constraint into its two one-sided inequalities. Each robust function is a maximum of affine functions over a compact convex set. A subgradient is a convex combination of active affine gradients. Since $q \mapsto (\theta(q), m(q))$ is affine and $\mathcal{Q}_\rho$ is convex, this combination is represented by one active prior. Thus the two bias subgradients, the variance subgradient, and the work subgradient may be represented respectively by q_L, q_H, q_v, q_w.

Let their nonnegative multipliers be $\alpha_L, \alpha_H, \beta, \eta$. The free offset forces $\alpha_L = \alpha_H = \alpha$. Minimizing $b^2 - 2\alpha b$ over $b \ge 0$ gives $-\alpha^2$, and minimizing $(1 - \beta)v$ over $v \ge 0$ requires $\beta \le 1$. The remaining expression on the flow cone is

$$\beta \langle m(q_v), t \rangle - \alpha \langle m(q_H) - m(q_L), z \rangle + \eta \langle m(q_w), c \rangle. \tag{172}$$

Homogeneity requires this expression to be nonnegative everywhere on $\mathcal{C}_N$; otherwise scaling a negative point gives infimum minus infinity. When it is nonnegative its infimum is zero. The dual objective is

$$\alpha[\theta(q_H) - \theta(q_L)] - \alpha^2 - \eta B. \tag{173}$$

We may take $\beta = 1$. Indeed, for any $q_0 \in \mathcal{Q}_\rho$, replace $\beta m(q_v)$ by

$$m(\beta q_v + (1-\beta)q_0) = \beta m(q_v) + (1-\beta)m(q_0).$$

The additional coefficient on t is nonnegative, so cone feasibility is preserved while the dual objective is unchanged. This also preserves attainment.

Eliminating outputs and the work price. For $\alpha > 0$, the perspective constraint gives, at fixed stopping flow,

$$\inf_{z_s^2 \le a_s t_s} \{v_s t_s - \alpha d_s z_s\} = -\frac{\alpha^2 a_s}{4}\mathfrak{r}(d_s, v_s).$$

For positive v_s this is completion of the square. If $a_s = 0$ the contribution is zero; if $a_s > 0$, $v_s = 0$, and $d_s \neq 0$, the infimum is minus infinity. The remaining zero case has value zero. Every nonzero stopping/continuation flow scales to root intensity one. At $\lambda = 0$, $a = c = z = 0$ and surplus t has a nonnegative contribution. Thus nonnegativity of (172) is exactly

$$\eta \ge \alpha^2 K_N/4.$$

For finite K_N, optimizing (173) yields

$$\sup_{\alpha \ge 0}\Big\{\alpha\Delta\theta - \alpha^2(1 + BK_N/4)\Big\} = \frac{(\Delta\theta)_+^2}{4 + BK_N}.$$

When $K_N = \infty$, only the zero-value choice $\alpha = \eta = 0$ remains. Weak duality applies to every quadruple, and the attained KKT dual above supplies an optimal quadruple. This proves the identity.

Finally take $q_L = 0$ and $q_H = q_v = \rho$ constant. The divergence of the latent sign contrast relative to q_v is $\rho/(1-\rho)$. Data processing bounds every stopped-transcript numerator by this quantity, and work is at least one. Thus, for any q_w,

$$P_N(B)^2 \ge \frac{\rho^2}{4 + B\rho/(1-\rho)} > 0. \tag{174}$$

An attained optimal dual therefore has $\alpha > 0$, finite K_N, and positive target separation, as asserted. □

J.2 A Bellman recursion with explicit null-state conventions

Divide each coefficient in (167) by $\binom{n}{k}$ and write $p_q(n,k)$ for the probability of one particular ordered prefix with that count. These coefficients satisfy

$$p_q(n,k) = p_q(n+1,k) + p_q(n+1,k+1).$$

In this subsection let $d_s = p_H(s) - p_L(s)$, $v_s = p_v(s)$, $w_s = p_w(s)$, and $r_s = \mathfrak{r}(d_s, v_s)$ denote the corresponding *prefix* coefficients. At the root $v_0 = w_0 = 1$ and $d_0 = r_0 = 0$.

Proposition J.2 (Bellman representation)**.** *Suppose $K_N < \infty$. For $\kappa \ge 0$, set*

$$\begin{aligned} U_{Nk}(\kappa) &= r_{Nk}, \\ U_{nk}(\kappa) &= \max\{r_{nk}, U_{n+1,k}(\kappa) + U_{n+1,k+1}(\kappa) - \kappa w_{nk}\}, \quad 1 \le n < N. \end{aligned} \tag{175}$$

Then

$$\Phi_N(\kappa) := U_{10}(\kappa) + U_{11}(\kappa) - \kappa = \max_{1 \le T_A \le N} \{\mathcal{I}(P_H^A, P_L^A; P_v^A) - \kappa \mathbb{E}_{q_w} T_A\}. \tag{176}$$

It is finite, continuous, piecewise affine, and strictly decreasing, and its unique nonnegative zero is K_N. A deterministic count-state policy attains the maximum at every price. If some prefix has $v_s = 0$ and $d_s \neq 0$, then $K_N = \infty$ instead.

Proof. At any observed prefix, stopping earns r_s and continuation earns the two child values minus the work charge κw_s. Backward induction therefore proves (175). Root activation is compulsory, which gives the final charge κ. All coefficients depend only on the count, so the maximizing choice may be deterministic and depend only on (n, k). Randomized or order-dependent choices cannot improve it. There are finitely many such deterministic policies, each with work in $[1, N]$. Their affine price values have slopes in $[-N, -1]$, proving the continuity and strict monotonicity claims. The zero-price value is nonnegative, the value tends to minus infinity, and the zero is precisely the maximal information-to-work ratio. If a prefix has zero reference likelihood and nonzero contrast, the policy stopping at its depth has infinite information. □

The recursion also admits an information-increment form. Put

$$j_{nk} = r_{n+1,k} + r_{n+1,k+1} - r_{nk} \ge 0. \tag{177}$$

Nonnegativity is Cauchy–Schwarz. At a zero-reference state, finite K_N forces both its contrast and its children's contrasts to vanish, so $j_s = 0$ there. If b_s^A is the unnormalized continuation-prefix flow, telescoping the flow identities gives

$$\mathcal{I}(P_H^A, P_L^A; P_v^A) = \sum_{n<N,k} b_s^A j_s, \qquad \mathbb{E}_{q_w} T_A = \sum_{n<N,k} b_s^A w_s. \tag{178}$$

Subtracting r_s from the Bellman values gives terminal values $V_N = 0$ and

$$V_s(\kappa) = \max\{0, j_s - \kappa w_s + V_{s0}(\kappa) + V_{s1}(\kappa)\}, \qquad \Phi_N(\kappa) = j_0 - \kappa + V_{10}(\kappa) + V_{11}(\kappa).$$

Thus the ratio accounts for both information gained at a state and the work needed to reach it. The pointwise inequalities $j_s \le \kappa w_s$ are sufficient, though generally not necessary, for $K_N \le \kappa$.

When q_v dominates q_w on all relevant prefixes, the posterior form is

$$\Phi_N(\kappa) = \sup_{1 \le T \le N} \mathbb{E}_{q_v} \left[L_T^2 - \kappa \sum_{n<T} W_n \right], \qquad L_n = d_n / v_n, \quad W_n = w_n / v_n.$$

The process L_n is a signed-likelihood martingale, initially zero. The unnormalized Bellman recursion itself requires no such domination assumption, and is the appropriate form at degenerate states.

K A midpoint modulus and a sharper adaptive comparison

Define the two-prior quantity

$$D_N(B)^2 = \sup_{q_-, q_+ \in \mathcal{Q}_\rho} \frac{|\theta(q_+) - \theta(q_-)|^2}{4 + B K_N(q_-, q_+; \bar{q}, \bar{q})}, \qquad \bar{q} = (q_- + q_+)/2. \tag{179}$$

Theorem K.1 (Absorbing the two references)**.** *For every law, budget, and finite cap in the stated range,*

$$D_N(B) \le P_N(B) \le 3 D_N(B). \tag{180}$$

Proof. The first inequality restricts the four-prior dual. For the second, start with any quadruple of finite K_N and set

$$q'_- = (q_L + q_v + q_w)/3, \qquad q'_+ = (q_H + q_v + q_w)/3.$$

Their midpoint dominates q_v and q_w in the following precise experiment sense: for every policy, $P^A_{\bar{q}'} \geq P^A_{q_v}/3$ and $\mathbb{E}_{\bar{q}'} T_A \geq \mathbb{E}_{q_w} T_A/3$. The signed transcript contrast is the old contrast divided by three. Consequently its information numerator is at most one third of the old numerator, and

$$K_N(q'_-, q'_+; \bar{q}', \bar{q}') \leq K_N(q_L, q_H; q_v, q_w).$$

The target gap is divided by three. Substitution in (179) and then (171) proves the result. □

For midpoint references, the numerator is twice triangular discrimination. With $h(P, Q) = 1 - \mathrm{Aff}(P, Q)$,

$$4h(P_+, P_-) \leq 2 \int \frac{(p_+ - p_-)^2}{p_+ + p_-} = \mathcal{I}(P_+, P_-; (P_+ + P_-)/2) \leq 8h(P_+, P_-). \tag{181}$$

These inequalities follow by factoring $(p_+ - p_-)^2 = (\sqrt{p_+} - \sqrt{p_-})^2(\sqrt{p_+} + \sqrt{p_-})^2$ and bounding the second factor divided by $p_+ + p_-$ between one and two. Hence the midpoint modulus is also a Hellinger-information-per-work modulus, up to universal constants. It still involves an optimal local stopping rule; it is not a closed tail-moment or entropy formula.

Lemma K.2 (Squared-loss testing by affinity)**.** *If two terminal laws have target gap Δ, every estimator has worst RMS at least $\Delta\,\mathrm{Aff}(P_-, P_+)/2$.*

Proof. The maximum MSE is at least the average MSE for equally likely endpoints. The minimum average MSE is

$$\frac{\Delta^2}{2} \int \frac{p_- p_+}{p_- + p_+}.$$

Cauchy–Schwarz gives $\mathrm{Aff}(P_-, P_+)^2 \leq 2 \int p_- p_+/(p_- + p_+)$, proving the claim. □

Let $R^{\mathrm{tot}}_N(B)$ denote population minimax RMS with at most N total observations and uniform expected work at most B. This total cap is different from the per-unit cap used in P_N.

Theorem K.3 (A direct adaptive comparison)**.**

$$R^{\mathrm{tot}}_N(B) \geq \frac{3}{8} D_N(B), \qquad P_N(B) \leq 8R^{\mathrm{tot}}_N(B). \tag{182}$$

Proof. Fix a pair in (179), with target gap Δ and information ratio K. Interpolate its endpoints toward their midpoint by $s \in [0, 1]$. The midpoint stays fixed, the target gap becomes $s\Delta$, and the information ratio becomes exactly $s^2 K$.

For every induced local policy of a cap-N population algorithm, (181) gives

$$h(P^A_-, P^A_+) \leq \frac{s^2 K}{4} \mathbb{E}_{\bar{q}} T_A.$$

This inequality also covers policies that sometimes take zero observations. Their zero-work event depends only on the seed and has zero signed contrast; conditioning on activation gives a root-active policy, and both information and expected work scale by the activation probability. Suppressing the seed can only decrease information.

Apply Lemma H.2. It realizes each hybrid as a one-genuine-unit policy, simulating other units from independent midpoint tapes in its auxiliary seed. The hybrid global transcript is a function of that local transcript, so data processing applies even if the seed reveals the other tapes. Summing the midpoint local work counts every population observation exactly once. Therefore

$$h_{\mathrm{pop}} \le s^2 BK/4.$$

If $K = 0$ take $s = 1$; otherwise take $s = \min\{1, (BK)^{-1/2}\}$. Then $h_{\mathrm{pop}} \le 1/4$. Lemma K.2 yields

$$R_N^{\mathrm{tot}}(B) \ge \frac{3}{8} s\Delta \ge \frac{3}{8} \frac{\Delta}{\sqrt{4 + BK}}.$$

Taking the pair supremum proves the first claim; the second follows from Theorem K.1. □

Corollary K.4 (An improved finite characterization)**.** *For every integer $N \ge 1$,*

$$\left(\frac{P_N(B)}{8} - \frac{\rho}{2} \sqrt{\frac{B}{N+1}} \right)_+ \le R_{\mu,\rho}(B) \le P_N(B). \tag{183}$$

At the explicit cap

$$N_* = \left\lceil 64 \max\{B, \rho B^2\} \right\rceil,$$

this gives

$$\boxed{R_{\mu,\rho}(B) \le P_{\mu,\rho,N_*}(B) \le 16 R_{\mu,\rho}(B).} \tag{184}$$

Proof. Clip any estimator to $[0, \rho]$ and, before requesting observation $N + 1$, returns $\rho/2$. Markov's inequality and the L^2 triangle inequality give

$$R_N^{\mathrm{tot}}(B) \le R_{\mu,\rho}(B) + \frac{\rho}{2} \sqrt{\frac{B}{N+1}}.$$

Combining this with Theorem K.3 and the constructive flow upper bound proves (183). Lemma H.8 gives

$$R_{\mu,\rho}(B) \ge \frac{1}{16} \min\{\rho, \sqrt{\rho/B}\}.$$

At $N = N_*$ the truncation remainder is at most that floor. Thus $P_{N_*}(B) \le 8(R+R) = 16R$. □

The program remains a uniform risk certificate rather than an exact formula for minimax risk, and its finite dimension does not imply an efficient algorithm at every budget. Theorem 13.2 supplies a complementary additive expression involving only the magnitude law, with its matching lower proof in Appendix E. Its conservative scalar constants are distinct from this program factor.

Remark K.5 (Scope of the factor-sixteen guarantee)**.** The comparison uses the sufficient cap in Corollary K.4, or a cap that passes the verified genie stopping condition in Section L. It is not a comparison for every fixed cap. In particular, an ordinary fixed-cap dual alone is not a lower certificate for unrestricted risk. With verified bounds $L \le G_N$ and $P_N \le U$, the orientation is

$$L \le G_N \le P_\infty \le P_N \le U, \qquad L/8 \le R \le U.$$

Thus $U \le 2L$ implies $U \le 16R$.

L Finite certificates for the unrestricted experiment

The universal cap in Corollary K.4 is a sufficient theoretical cap. This section gives a different computational route: upper and lower programs at a chosen cap, with an a posteriori stopping criterion. The lower program strengthens the one-unit experiment by revealing its latent magnitude and sign if it reaches that cap. This reveal-at-cap device has a direct precedent in Lee–Valiant [1, Section 5.2]; the contribution here is its convergent robust dual bracket and exact stopping certificate. The extra information is used only in a lower-bound argument. The upper estimator continues to observe ordinary Bernoulli samples alone.

L.1 The infinite-cap dual and its compactness argument

Define

$$P_\infty(B) = \inf_{N\geq 1} P_N(B), \qquad K_\infty(q_L, q_H; q_v, q_w) = \sup_{N\geq 1} K_N(q_L, q_H; q_v, q_w).$$

The latter definition uses only deterministically capped local policies. No assertion about a general unbounded policy is needed for the following interchange.

Lemma L.1 (The exact dual persists in the limit)**.** *For every law and every $B > 0$,*

$$P_\infty(B)^2 = \max_{q_L,q_H,q_v,q_w\in\mathcal{Q}_\rho} \frac{[\theta(q_H) - \theta(q_L)]_+^2}{4 + BK_\infty(q_L, q_H; q_v, q_w)}. \tag{185}$$

The maximum is attained and is positive. If D_∞ is the corresponding midpoint-reference modulus, then

$$D_\infty \leq P_\infty \leq 3D_\infty, \qquad R(B) \leq P_\infty(B) \leq 8R(B). \tag{186}$$

Proof. The space $\mathcal{Q}_\rho^4$ is weak-star compact. For each fixed cap, K_N is lower semicontinuous: every fixed policy has a lower semicontinuous closed-perspective numerator and a positive continuous work denominator, and taking their supremum preserves lower semicontinuity. Its dual ratio $f_N = (\Delta\theta)_+^2/(4 + BK_N)$ is upper semicontinuous, with value zero at an infinite denominator. Indeed, the numerator is continuous and bounded, and lower semicontinuity forces a diverging denominator when approaching a point with $K_N = \infty$. The functions f_N decrease with N. If $M = \inf_N \max f_N$, the nonempty compact sets $\{f_N \geq M\}$ are nested. Their intersection gives a quadruple with $\inf_N f_N \geq M$. The reverse inequality is immediate. This proves (185) and attainment. Positivity follows from (174).

The equal-third reference absorption in Theorem K.1 holds simultaneously at every cap, hence also for K_∞. It proves the first two comparisons. For a fixed midpoint pair, the hybrid Hellinger bound holds at every finite population cap, with the same coefficient K_∞. Increasing capped prefixes extends the affinity bound to the full terminal experiment of any finite-expected-work algorithm. The testing and interpolation argument of Theorem K.3 therefore gives $R \geq 3D_\infty/8$. The constructive inequality $R \leq P_N$ holds for every N. These facts prove (186). □

L.2 Revealing the latent state at the cap

Fix four priors and write $\delta = q_H - q_L$. Continue to use the prefix coefficients from Proposition J.2. For a prefix $s = (n, k)$, let

$$\ell_s^\pm(d) = (1/2 \pm d)^k (1/2 \mp d)^{n-k}.$$

The ordinary stopping reward is $r_s = \mathfrak{r}(p_H(s) - p_L(s), p_v(s))$. Define the latent-reveal reward

$$g_s = \int \left\{ \frac{\delta(d)^2 \ell_s^+(d)}{q_v(d)} + \frac{\delta(d)^2 \ell_s^-(d)}{1 - q_v(d)} \right\} \mu(dd). \tag{187}$$

Terms with zero numerator and denominator are zero; a nonzero numerator over zero has value infinity. In normalized count coordinates, multiply $r_s, g_s, p_w(s)$ by $\binom{n}{k}$ and use the corresponding Pascal factors.

Let K_N^+ be the optimal information/work ratio in the experiment which uses ordinary observations until it stops before depth N, but receives (D, S) at no extra cost if it reaches depth N. Its Bellman recursion is exactly (175), with terminal value g_{Nk} in place of r_{Nk}. The root condition is unchanged. Thus $\kappa \geq K_N^+$ exactly when this modified recursion has root surplus at most zero.

For a finite latent divergence, set

$$\begin{aligned} C_v &= \int \frac{\delta(d)^2}{q_v(d)(1 - q_v(d))} \, \mu(dd), \\ e_N^v &= C_v - \sum_{k=0}^{N} \binom{N}{k} \mathfrak{r}(p_H(N,k) - p_L(N,k), p_v(N,k)). \end{aligned} \tag{188}$$

Theorem L.2 (Monotone stopping-ratio brackets)**.** *For every admissible quadruple of priors, with information ratios interpreted in the extended nonnegative reals,*

$$K_N \uparrow K_\infty \leq K_N^+, \qquad K_{N+1}^+ \leq K_N^+. \tag{189}$$

If, in addition, $C_v < \infty$, then

$$0 \leq K_N^+ - K_N \leq e_N^v, \qquad e_N^v \longrightarrow 0. \tag{190}$$

Consequently $K_N^+ \downarrow K_\infty$ under this finite-divergence hypothesis. Without it, the monotone brackets remain valid, but pointwise convergence of the upper bracket need not hold.

Proof. Under the variance-reference latent law, define the score

$$Z(D,S) = \begin{cases} \delta(D)/q_v(D), & S = 1, \\ -\delta(D)/(1 - q_v(D)), & S = 0. \end{cases}$$

If $C_v < \infty$, it is square integrable, has mean zero, and has second moment C_v. The score of any observed transcript is its conditional expectation. Thus its signed-likelihood information is $\mathbb{E}_v[\mathbb{E}_v(Z \mid \text{transcript})^2]$.

Take any capped ordinary policy with cap larger than N and stop its paid observations at $T \wedge N$. On reaching N, the genie reveals (D, S); these values allow simulation of every future Bernoulli observation and therefore of the original terminal transcript. Data processing cannot decrease information when passing to the stronger genie transcript, while the work under q_w can only decrease. This proves $K_\infty \leq K_N^+$ by taking the supremum over finite ordinary caps. The same simulation shows that an earlier reveal is stronger than a later reveal, so K_N^+ is nonincreasing. The other monotonicity and $K_N \leq K_\infty$ are immediate. These arguments also apply to infinite divergences using the extended-value definition.

For a cap-N genie policy, erase the reveal. The resulting ordinary policy has exactly the same paid work. Its lost information is

$$\mathbb{E}_v \left[\mathbf{1}_{\{T=N\}} \operatorname{Var}_v(Z \mid \mathcal{F}_N) \right] \leq \mathbb{E}_v \operatorname{Var}_v(Z \mid \mathcal{F}_N) = e_N^v.$$

Here $\mathcal{F}_N$ is the first N observations of the completed counterfactual stream. An independent policy seed can be included; it adds no latent information. Since root-active work under q_w is at least one, dividing by work and taking suprema gives $K_N^+ \le K_N + e_N^v$.

The infinite Bernoulli stream identifies its mean almost surely and hence identifies (D, S) because $D > 0$. The martingale $\mathbb{E}_v(Z \mid \mathcal{F}_N)$ therefore converges to Z in L^2. This proves $e_N^v \to 0$. Its explicit formula is precisely (188). □

Remark L.3 (Pointwise convergence can fail at infinite latent divergence)**.** Let $\mu(dd) = 2\,dd$ on $(0, 1/2]$, $\rho = 1/4$, and take $q_L = 0$, $q_H = q_w = 1/4$, and $q_v(d) = 1/2 - d$. All four priors are admissible. Here

$$C_v = \frac{1}{8}\int_0^{1/2} \frac{dd}{1/4 - d^2} = \infty.$$

The policy which always reaches depth N and accepts the latent reveal has infinite information and work N, so $K_N^+ = \infty$ for every finite N.

Nevertheless $K_\infty \le 3/4$. To verify this for every capped policy, parameterize the latent state by its Bernoulli mean p. The variance reference has density $2(1-p)$ on $(0,1)$, the work reference has density $3/2$ below $1/2$ and $1/2$ above $1/2$, and the signed endpoint difference has density $-1/2$ below $1/2$ and $1/2$ above $1/2$. For an ordered prefix s of length $n \ge 1$ with k heads, put $u_s = \int_0^1 p^k(1-p)^{n-k}\,dp$. Then

$$|p_H(s) - p_L(s)| \le u_s/2, \qquad p_v(s) = \frac{2(n-k+1)}{n+2}u_s,$$

and hence

$$r_s \le \frac{n+2}{8(n-k+1)}u_s \le \frac{3n}{8}u_s.$$

Sum over stopped prefixes and the independent seed. If U denotes the auxiliary uniform law of p, the information of any capped policy is at most $(3/8)\mathbb{E}_U T$. The work-reference density dominates half the uniform density, so this is at most $(3/4)\mathbb{E}_{q_w} T$. Taking suprema gives the claimed bound. The auxiliary law U need not be admissible; it is used only for this domination calculation.

This example does not contradict the optimized convergence in the next theorem, whose proof first regularizes the variance reference.

L.3 A convergent lower program and a stopping rule

Define the strengthened-experiment dual value

$$G_N(B)^2 = \sup_{q_L, q_H, q_v, q_w \in \mathcal{Q}_\rho} \frac{[\theta(q_H) - \theta(q_L)]_+^2}{4 + BK_N^+(q_L, q_H; q_v, q_w)}. \tag{191}$$

As before, a ratio with infinite denominator is zero.

Theorem L.4 (A posteriori minimax brackets)**.** *For every law, rarity and budget in the model,*

$$0 < G_N(B) \uparrow P_\infty(B), \qquad P_N(B) \downarrow P_\infty(B), \tag{192}$$

and at every integer $N \ge 1$,

$$\boxed{G_N(B)/8 \le R_{\mu,\rho}(B) \le P_N(B).} \tag{193}$$

For finite-support laws both programs have finite conic formulations. The condition $P_N(B) \le 2G_N(B)$ eventually holds and certifies $R(B) \le P_N(B) \le 16R(B)$.

Proof. The preceding theorem gives $G_N \le P_\infty$ and monotonicity. The constant priors $q_L = 0, q_H = q_v = \rho$ have finite latent divergence and give a strictly positive lower bound for every G_N.

To prove convergence, fix a quadruple with finite K_∞ and positive target gap. Put $m = (q_L + q_H)/2$ and regularize only the variance reference:

$$q_v^\epsilon = (1-\epsilon)q_v + \epsilon m, \qquad 0 < \epsilon < 1.$$

The new transcript reference dominates $(1-\epsilon)$ times the old one, so

$$K_\infty(q_L, q_H; q_v^\epsilon, q_w) \le \frac{K_\infty(q_L, q_H; q_v, q_w)}{1-\epsilon}.$$

Also $|\delta| \le 2m$ and $|\delta| \le 2(1-m)$, so the new latent score has absolute value at most $2/\epsilon$. Its latent divergence is finite. Theorem L.2 now makes its genie ratio converge to its ordinary infinite-cap ratio. Keep $\epsilon > 0$ fixed throughout this limit in N; no convergence of the unregularized quadruple is used. Hence

$$\lim_N G_N(B)^2 \ge \frac{(\Delta\theta)^2}{4 + BK_\infty/(1-\epsilon)}.$$

Let $\epsilon \downarrow 0$ and maximize using Lemma L.1. The limits are taken in this order; no convergence uniform in ϵ is asserted. This proves $G_N \uparrow P_\infty$. Equation (186) gives the risk bracket. Since $P_\infty > 0$, the ratio P_N/G_N tends to one. The stated stopping criterion therefore succeeds at a finite cap, with no claimed uniform rate for that cap. □

For an explicitly given law $\mu = \sum_i w_i \delta_{d_i}$, the genie dual is a finite SOCP. In the price formulation, use endpoint variables $u_H = \alpha q_H$, $u_L = \alpha q_L$, work-reference variables $e = \eta q_w$, and a variance reference q_v. Their rarity and box constraints are linear perspective constraints. For each atom introduce

$$(u_{H,i} - u_{L,i})^2 \le q_{v,i} r_i^+, \qquad (u_{H,i} - u_{L,i})^2 \le (1 - q_{v,i}) r_i^-.$$

At terminal states replace the ordinary quadratic-over-linear stopping cone by the linear bound

$$X_s \ge \frac{1}{4} \sum_i w_i \{\ell_s^+(d_i) r_i^+ + \ell_s^-(d_i) r_i^-\}.$$

Keep the ordinary stopping cones at earlier states and the same linear continuation/root inequalities. Maximize $\sum_i w_i(u_{H,i} - u_{L,i}) - \alpha^2 - \eta B$. This is exactly the squared value in (191) after eliminating the price and scale. It uses $O(N^2 + m)$ variables, apart from the coefficient matrices. For a general measure the result remains a variational statement; a finite-support computational representation is not silently assumed for an arbitrary law.

L.4 Certificates that can be checked independently

In practice a feasible primal policy gives an upper bound $U \ge R$, and a feasible genie dual gives a square-root lower bound $L \le G_N$. Consequently the *verified* inequality $U \le 2L$ is sufficient for a factor-16 risk certificate. An approximate solver status or an uncertified objective is not sufficient. Section N uses exact rational checks for the finite rational examples.

A selected dual quadruple can also yield a stronger direct risk lower bound than $L/8$. Absorb its two references into its endpoints with equal thirds, and let t be the new target

separation. If a Bellman certificate gives $\kappa \geq K_N^+$ for the original quadruple, the absorbed midpoint pair has $K_\infty \leq \kappa$. With $x = B\kappa$, affinity testing gives

$$R(B) \geq \max_{0 \leq s \leq 1} \frac{st}{2} \left(1 - \frac{s^2 x}{4}\right)_+ . \tag{194}$$

The square of this bound is the rational expression

$$\begin{cases} \dfrac{t^2}{4}(1 - x/4)^2, & 0 \leq x \leq 4/3, \\ \dfrac{4t^2}{27x}, & x \geq 4/3. \end{cases} \tag{195}$$

This supplies an exact squared lower certificate when the inputs and the certified price are rational.

M What a fixed observation cap can identify

The exact dual also identifies an obstruction that is invisible in a small-cap numerical upper bound: increasing the number of units cannot resolve two priors with identical cap-N one-unit distributions.

Let $L = \lceil N/2 \rceil$ and define the moment-matching diameter

$$\Delta_N = \max \left\{ \theta(q_H) - \theta(q_L) : q_L, q_H \in \mathcal{Q}_\rho,\ \int (q_H - q_L) d^{2j+1}\, d\mu = 0\ (0 \le j < L) \right\}. \tag{196}$$

Compactness ensures that this maximum is attained.

Theorem M.1 (Exact fixed-cap bias floor)**.** *For every magnitude law,*

$$\lim_{B \to \infty} P_N(B) = \Delta_N/2. \tag{197}$$

If μ has exactly m distinct positive support points, then $\Delta_N = 0$ if and only if $N \ge 2m-1$. If μ has infinite support, then $\Delta_N > 0$ for every finite N.

Proof. The distribution of N Bernoulli observations from one unit determines the first N moments of its latent Bernoulli mean. The difference of the kth such moments under two priors is

$$\int (q_H - q_L)\{(1/2+d)^k - (1/2-d)^k\}\, d\mu.$$

These polynomials span exactly the odd monomials $d, d^3, \dots, d^{2L-1}$. Thus the constraints in (196) are equivalent to equality of the full depth-N count distributions. Equality at depth N implies equality for every stopped experiment with that cap, by marginalization and independent randomization. Conversely, the policy that always observes exactly N samples witnesses a positive information numerator whenever those count distributions differ. Therefore $K_N = 0$ exactly for pairs satisfying (196); otherwise $K_N > 0$ or $K_N = \infty$.

Apply the decreasing compact-set argument from Lemma L.1 to the dual ratios as $B \to \infty$. Their pointwise limit is $(\Delta\theta)_+^2/4$ when $K_N = 0$, and zero otherwise. Theorem J.1 gives (197).

For the support assertion, consider the finite-dimensional subspace

$$\mathcal{V}_L = \operatorname{span}\{d, d^3, \dots, d^{2L-1}\} \subset L^2(\mu).$$

If the support has $m \le L$ points, polynomial interpolation gives $p(d) = dQ(d^2) = 1$ on that support, with $\deg Q \le m-1$. Then $1 \in \mathcal{V}_L$, so moment matching forces zero target gap.

If the support has more than L points, $1 \notin \mathcal{V}_L$. Indeed, a polynomial Q of degree at most $L-1$ cannot agree with $x^{-1/2}$ at $L+1$ distinct positive values: otherwise repeated Rolle's theorem would force the nonvanishing Lth derivative of $x^{-1/2}$ to vanish somewhere. The same argument covers infinite support, since an almost-everywhere polynomial identity extends to the support.

Let $h = 1 - \operatorname{proj}_{\mathcal{V}_L} 1$. It is a nonzero bounded function, orthogonal to every required odd monomial, and

$$\int h\, d\mu = \int h^2\, d\mu > 0.$$

Set $\epsilon = \rho/(4\|h\|_\infty)$ and $q_H = \rho/2 + \epsilon h$, $q_L = \rho/2 - \epsilon h$. Both priors lie pointwise between $\rho/4$ and $3\rho/4$ and hence belong to $\mathcal{Q}_\rho$. They match the odd moments but have positive target gap $2\epsilon \int h^2\, d\mu$. This proves strict positivity. Finally $L \ge m$ is equivalent to $N \ge 2m-1$. □

The proof also yields the explicit lower estimate

$$\Delta_N/2 \geq \frac{\rho\|h\|^2_{L^2(\mu)}}{4\|h\|_{L^\infty(\mu)}} \quad \text{when } 1 \notin \mathcal{V}_L.$$

This theorem identifies the target functional from cap-limited laws; it does not require recovery of the entire allocation q. This theorem concerns the fixed-cap program and its infinite-budget limit. It does not impose a bias floor on the unrestricted experiment. The monotone brackets in Section L increase the cap when needed, and their stopping criterion tests the remaining gap directly.

N Exact numerical certificates on finite laws

The computations in this section produce two-sided certificates for eight finite rational instances. Numerical conic optimization selects candidate policies and dual priors. Separate rational-arithmetic verifiers then check the resulting bounds, including the observation budget and the stopping criterion. An approximate solver objective is never used as a certified endpoint.

N.1 What the two endpoints mean

Write U_N for the risk bound of a verified observable cap-N policy, and L_N for the value of a verified feasible genie dual. Their roles are

$$L_N \le G_N \le P_\infty \le P_N \le U_N, \qquad L_N/8 \le R_{\mu,\rho}(B) \le U_N.$$

Thus L_N is a lower bound on a program value. It is not, by itself, the reported lower bound on fully adaptive minimax risk. We also compute the stronger pair-specific adaptive-risk lower bound $\underline{r}_N$ from (194)–(195). Its target separation is that of the absorbed midpoint pair.

The cap search stops only after the exact rational comparison

$$U_N^2 \le 4L_N^2 \tag{198}$$

has passed. This implies $U_N \le 16R_{\mu,\rho}(B)$. The comparison uses the stored rational squared values, not the rounded decimals in the tables.

N.2 The benchmark laws and results

Every instance has $\rho = 1/10$. The four laws are

$$\begin{aligned}
\mu_{\text{one}} &= \delta_{1/4},\\
\mu_{\text{three}} &= \tfrac{3}{4}\delta_{2/5} + \tfrac{1}{5}\delta_{1/5} + \tfrac{1}{20}\delta_{2/25},\\
\mu_{\text{hard}} &= \tfrac{99}{100}\delta_{9/20} + \tfrac{1}{100}\delta_{2/25},\\
\mu_{\text{label}} &= \delta_{1/2}.
\end{aligned}$$

Each was run at $B = 100$ and $B = 1000$. The candidate caps were $4, 8, 16, 32$, with the perfect-label cases starting at $N = 1$. Seven instances passed (198) at their first cap. The three-scale law at $B = 1000$ failed at $N = 4$ and passed at $N = 8$. There were therefore nine attempted brackets and eight successful ones. The failed bracket is retained in the source bundle; it remains a valid bracket but does not meet the requested factor-two program criterion.

For the failed three-scale attempt at $B = 1000, N = 4$, the verified program endpoints, rounded outward, are 0.01359364 and 0.02989647, and the exact ratio exceeds two. At $N = 8$ the program ratio is about 1.6272, so the exact test passes. Across the eight successful instances, the program bracket ratios range from essentially one to 1.6272. The sharper adaptive-risk bracket ratios range from about 6.27 to 12.05, all below sixteen.

These are small benchmark instances. Their modest successful caps provide no uniform bound on the cap or runtime needed for another law, rarity, or budget. The convergence theorem is qualitative and the certificates concern the particular rational inputs displayed above.

Law	B	N	L_N	$\underline{r}_N$	U_N
One gap	100	4	0.03474453	0.00619890	0.03891866
One gap	1000	4	0.01461177	0.00196026	0.01825093
Three scales	100	4	0.03331182	0.00573109	0.04551457
Three scales	1000	8	0.01569090	0.00212025	0.02553208
Rare hard atom	100	4	0.02912416	0.00459699	0.03009854
Rare hard atom	1000	4	0.01105026	0.00145369	0.01166586
Perfect labels	100	1	0.02572478	0.00384900	0.02572479
Perfect labels	1000	1	0.00932054	0.00121716	0.00932055

Table 3: Successful exact brackets. Both lower columns are rounded downward and the upper column upward. The adaptive minimax risk lies in $[\underline{r}_N, U_N]$; the unrestricted flow-program value lies in $[L_N, U_N]$. These two intervals have different meanings.

N.3 An analytic check with perfect labels

For $\mu = \delta_{1/2}$, the first observation reveals the sign exactly. Further observations add no information. For every cap $N \geq 1$,

$$P_N(B)^2 = P_\infty(B)^2 = G_N(B)^2 = \frac{ac}{a+c}, \qquad a = \frac{\rho^2}{4}, \quad c = \frac{\rho(1-\rho)}{B}. \tag{199}$$

This is an analytic identity for the flow program, without an assertion that it is the exact fully adaptive minimax risk.

To verify (199), conditional averaging removes randomness in the output given the revealed sign and cannot increase its second moment. Let $\lambda \leq B$ be the active Poisson intensity, let y_0, y_1 be the two outputs, and put $s = \lambda(y_1 - y_0)$. Optimizing the deterministic offset gives worst bias $\rho|1-s|/2$. At the prior $q = \rho$, centering the outputs minimizes the raw second moment, giving the variance lower bound $s^2\rho(1-\rho)/\lambda$. It is attained by

$$y_0 = -\rho s/\lambda, \qquad y_1 = (1-\rho)s/\lambda.$$

Since $\rho \leq 1/4$, their second moment is increasing in q on $[0, \rho]$, so its worst value occurs at ρ. Choosing $\lambda = B$ therefore reduces the objective exactly to $a(1-s)^2 + cs^2$, whose minimum is $ac/(a+c)$. The genie already reveals nothing new after the first observation, which proves the same equality for G_N.

For the two stored perfect-label instances, the exact rational analytic value lies between the exact rational lower and upper certificates. This was checked directly, without rounded decimal comparisons.

N.4 How the certificates are repaired and checked

The conic solver may return a status indicating limited numerical accuracy. Such a status neither establishes nor invalidates a certificate. The following separate checks determine feasibility.

For a lower certificate, the selected priors are quantized to rational numbers, clipped to $[0, 1]$, and, if needed, scaled to satisfy the rarity constraint exactly. Variance and work references are regularized by mixing a weight 10^{-8} of the constant prior $\rho/2$. All binomial kernels and latent rewards are then recomputed exactly. A numerical bisection proposes the work price. A new backward Bellman pass uses rational arithmetic and rounds each potential upward to a dyadic grid of denominator 2^{96}. The price is increased if the exact root inequality does not hold. An

independent verifier reconstructs the kernels and checks every rarity, stopping, continuation, terminal, and root inequality.

For an upper certificate, a separate verifier works with the actual observable policy: its stopping probabilities at each count state, terminal outputs, Poisson intensity, and deterministic offset. It computes exact conditional means, second moments, and observation counts for every atom and sign. Total stopping probability must equal one exactly. The worst-prior linear expressions are evaluated by an exact fractional-knapsack calculation over $\mathcal{Q}_\rho$. If rationalization raises expected work above B, the intensity is reduced by the exact required factor and outputs are increased by its reciprocal. This preserves means, and the variance bound is recomputed. The reported risk upper is

$$U_N^2 = \left(\sup_{q \in \mathcal{Q}_\rho} |\operatorname{Bias}_q| \right)^2 + \sup_{q \in \mathcal{Q}_\rho} \operatorname{Var}_q .$$

The two maxima need not occur at the same prior; their sum remains a valid upper bound. Integer square-root calculations round every displayed lower RMS downward and every displayed upper RMS upward. The verifier also checks the direction of the stored decimal rounding.

In addition to the optimized genie brackets, the bundle includes eight ordinary finite-cap dual checks and sixteen fixed-quartet midpoint-genie checks. Standard-library-only re-verification accepted all 33 stored lower certificates and nine upper policies. Six targeted corruptions were rejected: a rarity violation, an insufficient Bellman price, an overstated lower value, an insufficient genie terminal reward, excessive actual policy work, and an understated upper risk. These checks remain active when Python is invoked with its optimization flag.

N.5 Standalone reproduction

The repository directory `paper/rare-event/v2/checks/` is self-contained. It includes a clearly attributed copy of the unchanged round-three primal solver, the independent dual solver, the exact verifiers, and all certificate files. It does not import a solver from a neighboring research directory. From that directory, verification needs only Python's standard library:

```
python self_check_certificates.py
python verify_finite_dual.py \
  optimized_genie/three_scales_N8_B1000_lower.json
python verify_policy_upper.py \
  optimized_genie/three_scales_N8_B1000_upper.json
```

The numerical search used Python 3.12 with CVXPY 1.9.3, NumPy 2.5.4, SciPy 1.18.1, and Clarabel 0.11.1. These observed versions are pinned in `requirements.txt`. The principal search is reproduced by

```
python -m pip install -r requirements.txt
python solve_genie_bracket.py
```

Re-running a numerical solver may select different feasible priors and slightly different bounds. The mathematical guarantee comes from the independently checked rational artifacts.

O A direct proof of the closed quantile law for finite laws

This appendix proves Theorem 14.1. Notation follows section 14: signs are $S \in \{-1, +1\}$, $h = B^{-1/2}$ is the budget scale, and $\mathfrak{h}$ is squared Hellinger distance. The proof uses the single-unit and transfer results of appendices F to H, the entropy-tier lemma of appendix E, and the upper construction of appendix D. It does not use Theorem 13.2.

O.1 Lower bound

Throughout the proof, $h = B^{-1/2}$ is the budget scale and $\mathfrak{h}(P, Q)$ is the squared Hellinger distance defined in Section 14.4.

O.1.1 An explicit posterior-martingale bound for the quantile tail

Fix a gap d and a capped one-coin policy. Under the equal sign prior, let $U_k = \mathbb{E}[S \mid \text{seed}, X_1, \dots, X_k]$, with $X_i \in \{-1, 1\}$ and $\mathbb{E}[X_i \mid S] = 2dS$. Then $U_0 = 0$ and U_k is a martingale. For a current posterior mean u,

$$U_{k+1} = \frac{u + 2dX_{k+1}}{1 + 2duX_{k+1}}, \qquad \mathbb{E}[(U_{k+1} - u)^2 \mid U_k = u] = \frac{4d^2(1-u^2)^2}{1 - 4d^2u^2} \le 4d^2.$$

At $d = 1/2$, the posterior becomes absorbing at ± 1 after the first observation; the undefined zero-over-zero expression there is interpreted through the zero increment. For the policy's capped stopping time N,

$$\mathbb{E}_{1/2} U_N^2 \le 4d^2 \mathbb{E}_{1/2} N. \tag{200}$$

The seed may be arbitrary and independent of the latent state.

Change the conditional positive-sign probability to $1/2 - \epsilon$ or $1/2 + \epsilon$, $0 \le \epsilon \le 1/4$. Relative to the midpoint transcript law the endpoint densities are $1 - 2\epsilon U_N$ and $1 + 2\epsilon U_N$. Since $1 - \sqrt{1-x} \le x$ on $[0, 1]$, (200) gives

$$\mathfrak{h}(P_{1/2-\epsilon}, P_{1/2+\epsilon}) \le 4\epsilon^2 \mathbb{E}_{1/2} U_N^2 \le 16\epsilon^2 d^2 \mathbb{E}_{1/2} N. \tag{201}$$

Lemma O.1 (Quantile-tail obstruction)**.** *For every allowed law, rarity and budget, $R_{\mu,\rho}(B) \ge T_h/32$.*

Proof. Use an oracle experiment that lets the algorithm choose any tail atom i as an observable type, then obtain a genuine coin with that fixed gap d_i and unknown sign. Each flip is still charged. The algorithm may interleave types and revisit coins. Set

$$\epsilon_i = \frac{1}{16} \min\{1, h/d_i\}, \qquad q_{i,\pm} = 1/2 \pm \epsilon_i.$$

Equation (201) bounds the local distance for every type by $(h^2/16)$ times its midpoint work. The midpoint decomposition, proved in Lemma H.2, therefore bounds global distance by $h^2 B/16 = 1/16$. It permits observable types chosen adaptively and arbitrary interleaving.

Every original algorithm can be simulated in this oracle experiment: for each fresh coin privately choose tail type i with probability τ_i; on the remaining mass simulate an all-negative coin with the residual magnitude law $\kappa/(1-\rho)$ for free. Keep that private choice on revisits. The resulting original endpoints are

$$q_\pm(d) = \frac{\mathrm{d}\nu}{\mathrm{d}\mu}(d) \left(\frac{1}{2} \pm \frac{1}{16} \min\{1, h/d\} \right).$$

The density uses total masses at a support point. At a distinct atom it equals τ_i/w_i; this remains well-defined even if an input lists identical gaps more than once. Both endpoints and their midpoint are admissible, since their total positive mass is at most $(9/16)\rho$. The oracle's genuine work is at most the original flip count, so its midpoint budget is at most B. No budget assumption over arbitrary oracle parameters is needed. The original target separation is $2\sum_i \tau_i\epsilon_i = T_h/8$. Data processing gives total variation at most $\sqrt{2/16} < 1/2$. Two-point midpoint testing gives RMS at least one quarter of the separation, namely $T_h/32$. For unbounded original stopping, cap the global experiment first. The same endpoint pair works at every cap; increasing prefixes give the terminal TV bound, as proved after Lemma H.2. □

Lemma O.2 (Rare Bernoulli-label floor)**.** $R_{\mu,\rho}(B) \geq \min\{\rho, h\sqrt{\rho}\}/32$.

Proof. Restrict to constant priors $q(d) = p$ and strengthen the experiment by revealing a fresh coin's sign for unit cost. Its magnitude and subsequent flips can then be simulated using independent randomness, so the number of purchased labels is at most the original number of flips. Take $p_- = \rho/2$, $p_+ = p_- + \delta$ with $\delta = \min\{\rho, h\sqrt{\rho}\}/8$. One Bernoulli-label divergence is at most $\delta^2/[p_+(1-p_+)] \leq 3\delta^2/\rho$. The stopped chain rule bounds total KL by $3\delta^2 B/\rho \leq 3/64$. Pinsker gives TV below $1/2$, and midpoint testing gives RMS at least $\delta/4$. Finite prefixes justify unbounded stopping. □

O.1.2 A deterministic entropy filter

Retain the canonical core notation a, A, F, λ from Section 14.5. If $h \geq t$, then $T_h = \rho$, so the tail lemma already proves the desired lower bound. Assume $h < t$. If $\lambda \geq t/4$, then

$$h\sqrt{\rho\Gamma_\kappa} \leq 2h\sqrt{\rho F} = 2h\rho/\lambda \leq 8h\rho/t \leq 8T_h,$$

and $\mathcal{Q} \leq 9T_h$. Thus it remains to treat $\lambda < t/4$.

Set $L_j = 1 + \log(A/a_j)$ on positive-cost bands. Declare a band *good* exactly when

$$b_j \geq \tfrac{\lambda}{2} L_j. \tag{202}$$

This is a direct test of the known weights and endpoints. Let F_g and F_b be the entropies of the good and bad cost vectors, each normalized by its own total cost.

Lemma O.3 (Filtering retains core entropy)**.** *If $\lambda < t/4$, then $F_b \leq F/8$ and $F_g \geq F/4$.*

Proof. Every core band has endpoint at least t. Number the bands upward from the one containing t, with endpoints $\beta_0 2^k$, $k \geq 0$. On a bad band, $L_j > x_j := 2b_j/\lambda$, and $x_j \geq 8$. Since $a_j/A = e^{1-L_j}$ and xe^{1-x} decreases for $x \geq 1$,

$$\sum_{j \text{ bad}} a_j L_j \leq A \sum_{k\geq 0} x_0 2^k e^{1-x_0 2^k}, \qquad x_0 = 2\beta_0/\lambda \geq 8.$$

The ratio of successive summands is at most $2e^{-8} < 1/2$. Their sum is at most $2x_0 e^{1-x_0} \leq 16e^{-7} < 1/8$. The bad vector's own entropy is at most this full-normalization sum, so $F_b \leq A/8 \leq F/8$. By (71), $F \leq 2(F_g + F_b)$; consequently $F_g \geq 3F/8 \geq F/4$. □

O.1.3 Packing the filtered entropy into actual experiments

We first group coordinates with comparable costs. Lemma E.3 in Appendix E gives the bound $\mathcal{G}(a) \leq 7\sum_k F_k$ over comparable-cost tiers that is used below.

Split the good bands into high bands $j < 6$ and low bands $j \geq 6$, with entropies F_H, F_L. Since $b_j \geq 1/64$ on the first six bands, $F_H \leq 4096(1 + \log 6) =: C_H$. If $F_H \geq F_L$, then

$$F \leq 4F_g \leq 8(F_H + F_L) \leq 16C_H,$$

and the Bernoulli-label floor implies $R \gtrsim \min\{\rho, h\sqrt{\rho F}\}$.

Otherwise $F_L > F_H$, so $F \leq 16F_L$. Split the low bands by index parity, then within each parity into the comparable-cost tiers of Lemma E.3. The sum of the resulting group entropies is at least $F_L/14$. Indeed $F_L \leq 2(F_{\text{even}} + F_{\text{odd}})$, and the tier lemma is applied separately within each parity. Both parities and all nonempty tiers are retained; choosing just one tier would not justify this bound. Each group has endpoint ratios at least four and costs in $(c, 2c]$ for some $c > 0$.

For a group of n bands, (202) and $A \geq nc$ give

$$b_j \geq \frac{\lambda}{2}[1 + \log(n/2)] \geq \frac{\lambda}{8}(1 + \log n).$$

Remove its $s = \lceil \log_4 n \rceil$ smallest-gap bands, with $s = 0$ for $n = 1$. For $n \geq 2$, $s \leq n/2$, so at least $n/2$ remain. Let G_i be the remaining group's entropy. Coordinate monotonicity yields

$$G_i \geq F_{\text{group}}/16, \qquad b_{\min,i} \geq \lambda n(1 + \log n)/8. \tag{203}$$

For the first bound, divide the lower bound $cn'(1 + \log n')$ by the upper bound $2cn(1 + \log n)$; the ratio is at least $(1 - \log 2)/4 > 1/16$. The second bound follows because deleting s smallest bands raises the minimum endpoint by at least $4^s \geq n$. The case $n = 1$ is immediate. The integer inequality for $n \geq 2$ can be checked directly at $n = 2, 3, 4$; when $4^{s-1} < n \leq 4^s$ and $s \geq 2$, we have $2s \leq 4^{s-1} + 1 \leq n$. Trimming is by increasing gap (equivalently, increasing endpoint), not by cost. The number n in (203) is the pre-trim cardinality.

Let σ_i be the restriction of the actual core measure to a remaining group, m_i its mass, u_i its largest endpoint, and $w_{*,i}$ its smallest band mass. Then

$$m_i/u_i^2 \geq c, \qquad w_{*,i} \geq cb_{\min,i}^2, \qquad G_i \leq 2cn(1 + \log n).$$

Consequently

$$\frac{m_i w_{*,i}}{u_i^2} \geq \lambda^2 G_i^2/256. \tag{204}$$

Assign original rarity $\theta_i = \lambda^2 G_i/1024$ and conditional rarity $r_i = \theta_i/m_i$. Equation (204) gives $\theta_i G_i \leq m_i w_{*,i}/(4u_i^2)$. Because $G_i \geq m_i/u_i^2$, this actually implies $\theta_i \leq w_{*,i}/4 \leq m_i/4$, hence $r_i \leq 1/4$. Here $w_{*,i}$ is a band mass, not a minimum individual atom mass. Under the conditional law σ_i/m_i, the entropy and minimum band mass are G_i/m_i and $w_{*,i}/m_i$. Its entropy-branch condition is

$$r_i u_i^2 (G_i/m_i) \leq w_{*,i}/m_i \quad \Longleftrightarrow \quad \theta_i G_i \leq m_i w_{*,i}/u_i^2.$$

Thus the preceding factor-four margin verifies the actual conditional condition. There is no unnormalized-to-conditional mass factor missing. Disjointness and superadditivity give

$$\sum_i \theta_i \leq \lambda^2 F/1024 = \rho/1024, \qquad \sum_i G_i \geq F_L/224 \geq F/3584.$$

All remaining bands have largest endpoint at most $1/128$ and endpoint ratios at least four. Apply the annular one-unit obstruction of Proposition G.4 to each normalized component σ_i/m_i, followed by the additive localization theorem (Theorem H.7). Their complete proofs

are included in Appendices G and H. The actual minimum-band-mass condition is verified by (204), so the entropy branch is active. We obtain

$$R(B) \gtrsim \sum_i \min\{\theta_i, h\sqrt{\theta_i G_i}\} = \left(\sum_i G_i\right) \min\{\lambda^2/1024, h\lambda/32\} \gtrsim \min\{\rho, h\sqrt{\rho F}\}.$$

The proportional allocation $\theta_i/G_i = \lambda^2/1024$ is what makes this estimate valid jointly at every budget.

O.1.4 Completion of the lower bound

The cases $h \geq t$ and $\lambda \geq t/4$ were handled by the tail lemma. In the remaining case, the floor or the packing argument gives $R \gtrsim \min\{\rho, h\sqrt{\rho F}\}$, while always $R \geq T_h/32$. Finally, by (72),

$$\mathcal{Q}(B) \leq T_h + 2\min\{\rho, h\sqrt{\rho F}\}.$$

The two lower bounds control this sum up to a universal constant. Thus $R(B) \geq c\mathcal{Q}(B)$, proving the lower half of Theorem 14.1. Appendix O.3 tracks the constants to obtain the stated value 10^{-20}.

O.2 Observable upper bound

We construct an estimator at the quantile split from Section 14. The classification and allocation lemmas used below are proved in Appendix F.

Theorem O.4 (Upper bound at the fixed quantile split)**.** *For every admissible (μ, ρ, B), there is an observable estimator with uniform expected work at most B and worst RMS error at most $30001\mathcal{Q}(B)$.*

If $\mathcal{Q}(B) = \rho$, use the constant estimator $\rho/2$. Otherwise $h < t$, because $h \geq t$ would imply $T_h = \rho$. Fix

$$\sigma = \kappa, \qquad \tau = \nu|_{[h,1/2]}, \qquad \zeta = \nu|_{(0,h)}, \qquad z = \zeta((0,1/2]), \quad H = \int d^{-1}\,\tau(dd).$$

Then $\mu = \sigma + \tau + \zeta$ and $T_h = z + hH$. These are analytical pieces only. The estimator never observes which piece produced its genuine coin. All likelihood evaluations and design parameters use the known law. The complexity measure counts physical coin observations, including those given to either side of the race; it does not assert a bound on arithmetic time, memory, or random-number generation.

O.2.1 Reduction to the general construction

Use the fixed decomposition above in the construction of Appendix D, with $z = \zeta((0,1/2])$. For a finite law, Γ_σ and H_τ are finite. The quantities $a_j(\sigma)$, A_σ, Γ_σ and H_τ there are the exact inverse-square and inverse costs of (60), not the rounded canonical costs used in the lower bound. Because ν denotes the removed quantile mass in this appendix, write $\chi = \sigma + \tau$ for the resolved law that Appendix D calls ν. Thus the caps satisfy $\eta_\chi(M_\ell) \leq \delta_\ell/8$, and $\eta_\chi(4) = 1 - z$. In the finite-law theorem the reference bank is finite; its stopping events are measurable and its positive reference is a finite probability mixture. No countable-support extension is needed here. The displays (97)–(111) therefore apply to this decomposition.

Completion of Theorem O.4. The shifted adjacent-level sum has a factor 1/4, so

$$S_J \le \frac{5\sqrt{17}}{4} \sum_{\ell=0}^{J} \sqrt{\delta_\ell C_\ell}.$$

Use (105), (109), (110), (111), and $A \le \Gamma_\sigma$. The elementary sums $\sum_{\ell\ge 0} 4^{-\ell}\sqrt{\ell+1} \le 16/9$ and $\sum_{\ell\ge 0} 4^{-\ell} = 4/3$ then give

$$S_J \le C_{\mathrm{sum}}[\sqrt{\rho\Gamma_\sigma} + H].$$

One valid explicit choice is $C_{\mathrm{sum}} = 30000$. Indeed, writing $R = \sqrt{\rho\Gamma_\sigma}$, the component bounds of Appendix D give

$$\sum_{\ell\le J} \sqrt{\delta_\ell C_\ell} \le 21R + \frac{15535}{3} H.$$

Here the core term is at most $4R$, the fixed tail surcharge is $2H/3$, the ordinary mixed term is at most $15R + 5007H$, and the unresolved term is at most $2R + 512H/3$. Use $\sqrt{21} < 5$, $\sqrt{2} < 3/2$ and $5\sqrt{17}/4 < 165/32$ to obtain

$$S_J \le \frac{3465}{32} R + \frac{854425}{32} H \le 30000(R + H).$$

For $z > 0$, (104) and (106) imply $R(B) \le 128z + C_{\mathrm{sum}}[\sqrt{\rho\Gamma_\sigma} + H]/\sqrt{B}$. For $z = 0$, put $D = \sqrt{\rho\Gamma_\sigma} + H > 0$ and choose the least finite $J \ge 0$ such that $\delta_J \le D/\sqrt{B}$. The bound on S_J holds at this finite terminal level. Equation (106) therefore gives RMS at most

$$\sqrt{1 + 30000^2}\, D/\sqrt{B} < 30001 D/\sqrt{B}.$$

Thus this branch has an actual finite-level estimator; no limiting or infinite-level construction is needed. The constant estimator covers $z > \rho/128$. For the fixed split, $z + hH = T_h$ and $\sigma = \kappa$. The preceding bound and the constant estimator therefore give $R(B) \le C \min\{\rho, T_h + h\sqrt{\rho\Gamma_\kappa}\} = C\mathcal{Q}(B)$. The preceding constants permit $C = 30001$ for this upper bound; no optimality of that constant is claimed. The number of levels and every local cap are finite, while the total Poisson replication count is unbounded but almost surely finite and has the stated expected budget. Since the law is finite, the reference bank is finite throughout. □

O.3 Completion of the main theorem

The same finite known law μ, rarity bound ρ, parameter class $\mathcal{Q}_\rho$, persistent latent-pair experiment, target $\theta(q)$, RMS loss, and uniform expected observation budget are used on both sides. For every fixed (μ, ρ, B), the upper proof gives one observable algorithm whose work and risk bounds hold for every $q \in \mathcal{Q}_\rho$. Its design may depend on (μ, ρ, B) but not on q or a genuine coin's latent state. The lower proof applies to every such algorithm, including nonlinear outputs and arbitrary interleaving. There is no claim of a single procedure simultaneously attaining every budget, or of a deterministic total observation cap.

O.3.1 Upper-bound branches and actual attainment

Put $D = \sqrt{\rho\Gamma_\kappa} + H$ for the split in Appendix O.2. The following exhaustive branches are selected using only known design parameters:

1. If $\mathcal{Q} = \rho$, output $\rho/2$ at zero cost.
2. If $\mathcal{Q} < \rho$ and $z > \rho/128$, the same constant has risk $\rho/2 < 64z \le 64\mathcal{Q}$.

3. If $\mathcal{Q} < \rho$ and $0 < z \le \rho/128$, then $\delta_0 = \rho/16 \ge 8z$, so the largest J with $\delta_J \ge 8z$ exists and is finite. The error is at most $128z + 30000hD \le 30000\mathcal{Q}$.

4. If $\mathcal{Q} < \rho$ and $z = 0$, then $D > 0$ since the core has positive mass. The least J with $\delta_J \le hD$ exists and is finite; the error is less than $30001hD = 30001\mathcal{Q}$.

In the last two branches every local cap exists: the finite resolved law has positive gaps and $\eta_\chi(M) \to 0$. Moreover $\eta_\chi(4) = 1 - z > \delta_\ell/8$, including at the boundary $z = \rho/128$. Poisson means are positive and finite. Independently drawn counts and the uniform per-replication certificates give, for every q,

$$\mathbb{E}_q N \le \sum_{\ell=0}^{J} n_\ell k_\ell = B.$$

The counts are almost surely finite and all local caps are finite. Thus an actual admissible finite-level procedure exists at every $B > 0$; no attainment of a minimax infimum or infinite-level limit is assumed. It follows that $R(B) \le 30001\mathcal{Q}(B)$.

O.3.2 Lower-bound constants and separate hard parameters

Put $M_B = \min\{\rho, h\sqrt{\rho F}\}$ and

$$c_{\mathrm{ann}} = \frac{e^{-28}}{2000000}, \qquad k = \frac{\sqrt{c_{\mathrm{ann}}}}{128 \cdot 1024 \cdot 3584}.$$

In the packing branch, additive localization gives

$$R(B) \ge \frac{1}{128} \sum_i \min\{\theta_i/64, \sqrt{c_{\mathrm{ann}}}\, h\sqrt{\theta_i G_i}\} \ge kM_B.$$

The last step uses $\sqrt{c_{\mathrm{ann}}} < 1/64$, the proportional rarity allocation, and $\sum_i G_i \ge F/3584$. In the high-band branch $F \le 16C_H$ and $C_H = 4096(1 + \log 6) < 16384$. Hence the Bernoulli-label floor yields $R(B) \ge M_B/16384 \ge kM_B$ there as well.

If $h \ge t$, the tail bound gives $R \ge \mathcal{Q}/32$. If $h < t$ and $\lambda \ge t/4$, it gives $R \ge \mathcal{Q}/288$. Otherwise both $R \ge T_h/32$ and $R \ge kM_B$ hold. Since $F \le \Gamma_\kappa \le 4F$,

$$\mathcal{Q} \le T_h + 2M_B \le (32 + 2/k)R, \qquad R \ge \frac{k}{32k + 2}\mathcal{Q} \ge \frac{k}{3}\mathcal{Q}.$$

Here $32k \le 1$. The other branches have larger constants, so this proves one lower constant uniformly over all laws, rarities and budgets. The tail and core obstructions need not be witnessed by the same parameter: a worst-case risk exceeds each obstruction, hence their maximum. No exchange of a supremum and a sum is being used.

The explicit bound in Theorem 14.1 is therefore

$$10^{-20}\mathcal{Q}(B) \le R(B) \le 30001\mathcal{Q}(B).$$

Indeed $k/3 = e^{-14}/[\sqrt{2000000}\,(3 \cdot 128 \cdot 1024 \cdot 3584)]$, and $e < 3$, $\sqrt{2000000} < 1500$ and

$$3^{14}\, 1500\, (3 \cdot 128 \cdot 1024 \cdot 3584) < 10^{20}$$

certify this deliberately loose lower constant.

O.3.3 Lower-bound quantifiers and budget transport

The half-prior components are subsets of the full rarity class after the fractional mixture is integrated. Each induced endpoint and its population midpoint is admissible, so the original uniform budget applies to all three. Auxiliary oracle types strengthen only the lower experiment; they are not observations available to the upper estimator. The original algorithm is simulated using persistent private labels, and genuine oracle observations never exceed its original work.

At each fixed budget, the common-pair result selects a pair simultaneously for all local policies and deterministic caps. That pair is held fixed while the cap tends to infinity. Population midpoint work is bounded by B, and the overflow probabilities at the two endpoints sum to at most $2B/L$ at cap L. Neither a null-only nor an endpoint-only budget is substituted for the uniform assumption. The population midpoint is the experiment with an independent midpoint latent pair for every fresh coin, not the mixture of the two full population endpoint transcripts.

This completes the proof of Theorem 14.1.

References

[1] J. C. H. Lee and P. Valiant. Uncertainty about uncertainty: Optimal adaptive algorithms for estimating mixtures of unknown coins. *Proceedings of SODA*, 2021. Extended version: arXiv:1904.09228, v3.

[2] M. B. Giles and A.-L. Haji-Ali. Multilevel nested simulation for efficient risk estimation. *SIAM/ASA Journal on Uncertainty Quantification*, 7(2):497–525, 2019. doi:10.1137/18M1173186.

[3] C.-H. Rhee and P. W. Glynn. A new approach to unbiased estimation for SDE's. arXiv:1207.2452, 2012.

[4] M. Vihola. Unbiased estimators and multilevel Monte Carlo. *Operations Research*, 66(2):448–462, 2018. doi:10.1287/opre.2017.1670. Preprint: arXiv:1512.01022, v4.

[5] Y. Syed and G. Wang. Optimal randomized multilevel Monte Carlo for repeatedly nested expectations. *Proceedings of ICML*, PMLR 202:33343–33364, 2023. PMLR article.

[6] J. Brennan, R. K. Vinayak, and K. Jamieson. Estimating the number and effect sizes of non-null hypotheses. *Proceedings of ICML*, PMLR 119, 2020. PMLR article.

[7] Z. Chen, M. Naslidnyk, and F.-X. Briol. Nested expectations with kernel quadrature. *Proceedings of ICML*, PMLR 267:8760–8793, 2025. PMLR article.

[8] J. B. S. Haldane. On a method of estimating frequencies. *Biometrika*, 33(3):222–225, 1945. doi:10.1093/biomet/33.3.222.

[9] M. Broadie, Y. Du, and C. C. Moallemi. Efficient risk estimation via nested sequential simulation. *Management Science*, 57(6):1172–1194, 2011. doi:10.1287/mnsc.1110.1330.

[10] S.-H. Lee and P. W. Glynn. Computing the distribution function of a conditional expectation via Monte Carlo: Discrete conditioning spaces. *ACM Transactions on Modeling and Computer Simulation*, 13(3):238–258, 2003. doi:10.1145/937332.937334.

[11] M. B. Gordy and S. Juneja. Nested simulation in portfolio risk measurement. *Management Science*, 56(10):1833–1848, 2010. doi:10.1287/mnsc.1100.1213.

[12] A.-L. Haji-Ali, J. Spence, and A. L. Teckentrup. Adaptive multilevel Monte Carlo for probabilities. *SIAM Journal on Numerical Analysis*, 60(4):2125–2149, 2022. doi:10.1137/21M1447064.

[13] M. B. Giles. MLMC techniques for discontinuous functions. arXiv:2301.02882, v2, 2023.

[14] E. Kaufmann, O. Cappé, and A. Garivier. On the complexity of best-arm identification in multi-armed bandit models. *Journal of Machine Learning Research*, 17(1):1–42, 2016. JMLR article.

[15] J. H. Blanchet and P. W. Glynn. Unbiased Monte Carlo for optimization and functions of expectations via multi-level randomization. *Proceedings of the Winter Simulation Conference*, pp. 3656–3667, 2015. Author's publication page.

[16] L. Mendo. Asymptotically Optimum Estimation of a Probability in Inverse Binomial Sampling under General Loss Functions. Author preprint, 25 April 2012. `https://oa.upm.es/10749/1/asympt-opt16completo.pdf`. Introduction and equation (1), explicitly attributing the inverse-count estimator to J. B. S. Haldane (1945).

[17] A. Juditsky and A. Nemirovski. *Nonparametric estimation by convex programming*. Annals of Statistics 37 (2009), 2278–2300. https://www2.isye.gatech.edu/~nemirovs/AnnStat_Saddle_2009.pdf.

[18] Y. Polyanskiy and Y. Wu. *Dualizing Le Cam's method for functional estimation, with applications to estimating the unseens*. arXiv:1902.05616v2 (2021), Theorem 1, Proposition 4 and Section 4.1. https://arxiv.org/abs/1902.05616.

[19] L. Chen, J. Li and M. Qiao. *Towards Instance Optimal Bounds for Best Arm Identification*. COLT 2017; arXiv:1608.06031v2. https://arxiv.org/html/1608.06031v2.

[20] M. Sion. *On general minimax theorems*. Pacific Journal of Mathematics 8(1):171–176, 1958. https://msp.org/pjm/1958/8-1/pjm-v8-n1-p14-s.pdf.

[21] R. H. Farrell. *Asymptotic Behavior of Expected Sample Size in Certain One Sided Tests*. Annals of Mathematical Statistics 35(1):36–72, 1964. https://doi.org/10.1214/aoms/1177703731.

[22] L. Chen and J. Li. *On the Optimal Sample Complexity for Best Arm Identification*. arXiv:1511.03774v3, 2016. https://arxiv.org/abs/1511.03774v3.

[23] S. Agrawal, A. Ram and A. Ramdas. *On Stopping Times of Power-one Sequential Tests: Tight Lower and Upper Bounds*. arXiv:2504.19952v2, 2026. https://arxiv.org/abs/2504.19952v2.

[24] J. Yao, J. Zhao and X. Zhou. *Gap Entropy and Almost Instance-Wise Optimal Best-Arm Identification*. arXiv:2609.13703v1, 2026. https://arxiv.org/abs/2609.13703v1.

[25] L. Chen and J. Li. *Open Problem: Best Arm Identification: Almost Instance-Wise Optimality and the Gap Entropy Conjecture*. Proceedings of COLT, PMLR 49:1643–1646, 2016. https://proceedings.mlr.press/v49/chen16b.html.

[26] W. Hoeffding. *Probability Inequalities for Sums of Bounded Random Variables*. Journal of the American Statistical Association 58(301):13–30, 1963. https://doi.org/10.1080/01621459.1963.10500830.

[27] S. Boyd and L. Vandenberghe. *Convex Optimization*. Cambridge University Press, 2004. Chapter 5. https://web.stanford.edu/~boyd/cvxbook/.

[28] M. J. Cho and R. H. Stockbridge. *Linear Programming Formulation for Optimal Stopping Problems*. SIAM Journal on Control and Optimization 40(6):1965–1982, 2002. https://doi.org/10.1137/S0363012900377663.

[29] P. M. Aronow, N. Kallus, and P. Lopatto. *A positive resolution of the gap-entropy conjecture*. arXiv:2609.10529v1, 2026. https://arxiv.org/abs/2609.10529v1.